

Instrumental Analysis

Bachelor Chemistry and Chemical Technology
and
Bachelor Biological Chemistry

(Winter Semester 2026/27)

- The lecture will **primarily be held with physical attendance, so it is strongly recommended to attend this lecture in person!**
- In addition the sound + the slides **(but no explanatory drawings or sketches from the blackboard)** will be available via a ZOOM link:

<https://jku.zoom.us/j/99338890405?pwd=ZnBtOHhN6QUw1eHZKQmZBWkxVaHFuUT09>

Meeting-ID: 993 3889 0405
Password: 367402

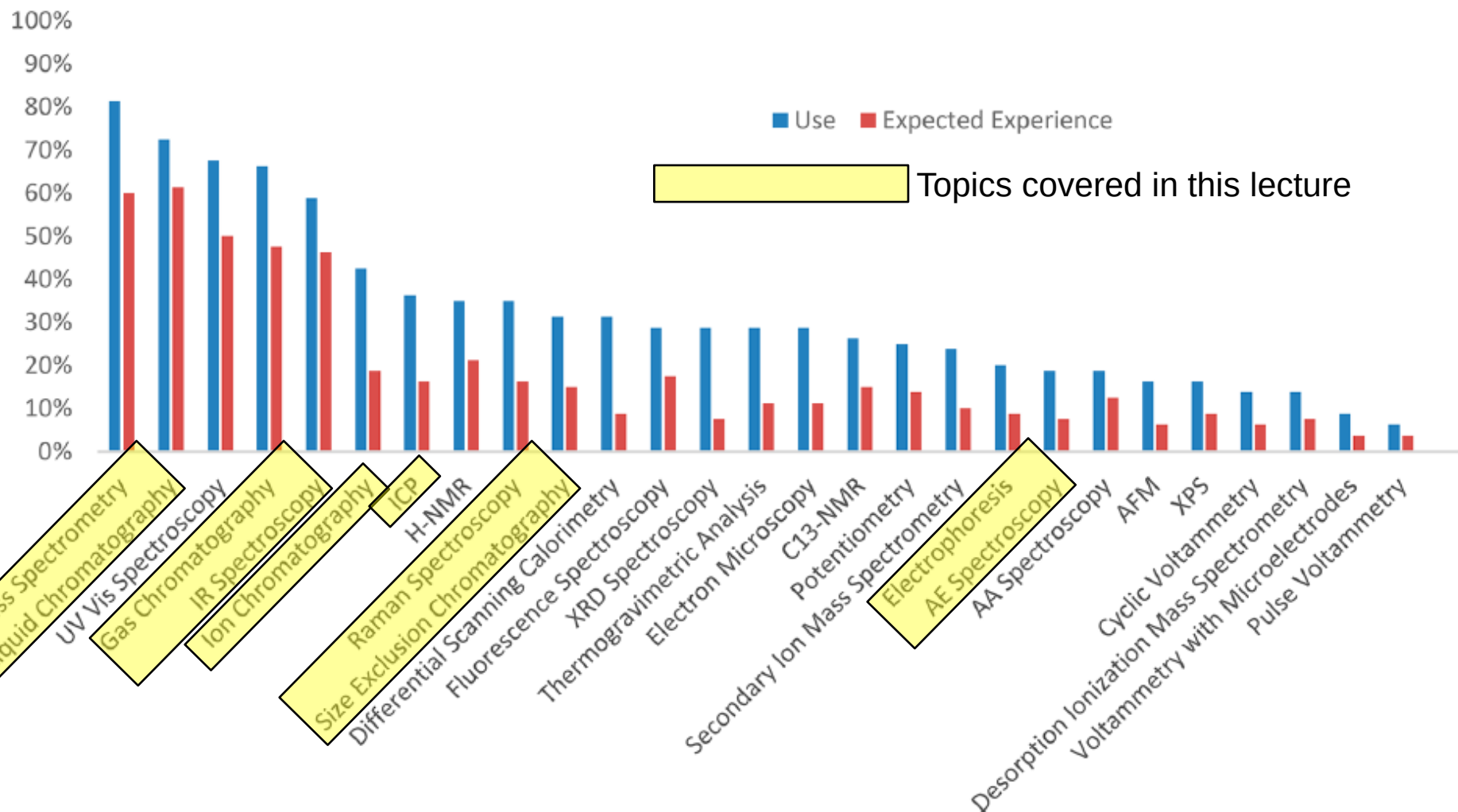
Exam:

- **Oral exam** – for an appointment (date/time-slot) for your exam please contact Birgit Sammer (birgit.sammer@jku.at)
 - For BC students (lecture = 5th semester) the connected lab course (345.006) is scheduled for the 6th semester (actually March) – so the exam has to be passed before end of February
 - For CCT students (lecture = 3rd semester) the connected lab course (345.119) is scheduled for the 5th semester – so there is plenty of time to do the exam

Survey of Instrumentation Use in Industry: What Does Industry Want New Chemists to Know?

David Hamilton, Alyssa Castillo, and Molly B. Atkinson*

JOURNAL OF
CHEMICALEDUCATION



Content:

1.Theory of Chromatography

- 1.1. Basics of chromatography
- 1.2. Peak broadening
- 1.3. Van Deemter equation
- 1.4. Chromatographic resolution
- 1.5. Peak capacity
- 1.6. Parameters for Column Performance

2. High-performance liquid chromatography

- 2.1. HPLC instrumentation
 - 2.1.1. On-line degasser
 - 2.1.2. HPLC pumps
 - 2.1.3. HPLC injector
 - 2.1.4. HPLC columns
 - 2.1.5. HPLC detectors
- 2.2. HPLC stationary phases
- 2.3. Modes of HPLC
 - 2.3.1. Normal phase HPLC
 - 2.3.2. Reversed phase HPLC
 - 2.3.3. Size exclusion chromatography
 - 2.3.4. Ion-exchange chromatography
 - 2.3.5. Ion-pair chromatography
 - 2.3.6. Affinity chromatography
 - 2.3.7. Chiral separations
 - 2.3.8. 2D Separation in liquid phase
- 2.4 Supercritical fluid chromatography

Content (continued)

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- 3.2. Mobile phases in GC
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- 3.4. Stationary phases in GC
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- 3.7. Two dimensional GC (GCxGC)

4. Electrophoresis

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- 4.4. Electrophoresis with (anticonvective) support

Content (continued)

5. Infrared Spectroscopy

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- 5.1.1. Radiation Source for IR
- 5.1.2. Monochromators for IR
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5.2. Principles of Vibrations in IR

- 5.2.1. The Harmonic Oscillator
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- 7.1.1. Electron (Impact) Ionization
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Content (continued)

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- 7.4.1. GC-MS
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- 7.5.3. Multiple Charging-Deconvolution
- 7.5.4. Protein Identification

8. Inductively Coupled Plasma Spectroscopy

8.1 ICP-AES

8.2 ICP-MS

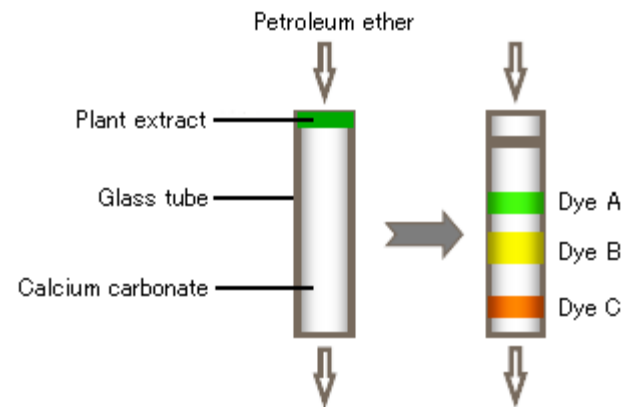
8.3 Comparison of Techniques

8.4 Laser Ablation ICP-MS

1 Theory of Chromatography

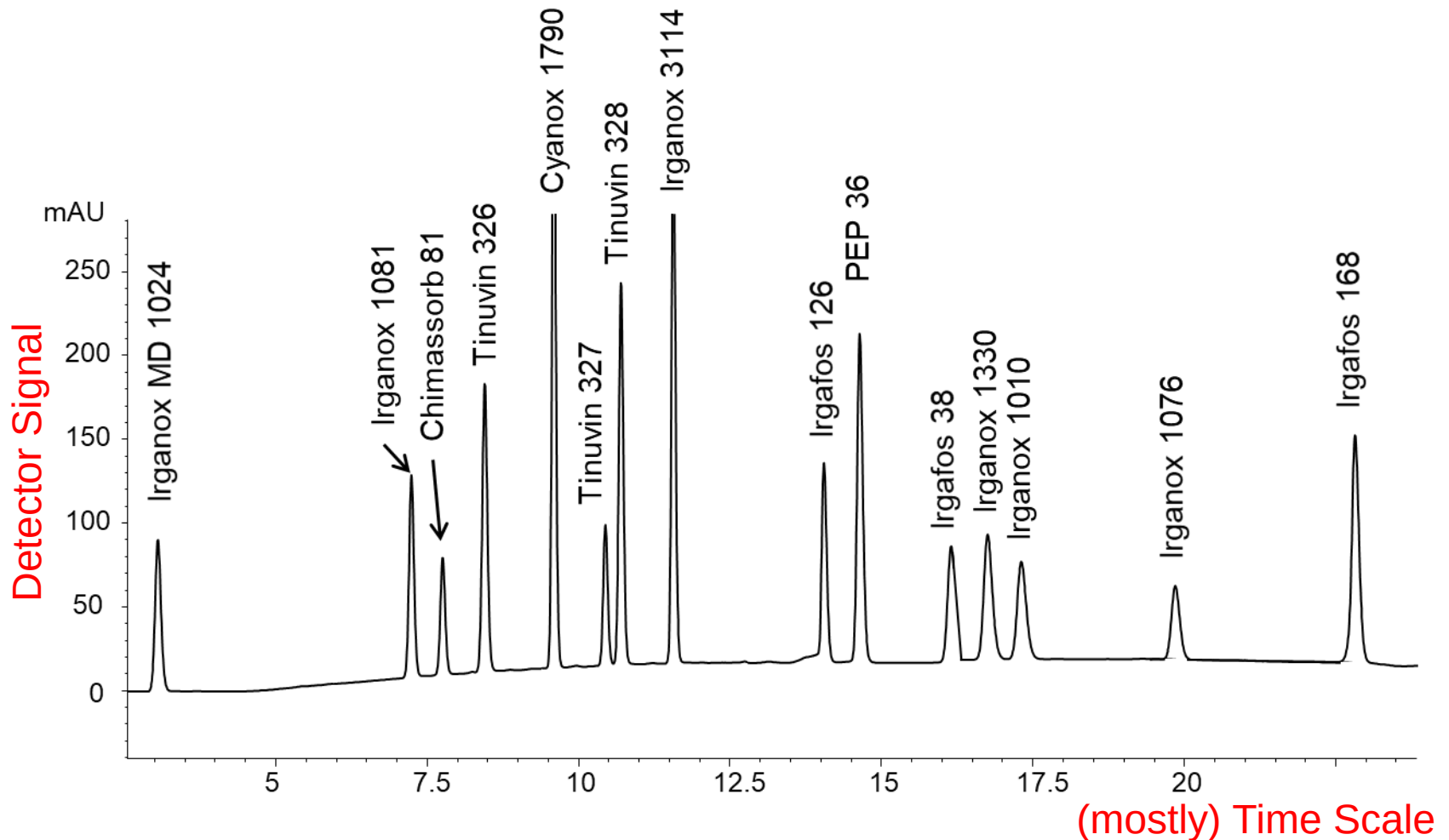


Mikhail Tswett (1872 – 1919)



1 Theory of Chromatography

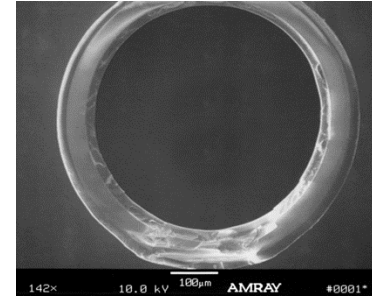
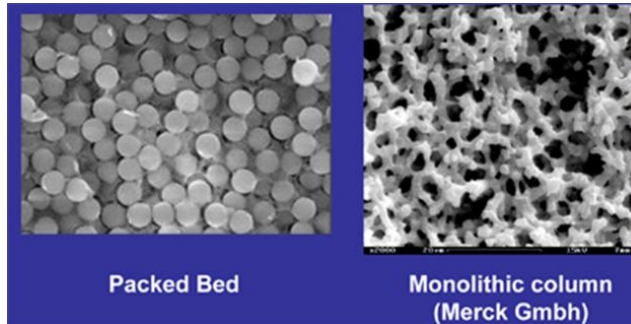
The Chromatogramm:



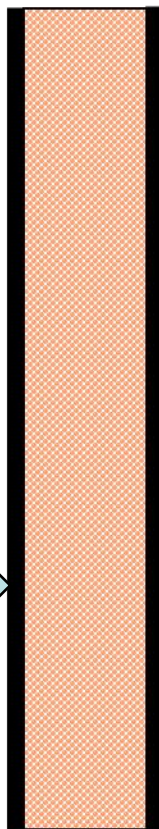
1 Theory of Chromatography

- Chromatography is a method to separate components in a mixture based on different distribution coefficients between a mobile phase and a stationary phases
- Mobile phases may be gaseous, supercritical fluids or liquids
- Stationary phases are most commonly coated or packed in a column (except for TLC – see next slides and Introductory Lecture)
- Stationary phases may be either liquid (polymer) or solid
- Interaction between solute and stationary phase:
 - * partition
 - * adsorption
 - * ion exchange
 - *

1 Theory of Chromatography



Column packed with particles as stationary phase



Packed columns
(typically used in liquid chromatography)

Column packed with monolith as stationary phase



Open tubular columns
(typically used in gas chromatography)



Capillary column coated on the inner surface with film of stationary phase

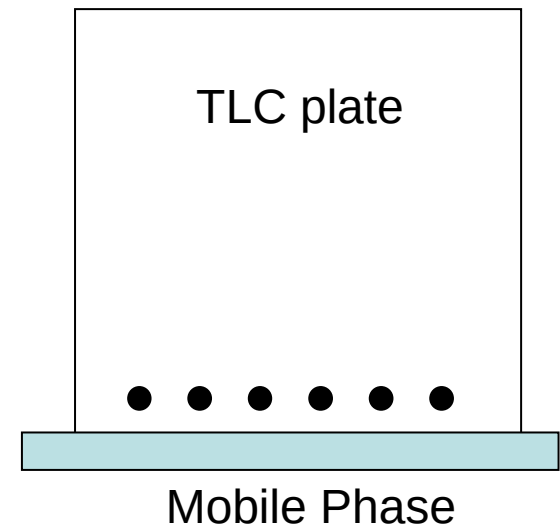
1.1 Basics of Chromatography

(already discussed in „Introduction to Analytical Chemistry“)

Thin layer chromatography (TLC)

The sample solution (2-10 μL) is applied to a thin layer plate (20x20cm or similar, with a thickness of stationary phase layer of 0.25 mm). In the same way various calibration solutions are applied.

Subsequently, the plate is put into the development chamber containing the mobile phase. Due to capillary force, the mobile phase flows upwards and transports the different components of the sample to different positions according to the partitioning of the components between stationary and mobile phase.



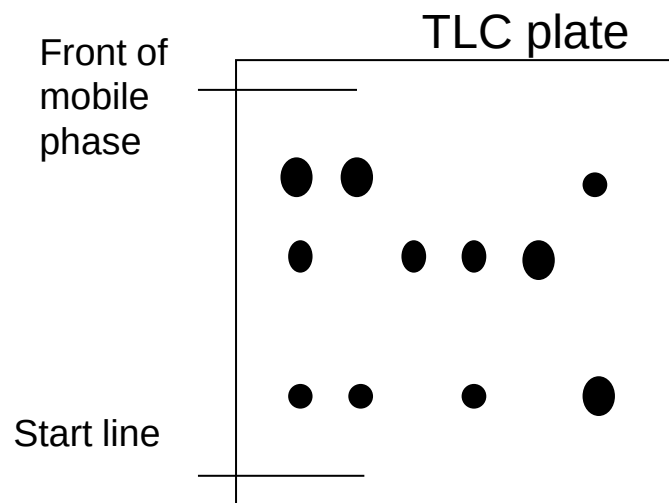
1.1 Basics of Chromatography

(already discussed in „Introduction to Analytical Chemistry“)

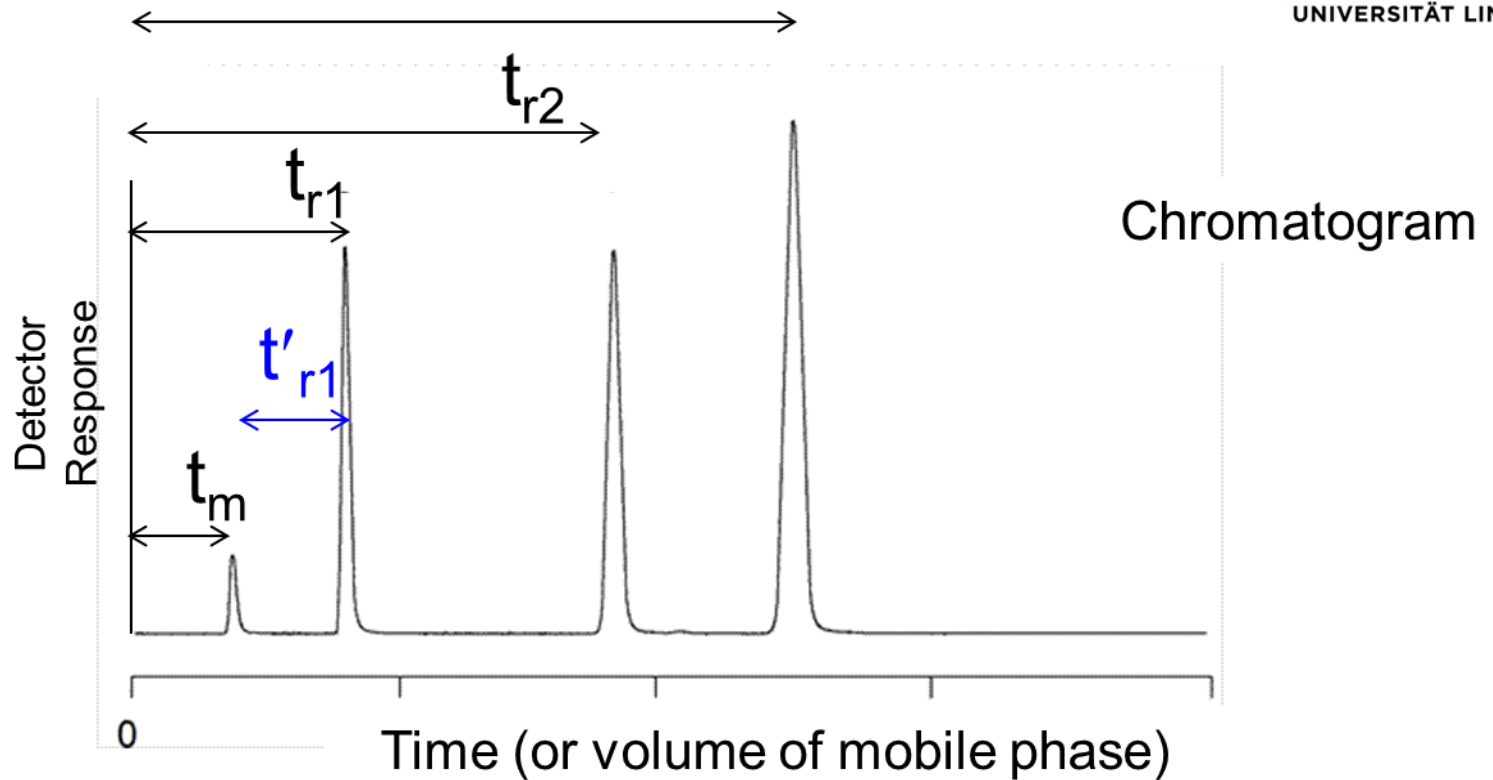
The **qualitative information** is given by the R_f -value of a separated compound:

$$R_f = \frac{\text{Distance } \textit{Start} - \textit{Spot}}{\text{Distance } \textit{Start} - \textit{Front}}$$

The R_f value can be seen as a parameter providing some **qualitative information**. It allows (to a limited degree) to identify a substance by comparing R_f values with those from standards of known identity. On the other hand it allows a judgement on the polarity of the substance.



1.1 Basics of Chromatography



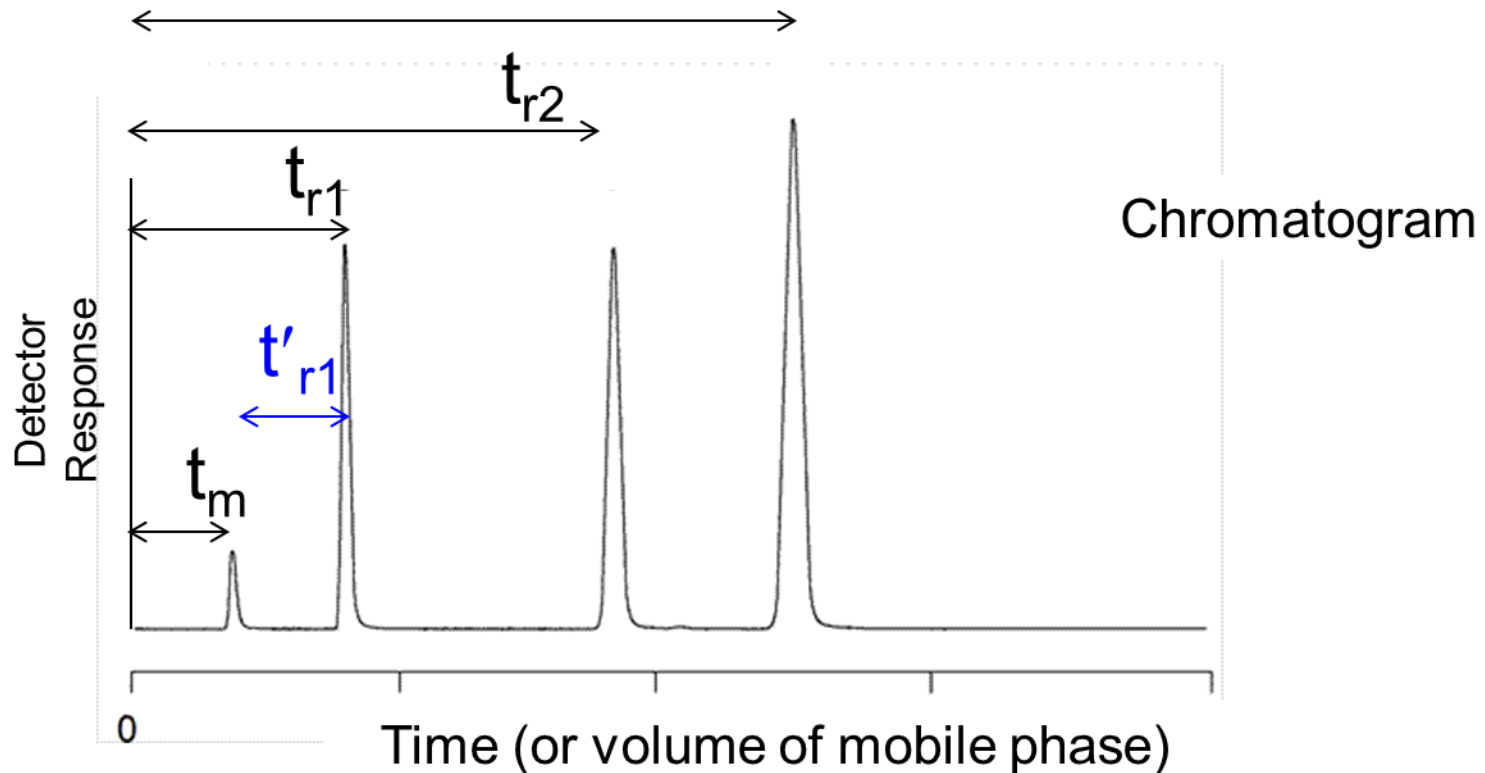
t_m = time for mobile phase to flow from top of column to end of column (dead time)

t_r = retention time

t'_r = adjusted retention time = $t_r - t_m$

$\alpha = t'_{r2}/t'_{r1}$ = relative retention separation factor

1.1 Basics of Chromatography

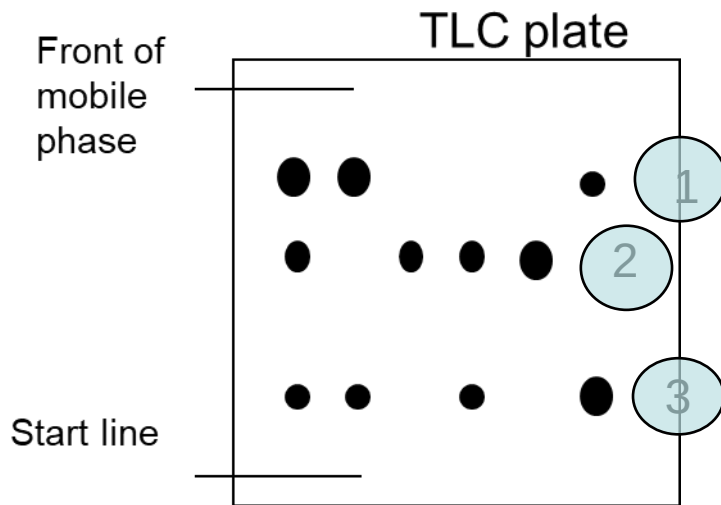


t_m = time period which the analyte spends in the mobile phase

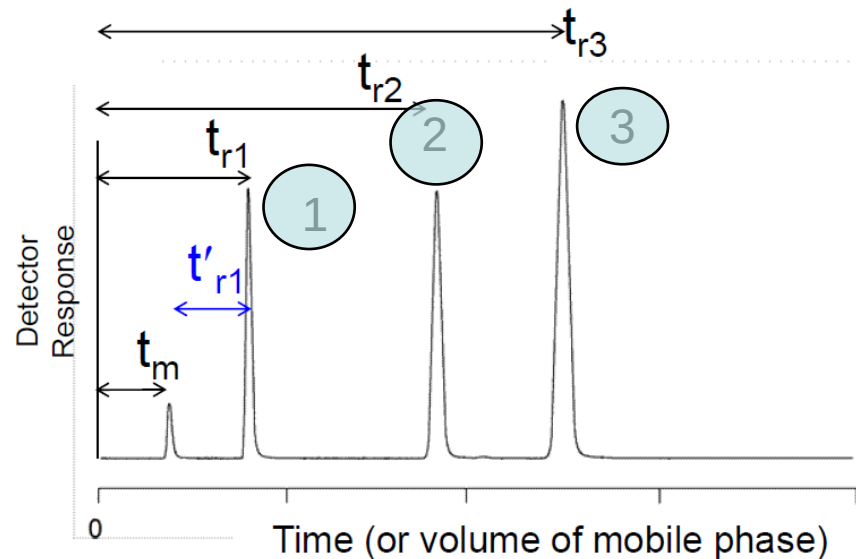
t'_r = time period which the analyte spends in the stationary phase

1.1 Basics of Chromatography

Comparison column-LC vs TLC:



$$R_f = \frac{\text{Distance Start - Spot}}{\text{Distance Start - Front}}$$



- High R_f values in TLC correspond to short retention times in column-LC
- The further up an analyte runs in TLC the earlier it elutes in column-LC
- t_m in column-LC corresponds to the mobile phase front in TLC

1.1 Basics of Chromatography

- Mobile phase flow rate:
 - volumetric flow rate (F): mL(or μ L)/min
 - linear flow rate (u): cm/min (mm/min)
- Ways to describe analyte “retention”
 - retention time, t_r
 - retention volume, V_r $V_r = Ft_r$
 - retention factor, k

1.1 Basics of Chromatography

Partition coefficient K of an analyte between stationary phase and mobile phase:

$$K = C_s / C_m$$

C_s = concentration of analyte in stationary phase

C_m = concentration of analyte in mobile phase

Retention factor

$$k = K \frac{V_s}{V_m} = \frac{\text{moles}_{\text{analyte in stationary phase}}}{\text{moles}_{\text{analyte in mobile phase}}}$$

V_s = volume of stationary phase

V_m = volume of mobile phase

1.1 Basics of Chromatography

If t'_r (time period which the analyte spends in the stationary phase) is for example three times t_m (time period which the analyte spends in the mobile phase) – meaning that the value for k' would equal three - then at any time the number of analyte molecules present in the stationary phase will be three times the number of analyte molecules in the mobile phases.

$$k = \frac{t'_r}{t_m} = \frac{t_r - t_m}{t_m}$$

$$t_r = t_m (1 + k)$$

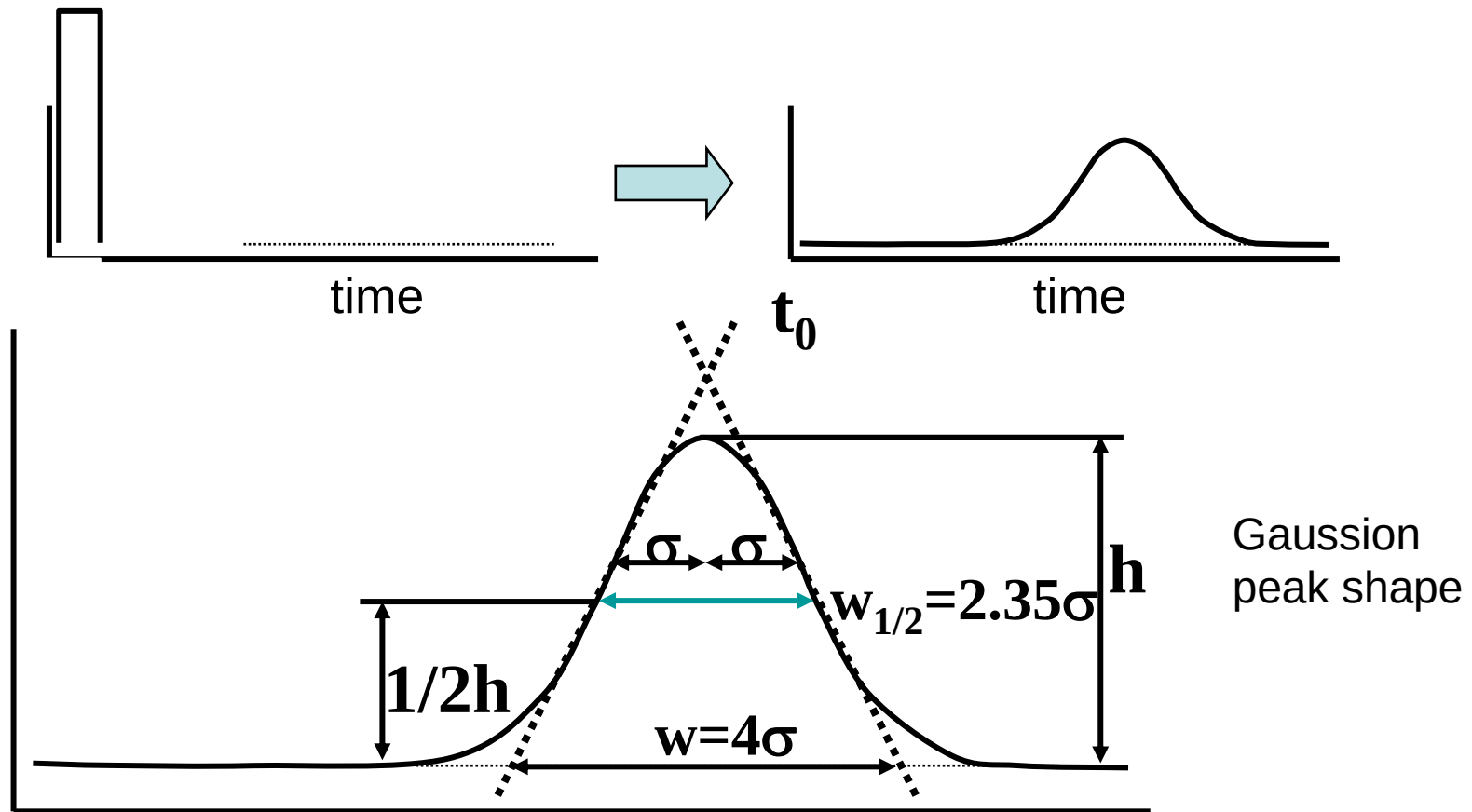
$$\alpha = t'_{r2}/t'_{r1} = \text{relative retention separation factor} \\ = k_2/k_1$$

1.2 Peak Broadening

Two factors affect how well the analytes are separated:

- * difference in retention time
- * peak widths

Injected analytes in a column spread into a Gaussian profile (if we assume that diffusion processes cause band broadening):



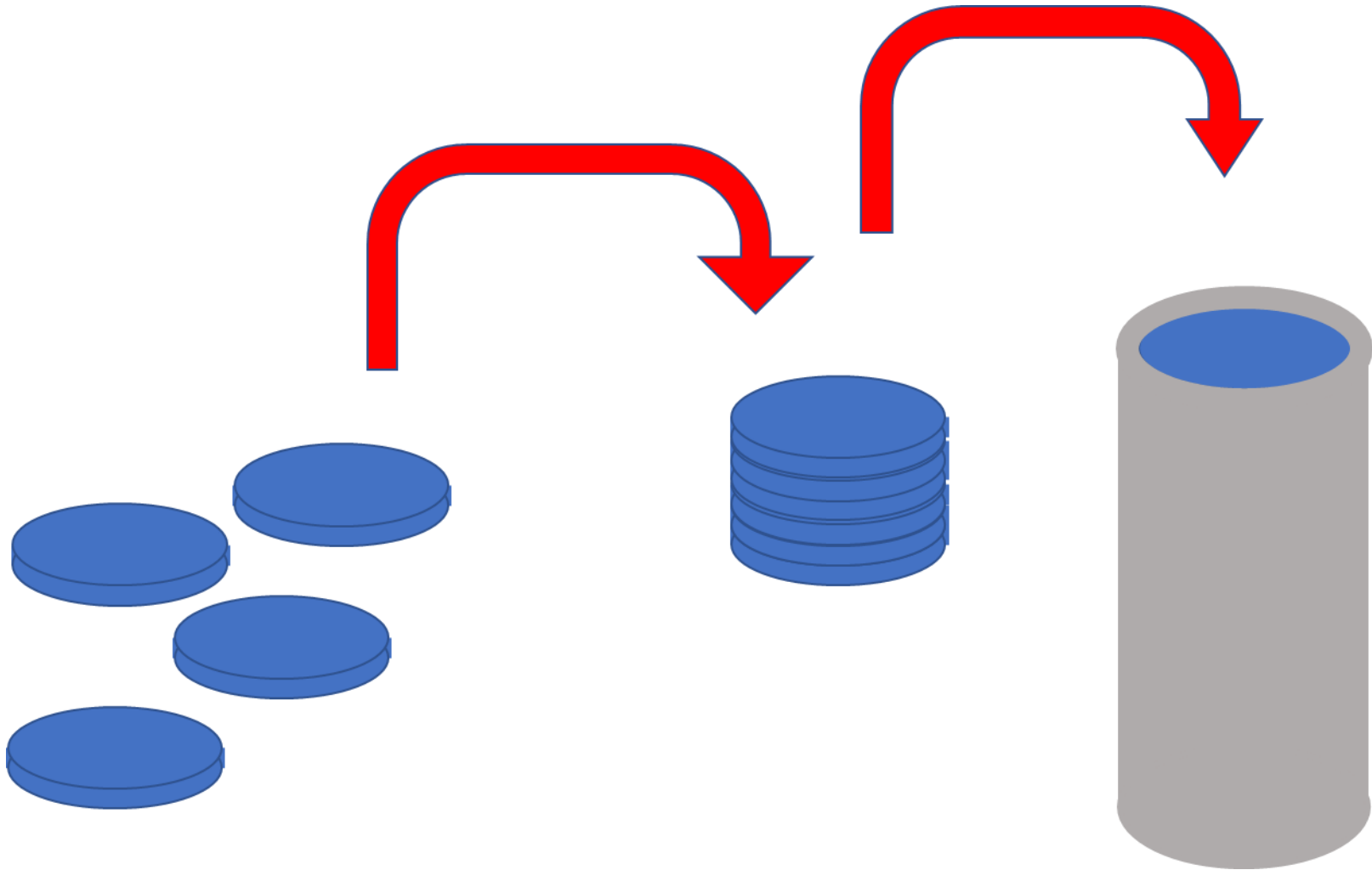
1.2 Peak Broadening

The concept of theoretical plates

Let us approximate the continuous chromatographic process by a stepwise process. For this purpose we divide the packing of the column into segments (that is, the packing can be seen as a stack of disks of stationary phase – see next slide)

A certain volume of mobile phase containing the analytes flows into the first segment (disk) of the column and stops. Within this segment the equilibrium of an analyte between mobile phase and stationary phase is established. Then this volume of mobile phase of this segment flows into the second segment of stationary phase, and fresh mobile phase flows into the first segment. At this point the flow again stops, and the equilibria in the first segment and in the second segment are established.... and so on.

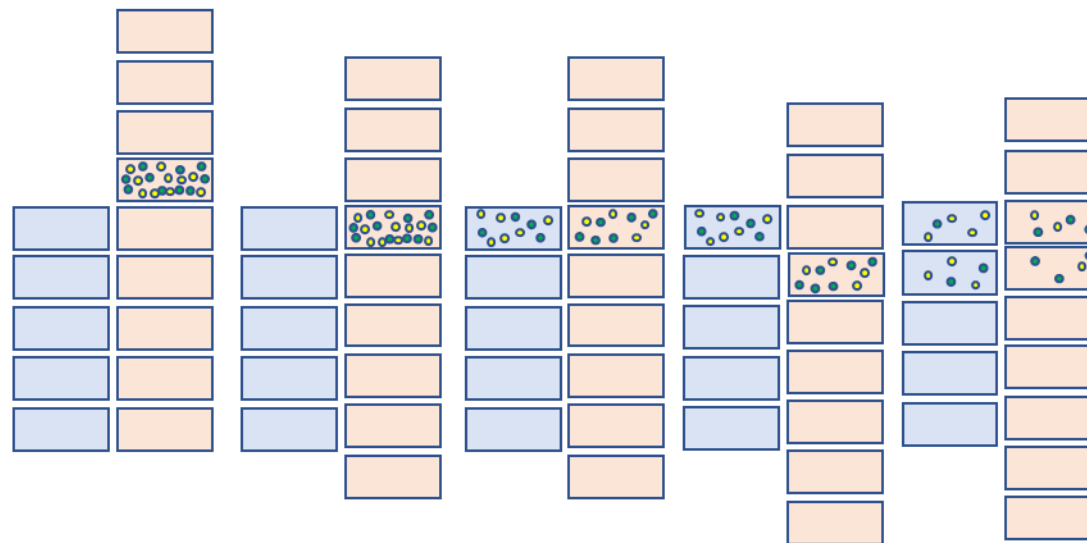
1.2 Peak Broadening



1.2 Peak Broadening

The concept of theoretical plates

stat. phase mob. phase



...and so on

Mobile phase with
analytes flows into
the 1. segment of
column

Equilibrium
is
established

Further flow of
mobile phase

Equilibrium is
established in 1. and 2.
segment

1.2 Peak Broadening

The concept of theoretical plates

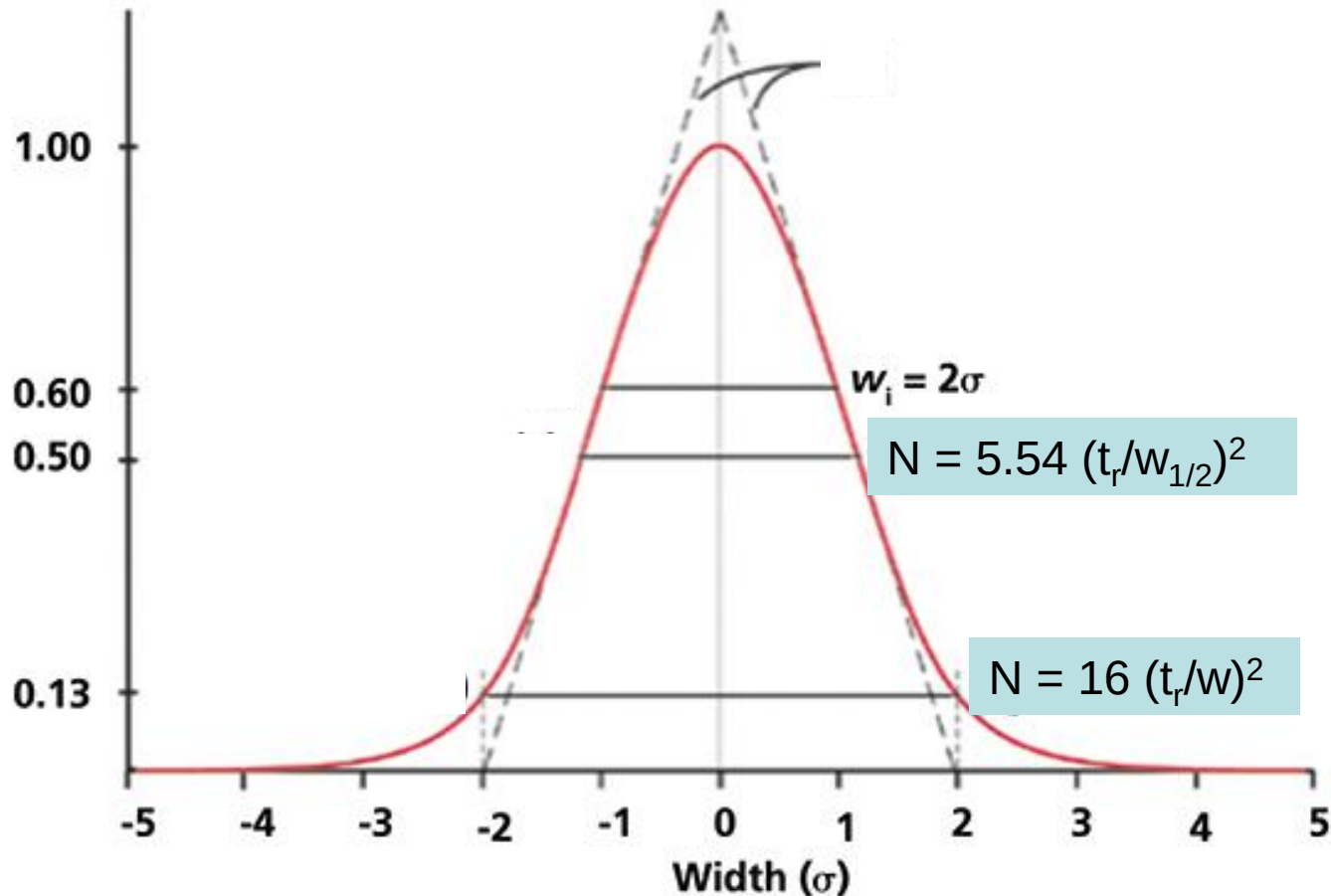
What can we learn?

1. If we increase the number of disks (segments) of the stationary phase in the column), the equilibrium of analytes between mobile phase and stationary phase is more often established. Thereby, the separation is improved.
2. The number of disks (segments) can be increased if we increase the length of the column (quite obvious!)
3. For a given length of column, the number of disks can be increased if the thickness of the disk required for establishment of an equilibrium can be decreased. This required thickness depends on the physical and chemical properties of the stationary phase, but also on the velocity of the mobile phase.

Segment (disk)	theoretical plate
Number of segments (disks) in a column.....	number of theoretical plates (N)
Thickness of segment (disk)	height equivalent of a theoretical plate (HETP)

1.2 Peak Broadening

N is a measure for “efficiency” of the separation
The higher N, the narrower the peaks

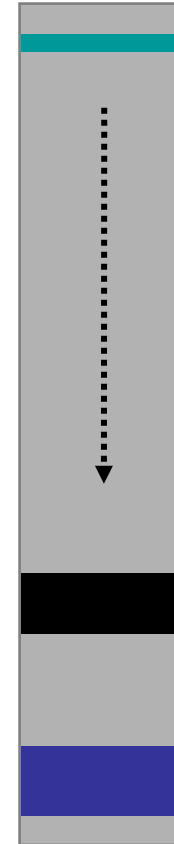


For chromatographic peaks use width in time scale (s or min)!

1.2 Peak Broadening

- Causes of peak broadening:
 - Eddy dispersion
 - Longitudinal diffusion
 - Resistance to mass transfer

column



1.3 The van Deemter Equation

$$H_{\text{Equivalent for a Theoretical Plate}} = H_A + H_B + H_C$$

Eddy dispersion

Longitudinal
diffusion

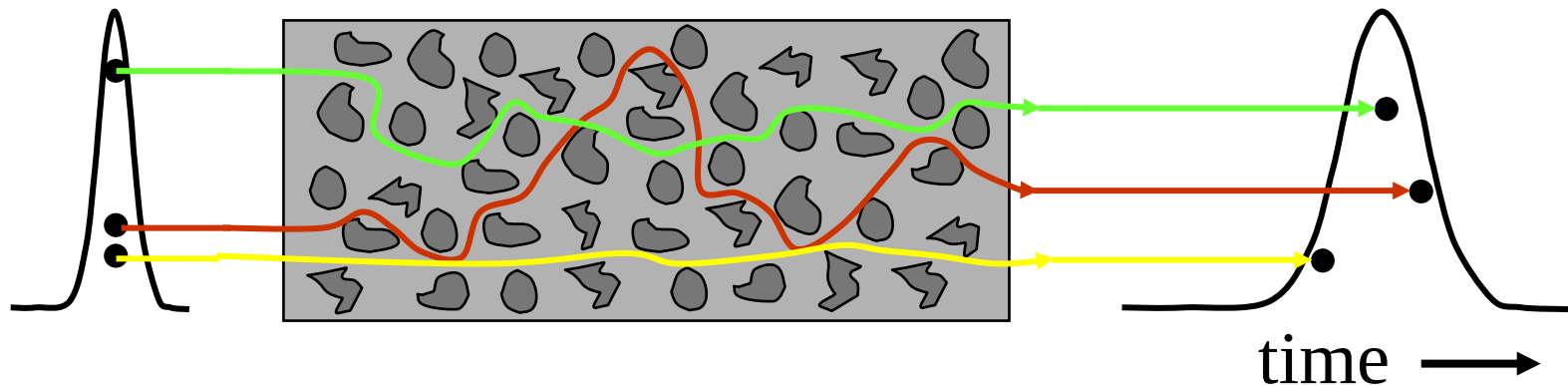
Resistance to
mass transfer



Jan van Deemter

1.3 The van Deemter Equation

- Eddy dispersion (not simple diffusion):

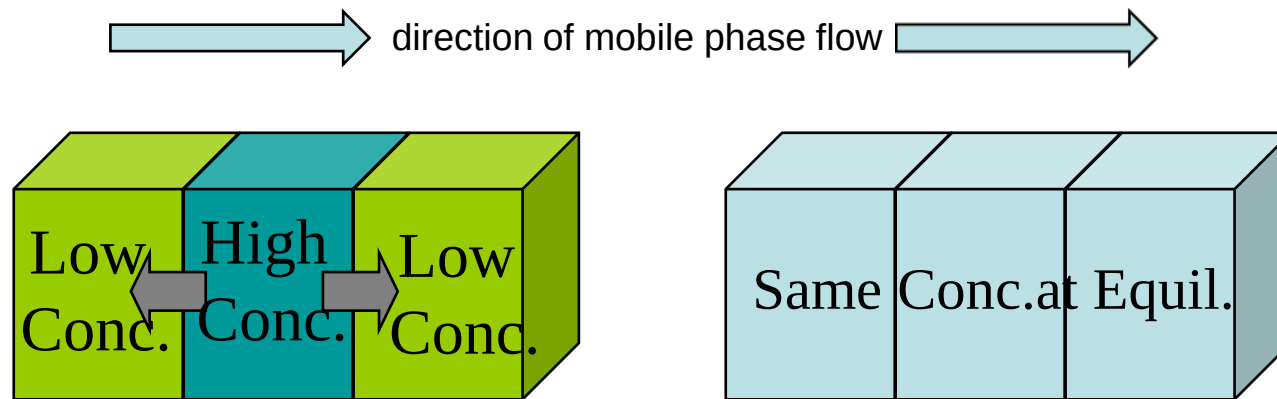


$$H_A = A$$

$A = \text{constant, depends on size of particles } d_p, \quad A = \text{prop } d_p$

1.3 The van Deemter Equation

- Longitudinal diffusion: solute concentration is lower at the edges of a band; solute diffuses to the edges



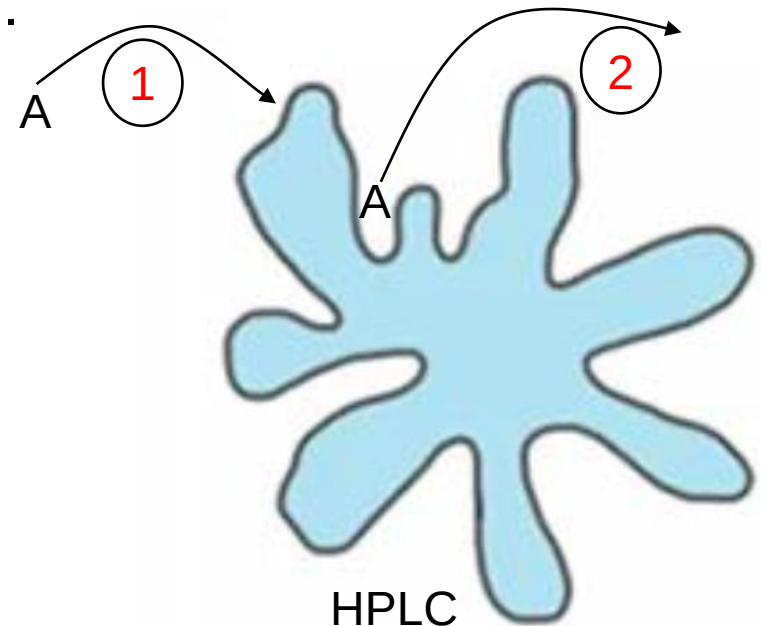
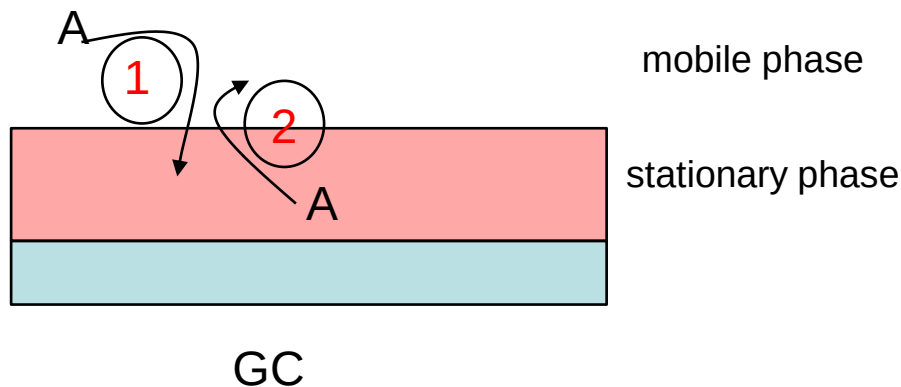
$H_B = B/u$; $B = \text{constant}$, $u = \text{linear velocity of mobile phase}$

$B = 2 D$ (D diffusion coefficient)

More exactly: $B = 2 \gamma D$ (γ obstruction factor)

1.3 The van Deemter Equation

- Resistance to mass transfer between mobile and stationary phase
- slow transport from the mobile phase to the stationary phase (1),
- slow diffusion in the layer of stationary phase (GC columns),
- slow diffusion in the stagnant liquid mobile phase in the pores of the stationary phase (2) (HPLC columns),....



1.3 The van Deemter Equation

- Resistance to mass transfer between mobile and stationary phase

slow transport from the mobile phase to the stationary phase,
slow diffusion in the layer of stationary phase (GC columns),
slow diffusion in the stagnant liquid mobile phase in the pores of the stationary phase (HPLC columns),....

$H_C = C u$; $C = \text{constant}$, $u = \text{linear velocity of mobile phase}$

H_C can be written as sum of two terms $H_{C(s)}$ and $H_{C(m)}$
referring to the stationary phase and to the mobile phase

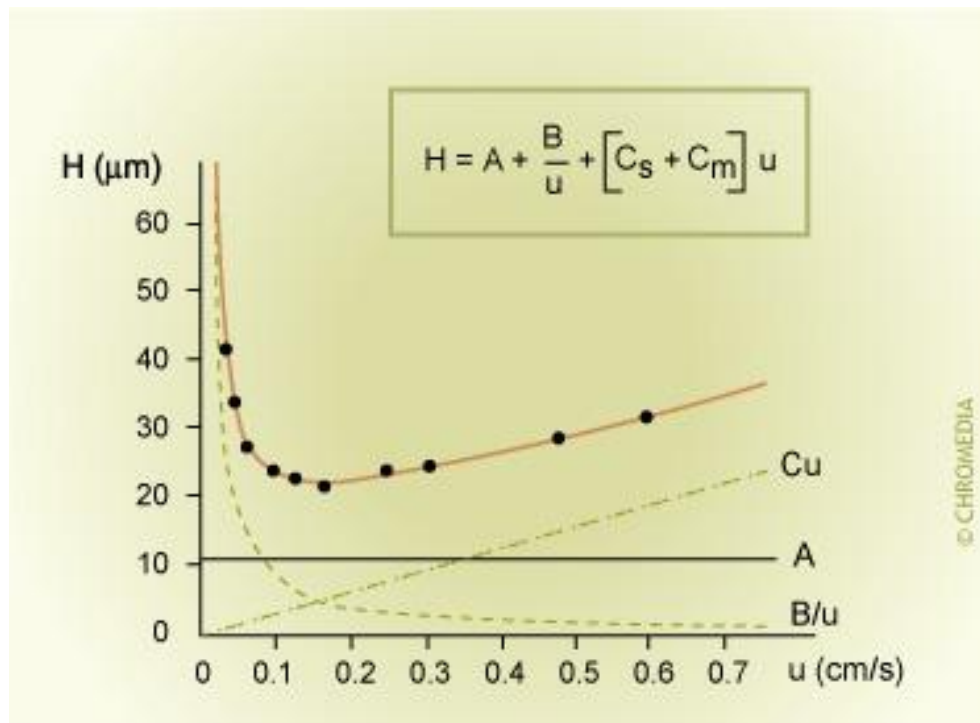
H_C also depends on particle size (or thickness) of stationary phase

1.3 The van Deemter Equation

- van Deemter Equation:

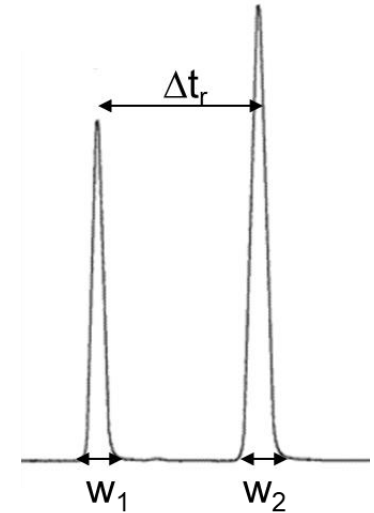
$$\text{HETP} = H_A + H_B + H_C$$

$$\text{HETP} = A + (B/u) + Cu$$



1.4 Chromatographic Resolution

- Resolution (R) of two peaks = $\frac{\Delta t_r}{w_{avg}}$

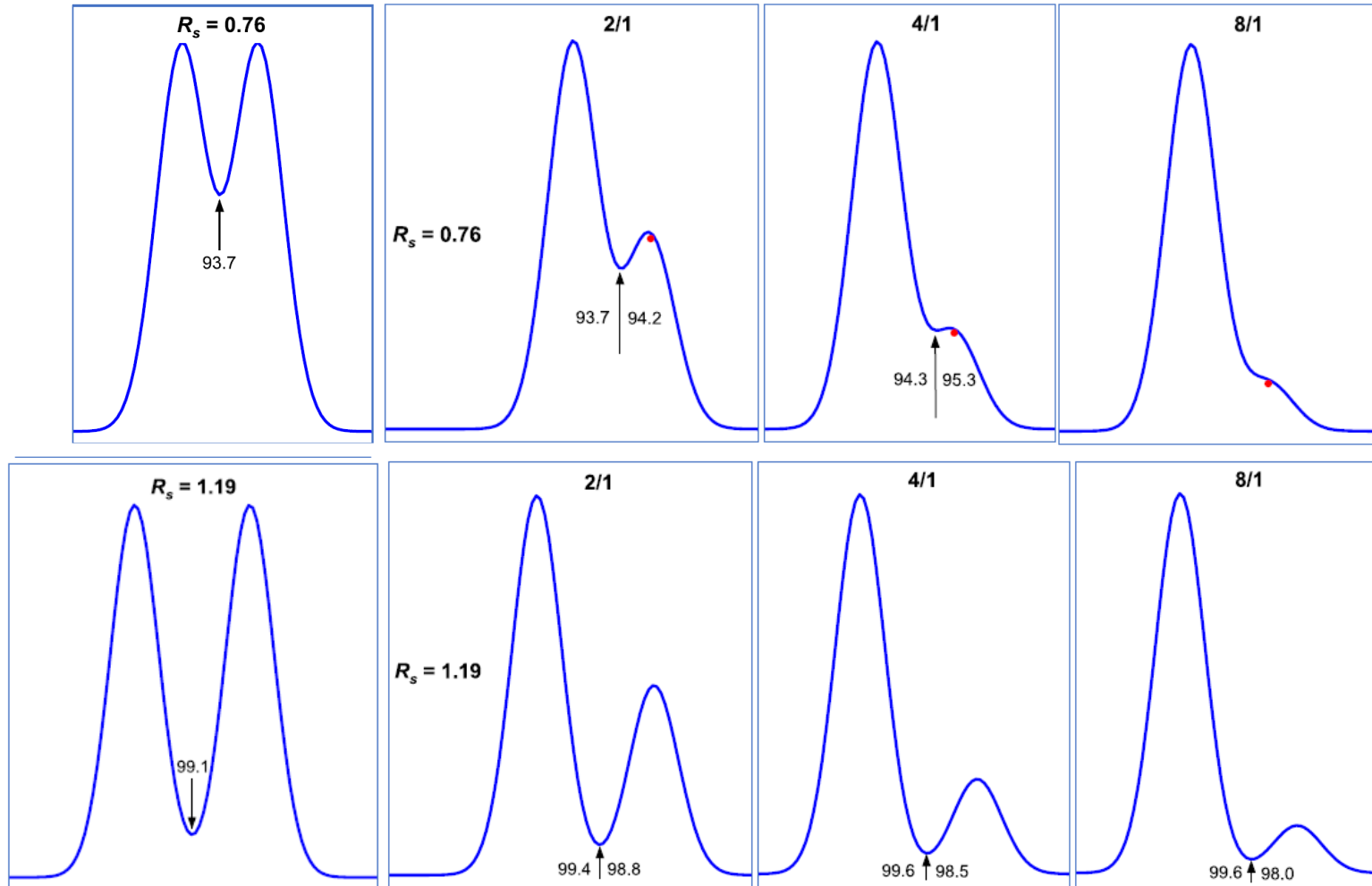


Δt_r = difference between retention times of two peaks = $(t_{r2} - t_{r1})$

w_{avg} = average of the peak widths at baseline ($\approx 4\sigma$)

$$= (w_1 + w_2) / 2$$

1.4 Chromatographic Resolution



Peak purity as function of resolution and peak ratio

1.4 Chromatographic Resolution

- Resolution depends on

N (efficiency)

α (selectivity)

k (retention)

$$R = \frac{2(t_{r(2)} - t_{r(1)})}{w_1 + w_2} = \frac{2(t_{r(2)} - t_{r(1)})}{4\left(\frac{t_{r(1)}}{\sqrt{N}} + \frac{t_{r(2)}}{\sqrt{N}}\right)} =$$

$$\frac{(t_{r(2)} - t_{r(1)})}{2\left(\frac{t_{r(1)}}{\sqrt{N}} + \frac{t_{r(2)}}{\sqrt{N}}\right)} = \frac{(t_{r(2)} - t_{r(1)})}{t_{r(1)} + t_{r(2)}} \frac{\sqrt{N}}{2} = \frac{(t'_{r(2)} - t'_{r(1)})}{t'_{r(1)} + t'_{r(2)} + 2t_m} \frac{\sqrt{N}}{2} =$$

$$\frac{(k_{(2)} - k_{(1)})}{k_{(1)} + k_{(2)} + 2} \frac{\sqrt{N}}{2} = \frac{(k_{(2)} - k_{(1)})}{k_{(1)} + k_{(2)}} \frac{(k_{(1)} + k_{(2)})}{k_{(1)} + k_{(2)} + 2} \frac{\sqrt{N}}{2} =$$

$\frac{a-1}{a+1}$	$\frac{\bar{k}}{1+\bar{k}}$	$\frac{\sqrt{N}}{2}$
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1.4 Chromatographic Resolution

$$R = \frac{\Delta t_r}{W_{\text{avg}}} \quad \text{so} \quad R = \frac{2(t_{r(2)} - t_{r(1)})}{w_1 + w_2} \quad \text{using} \quad N = \left(\frac{4t_r}{w}\right)^2 \Rightarrow w = \frac{4t_r}{\sqrt{N}}$$

leads to:

$$\frac{\cancel{2}(t_{r(2)} - t_{r(1)})}{\cancel{4}\left(\frac{t_{r(1)}}{\sqrt{N}} + \frac{t_{r(2)}}{\sqrt{N}}\right)} \Rightarrow \frac{(t_{r(2)} - t_{r(1)})}{t_{r(1)} + t_{r(2)}} \boxed{\frac{\sqrt{N}}{2}}$$

using: $t_r' = t_r - t_m$ and $t_r = t_r' + t_m$

$$\Rightarrow \frac{(t'_{r(2)} - t'_{r(1)})}{t'_{r(1)} + t'_{r(2)} + 2t_m} \boxed{\frac{\sqrt{N}}{2}}$$

1.4 Chromatographic Resolution

$$\frac{(t'_{r(2)} - t'_{r(1)})}{t'_{r(1)} + t'_{r(2)} + 2t_m} \frac{\sqrt{N}}{2} / \quad \text{and using } k = \frac{t'_r}{t_m} \quad \longrightarrow$$

$$\frac{(k_{(2)} - k_{(1)})}{k_{(1)} + k_{(2)} + 2} \frac{\sqrt{N}}{2} \times \frac{(k_1 + k_2)}{(k_1 + k_2)} \quad \longrightarrow$$

$$\frac{(k_{(2)} - k_{(1)})}{k_{(1)} + k_{(2)}} \frac{(k_{(1)} + k_{(2)})}{k_{(1)} + k_{(2)} + 2} \frac{\sqrt{N}}{2}$$

1.4 Chromatographic Resolution

$$\frac{(k_{(2)} - k_{(1)})}{k_{(1)} + k_{(2)}} \bigg/ : k_1 \text{ and using } \alpha = k_2/k_1 \longrightarrow \boxed{\frac{a-1}{a+1}}$$

$\uparrow$
both – numerator and denominator

$$\frac{(k_{(1)} + k_{(2)})}{k_{(1)} + k_{(2)} + 2} \bigg/ : 2 \text{ and using } \bar{k} = (k_1 + k_2)/2 \longrightarrow \boxed{\frac{\bar{k}}{1 + \bar{k}}}$$

$$\boxed{\frac{\sqrt{N}}{2}}$$

1.4 Chromatographic Resolution

- Resolution depends on

N (efficiency)

α (selectivity)

k (retention)

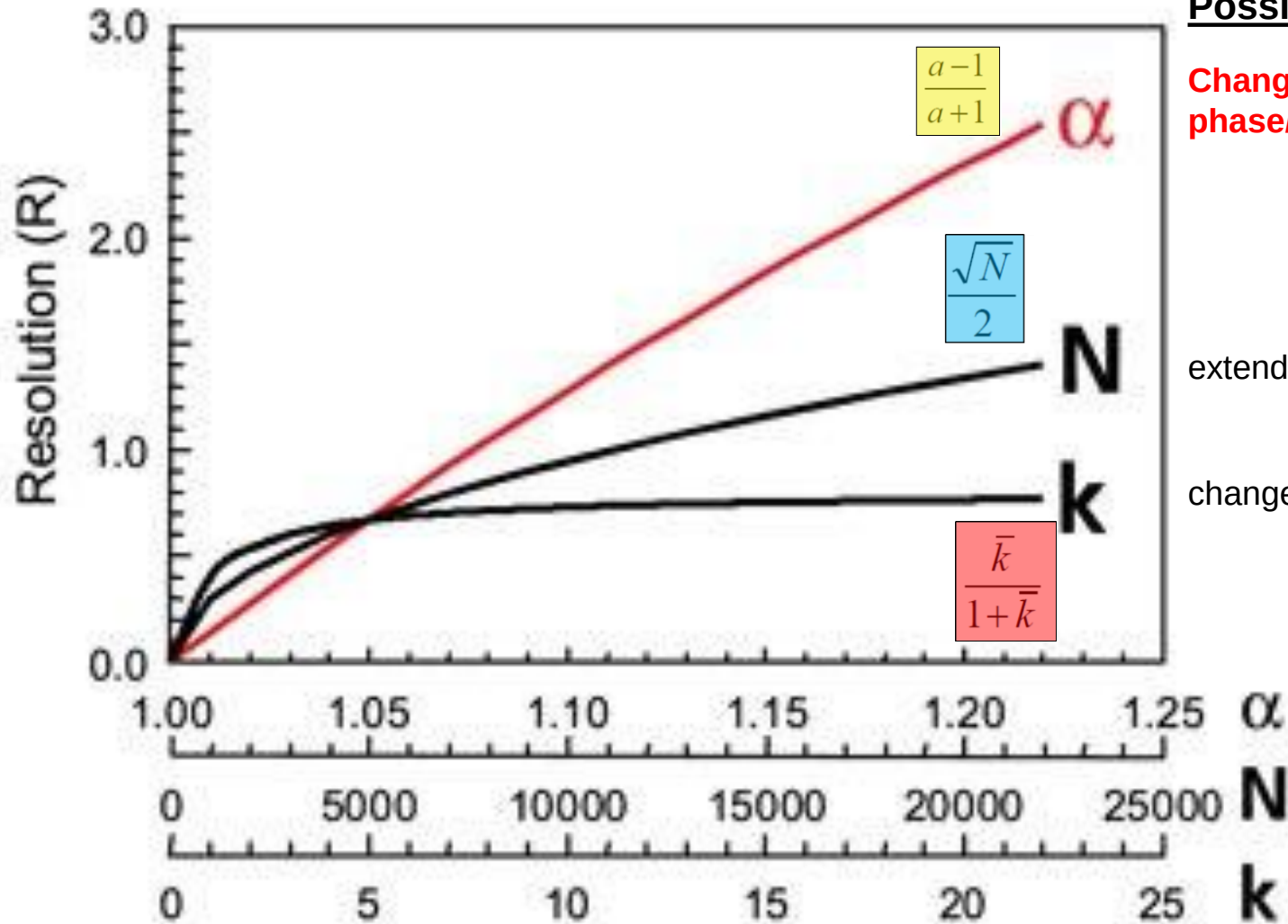
$$R = \frac{2(t_{r(2)} - t_{r(1)})}{w_1 + w_2} = \frac{2(t_{r(2)} - t_{r(1)})}{4\left(\frac{t_{r(1)}}{\sqrt{N}} + \frac{t_{r(2)}}{\sqrt{N}}\right)} =$$

$$\frac{(t_{r(2)} - t_{r(1)})}{2\left(\frac{t_{r(1)}}{\sqrt{N}} + \frac{t_{r(2)}}{\sqrt{N}}\right)} = \frac{(t_{r(2)} - t_{r(1)})}{t_{r(1)} + t_{r(2)}} \frac{\sqrt{N}}{2} = \frac{(t'_{r(2)} - t'_{r(1)})}{t'_{r(1)} + t'_{r(2)} + 2t_m} \frac{\sqrt{N}}{2} =$$

$$\frac{(k_{(2)} - k_{(1)})}{k_{(1)} + k_{(2)} + 2} \frac{\sqrt{N}}{2} = \frac{(k_{(2)} - k_{(1)})}{k_{(1)} + k_{(2)}} \frac{(k_{(1)} + k_{(2)})}{k_{(1)} + k_{(2)} + 2} \frac{\sqrt{N}}{2} =$$

$\frac{a-1}{a+1}$	$\frac{\bar{k}}{1+\bar{k}}$	$\frac{\sqrt{N}}{2}$
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1.4 Chromatographic Resolution



Possible action:

Change stationary phase/ (mobile phase)

extend column

change mobile phase

1.4 Chromatographic Resolution



Increase k



- 
- Long retention times
 - Peak broadening

Increase α



- 
- Availability of stationary phases with sufficiently different properties

Increase N

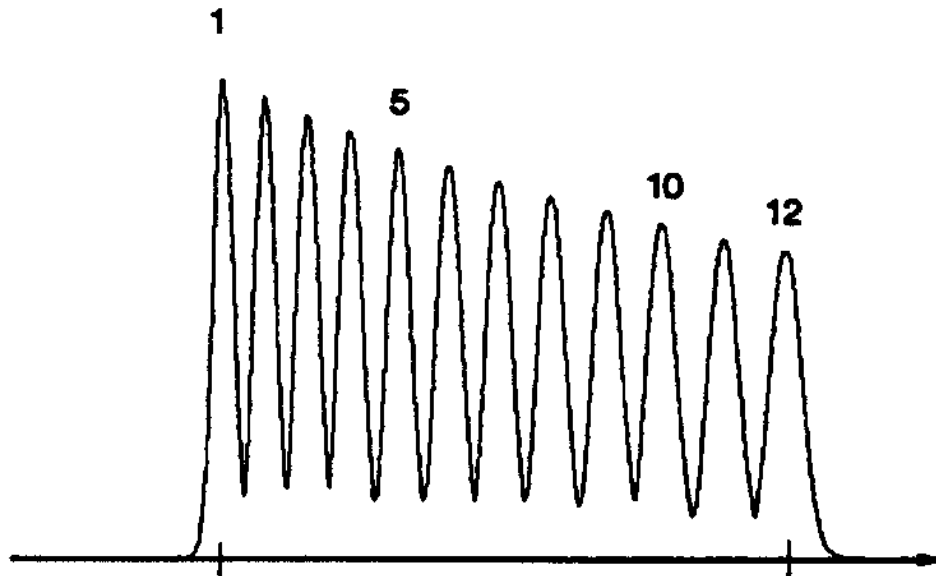


- 
- Longer columns also mean longer retention times
 - Pressure issues with very small particles

1.5 Peak Capacity

$$n_p = \frac{\sqrt{N}}{4R} \ln \left(\frac{1+k_\omega}{1+k_\alpha} \right) + 1$$

α, ω first and last peak

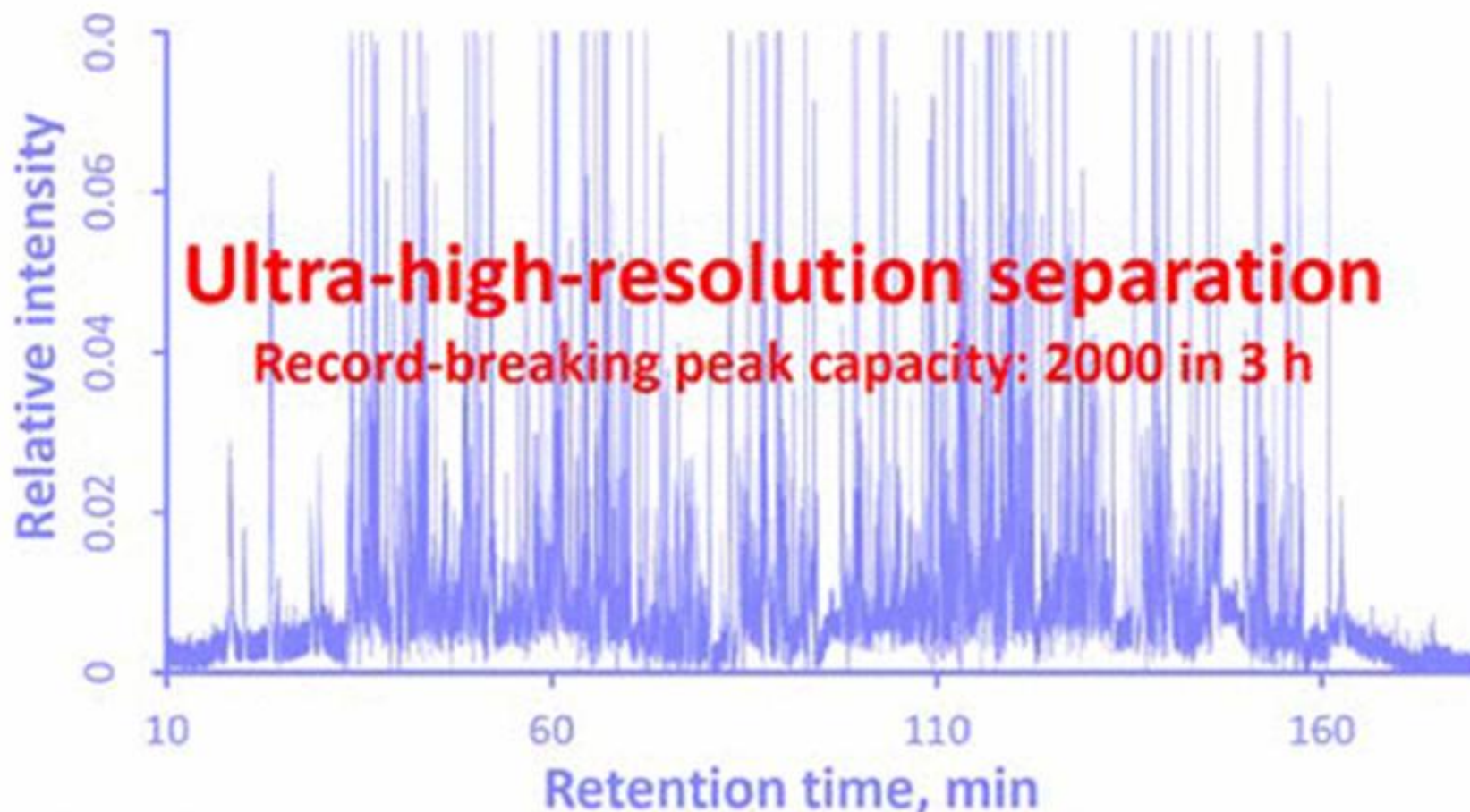


In multidimensional chromatography (see chapter on GCxGC later), peak capacities (theoretically) multiply

Simulation diagram of the column peak capacity

1.5 Peak Capacity

- But also might look like this:



1.6 Parameters for Column Performance

LC Column Performance Report



SERIAL NUMBER: USFBQ01763

PART NUMBER: 695968-301
COLUMN TYPE: Poroshell 120 Bonus-RP 3 x 100 mm, 2.7 µm
PACKING LOT #: B22008

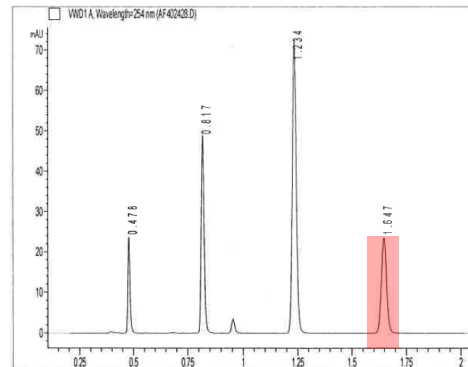
TEST CONDITIONS

MOBILE PHASE = 60% Acetonitrile / 40% Water
COLUMN PRESSURE = 223.0 Bar
COLUMN FLOW = 0.80 ml / min
LINEAR VELOCITY = 0.348 cm / sec
TEMPERATURE = AMBIENT (Nominally 23 °C)
INJECTION VOLUME = 2 µl

- N is a measure for “efficiency” of the separation
- The higher N, the narrower the peaks
- N is also used to describe column performance

QUALITY CONTROL PERFORMANCE RESULTS FOR NAPHTHALENE

TEST VALUES	SPECIFICATIONS
THEORETICAL PLATES = 22379	MIN = 20000
SELECTIVITY = 1.55	RANGE = 1.50 - 1.60
USP TAILING FACTOR = 1.06 (@ 5% Peak Height)	RANGE = 0.98 - 1.20
k' = 2.44	



Sample components with concentrations diluted in mobile phase in the following elution order.

Peak #	Conc (ug/ml)	Sample Component
1	10	Uracil
2	400	Phenol
3	20	N,N-Dimethylaniline
4	80	Naphthalene

THIS COLUMN WAS SHIPPED CONTAINING ACETONITRILE AND WATER.
MATERIAL SAFETY DATA SHEETS ARE AVAILABLE UPON REQUEST.

1.6 Parameters for Column Performance

LC Column Performance Report



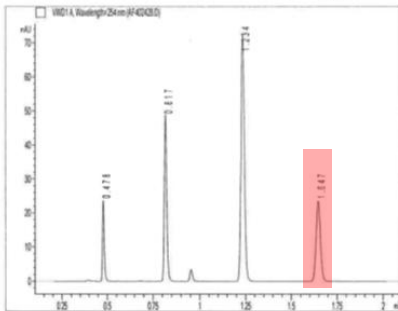
SERIAL NUMBER: USFBQ01763

PART NUMBER: 695968-301
COLUMN TYPE: Poroshell 120 Bonus-RP 3 x 100 mm, 2.7 µm
PACKING LOT #: B22008

TEST CONDITIONS
MOBILE PHASE = 60% Acetonitrile / 40% Water
COLUMN PRESSURE = 223.0 Bar
COLUMN FLOW = 0.80 ml / min
LINEAR VELOCITY = 0.348 cm / sec
TEMPERATURE = AMBIENT (Nominally 23 °C)
INJECTION VOLUME = 2 µl

QUALITY CONTROL PERFORMANCE RESULTS FOR NAPHTHALENE

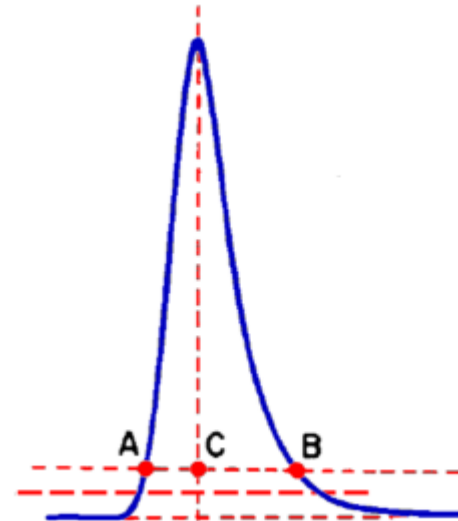
TEST VALUES	SPECIFICATIONS
THEORETICAL PLATES = 22379	MIN = 20000
SELECTIVITY = 1.55	RANGE = 1.50 - 1.60
USP TAILING FACTOR = 1.06 (@ 5% Peak Height)	RANGE = 0.98 - 1.20
$k' = 2.44$	



Sample components with concentrations diluted in mobile phase in the following elution order.

Peak #	Conc (ug/ml)	Sample Component
1	10	Uracil
2	400	Phenol
3	20	N,N-Dimethylaniline
4	80	Naphthalene

THIS COLUMN WAS SHIPPED CONTAINING ACETONITRILE AND WATER.
MATERIAL SAFETY DATA SHEETS ARE AVAILABLE UPON REQUEST.



United States Pharmacopeia
Tailing Factor:

$$\frac{AB}{2AC} \text{ (@ 5\% peak height)}$$

2 High Performance Liquid Chromatography

Columns packed with small particles (stationary phase)

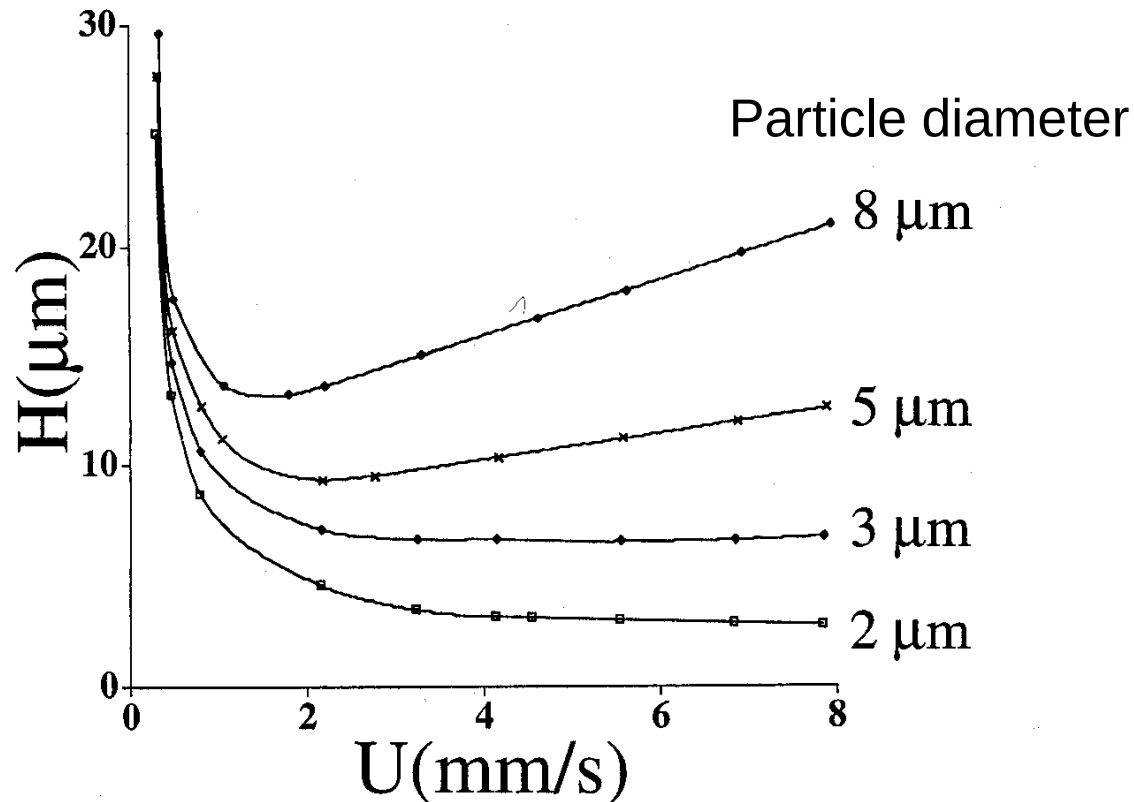
Why use small particles?

- provides more uniform flow ($\downarrow A$)
- less distance for solutes to be transported in the mobile phase to interact with stationary phase ($\downarrow C$)
- comes at a cost: much higher pressure is required to “push” mobile phase through column ($\sim$ several hundreds atmospheres)

(Alternative: monolithic columns)

2 High Performance Liquid Chromatography

High Performance Liquid Chromatography VS Ultra High Performance Liquid Chromatography



HPLC

UHPLC

Ultra high performance liquid chromatography (particle size of 2 μm and less)

Van Deemter plot for different particle size

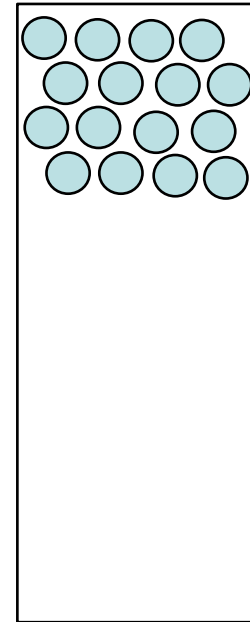
2 High Performance Liquid Chromatography

Small particles



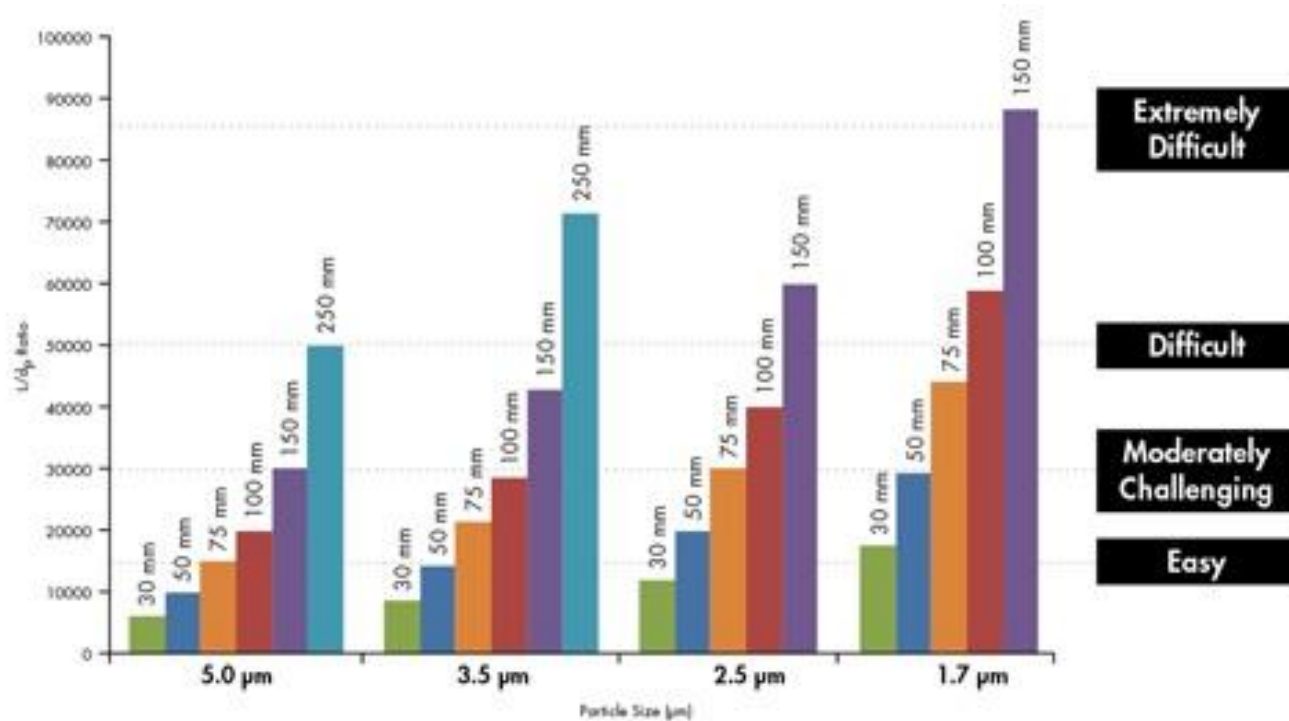
- Smaller particles lead to reduced void volume between particles
- Analyte molecules have shorter distance to travel when changing from mobile to stationary phase and vice versa – beneficial for the C-term of the van Deemter equation!

Large particles



Effect of particle size on peak broadening

2 High Performance Liquid Chromatography



Column resolving power:
Length/particle diameter
ratio (L/d_p)

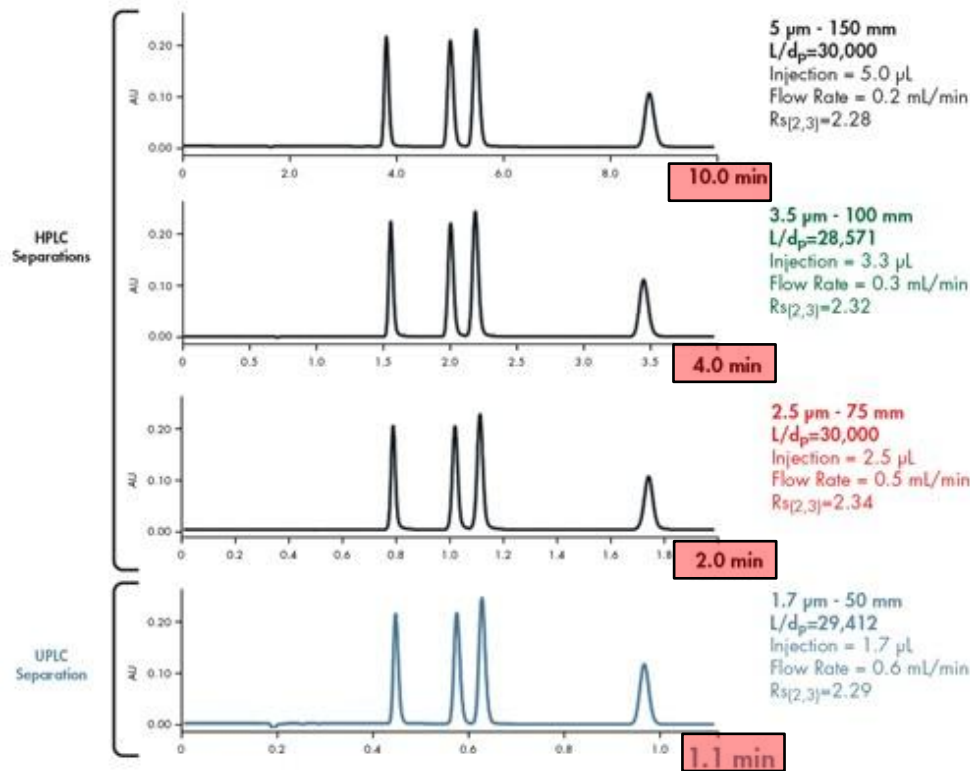


Separation Index	Application Example	Efficiency (N)	L/d_p
Easy	Content uniformity	5,000	15,000
Moderately Challenging	Related compound assay	12,000	30,000
Difficult	Impurity profiling	20,000	50,000
Extremely Difficult	Metabolite identification	35,000	85,000



2 High Performance Liquid Chromatography

Constant L/d_p while changing L and d_p



Typical HPLC Column 4.6 x 150 mm, 5 μm

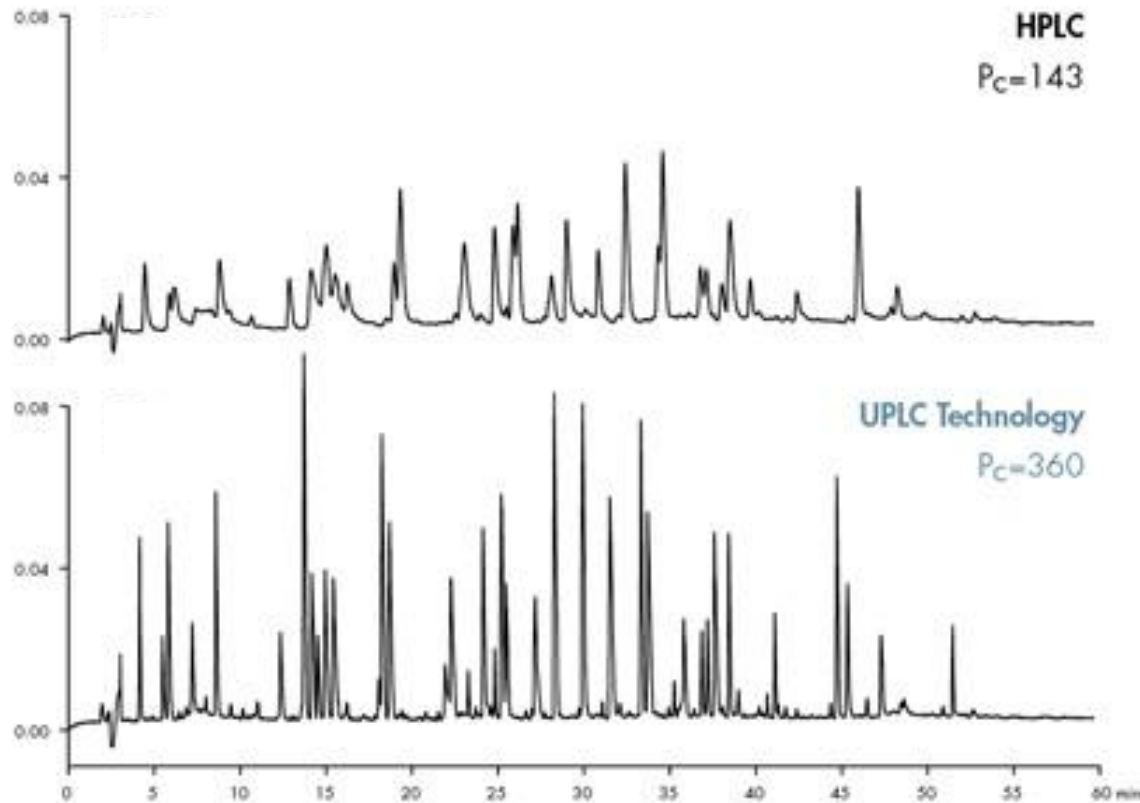
Column Length (L) = 150 mm = 150,000 μm

$d_p = 5 \mu\text{m}$

$$\frac{L}{d_p} = \frac{150,000}{5} = 30,000$$

Separation Index	Application Example	Efficiency (N)	L/d_p
Easy	Content uniformity	5,000	15,000
Moderately Challenging	Related compound assay	12,000	30,000
Difficult	Impurity profiling	20,000	50,000
Extremely Difficult	Metabolite identification	35,000	85,000

2 High Performance Liquid Chromatography



Separation of a tryptic digest of phosphorylase b – comparison of peak capacity (P_C) in HPLC and U(H)PLC

2.1 HPLC - Instrumentation



Degasser

Pump

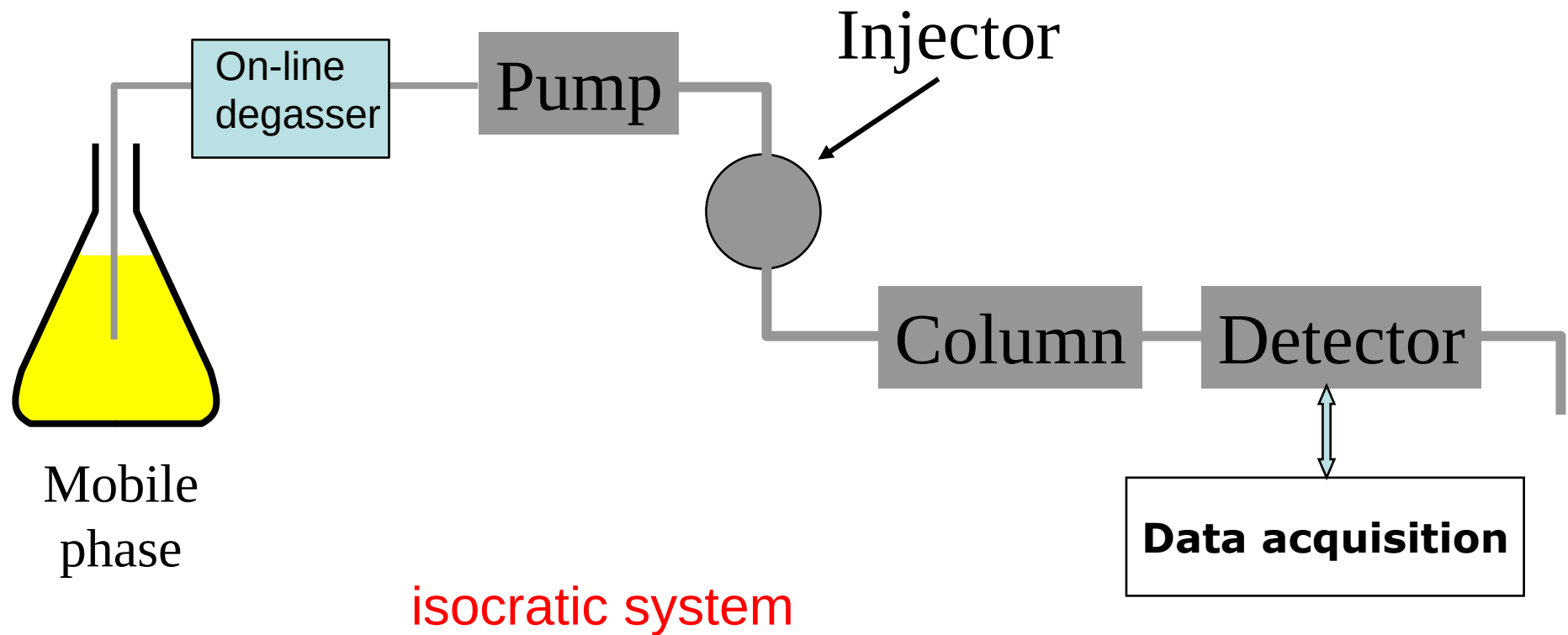
Autosampler with Injector

Column Compartment with Heater/Cooler

Detector

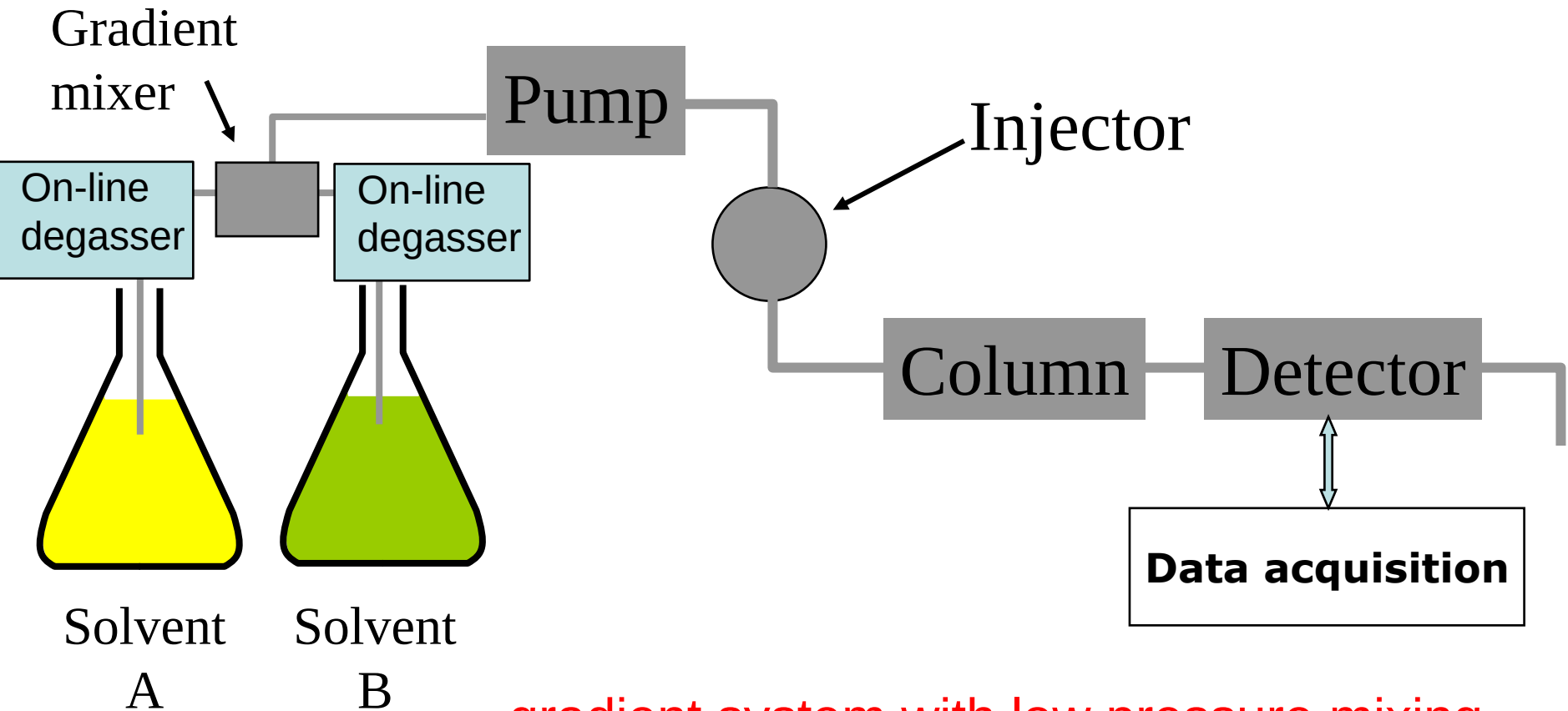
2.1 HPLC - Instrumentation

Schematic Layout:



2.1 HPLC - Instrumentation

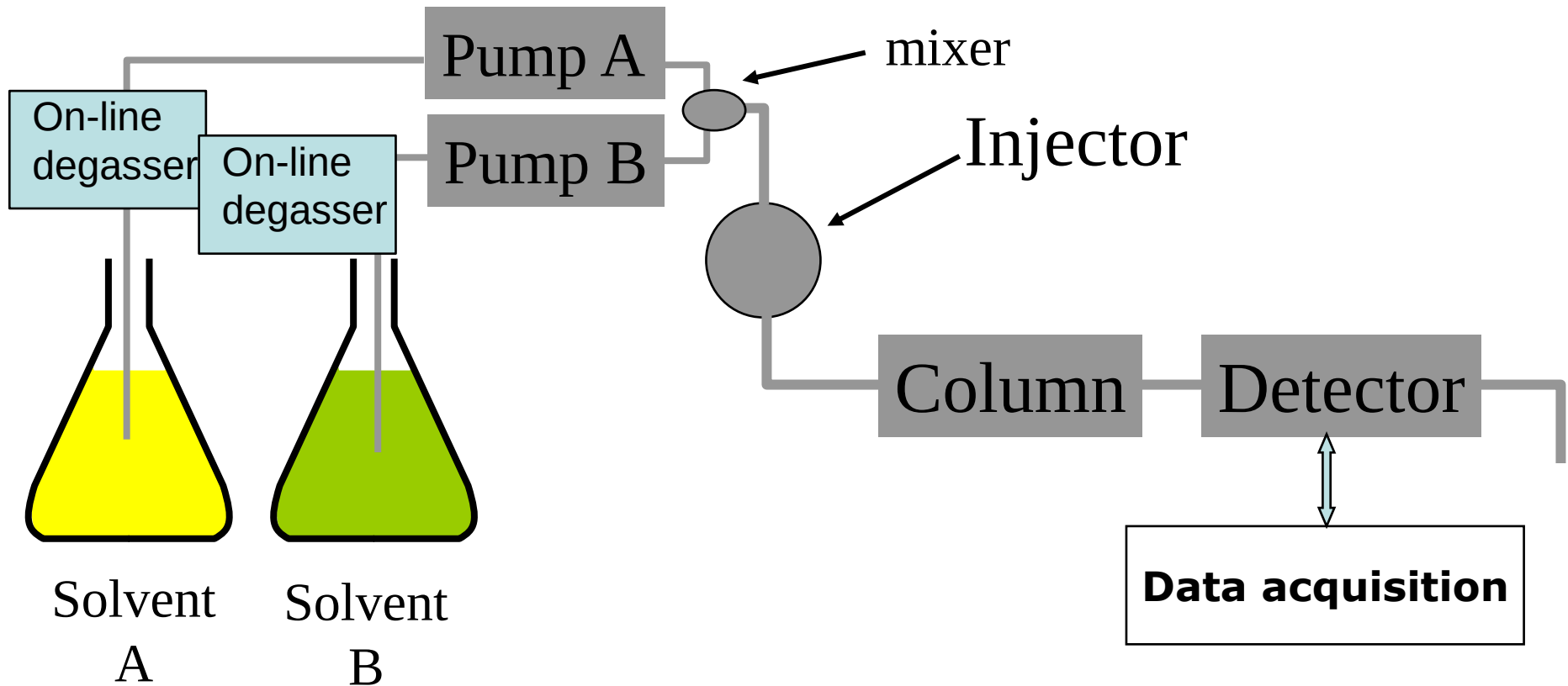
Schematic Layout:



gradient system with low pressure mixing

2.1 HPLC - Instrumentation

Schematic Layout:



gradient system with high pressure mixing

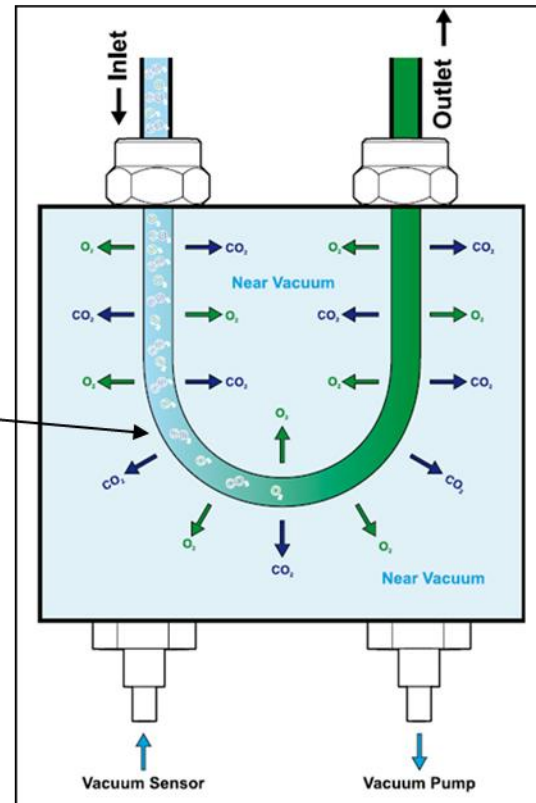
2.1.1 On-line Degasser

Mobile phases

must be degassed to avoid bubble formation
after the column in the detector

On-line degasser

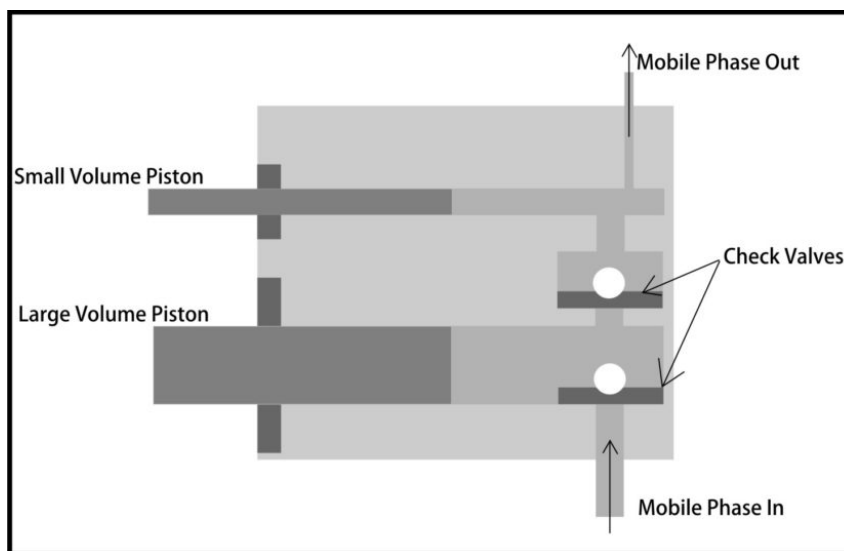
Gas-permeable tube



Alternative: Purging of mobile
phase in the reservoir with
helium

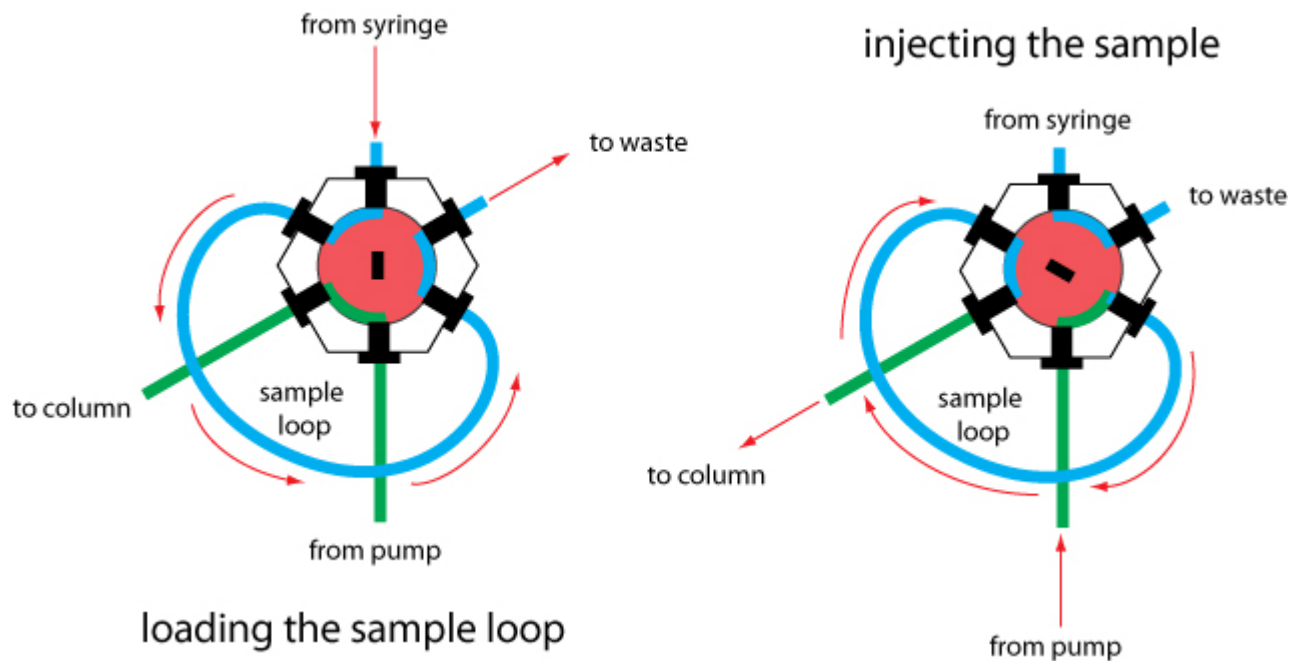
2.1.2 HPLC Pumps

- Historically one and two piston design in use - but today most instruments are built with a larger (main) and a smaller (auxiliary) piston



Both the small and large volume pistons operate 180° out of phase. The bigger piston is sucking the mobile phase while the smaller one is pushing it out. When the pistons change direction, the bigger piston simultaneously refills the smaller chamber and dispenses the eluent into the HPLC system.

2.1.3 HPLC Injector



Loop volumes can vary from less than 1 μL to more than 1 mL. Small injection volumes are commonly used for small-diameter columns (capillary columns) whereas large injection volumes require larger columns. In general injection volume needs to be adapted to stationary phase volume!

2.1.4 HPLC Column

Columns packed with the stationary phase (particles with diameters between 1.5 and 5 μm , or monolithic columns)

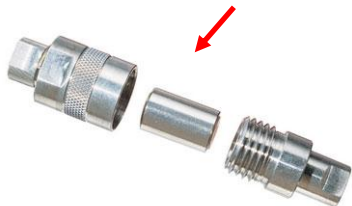
3-25 cm long (typically stainless steel columns)

0.5-5 mm inner diameter (i.d.)



typical flow rates: 1 mm i.d.: 100 $\mu\text{L}/\text{min}$
 4 mm i.d.: 1 mL/min

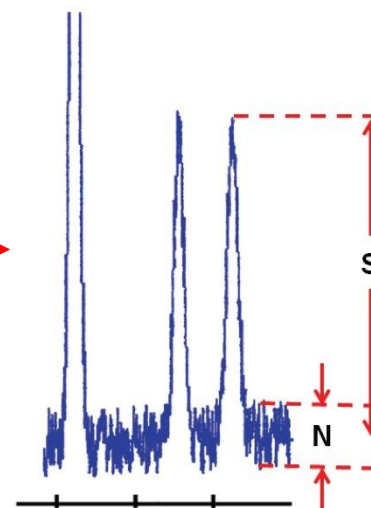
guard column should be used to protect main column



2.1.5 HPLC Detectors

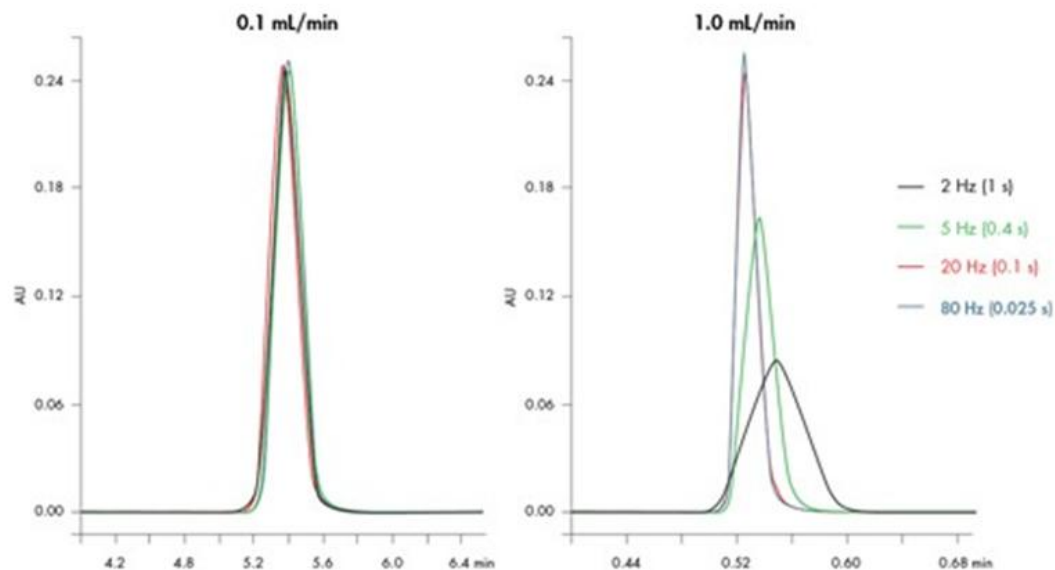
Characterization of Detectors:

- Limit of detection (Signal/Noise ratio of 3) →
- Sensitivity (slope of calibration line)
- Linearity
- Compatibility with mobile phase (gradients)
- selective / universal
- concentration-dependent / mass flow–dependent
- bulk property response / solute property response
- Robustness, simple handling
- Price
- Response time/frequency



Frequency („speed“) of detector needs to be sufficient for chromatographic separation (i.e. must be able to describe the narrowest peak properly)

@ Waters



2.1.5 HPLC Detectors

UV-visible absorbance detector

Fluorescence detector

Refractive index detector

Evaporative light scattering detector

Charged aerosol detector

Amperometric detector / coulometric detector

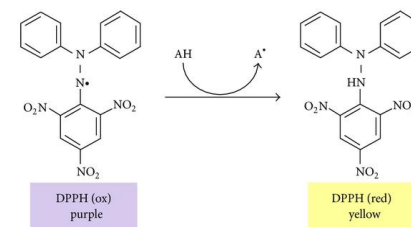
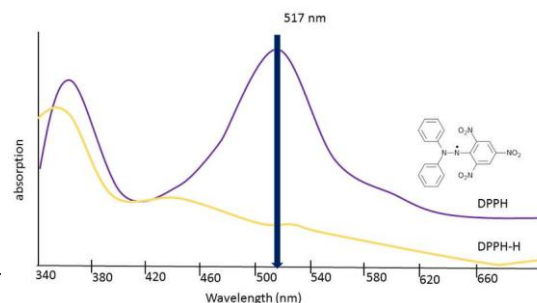
Pulsed amperometric detector

Conductivity detector

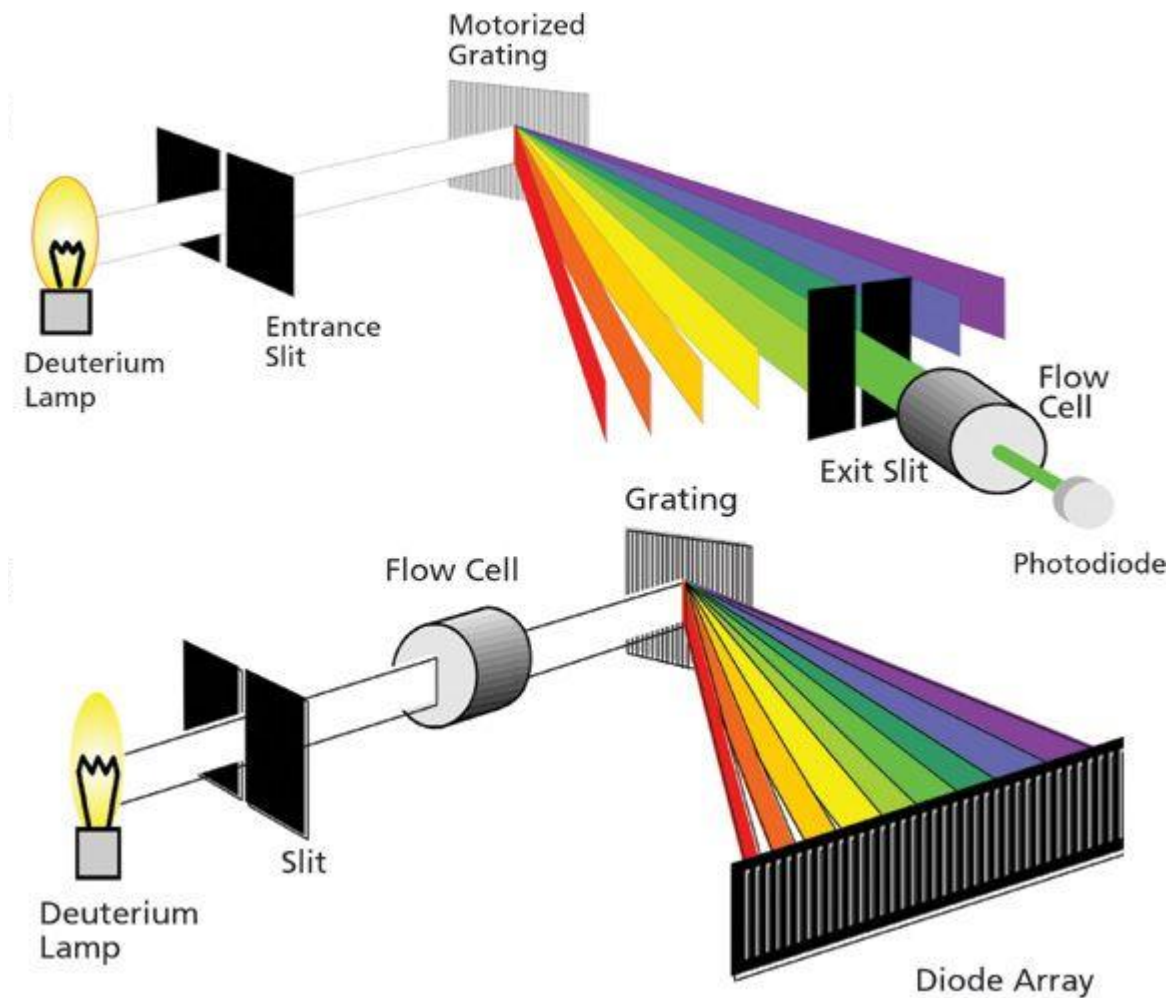
(will be discussed with ion chromatography)

Post-column reaction detector ⇒

Mass spectrometric detector (will be discussed in detail in the second part of the lecture)

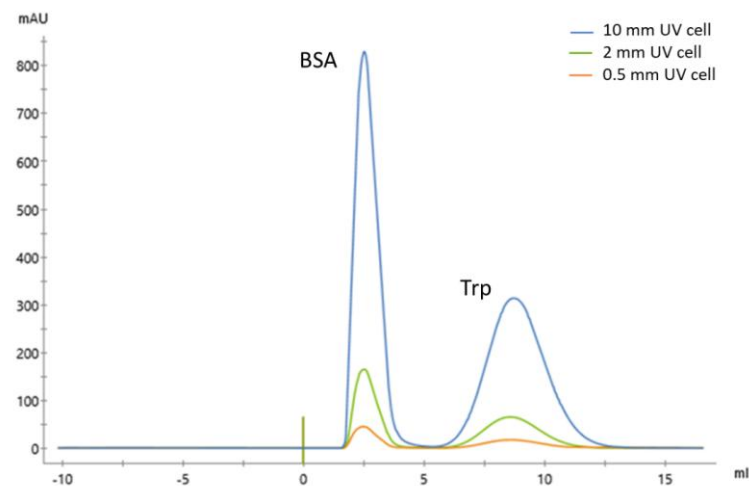
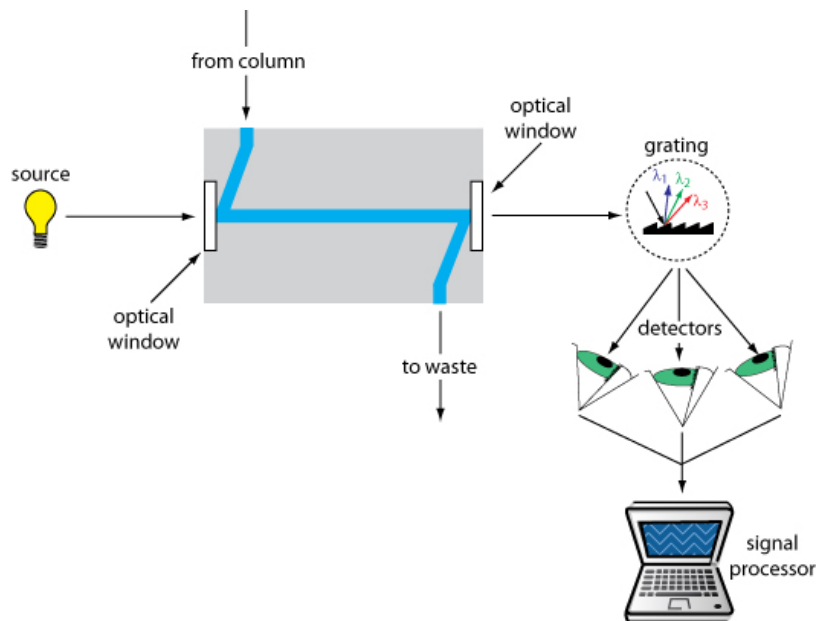


2.1.5 HPLC Detectors: UV-vis



For relevant theory of operation see „Introduction to Analytical Chemistry“ lecture

2.1.5 HPLC Detectors: UV-vis

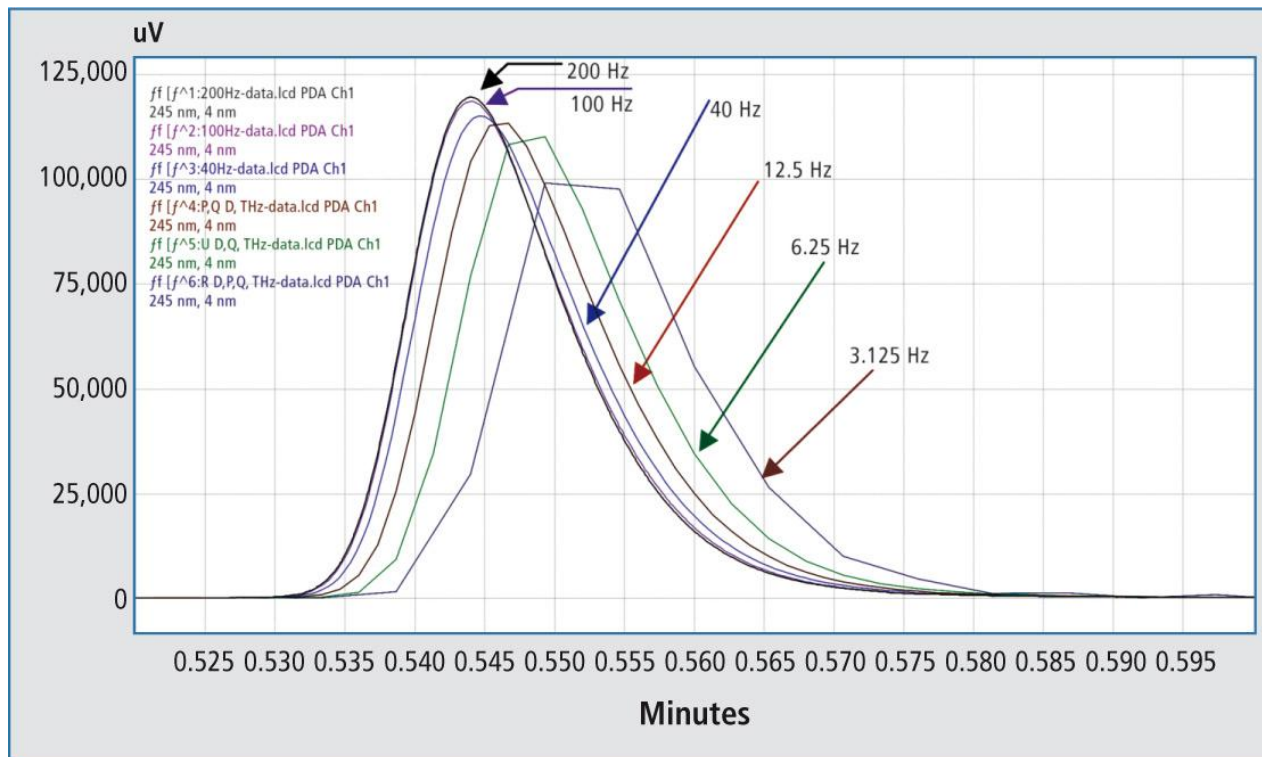


For relevant theory of operation see „Introduction to Analytical Chemistry“ lecture

<https://asplib.org/imageandvideoexchangeforum/absorbance-and-fluorescence-detection-for-hplc-separations/>

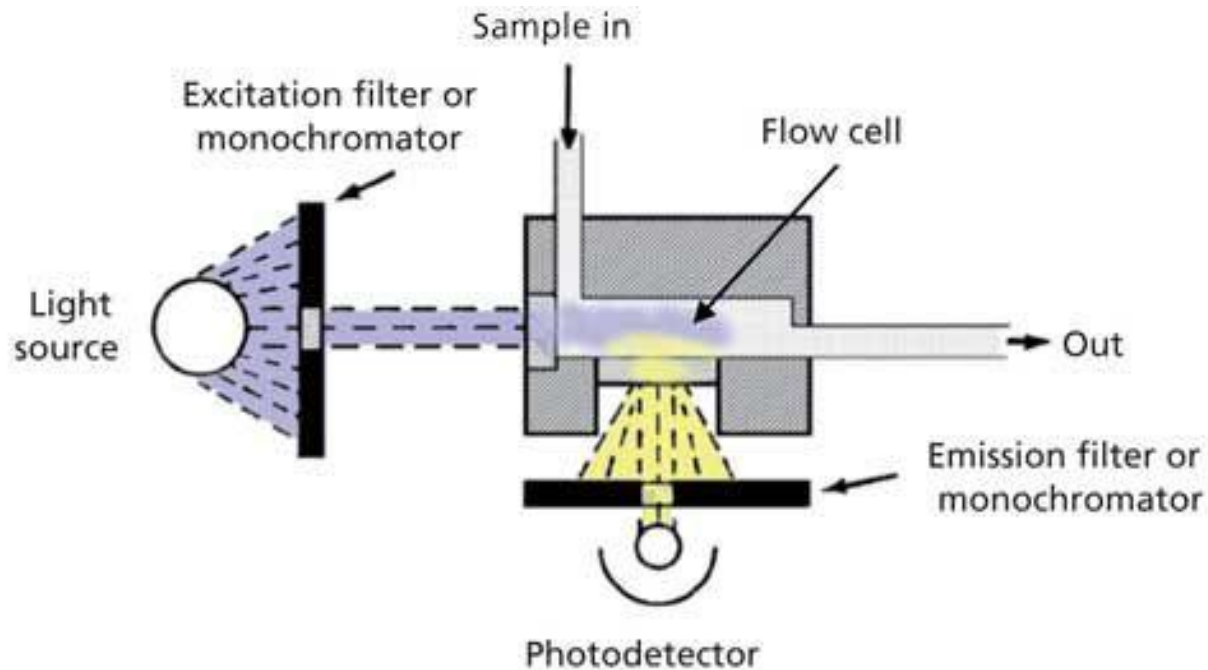
<https://www.cytivalifesciences.com/en/us/news-center/the-effect-of-uv-flow-cell-pathlength-in-chromatography-10001>

2.1.5 HPLC Detectors: UV-vis



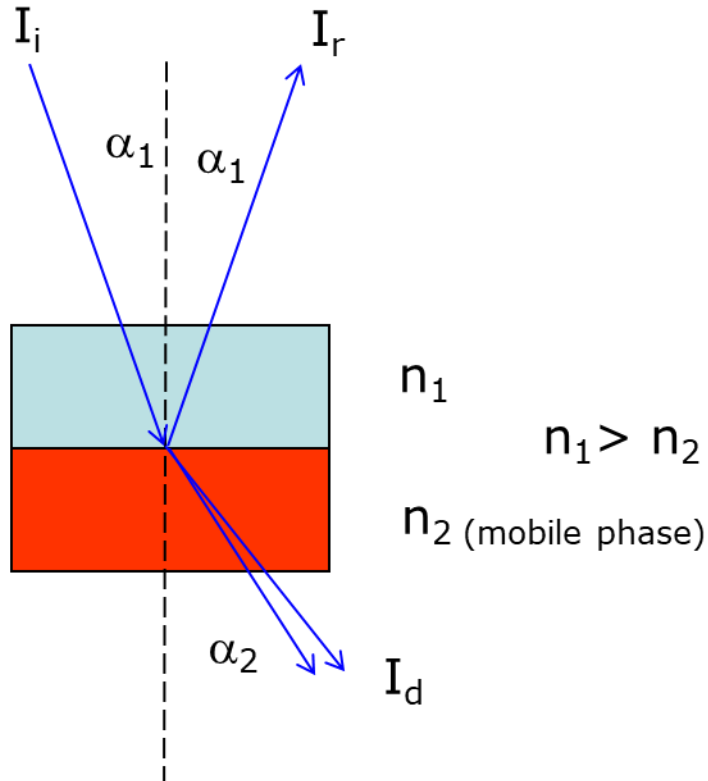
Effect of “Data Acquisition Rate” on peak shapes
(applies to all detectors not only UV-vis)

2.1.5 HPLC Detectors: Fluorescence



For relevant theory of operation see „Analytical Chemistry“ lecture

2.1.5 HPLC Detectors: Refractive Index



$$\sin \alpha_1 / \sin \alpha_2 = n_2 / n_1 \quad (\text{Snellius})$$

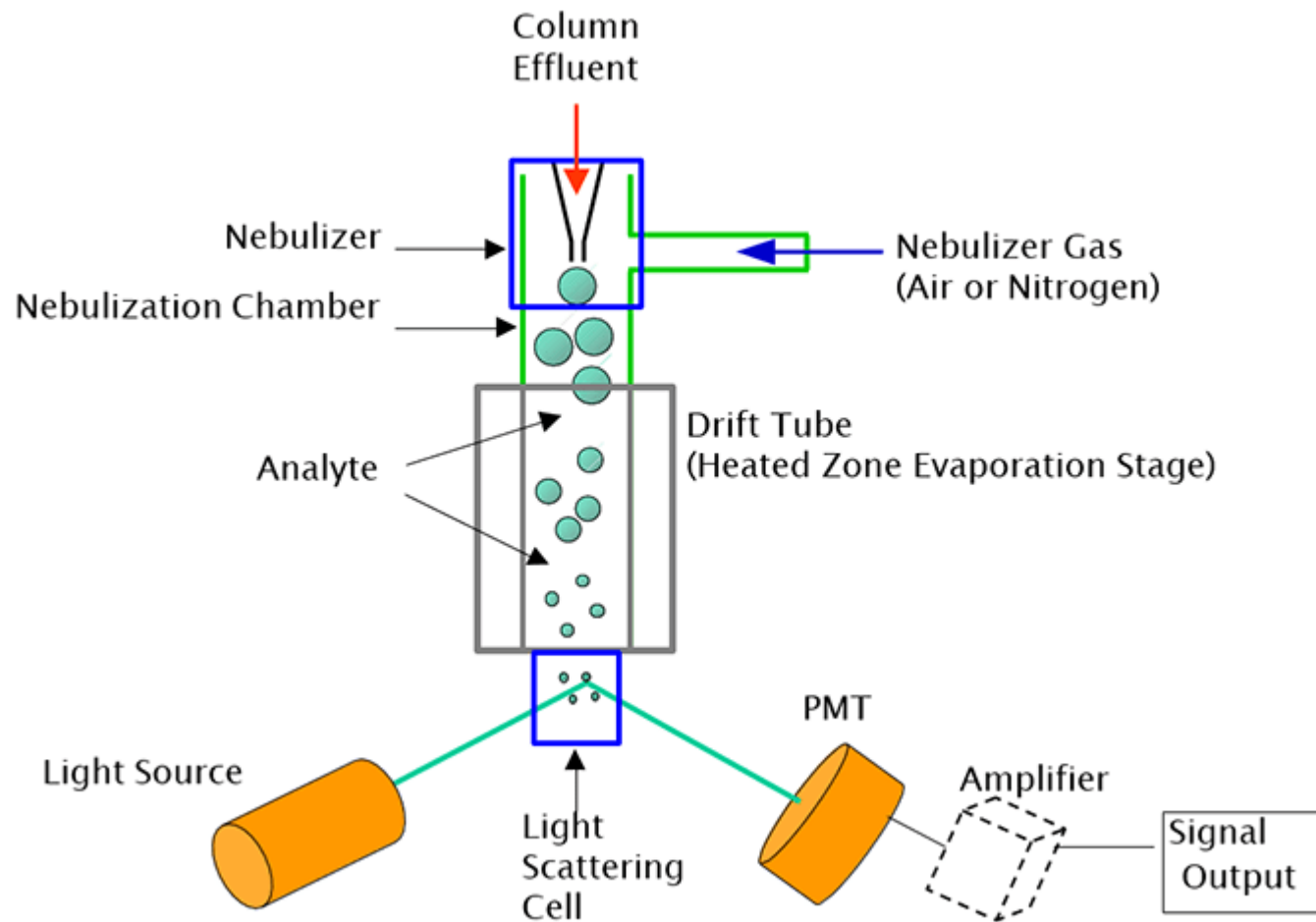
(measurement of refracted light at a certain position)

$$I_r / I_i = f(\alpha_1, n_1, n_2) \quad (\text{Fresnel})$$

(measurement of intensity of reflected light)

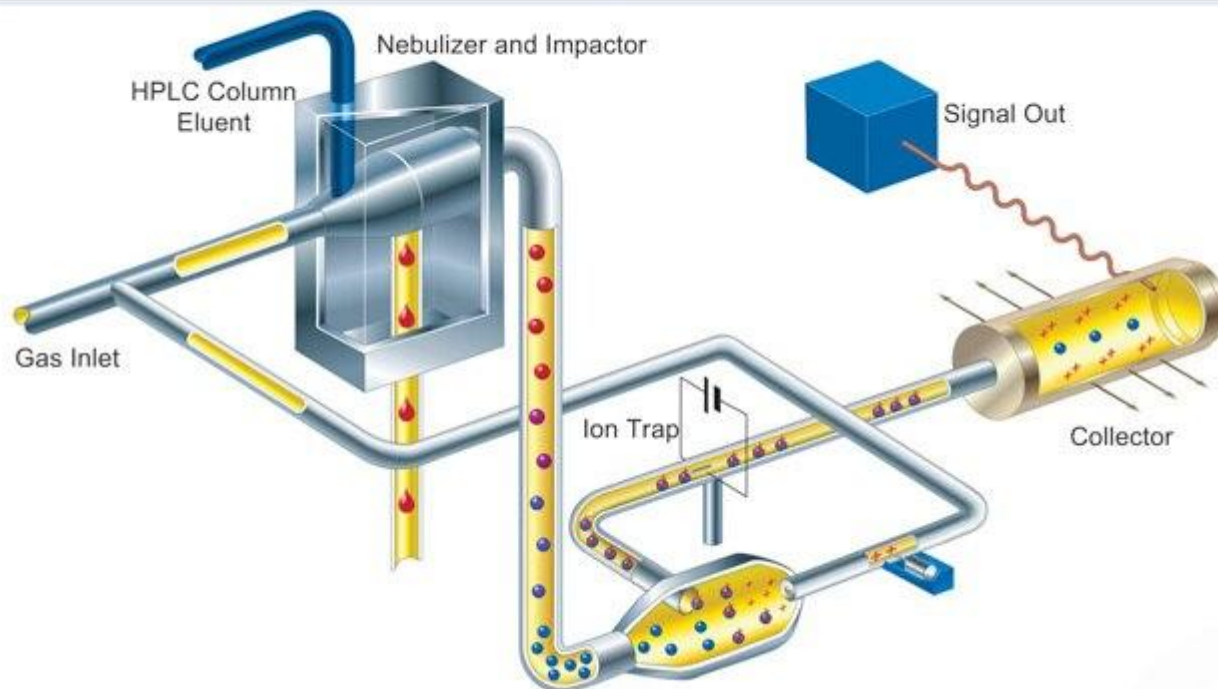
Two options for measurement

2.1.5 HPLC Detectors: Evaporative light scattering detector (ELSD)



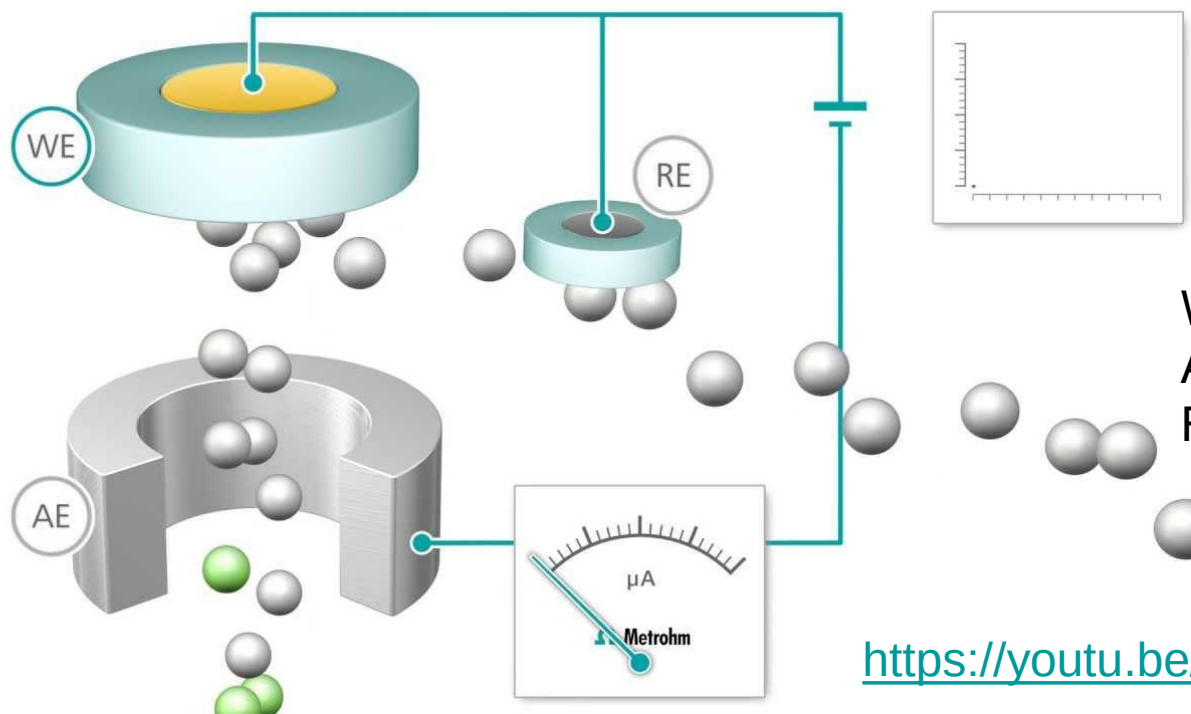
2.1.5 HPLC Detectors: Charged aerosol detector (CAD)

Charged Aerosol Detection measures charge not analytes



STEP ONE	STEP TWO	STEP THREE
The detector converts the analyte molecules into dry particles. The number of particles increases proportionally with the amount of analyte.	A stream of positively charged gas collides with the analyte particles. The charge is then transferred to the particles—the larger the particles, the greater the charge.	The particles are transferred to a collector where the charge is measured by a highly-sensitive electrometer. This generates a signal in direct proportion to the quantity of analyte present.

2.1.5 HPLC Detectors: Amperometric detection

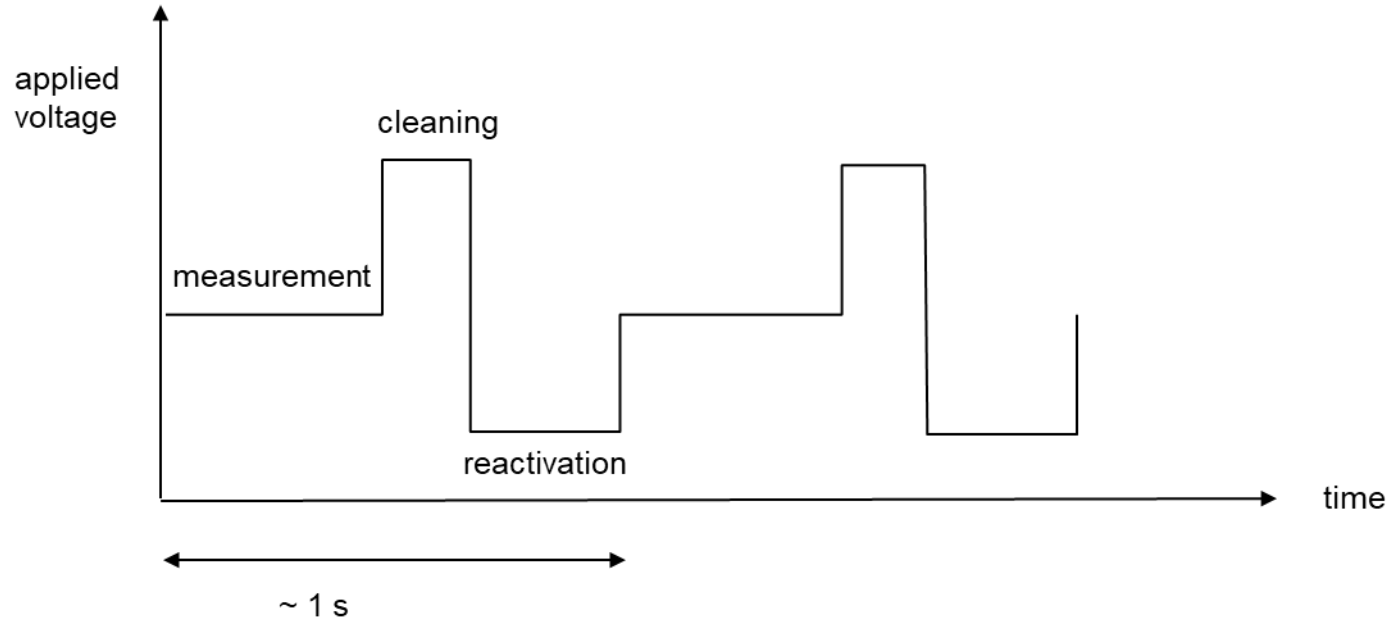


WE....Working electrode
AE.....Auxilliary electrode
RE.....Reference electrode

<https://youtu.be/IXJyvTCdyxk>

Analytes that can be oxidized or reduced are suitable for amperometric detection. A simple form of amperometric detection is single-potential, or direct current amperometry. A voltage (potential) is applied between two electrodes positioned in the column effluent. The measured current changes as an electroactive analyte is oxidized at the anode or reduced at the cathode. The applied potential can be adjusted to improve the response for the analyte of interest while minimizing the response for interfering substances.

2.1.5 HPLC Detectors: Pulsed amperometric detection



Pulsed Amperometric Detection (PAD) is a technique used to detect certain classes of compounds, notably **sugars** and polyalcohols, among others. These compounds tend to foul the surface of an electrode, making ordinary constant-potential amperometric detection difficult.

In PAD, cleaning potentials are applied to the electrode roughly once per second, interspersed with the detecting potential. The result is a waveform like the one pictured above.

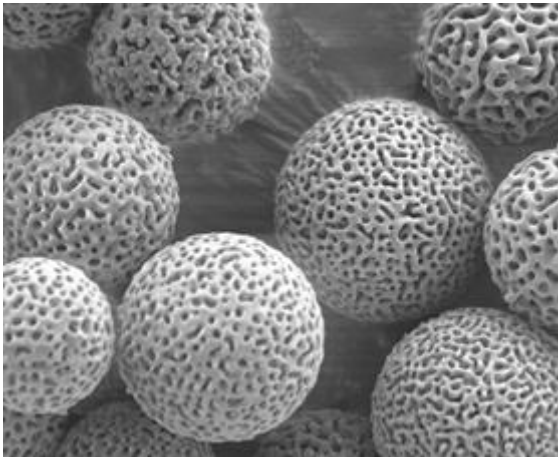
2.1.5 HPLC Detectors: Overview

Detector	LOD	Linear range	Compatibility with Gradient elution	Selectivity	Bulk/Solute-property
UV-vis (variable wavelength)	+	+++	+++	+	S
UV-vis (Diode Array)	+	+++	+++	+	S
Flourescence	+++	+++	++	++	S
Refractive Index	--	+	---	---	B
ELSD	0	++	+	---	S
Electrochemical Detection (amperometric)	+++	++	---	++	S
Conductivity Detector	++	++	-	+/-	B

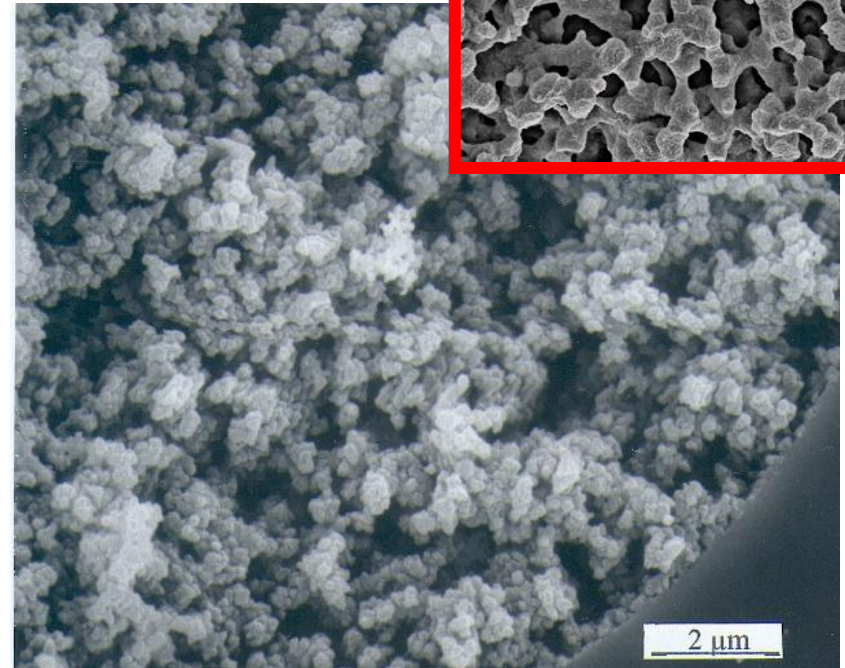
Will be discussed with Ion Chromatography!

2.2 HPLC: Stationary phases

Particles (widely used) or monoliths (special applications)



HPLC particles



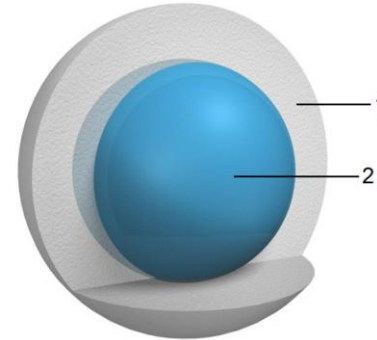
Monolith
(macropores and micropores)

2.2 HPLC: Stationary phases

Physical properties of stationary phases

- * particle diameter (1.5 to 5 micrometer, smaller particles yield larger N !)
- * particle size distribution (should be very narrow)
- * shape (spherical)
- porosity (non-porous, totally porous, porous thin-layer particles / core-shell particles)

Core-shell particles provide advantages of very small particles while still maintaining an acceptable back pressure



1 porous shell
2 nonporous core

- * pore size (5 to 30 nm and larger)
- * pore size distribution (should be narrow)
- * surface area (including inner surface of porous particles)
- * pressure stability – for UHPLC!
- * chemical stability (pH stability, organic solvents,...) - determines compatibility with mobile phase

2.3 Modes of HPLC

Normal-phase chromatography NP
(including Hydrophilic Interaction Liquid Chromatography HILIC)

Reversed-phase chromatography RP

Size exclusion chromatography SEC
(gel permeation chromatography GPC,
gel filtration chromatography)

Ion-exchange chromatography

Ion-pair chromatography

Ion-exclusion chromatography

Affinity chromatography

Chiral separations

Hydrophobic interaction chromatography

2.3.1 Normal phase HPLC

Stationary phases: polar

Mobile phases: non-polar (for example hexane)

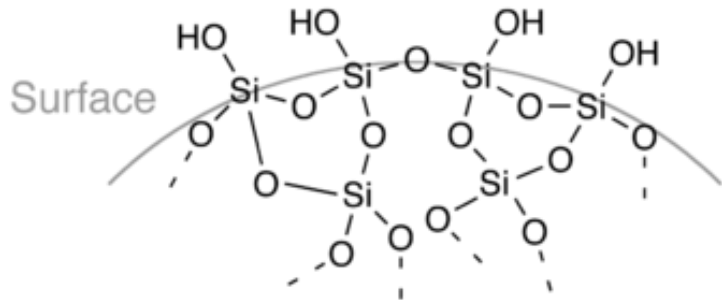
Retention increases with increasing polarity of the analyte

Retention decreases with increasing polarity of the mobile phase

- silica gel: $\text{SiO}_2 \cdot x\text{H}_2\text{O}$ (not stable at $\text{pH} > 8.5$!)
- alumina: $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- silica gel with polar surface modifications (diol, cyanopropyl, amino)

2.3.1 Normal phase HPLC

Silanol groups on a silica surface are polar



Liquid/solid chromatography, retention of analytes due to **adsorption**

Solvent strength (elution strength) of mobile phase: a parameter for the solvent's ability to cause elution of an analyte

Eluotropic series: relative ability of solvent to elute analyte

In case of gradient elution, the solvent strength of the mobile phase is increased during the chromatographic run

2.3.1 Normal phase HPLC

Eluotropic series: relative ability of solvent to elute analyte

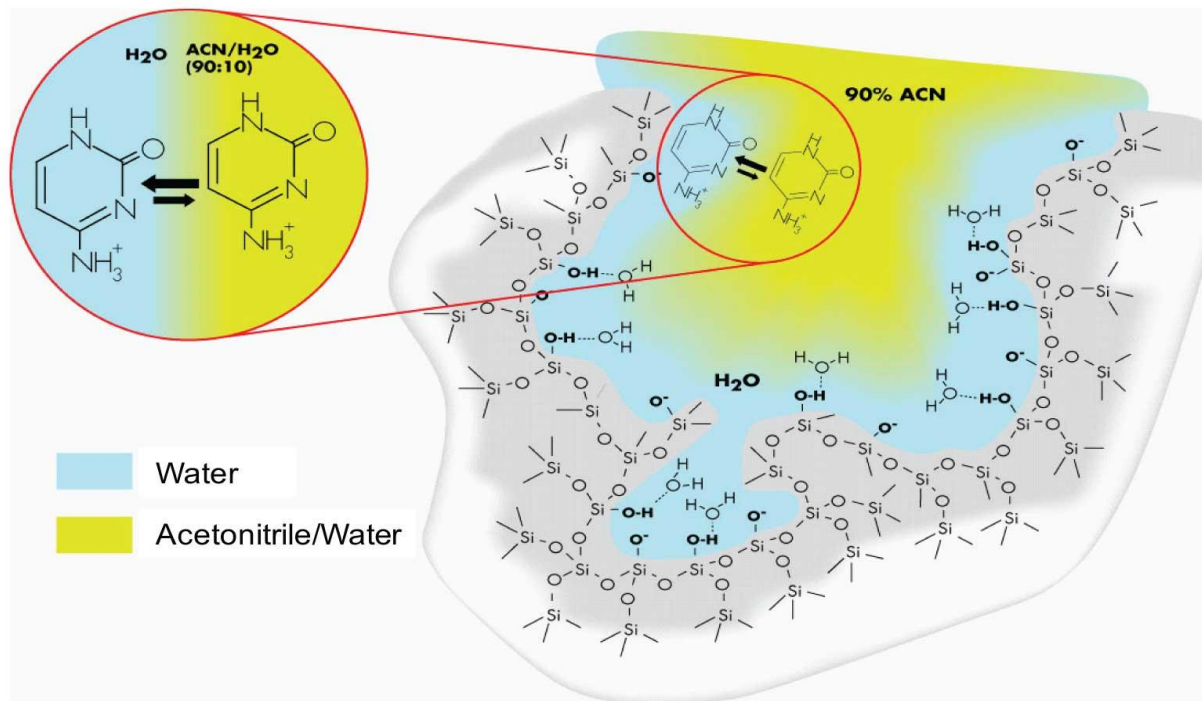
Solvent	Solvent strength	
	Silica	Alumina
Fluorochemical FC-78	-0.2	-0.25
<i>n</i> -Hexane		
<i>n</i> -Heptane	0.01	0.01
Iso-octane		
1-Chlorobutane	0.20	0.26
Chloroform	0.26	0.40
Methylene chloride	0.32	0.42
Isopropyl ether	0.34	0.28
Ethyl acetate	0.38	0.58
Tetrahydrofuran	0.44	0.57
Acetonitrile	0.50	0.65
Methanol	0.7	0.95

2.3.1 Normal phase HPLC: HILIC

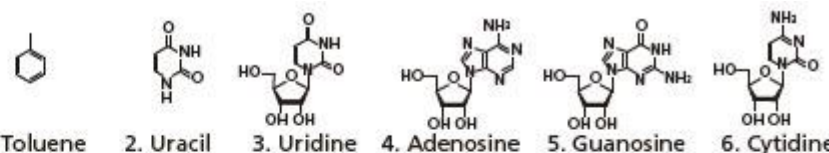
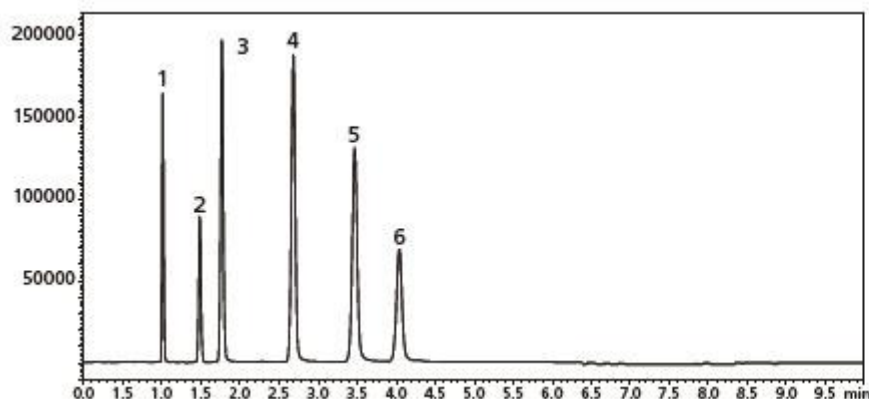
Hydorphilic **I**nteraction **L**iquid **C**hromatography - a variant of normal phase HPLC
(stationary phase: silica, ...)

Mobile phase
for example:

acetonitrile / water 90:10
gradient by increasing water content



2.3.1 Normal phase HPLC: HILIC



LC Conditions

Column: Shim-pack Velox HILIC, 2.7 μ m, 3.0x100 mm (PN: 227-32026-02)

Column Temp.: 30°C

Flow Rate : 0.4 mL/min

Mobile Phase : Acetonitrile/20mM AcONH₄aq.=9/1

Detection : UV 254 nm

Sample1: Toluene, 2. Uracil, 3. Uridine, 4. Adenosine, 5. Guanosine, 6. Cytidine

Injection Volume : 1 μ L

- Least polar analyte (toluene) elutes first
- Polar analytes tend to show a stronger interaction with the stationary phase or the water-rich layer near the particles

2.3.2 Reversed phase HPLC

Stationary phases: non-polar

Mobile phases: polar — mostly mixture of an organic solvent + water/aqueous buffer

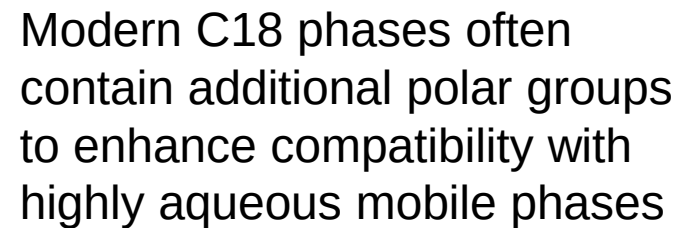
Retention decreases with increasing polarity of the analyte

Retention increases with increasing polarity of the mobile phase

- silica gel with surface modifications (C8 , **C18**, pentafluorophenyl, phenyl, cyano,),
often not stable at $\text{pH} > 8.5$!
- poly(styrene/divinylbenzene) - pH stable!
- porous graphite

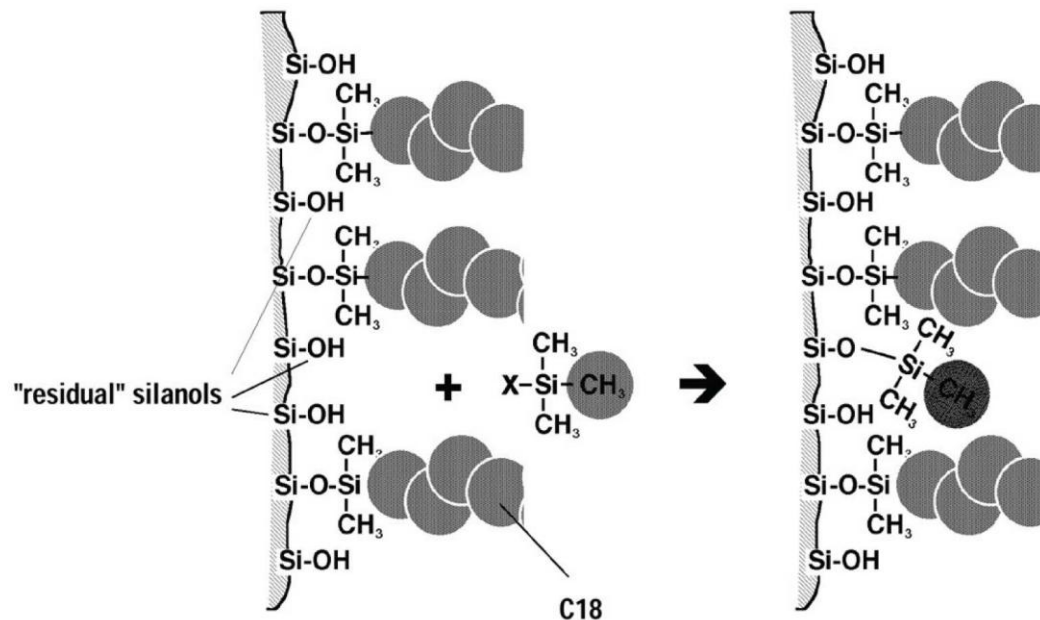
JKU
JOHANNES KEPLER
UNIVERSITÄT LINZ

silica + chlorosilane → C18 silica

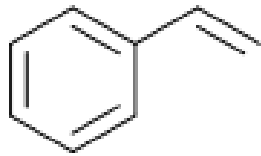


2.3.2 Reversed phase HPLC

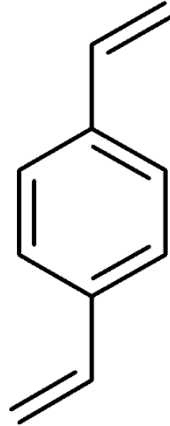
The number of residual free silanol groups can be minimized by a small „endcapping“ reagent (such as trimethylchlorosilane)



2.3.2 Reversed phase HPLC

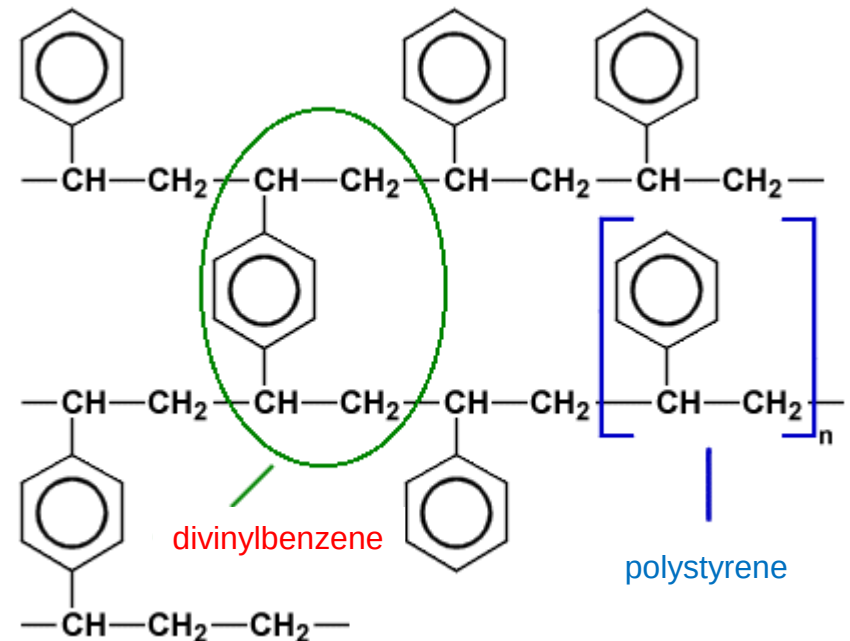


styrene

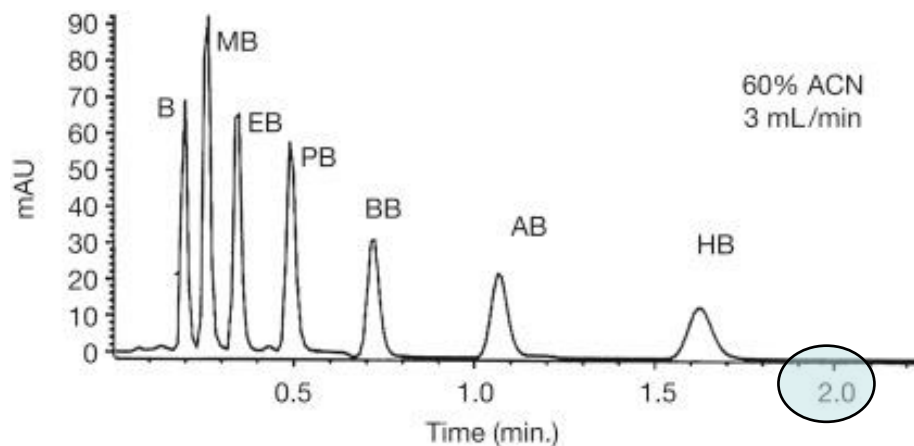


divinylbenzene

amount of divinyl benzene changes
pore size, rigidity,...

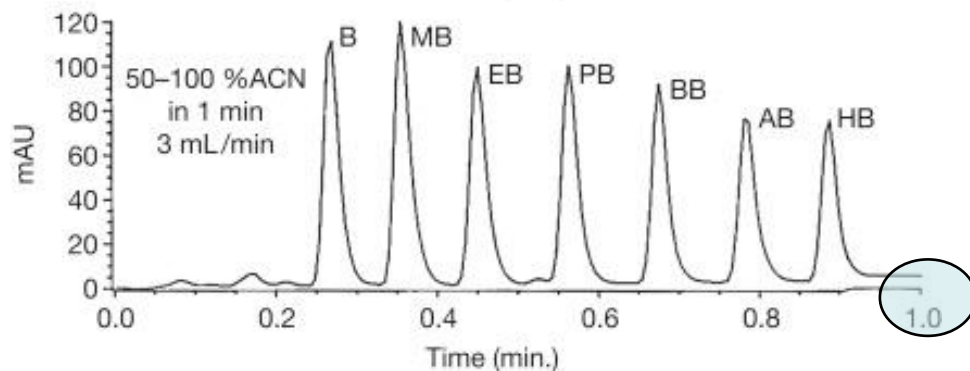


2.3.2 Reversed phase HPLC



Common mobile phases in RP-HPLC are water or aqueous buffers mixed with organic solvents.

If gradient elution is performed the %age of the stronger eluting solvent (the organic solvent) is increased over time



Isocratic (top) and gradient (bottom) separation of alkylbenzenes on a short C18 column (30 mm × 3.0 mm I.D., particle size 3 μ m) with acetonitrile - water mobile phases. Sample: benzene (B), methyl- (MB), ethyl- (EB), propyl- (PB), butyl- (BB), amyl- (AB) and hexyl- (HB) benzenes.

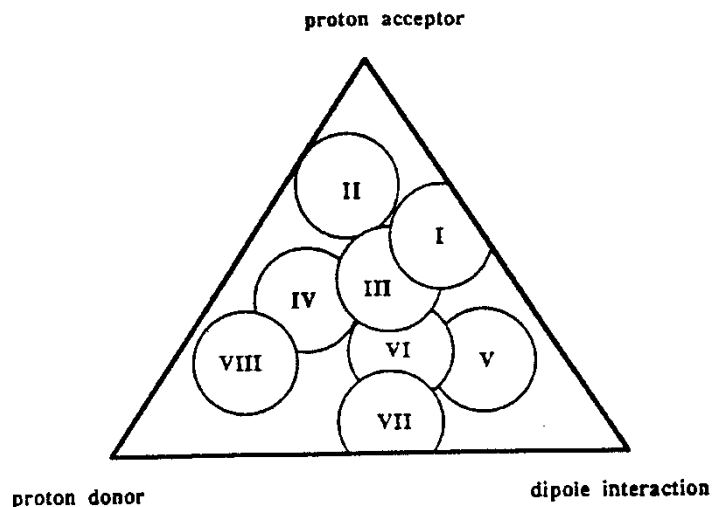
2.3.2 Reversed phase HPLC

Optimization of mobile phases in RP-HPLC

Mobile phase: Mixtures of organic solvent / water

Percentage of water determines elution strength (k), nature of organic solvent determines separation selectivity (α)

Solvent classification by Snyder
(proton acceptor, proton donor,
dipole interaction)



I: aliphatic ethers, trialkylamines,...

II: aliphatic alcohols, **methanol**,...

III: pyrine derivatives, amides, glycol ethers, sulfoxides, **tetrahydrofuran**,...

IV: glycols, benzyl alcohol, formamide,...

V: methylene chloride, ethylene chloride,...

VI: aliphatic ketones and esters, polyethers, dioxane, sulfones, nitriles, **acetonitrile**,...

VII: aromatic hydrocarbons, nitro compounds, aromatic ethers,...

VIII: fluoroalkanols, chloroform, water

2.3.3 Size exclusion chromatography JKU

JOHANNES KEPLER
UNIVERSITÄT LINZ

Size exclusion chromatography (SEC) is a technique for separation of high-molecular-mass analytes

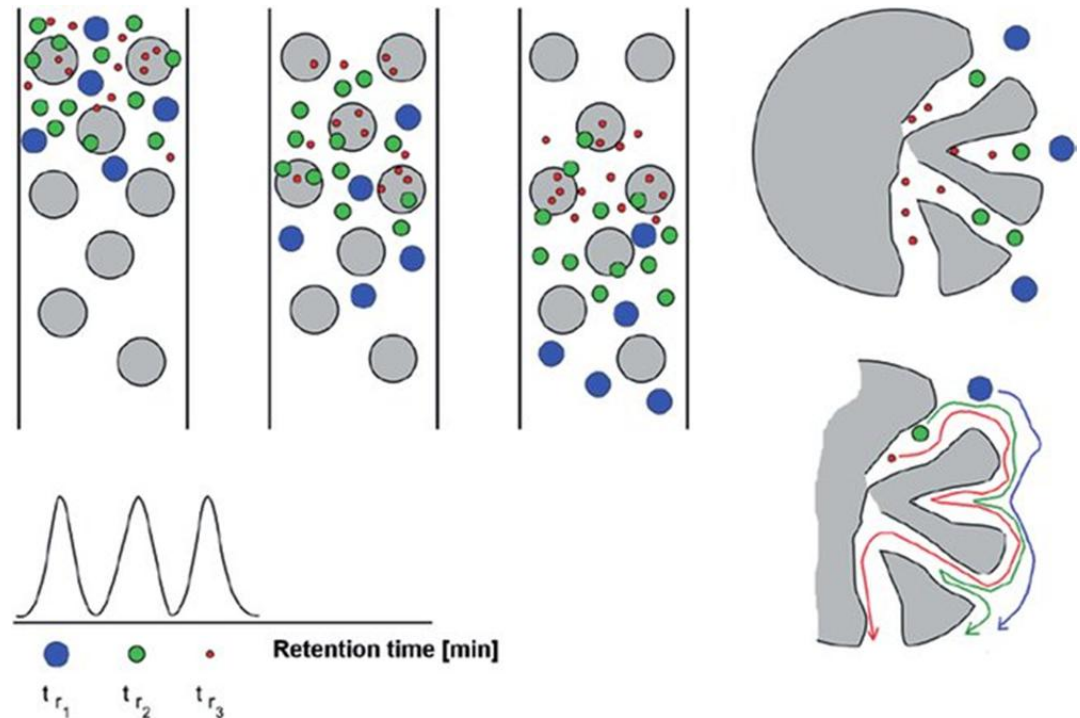
Stationary phase consists of porous particles with well-defined pore size

Retention differences are controlled by the extent to which analytes can diffuse into the pore structure of the stationary phase, which depends on the ratio of molecular dimensions to the pore-size diameters.

Compared with other LC retention mechanisms, a unique feature of SEC is that there are no true interactions of the analytes with the stationary phase

Typical stationary phases:

- poly(styrene/divinylbenzene)
- polymethacrylate
- silica gel and others



2.3.3 Size exclusion chromatography

- “Stationary Phase” - solvent trapped in pores
- “Mobile Phase” - solvent outside of pores

- $V_R = V_I + K_{SEC} V_P$

V_R retention volume

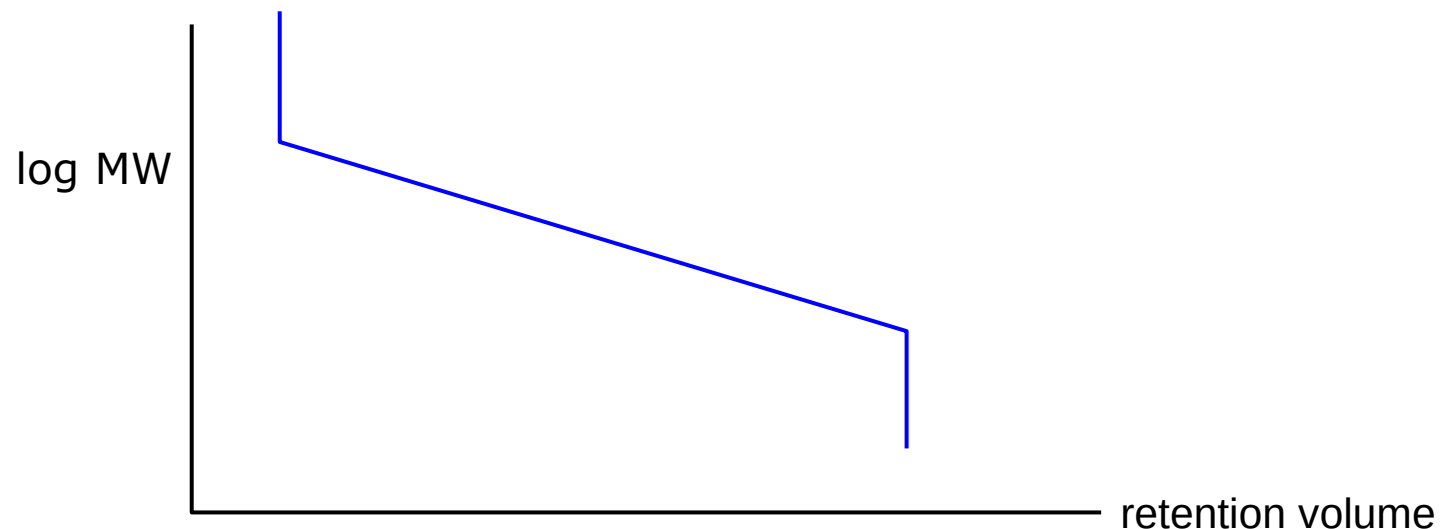
V_I interstitial (interparticle) volume

V_P stationary phase pore volume

$$0 < K_{SEC} < 1$$

if $K > 1$, analyte has adsorbed to sites on stationary phase

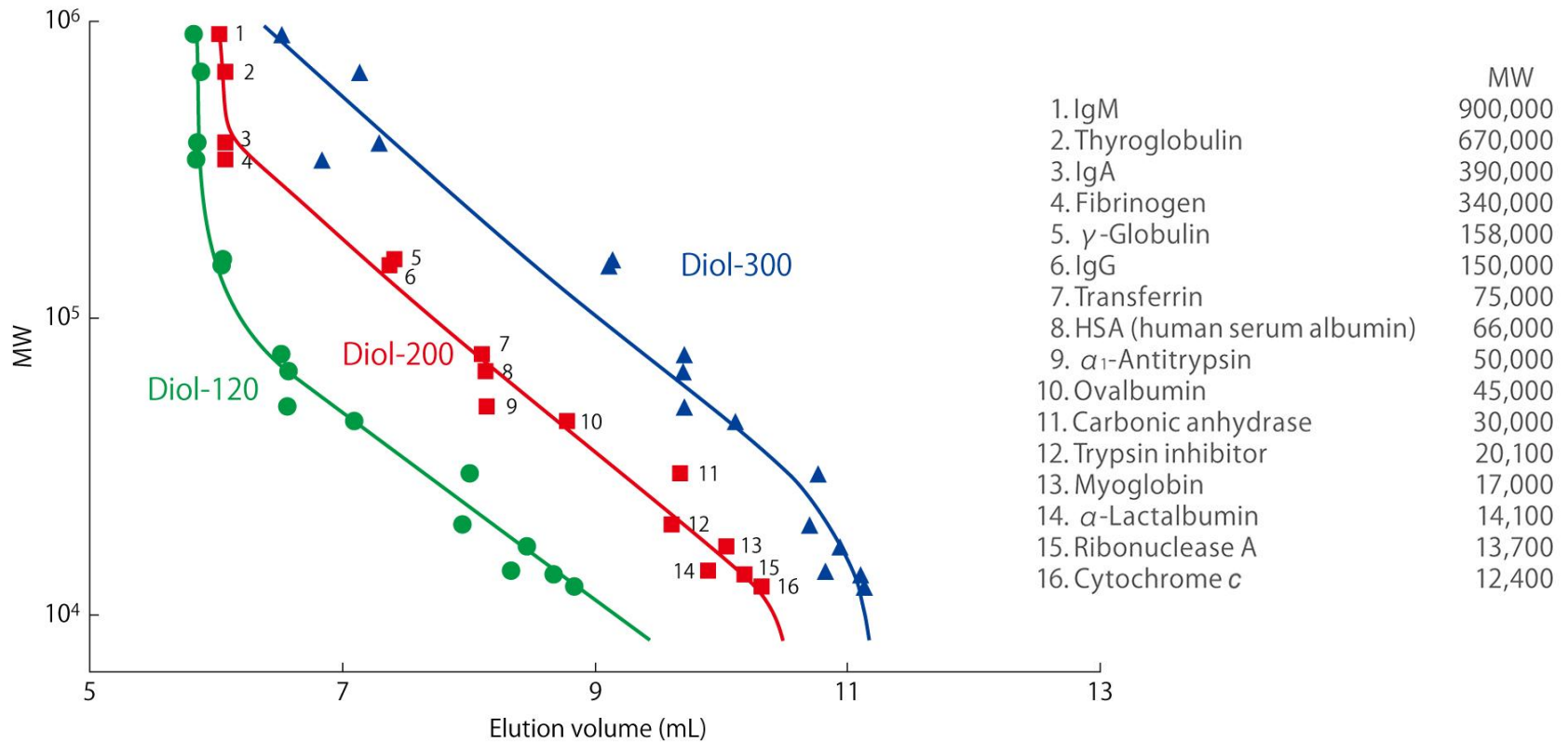
Dependence of elution volume on molecular weight (MW):



2.3.3 Size exclusion chromatography



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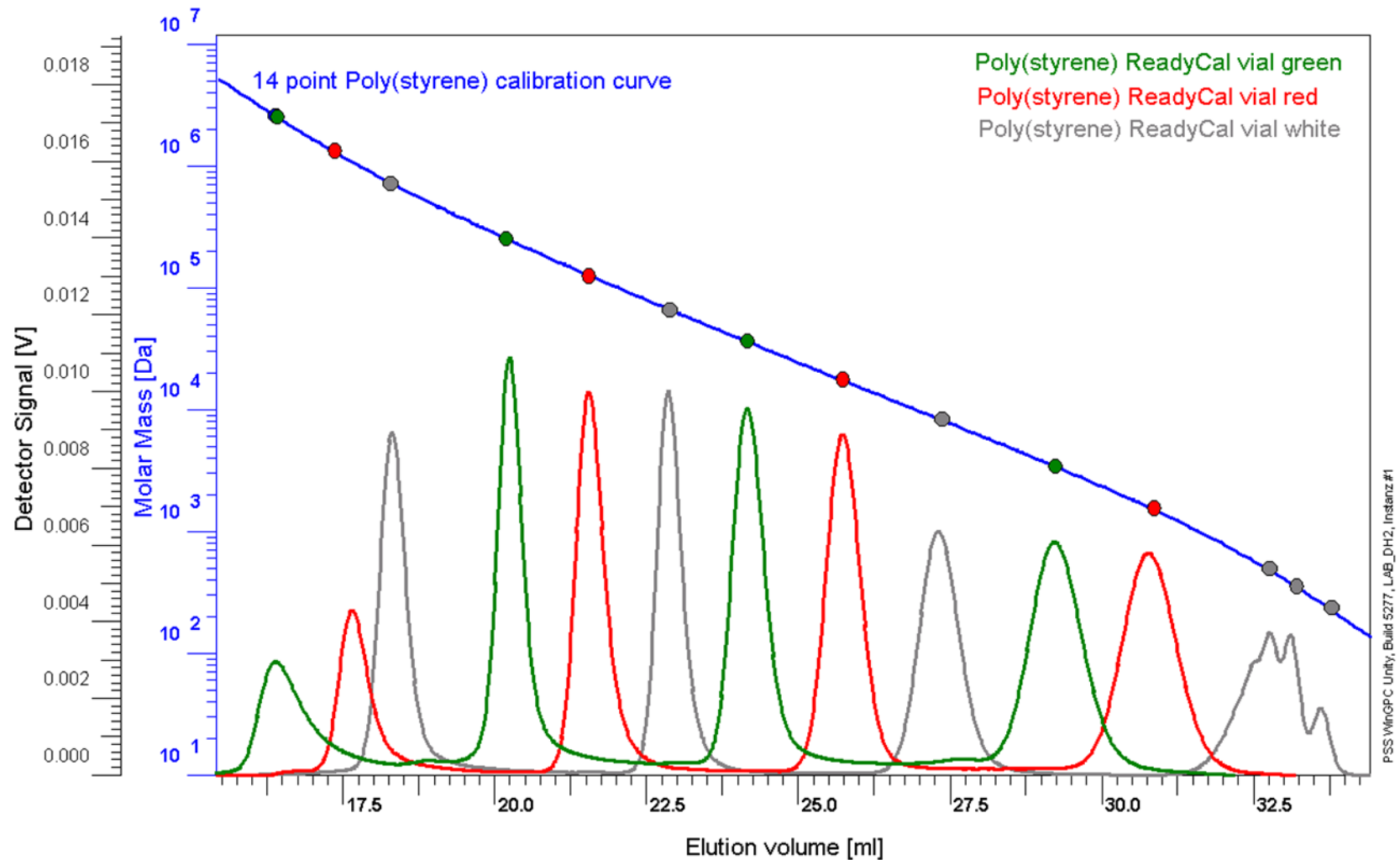


Separation of proteins on columns with different pore size

2.3.3 Size exclusion chromatography



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2.3.3 Size exclusion chromatography JKU

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UNIVERSITÄT LINZ

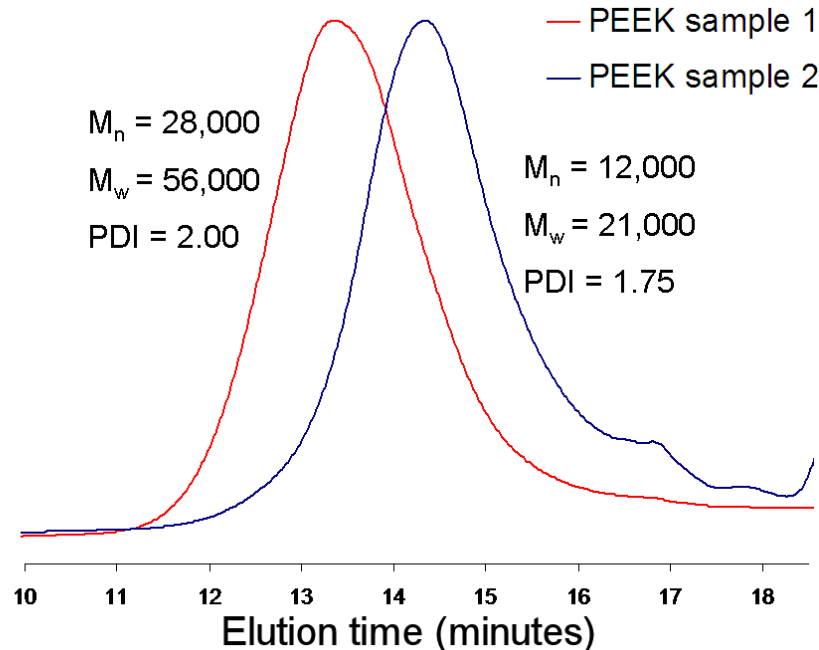
GPC is the same as SEC, but the name is sometimes used specifically for separation of technical polymers (such as polypropylene, polyethylene, polystyrene,

Dissolution of such polymers may be only possible under elevated temperature (PP: mobile phase trichlorobenzene at 160 °C)

$$M_n = \sum x_i \times M_i$$

$$M_w = \sum w_i \times M_i$$

$$w_i = (x_i \times M_i) / M_n$$

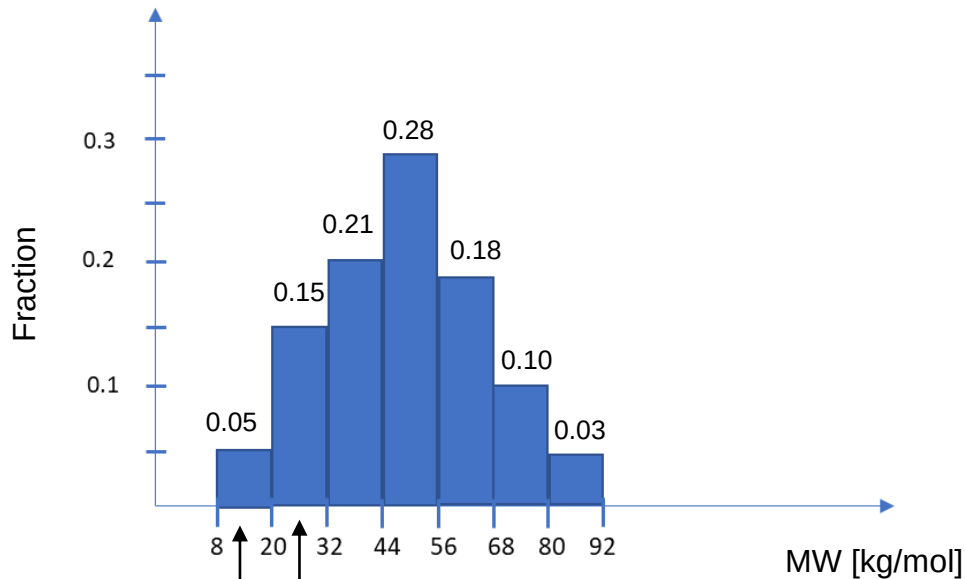


M_nNumber average molecular weight

M_wWeight average molecular weight

2.3.3 Size exclusion chromatography JKU

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$$M_n = \sum x_i \times M_i$$

$$M_w = \sum w_i \times M_i$$

$$w_i = (x_i \times M_i) / M_n$$

x_i Fraction of polymer chains

M_i(average) MW of polymer chains

w_i Weight fraction

$$M_n = 0.05 \times 14 + 0.15 \times 26 + \dots + 0.03 \times 86 = 47.7 \text{ kg/mol}$$

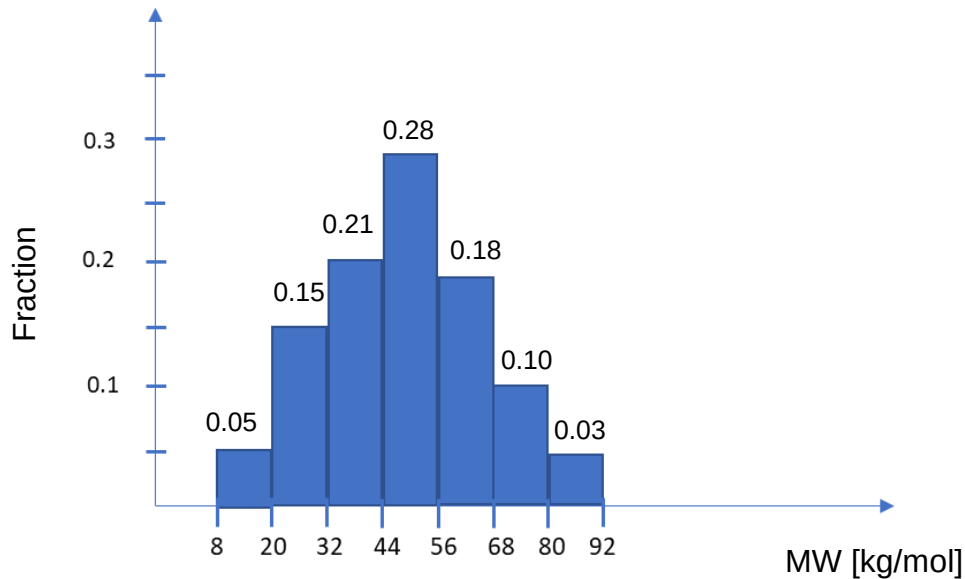
$$w_i = (0.05 \times 14) / 47.7 = 0.0147 \text{ and so on}$$

$$M_w = 0.0147 \times 14 + \dots + 0.054 \times 86 = 53.9 \text{ kg/mol}$$

$$M_w / M_n = \text{Polydispersity Index} \sim 1.13$$

2.3.3 Size exclusion chromatography JKU

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$$M_n = \sum x_i \times M_i$$

$$M_w = \sum w_i \times M_i$$

$$w_i = (x_i \times M_i) / M_n$$

x_i Fraction of polymer chains
 M_i (average) MW of polymer chains
 w_i Weight fraction

$$M_n = 0.05 \times 14 + 0.15 \times 26 + \dots + 0.03 \times 86 = 47.7 \text{ kg/mol}$$

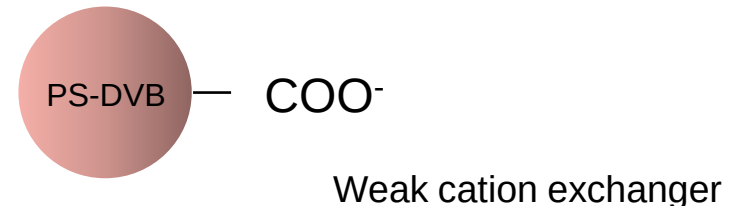
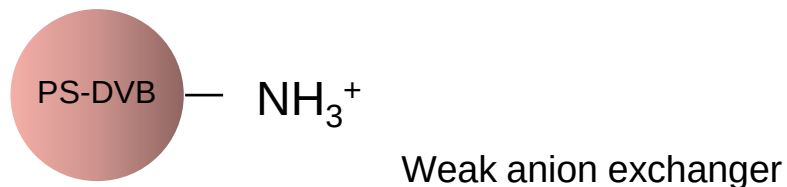
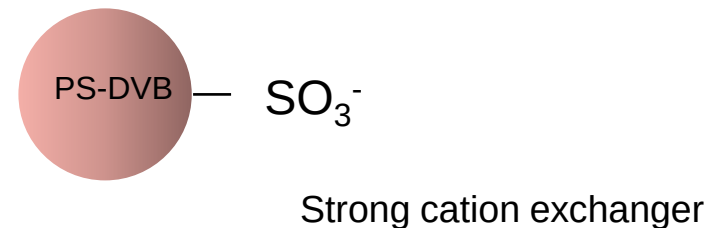
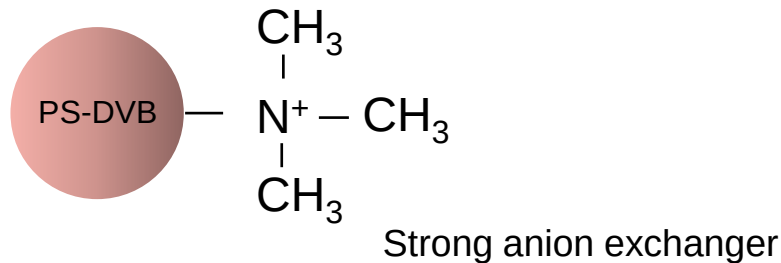
Assumption: the MW of a single unit used for polymerization equals 62 g/mol.
 So the degree of polymerization (Dp) can be calculated as:

$$Dp = M_n / M_{(\text{unit})} = 47.700 \text{ g/mol} / 62 \text{ g/mol} = 769$$

So polymer chains consist of 769 repeating units on average

2.3.4 Ion exchange chromatography

Stationary phases in ion-exchange chromatography



Particles mostly made from a polymeric resin

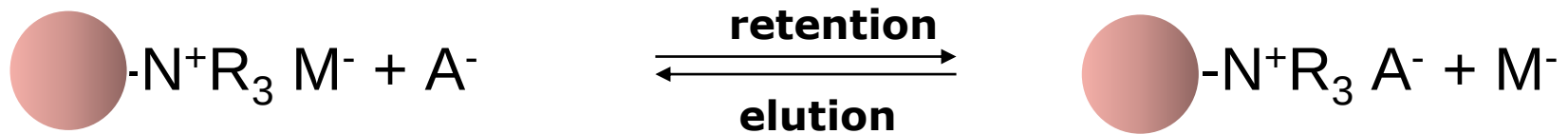
Mobile phase for

cation exchange chromatography: acids

anion exchange chromatography: bases

2.3.4 Ion exchange chromatography

Example: Separation of anions (A^- analyte, M^- mobile phase anion)



the equilibrium constant of this reaction is often called selectivity constant of A^-

Retention time of A^- increases

- with decreasing concentration of the mobile phase
($\log k = \text{invers prop } [M^-]$)
So gradient elution is performed by increasing the concentration of the mobile phase over time
- with increasing capacity of the stationary phase
(mmoles functional group per gram)
- with increasing selectivity constant

2.3.4 Ion exchange chromatography

Ion chromatography (IC) is a special case of ion-exchange chromatography....

...separation of low-molecular-mass ions with conductivity detection



Signal
(change of conductivity)

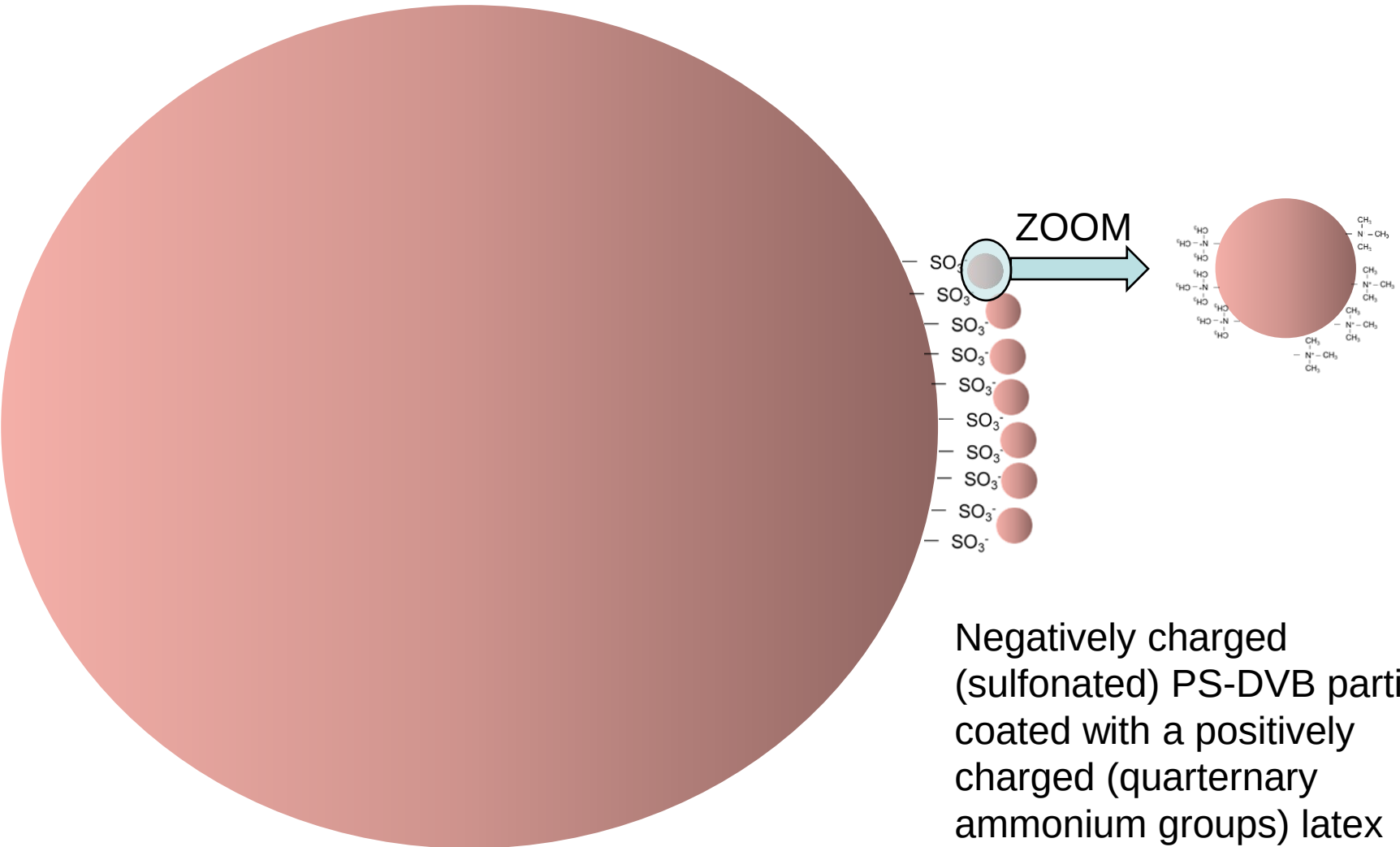
$$\Delta\kappa = \frac{(\lambda_{\text{A}^-} - \lambda_{\text{M}^-}) \cdot c_{\text{eq}}^{\text{A}} \cdot \alpha_{\text{A}}}{10^{-3}}$$

In the worst case, the difference between λ_{A^-} and λ_{M^-} may be zero !

(in the same way, indirect UV detection can be used, if a UV-active ion is employed in the mobile phase)

2.3.4 Ion exchange chromatography

Modern stationary phases for **anion-** chromatography



2.3.4 Ion exchange chromatography

Ion chromatography with suppressed conductivity detection

suitable mobile phases for IC of anions:

NaOH

$\text{NaHCO}_3/\text{Na}_2\text{CO}_3$

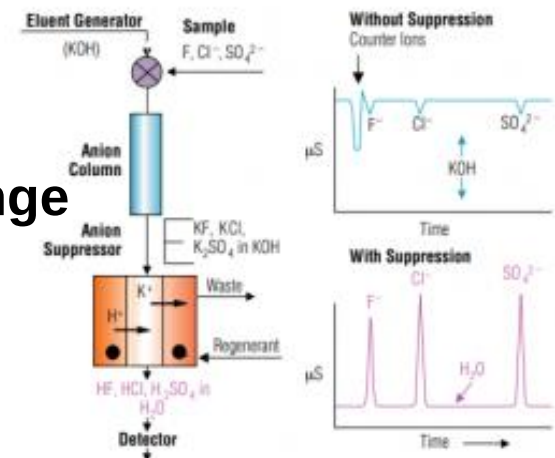
After the separation column a suppressor converts the mobile phase into species of low conductivity and enhances the conductivity of the analyte

Suppressors (for anion separations):

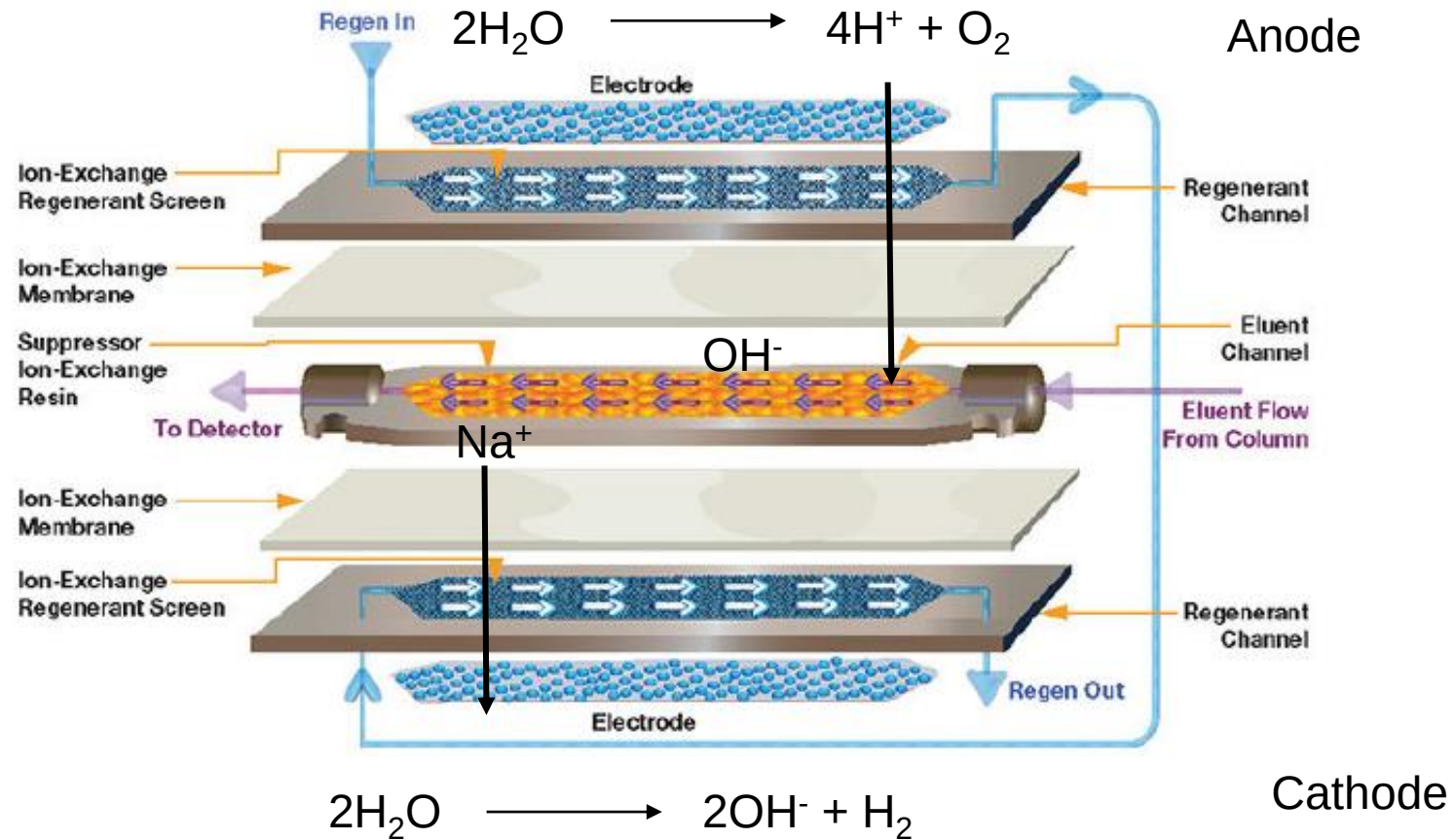
- cation exchange column in the H^+ form
- **membrane suppressor with a cation-exchange membrane**

NaOH (Na carbonate) is converted into water (CO_2)

Na^+A^- is converted into H^+A^-



2.3.4 Ion exchange chromatography



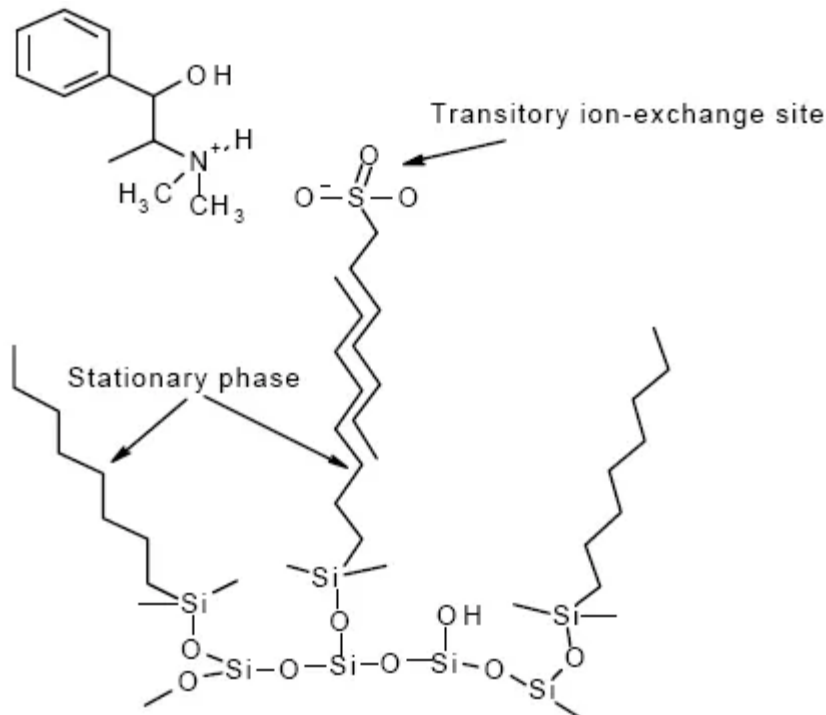
Electrolytically regenerated suppressor

2.3.5 Ion pair chromatography

Ionic analytes can be separated on RP-materials, if a hydrophobic counterion is added to the mobile phase

for anionic analytes: hydrophobic alkyltrimethylammonium salts

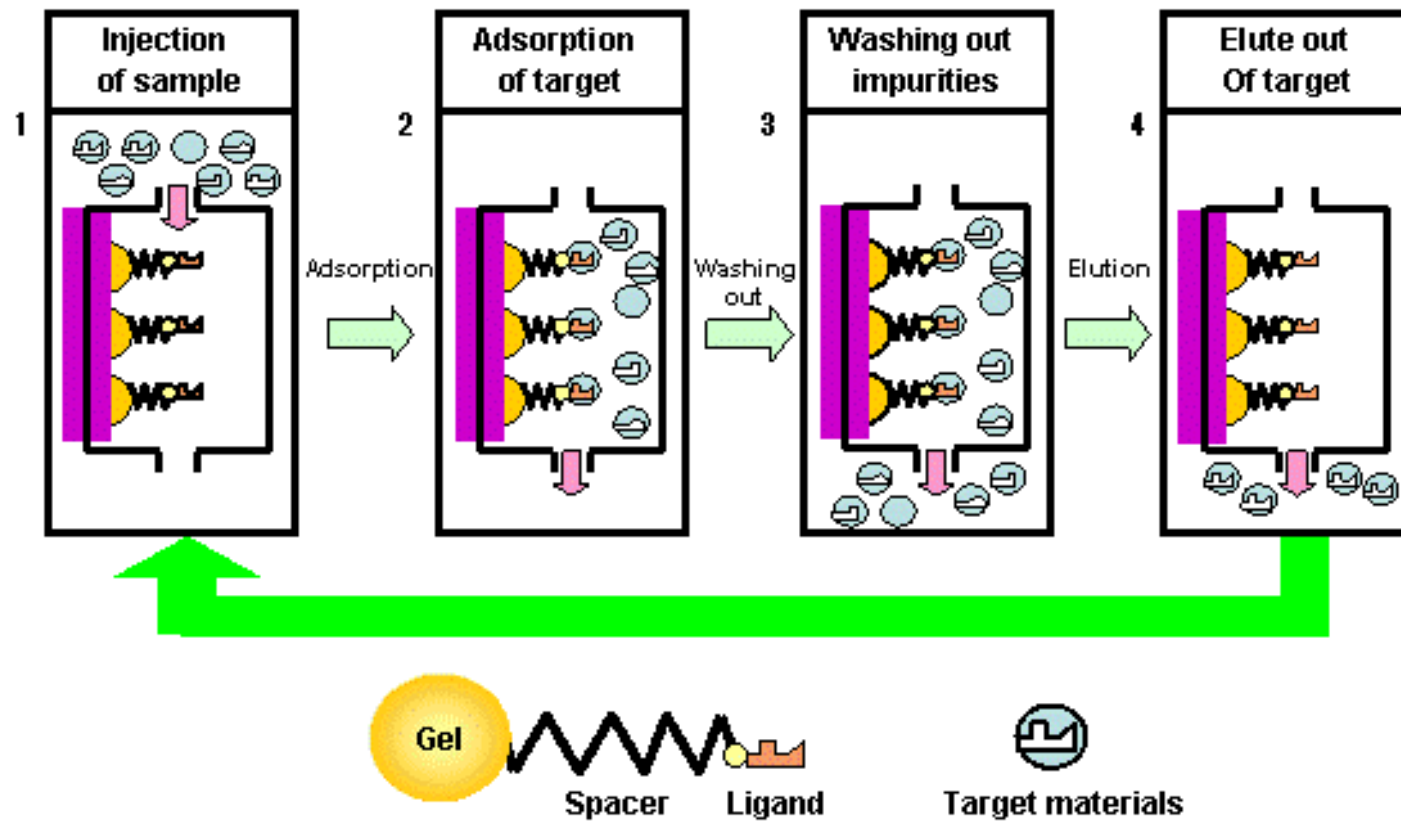
for cationic analytes: hydrophobic alkylsulfonates



Phenomenex

- An alternative way for explaining Ion pair chromatography would be the formation of ion-pairs in the mobile phase that subsequently interact with the non-polar groups of the stationary phase

2.3.6 Affinity chromatography

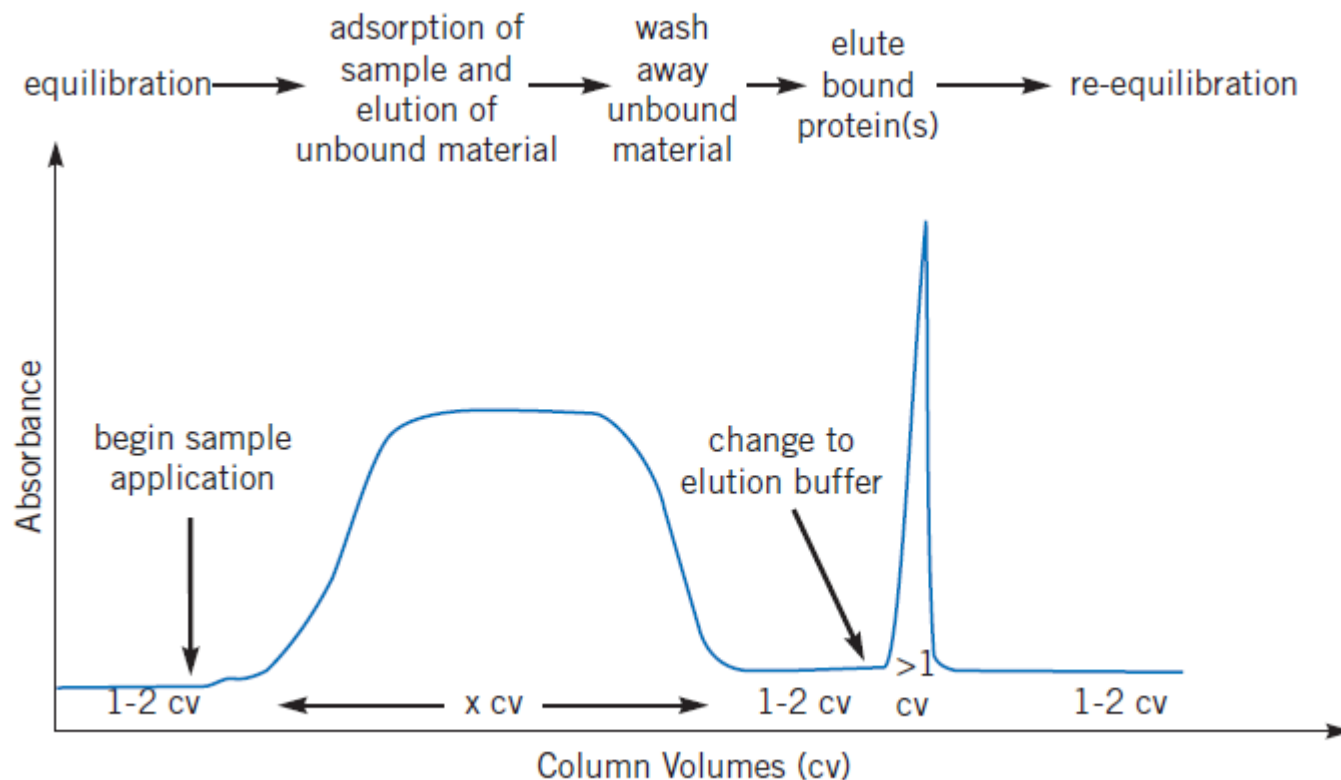


Interactions: antigen and antibody, enzyme and substrate, receptor and ligand, or protein and nucleic acid

2.3.6 Affinity chromatography

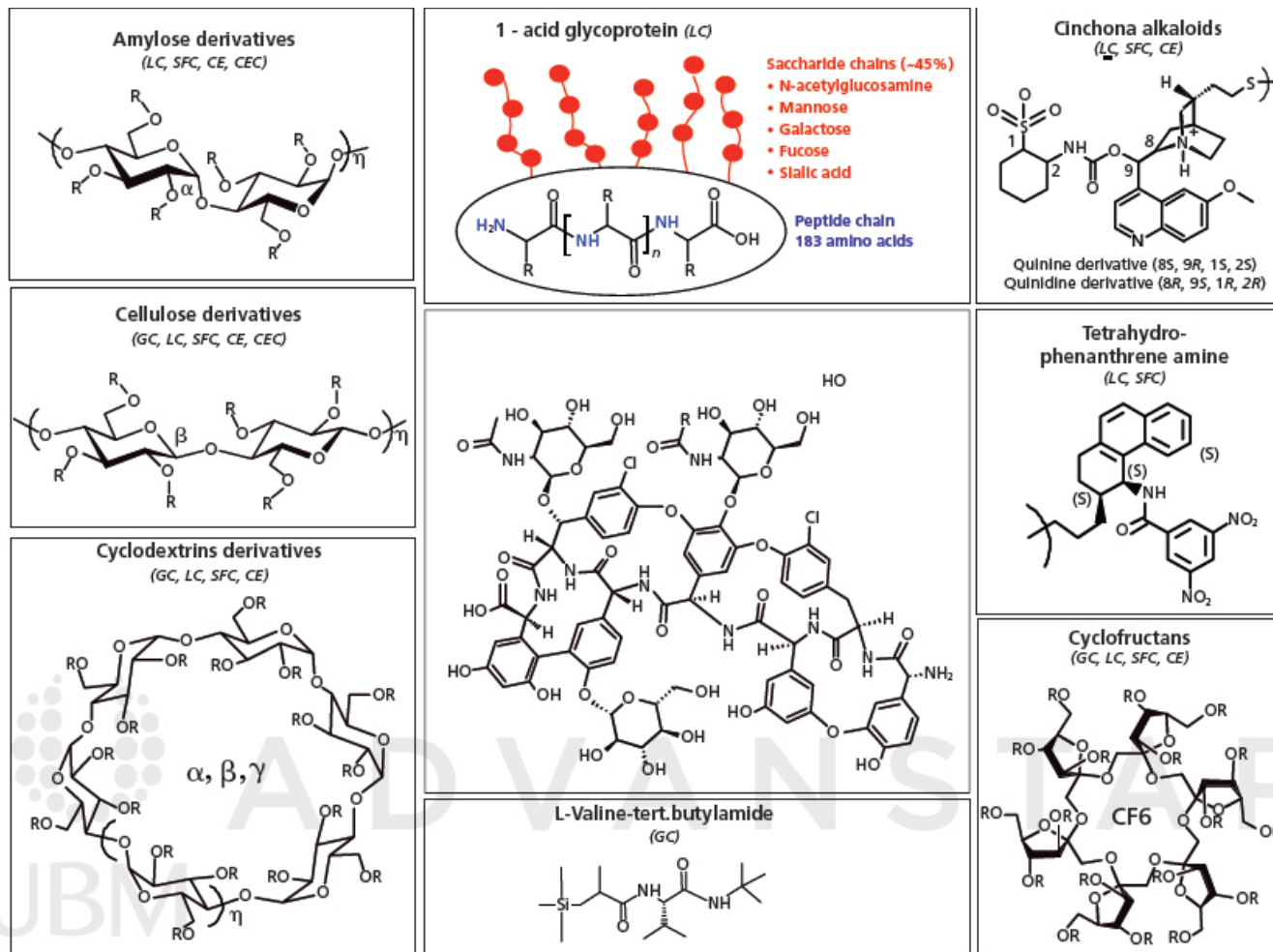
bound species can be released by changing pH, temp, ionic strength, etc.

Example chromatogram:

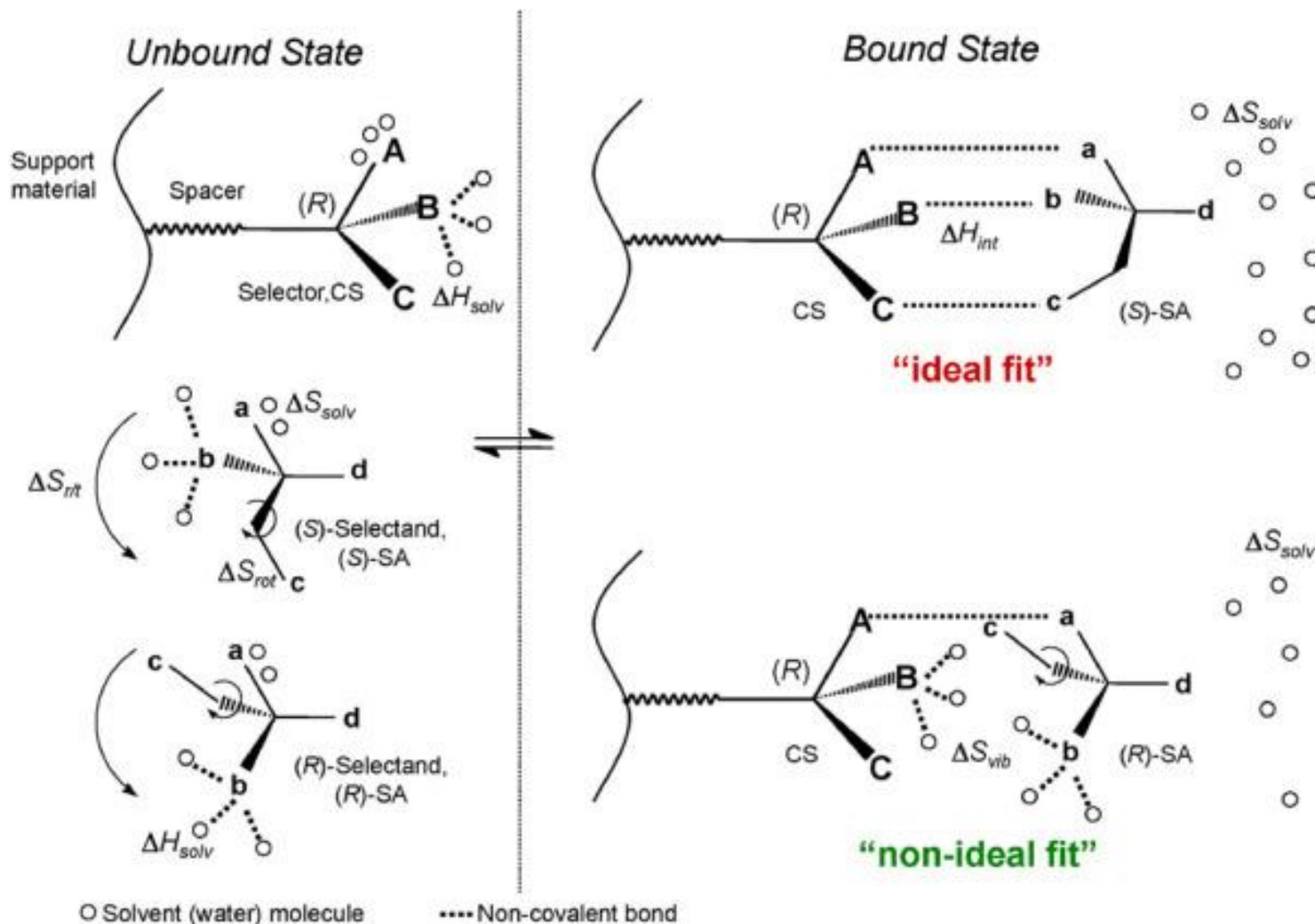


2.3.7 Chiral separations

- Stationary phase with chiral groups



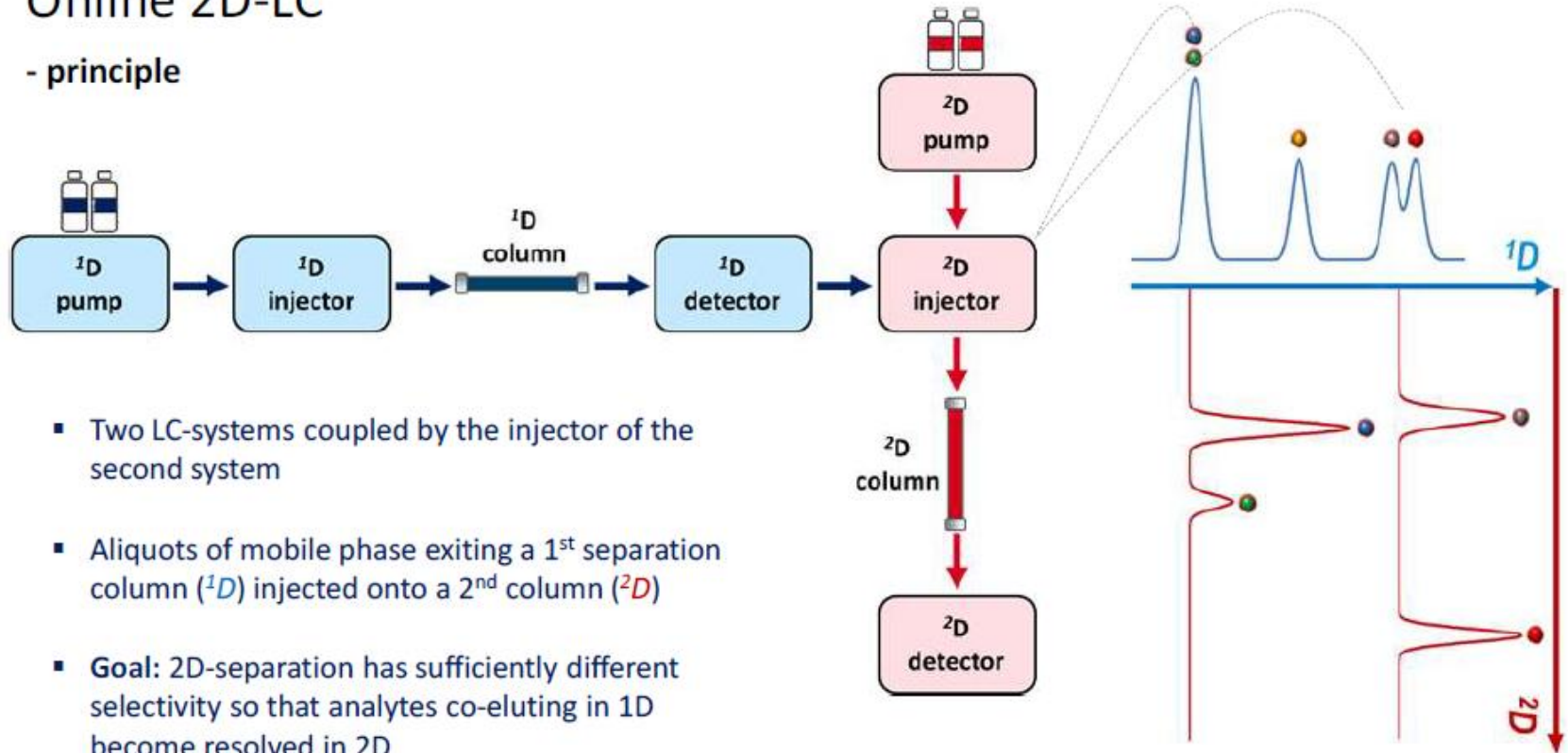
2.3.7 Chiral separations



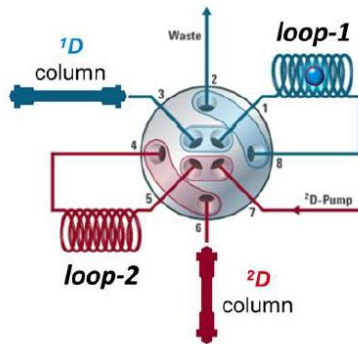
2.3.8 2D-Separations in Liquid Phase

Online 2D-LC

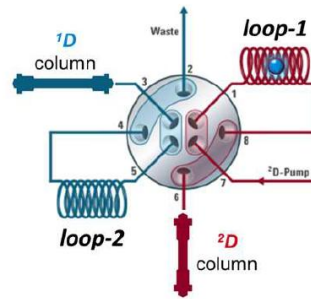
- principle



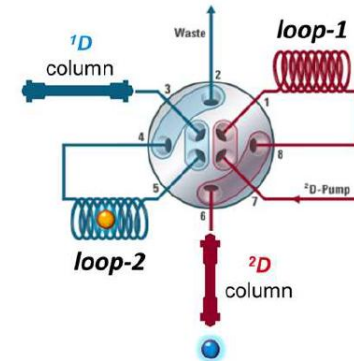
2.3.8 2D-Separations in Liquid Phase



Eluate from column 1 is filled into Loop 1



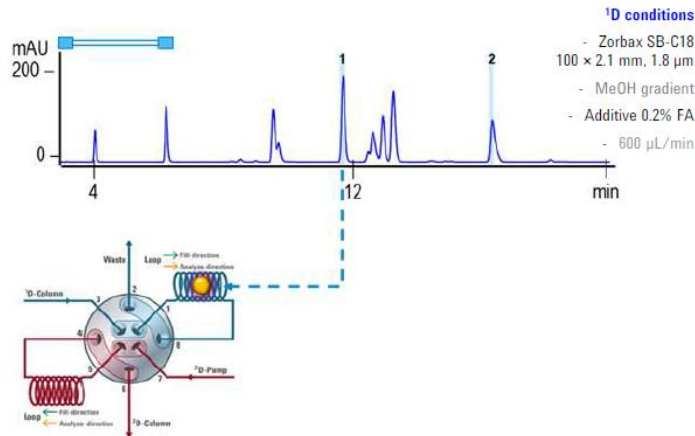
Loop 1 is eluted to column 2 while eluate from column 1 is filled into Loop 2



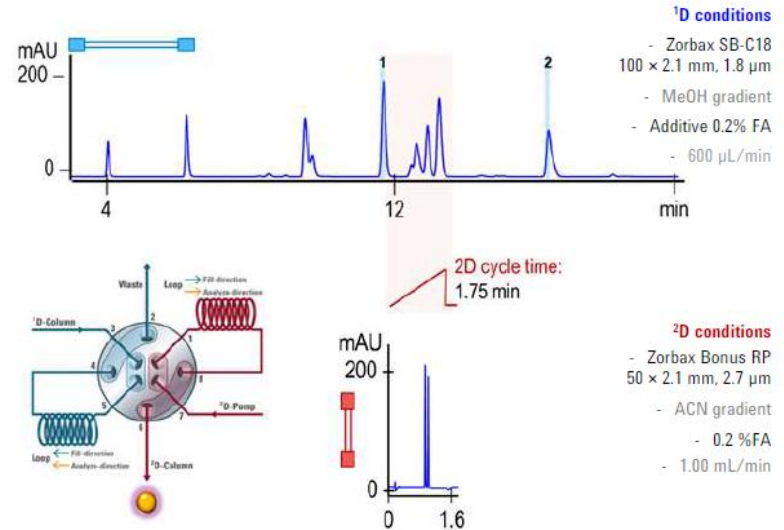
Separation in column 2 performed –

2.3.8 2D-Separations in Liquid Phase

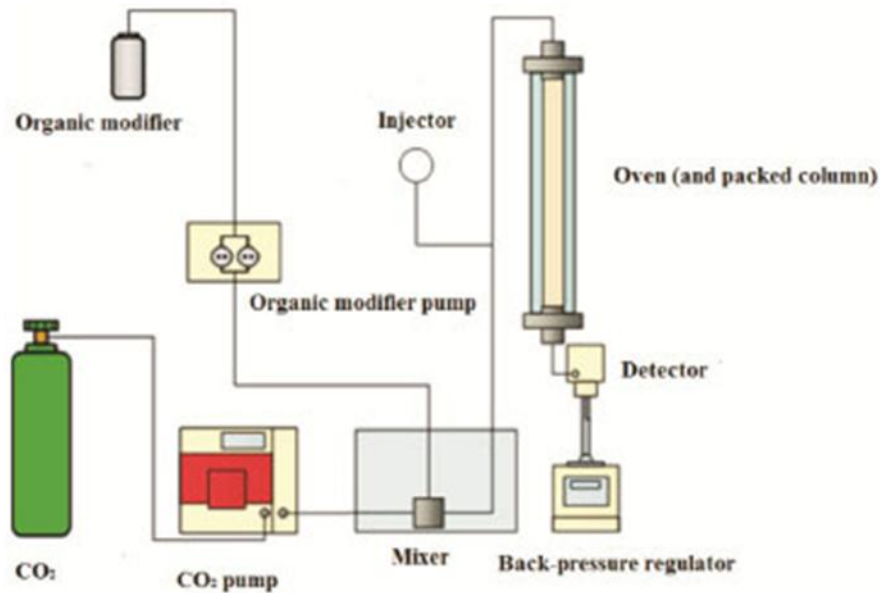
Heart-Cutting 2D-LC



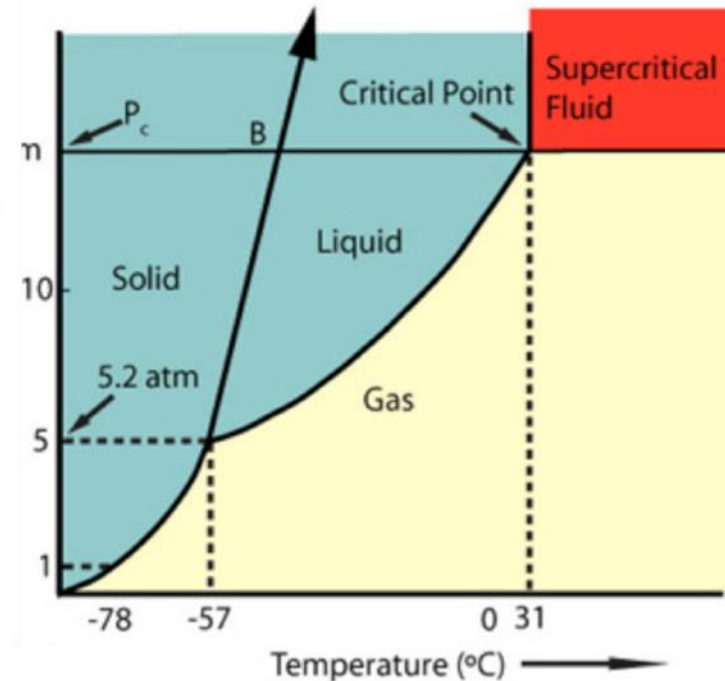
Heart-Cutting 2D-LC



2.4 Supercritical fluid chromatography



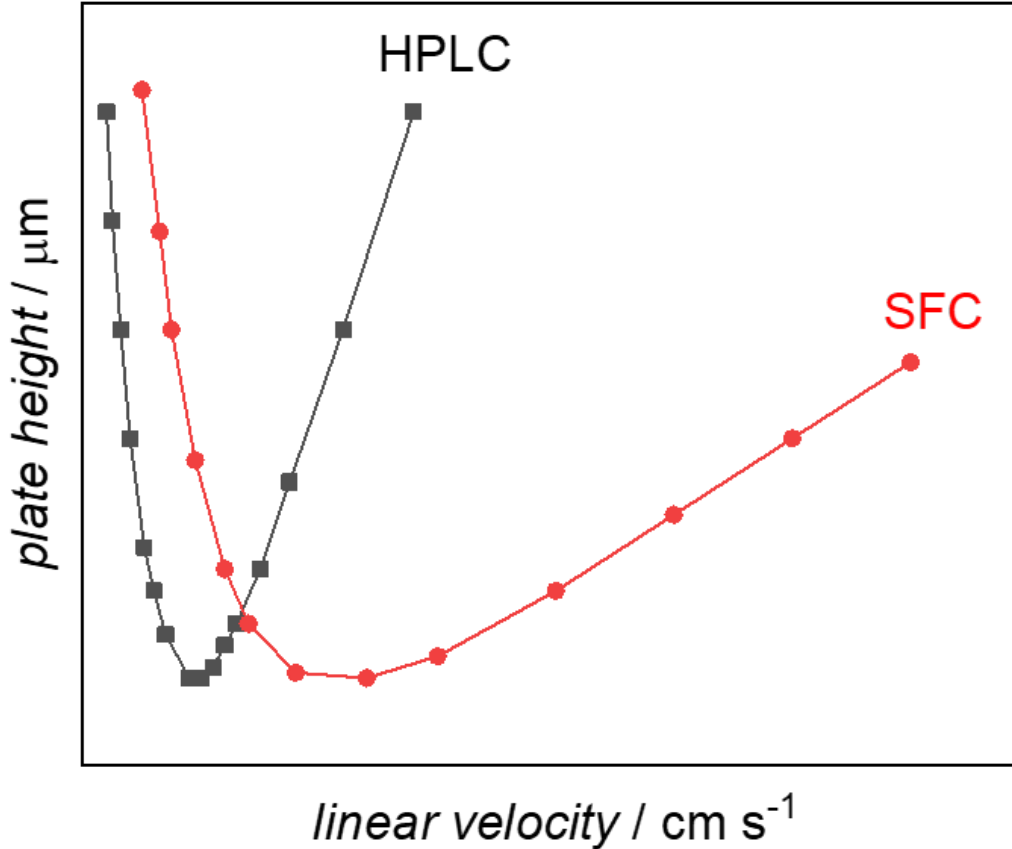
Instrumentation



Phase diagram of CO₂

Particularly useful for preparative chromatography – for recovering the analyte no evaporation of mobile phase needed!

2.4 Supercritical fluid chromatography

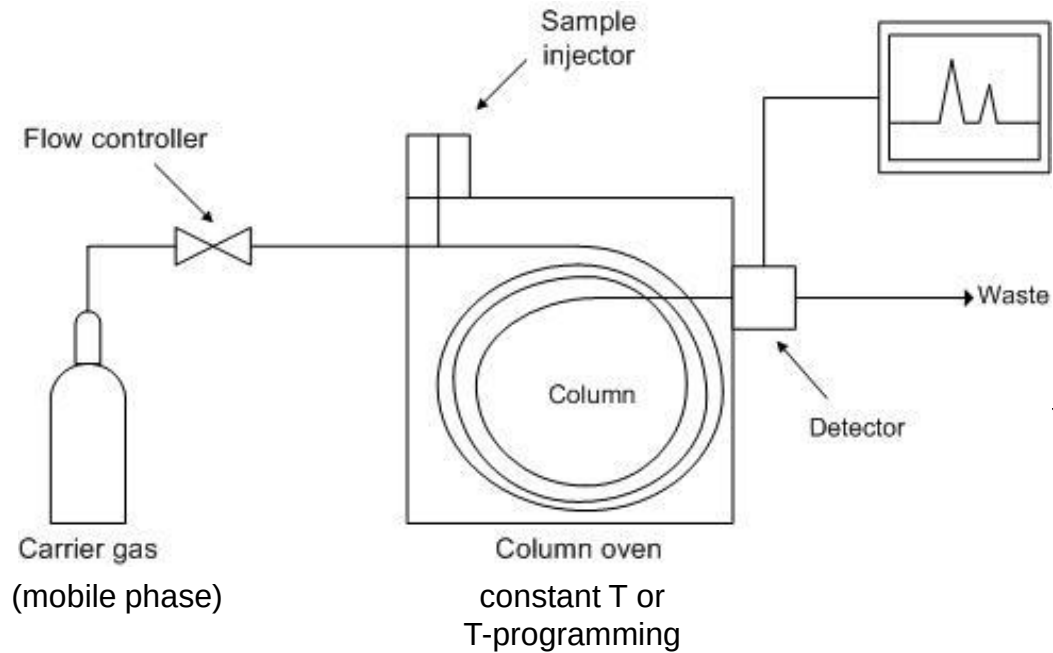


Can be used with much higher linear flow velocity of mobile phase!

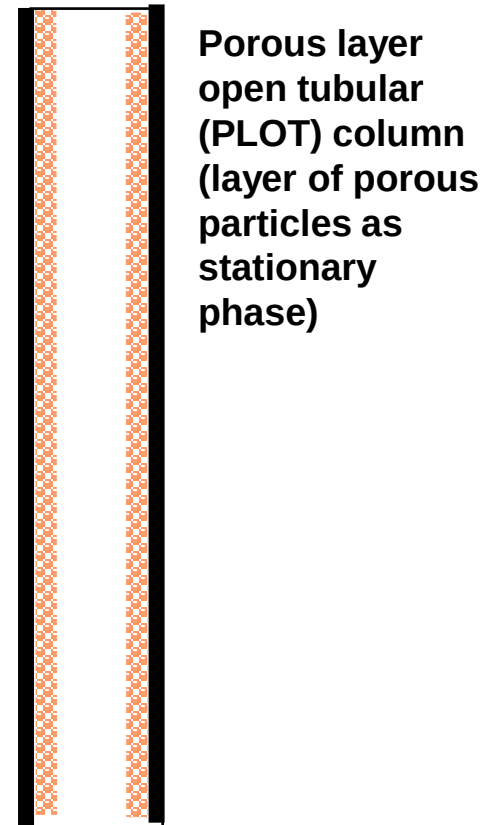
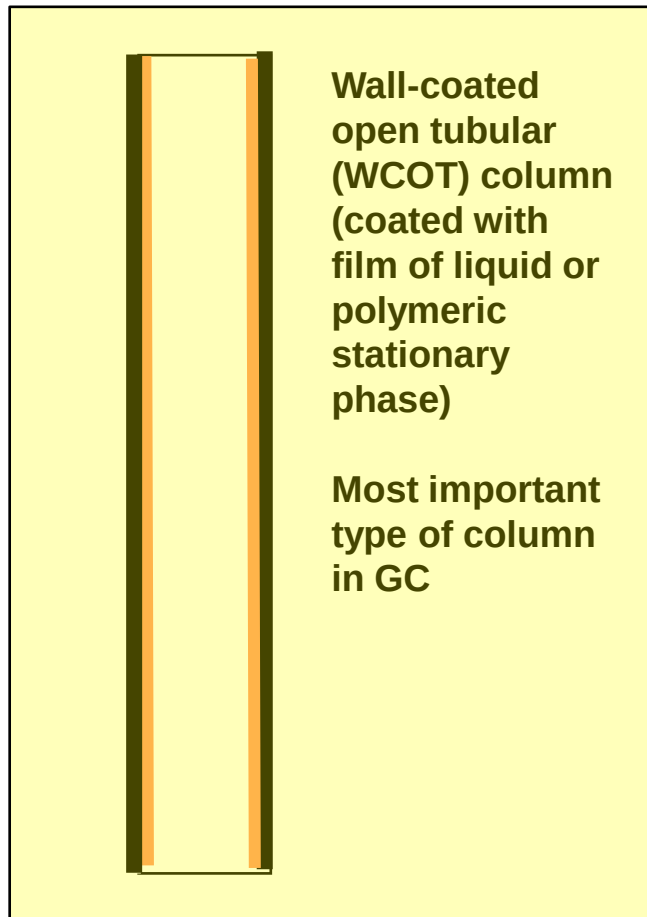
3 Gas chromatography

- Gaseous mobile phase: (He, N₂, or H₂)
- Stationary phase:
 - non-volatile liquid (polymer)
(gas/liquid partition chromatography)
 - solid particles
(gas/solid adsorption chromatography)
- Analytes must be vaporized prior to entering the separation column

3 Gas chromatography



3.1 Gas chromatography: Columns



3.1 Gas chromatography: Columns

Fused silica capillaries

(protective outer coating of polyimide)

10-100 m length

Inner diameter

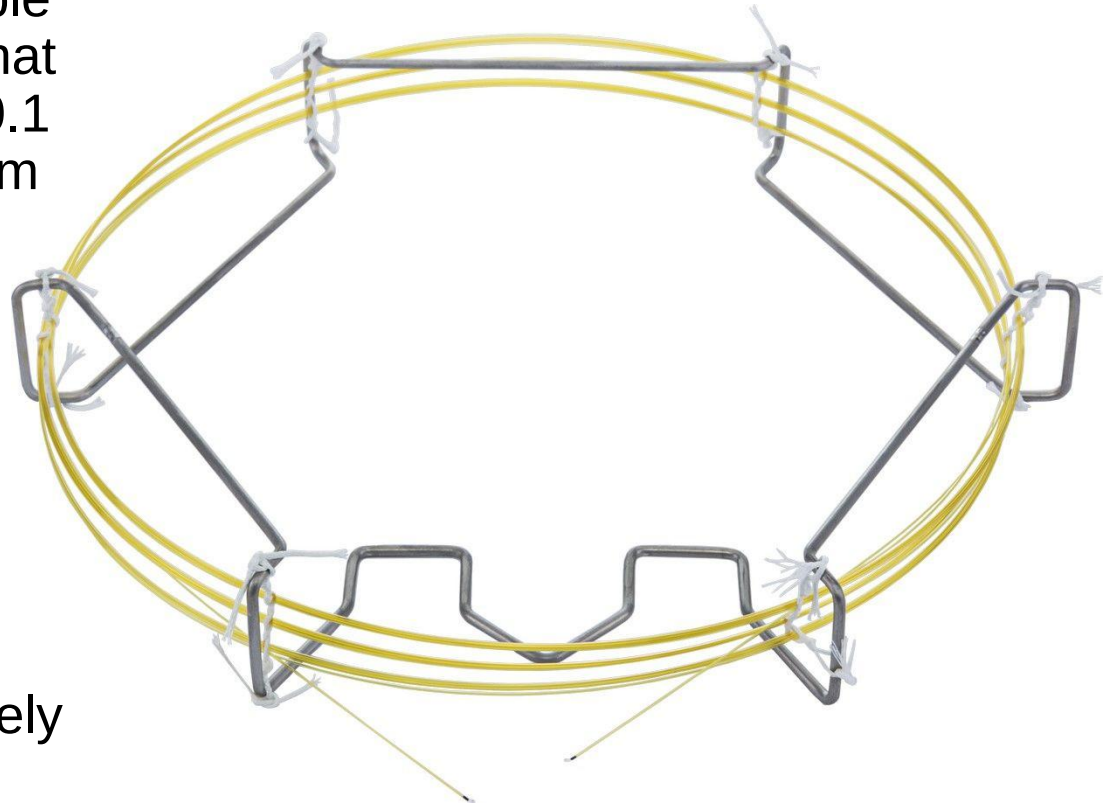
0.10 mm	microbore columns
0.18 mm	minibore columns
0.25 mm	narrow bore columns
0.32 mm	wide bore columns
0.53 mm	megabore columns



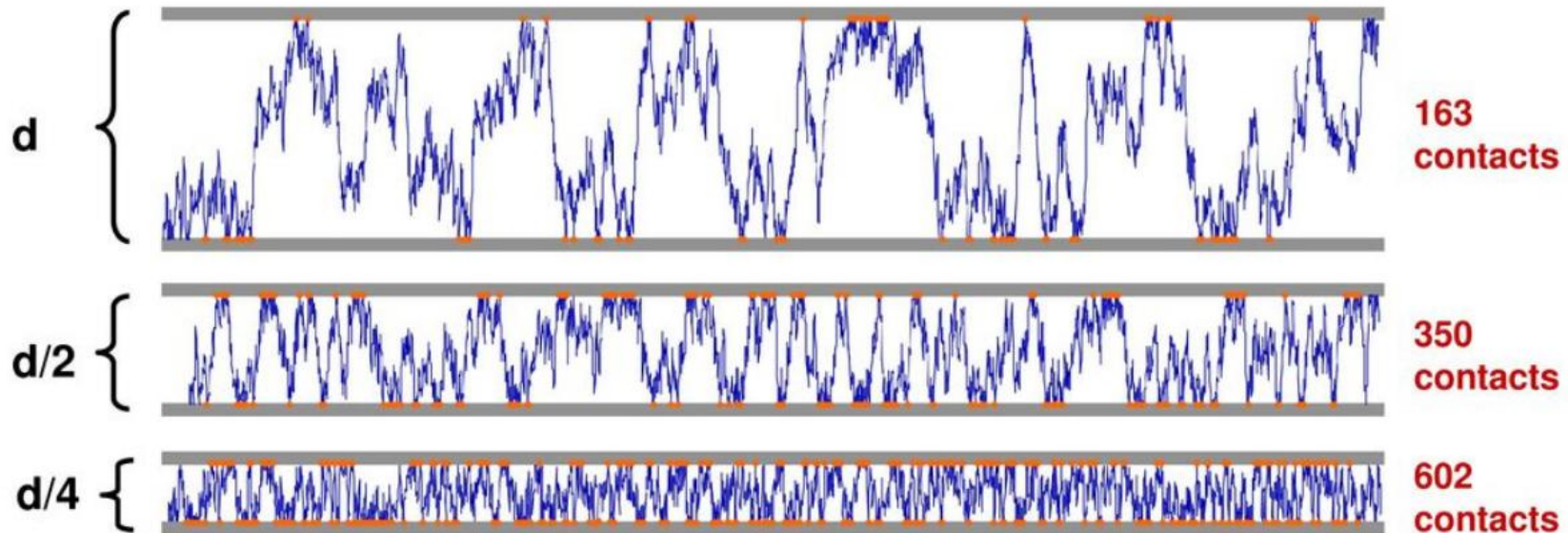
Film thickness of
WCOT columns:
0.1 ... 5 μm

3.1 Gas chromatography: Columns

- small film thickness and small inner diameter result in high separation efficiency, but sample capacity (amount of analytes that can be injected) is small - i.d. 0.1 mm and film thickness of 0.1 μm allow a sample capacity of approximately 1 ng
- increased sample capacity is achieved with increased film thickness and larger inner diameter - (i.d. 0.53 mm and film thickness of 5 μm allow a sample capacity of approximately 2500 ng)



3.1 Gas chromatography: Columns



Small film thickness and **small inner diameter** result in high separation efficiency. **Small inner diameter** increases the number of contacts that will be established between the analyte and the stationary phase.

Small film thickness is beneficial for a low C-term in the van Deemter equation – but it may lead to insufficient k values for fast eluting analytes.

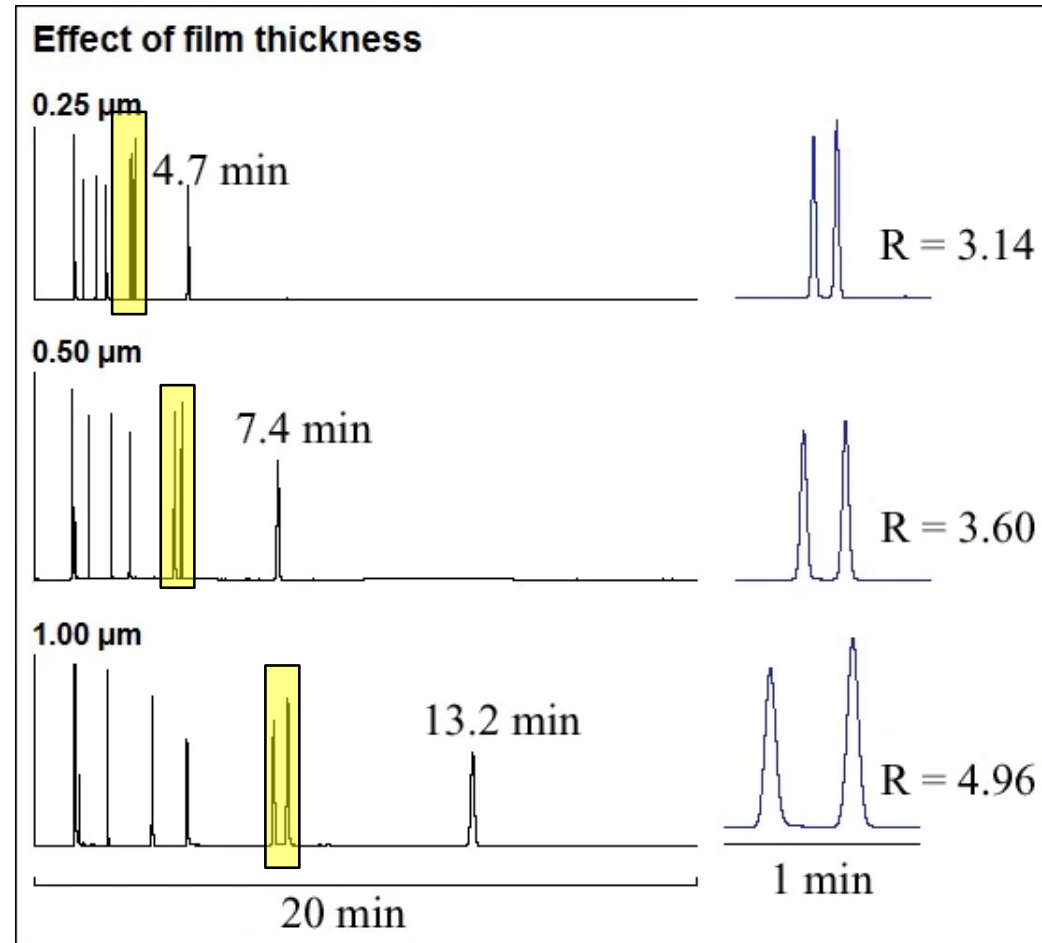
3.1 Gas chromatography: Columns

In gas/liquid partition chromatography, k values depend on the vapour pressure of the analytes (which depends on the temperature)

Increase of film thickness of stationary phase results in higher k

$$k = K \frac{V_s}{V_m} = \frac{\text{moles}_s}{\text{moles}_m}$$

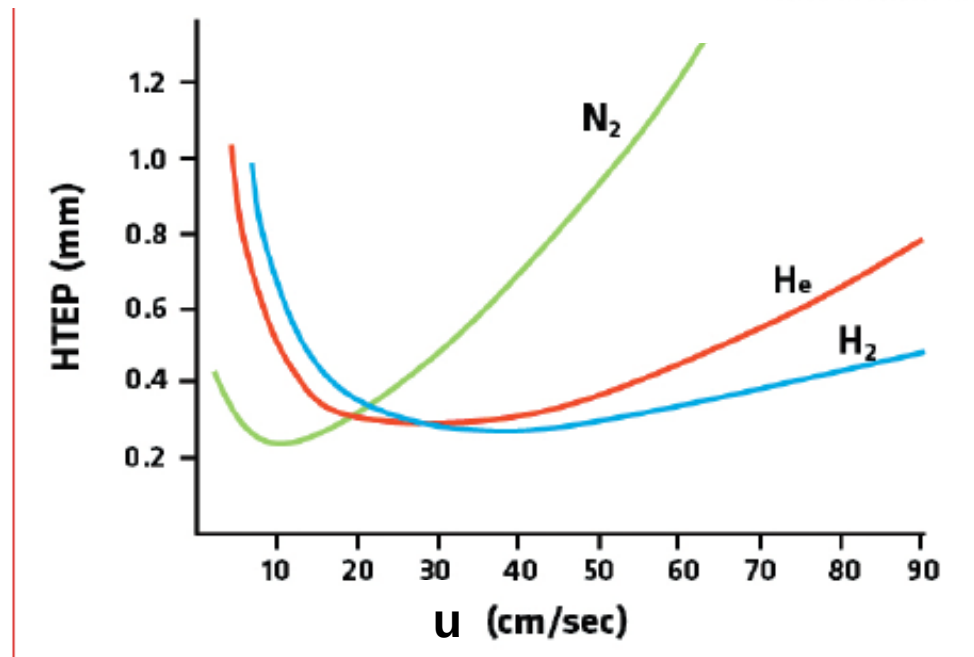
(see „Theory of Chromatography“)



3.2 Gas chromatography: Mobile Phase

Helium
Nitrogen
Hydrogen

Gas purification systems (cartridges) may be necessary:
Mole sieves (removal of water)
Carbon (removal of hydrocarbons)
Dispersed metallic copper (removal of oxygen)



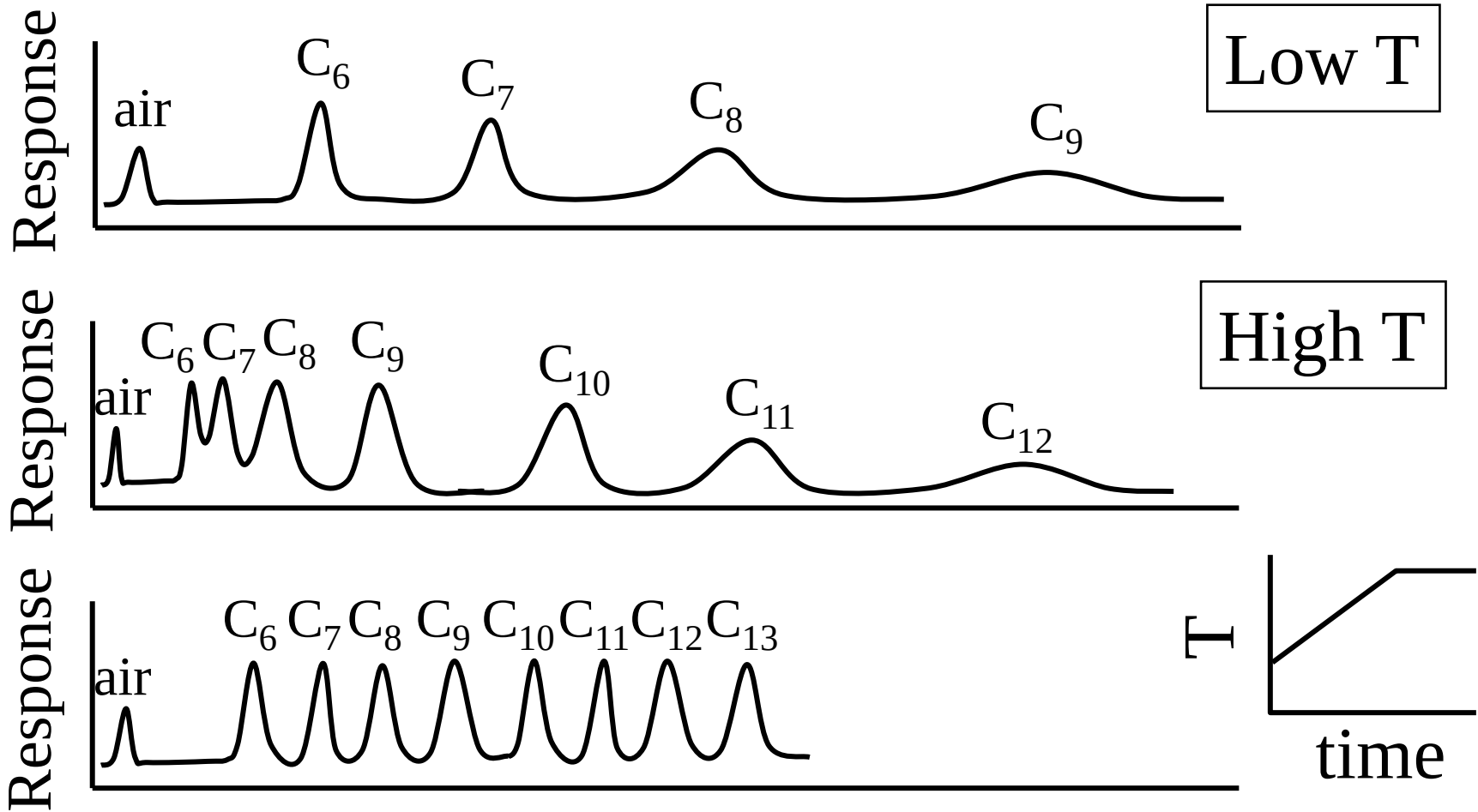
Optimum of linear velocity at about 30 cm/s

Suggestions for flow rates (ml / min):

Inner diameter	Hydrogen	Helium
0.10 mm	1.0	0.5
0.25 mm	2.5	1.3
0.32 mm	3.2	1.9
0.53 mm	5.3	4.2

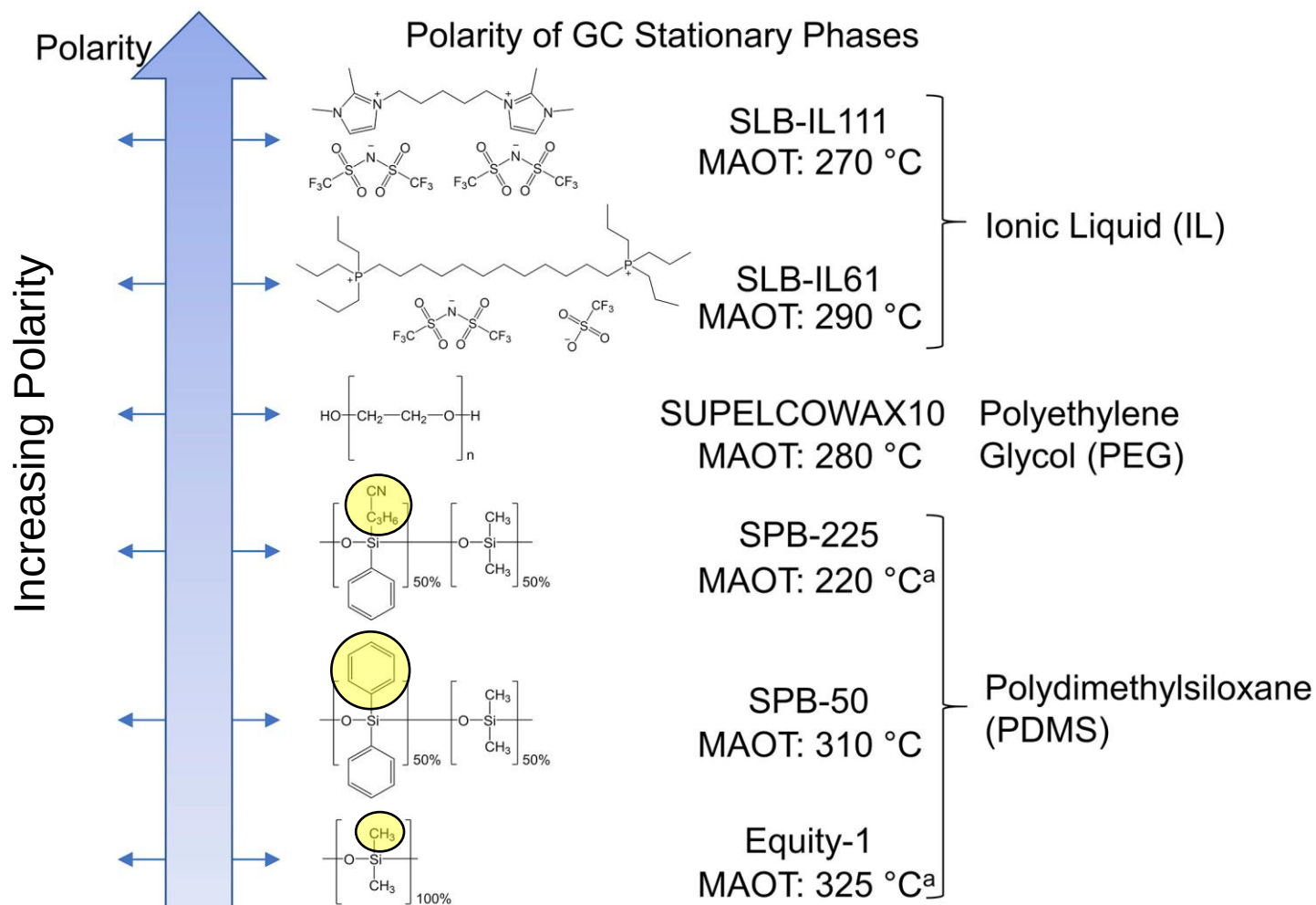
3.3 Gas chromatography: Temperature programming

Column-oven designed for constant temperature or temperature-programming



3.4 GC: Stationary Phase

- Liquid polymeric stationary phase for **WCOT** columns



3.4 GC: Stationary Phase

- Solid stationary phases (for **PLOT** columns):

alumina (also mixed with KCl or Na₂SO₄)

- for separation of very volatile hydrocarbons such as methane, ethane, propane, ethylene, ...

silica

- for separation of CO₂, H₂S, CS₂, SO₂, ...

porous organic polymers (Poly(styrene-divinylbenzene, or Tenax (polymer of 2,6-diphenyl-phenyleneoxide))

- for separation of gases, fluorochlorohydrocarbons,...

mole sieves (zeolithes)

for separation of H₂, O₂, N₂, CH₄, CO, ...

graphitized carbon

for separation of very volatile hydrocarbons

3.4 GC: Stationary Phase

Some commercially available (WCOT) GC columns

Polysiloxanes:

100% methyl

DB¹, HP¹, Rtx-¹, BP-¹, SPB-¹,

5 % phenyl

DB⁵, HP⁵, Rtx-⁵, BP-⁵, SPB-⁵,

50 % phenyl

DB17, HP17, HP⁵⁰, Rtx-⁵⁰, SPB-⁵⁰,

14 % cyanopropylphenyl

DB1701, HP1701, Rtx1701,

50 % cyanopropylphenyl

DB225, HP225, Rtx225,

PEG:

DB-^{Wax}, Stabil^{wax}, Permabond Carbo^{wax}20M, ...

**choose stationary phase to match polarity
of analyte (like dissolves like) !!!**

3.5 GC: Injectors

Split/splitless injector

most common injector

PTV injector

allows temperature programming

On-Column injector

for thermally labile compounds

Head-space analysis

for volatiles

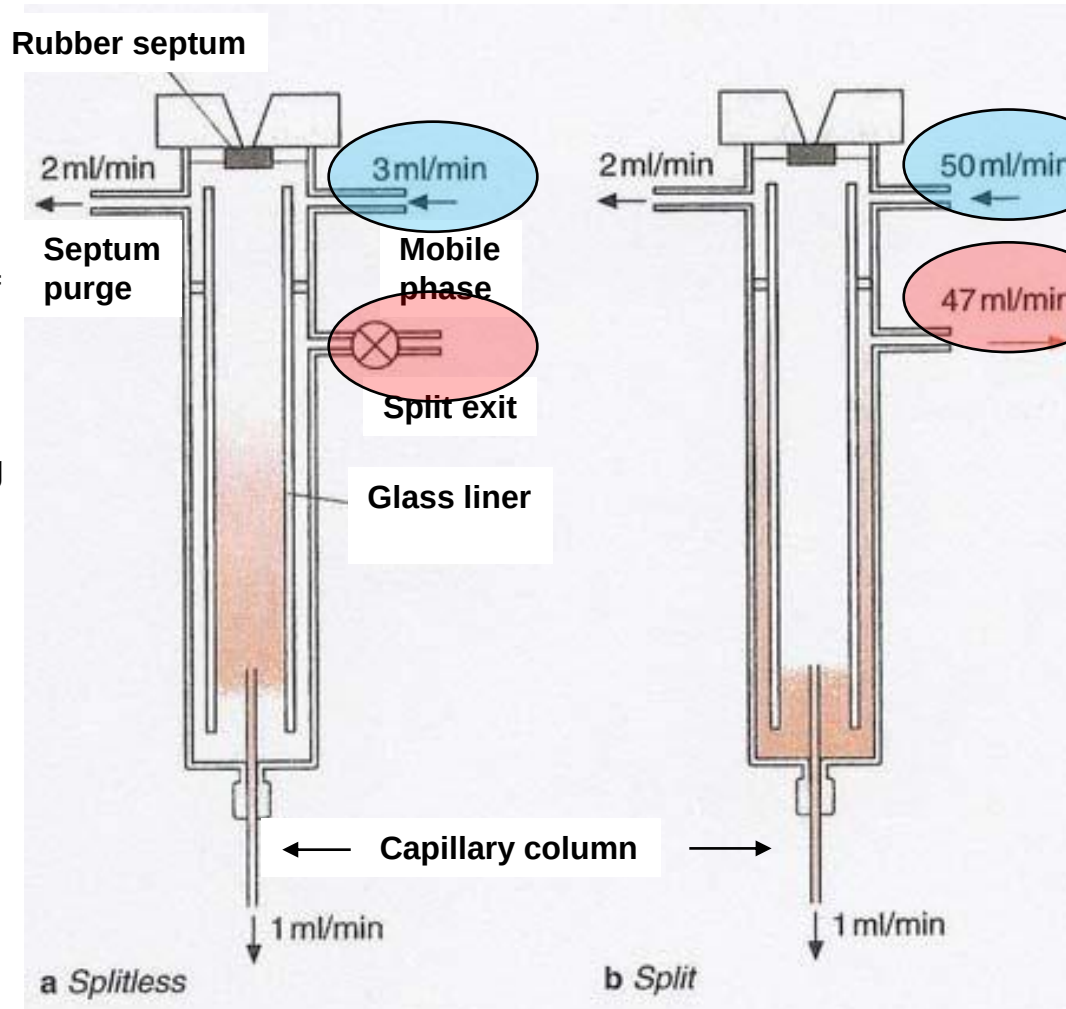
Solid-phase micro-extraction

for sample pre-concentration

Thermodesorption

for thermal extraction

3.5.1 Split/splitless injector



- Low mobile phase flow rate
- Slow transport of analytes to the column head
- Peak broadening
- Need for re-focusing at column head

- High mobile phase flow rate
- Fast transport of analytes to the column head
- Only very little peak broadening

3.5.1 Split/splitless injector

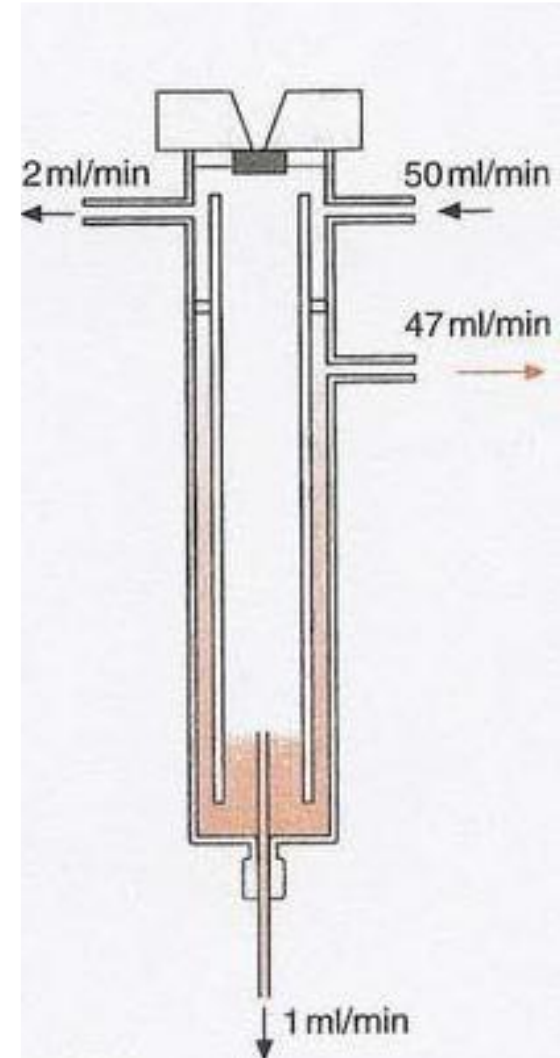
3.5.1 Split/splitless injector

Split injection

- ➔ most of the carrier gas leaves via split valve
- ➔ **hot** injector
- ➔ **hot** column – as no re-focusing is needed

- 0.5 ... 1 μL of sample solution is injected, droplets are immediately vaporized
- high mobile phase flow rate leads to sharp peaks!

Problems with split injection:
Fractionation effects !!



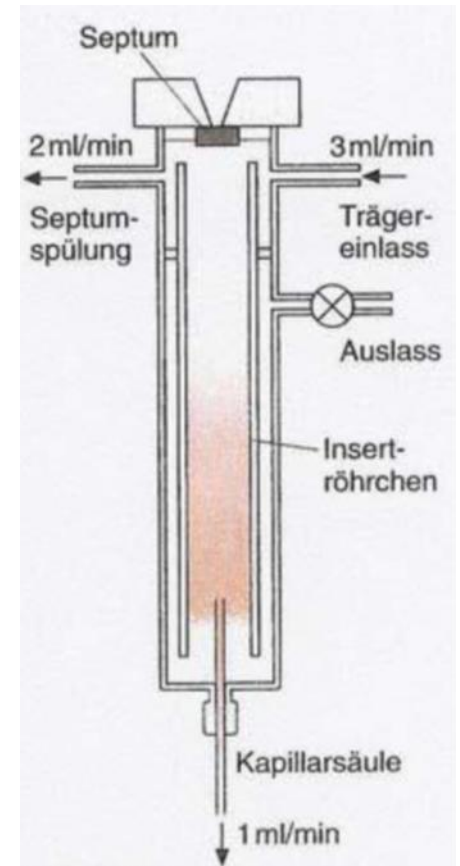
3.5.1 Split/splitless injector

Splitless injection

- split valve is closed
- hot injector
- „cold“ column (approximately 10°C below boiling point of solvent), followed by temperature program

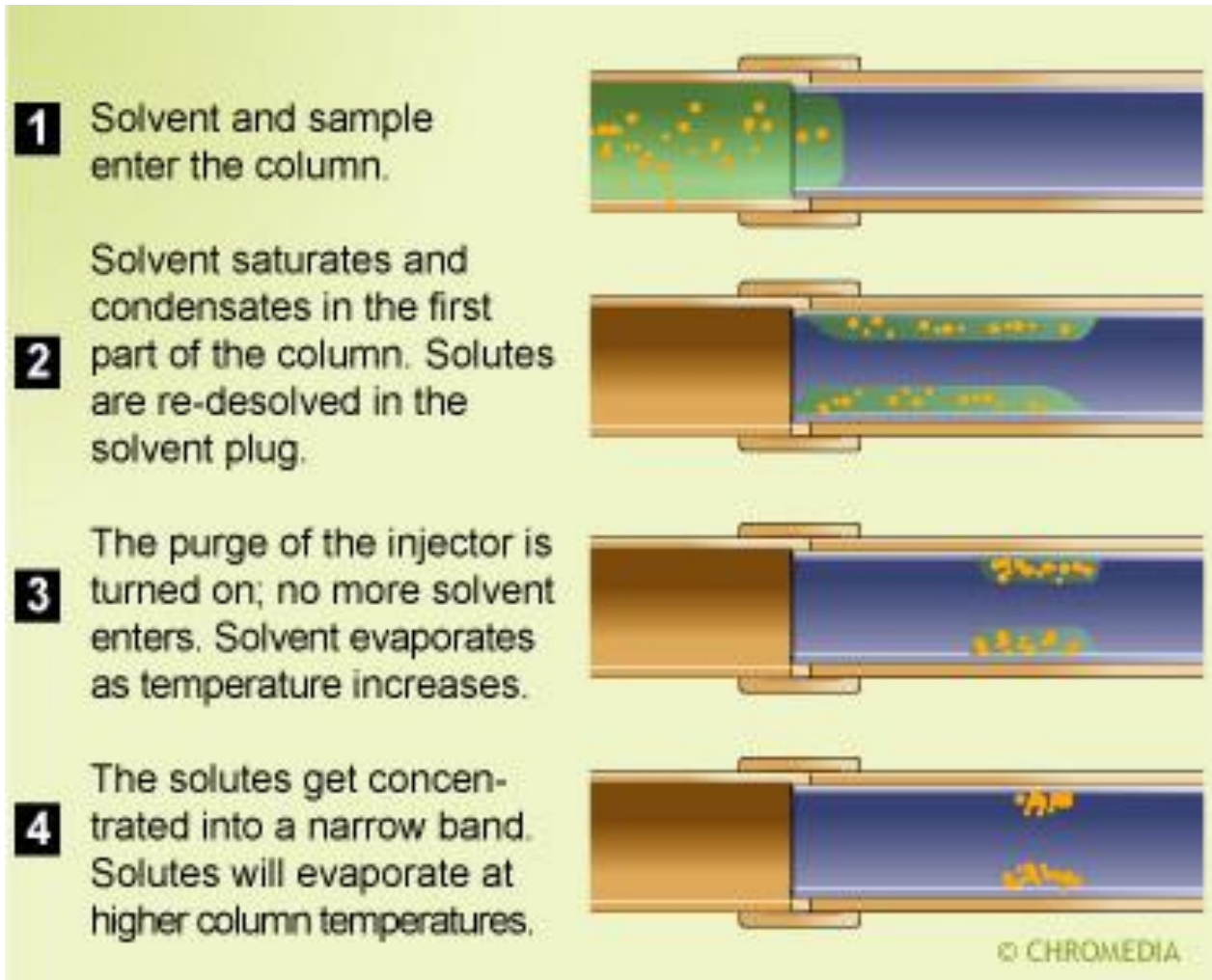
0.5 ... 2 µL of sample solution is injected with the split valve closed, after 30 ... 90 s the split valve is opened

- Low mobile phase flow rate leads to broad peaks and the long time within the hot injector may lead to analyte degradation
- Re-focussing of the injected sample band is achieved at the column head by solvent trapping, cold trapping



3.5.1 Split/splitless injector

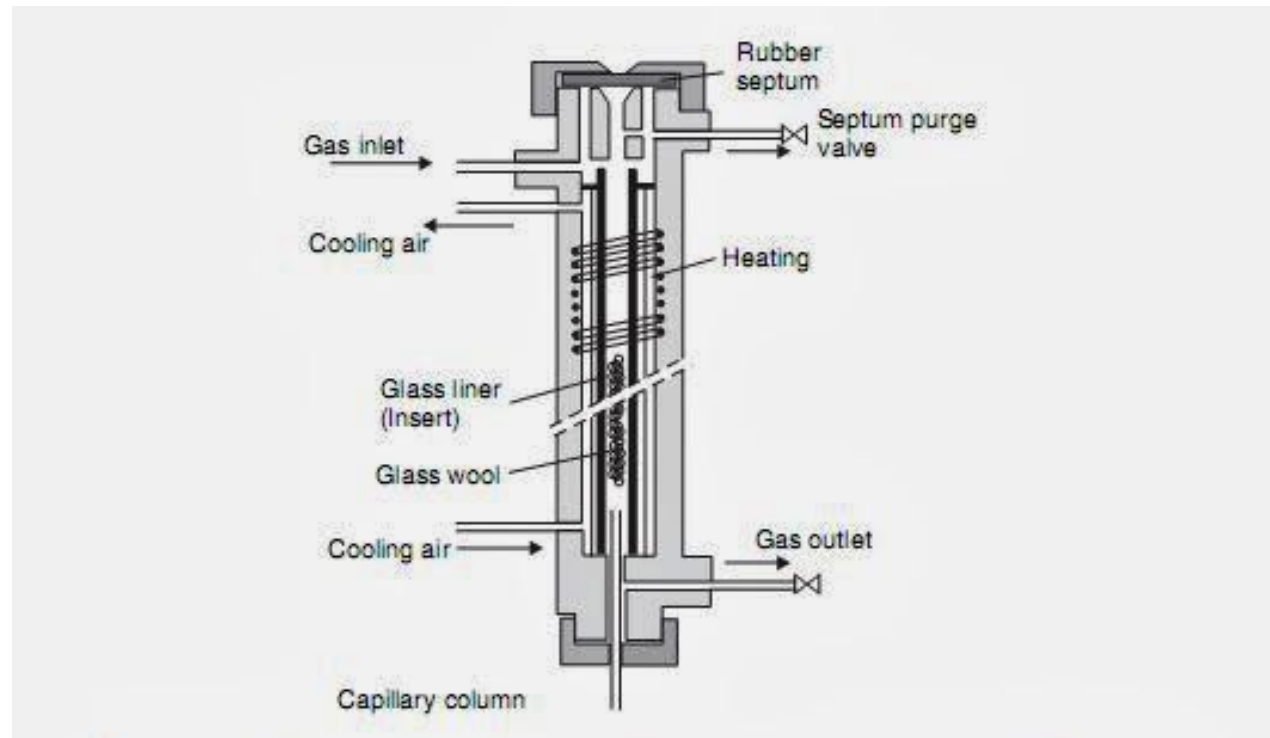
Solvent/cold trapping during splitless injection



Often a „retention gap“ (piece of fused silica capillary without stationary phase) is used between injector and separation column

3.5.2 PTV injector

Programmed Temperature Vaporization



Injector can be heated or cooled using programmable temperature ramps

3.5.2 PTV injector

- Step 1: solvent is evaporated and leaves via split valve
- Step2: temperature is elevated (controlled heating) and analytes are transferred to the column

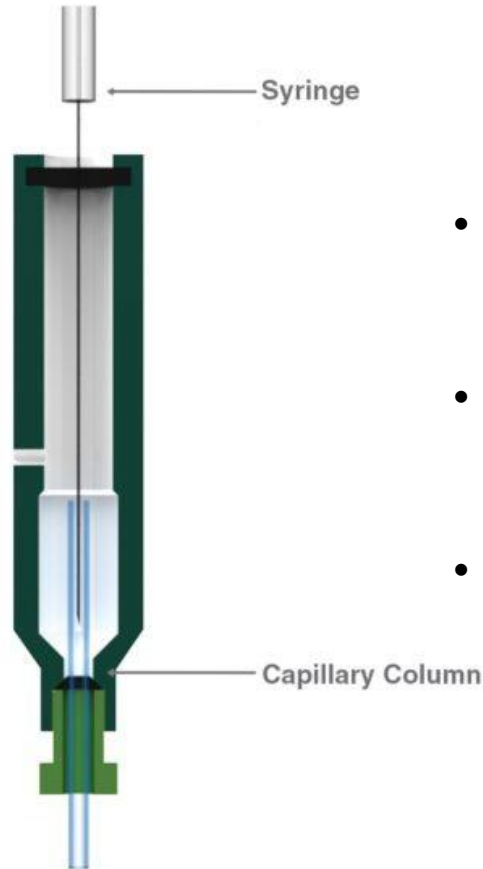
Advantages

- No discrimination
- No thermal decomposition of analytes
- Larger volumes can be injected
- No overloading as solvent can be selectively removed
- Perfect for trace analysis

Disadvantages

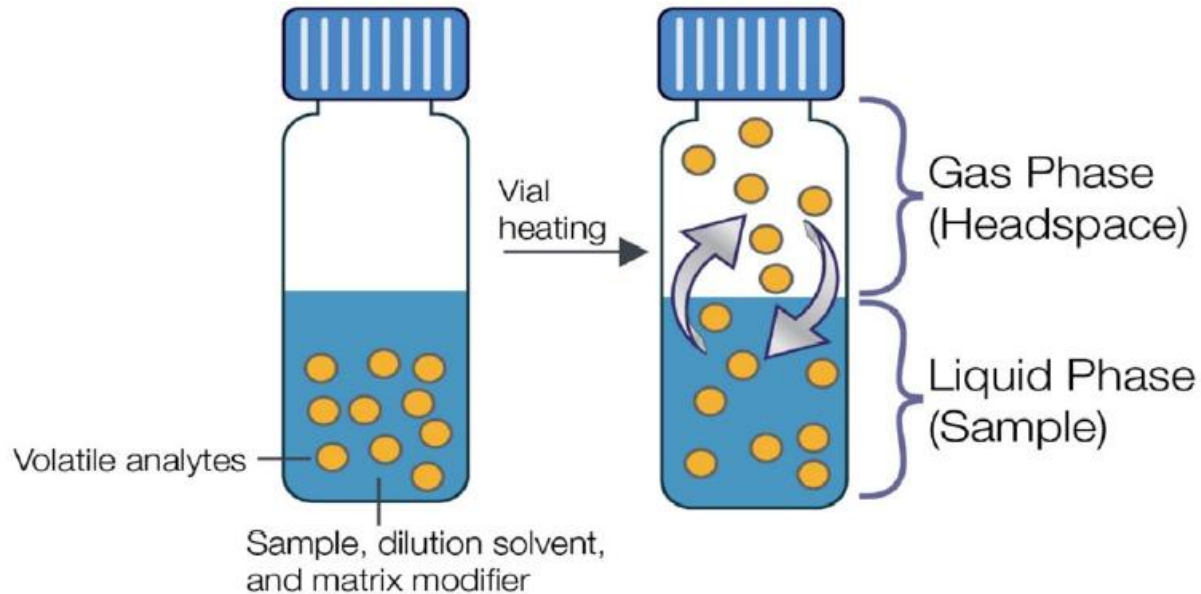
- Costs
- Adjustment of parameters requires experience

3.5.3 Cold on column injector



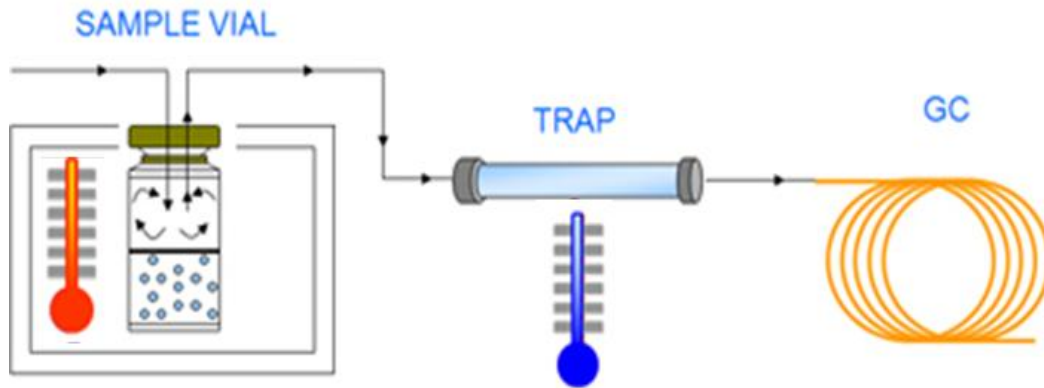
- Analyte is brought directly to the column by a syringe with a long thin needle
- Column is kept at a relative low temperature (starting point of GC-oven temperature programm)
- Also thermally labile compounds can be injected

3.5.4 Head space analysis



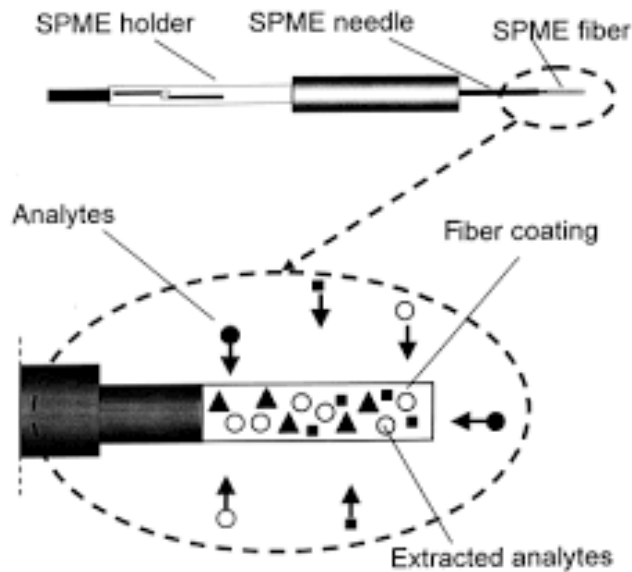
- In a closed sample vial, the concentration of a volatile analyte in the gas phase is proportional to its concentration in the sample solution
- Therefore, the gas phase can be injected instead of the liquid phase (sample must be kept at constant elevated temperature)
- This procedure is called **static** head space analysis

3.5.4 Head space analysis

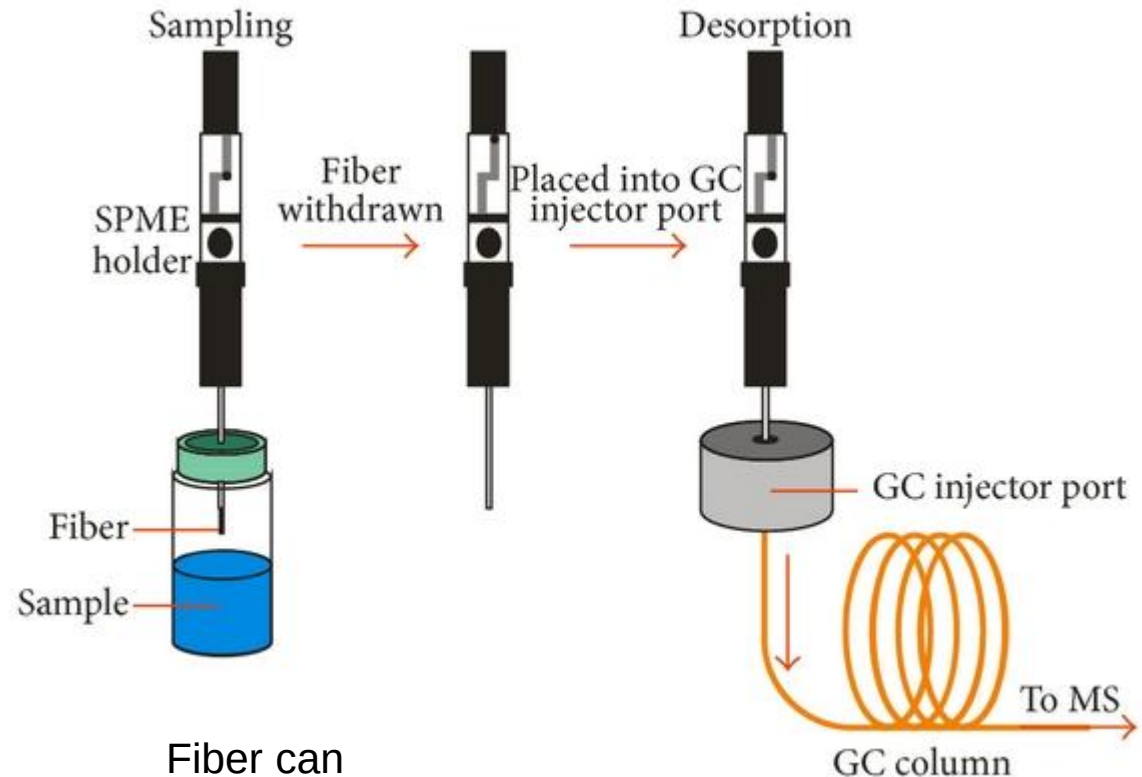


- A closed sample vial is kept at a constant temperature while stirring for a specified time
- Gas is blown through the head space (for a specified time) and gas from the head space is transferred to a cooled trap
- There the analyte molecules are concentrated by condensation at low temperature
- The trap is quickly heated and the sample is brought onto the GC-column for analysis
- This procedure is called **dynamic** head space analysis

3.5.5 Solid Phase Micro Extraction



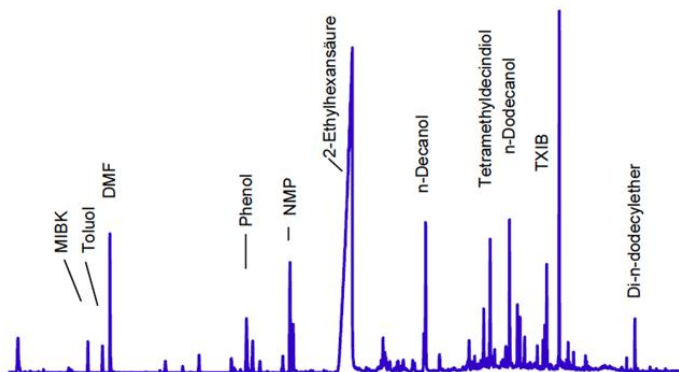
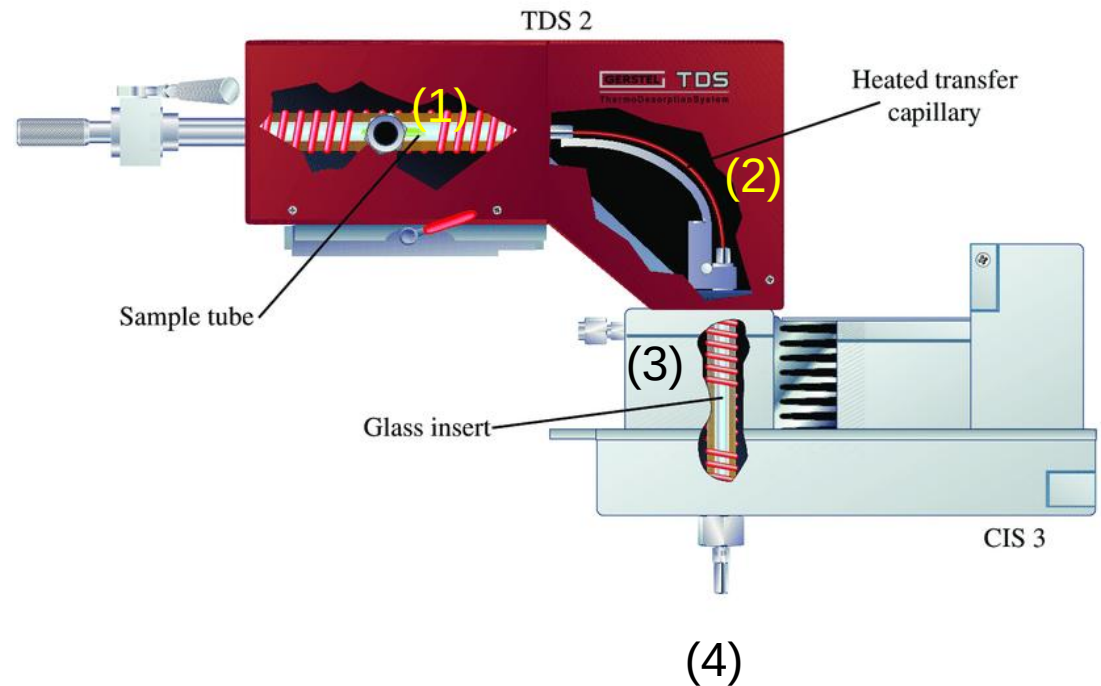
Fiber is
(usually)
coated with a
hydrophobic
layer (PDMS)



Fiber can
sample head
space or can
be immersed
into liquid

3.5.6 Thermodesorption analysis

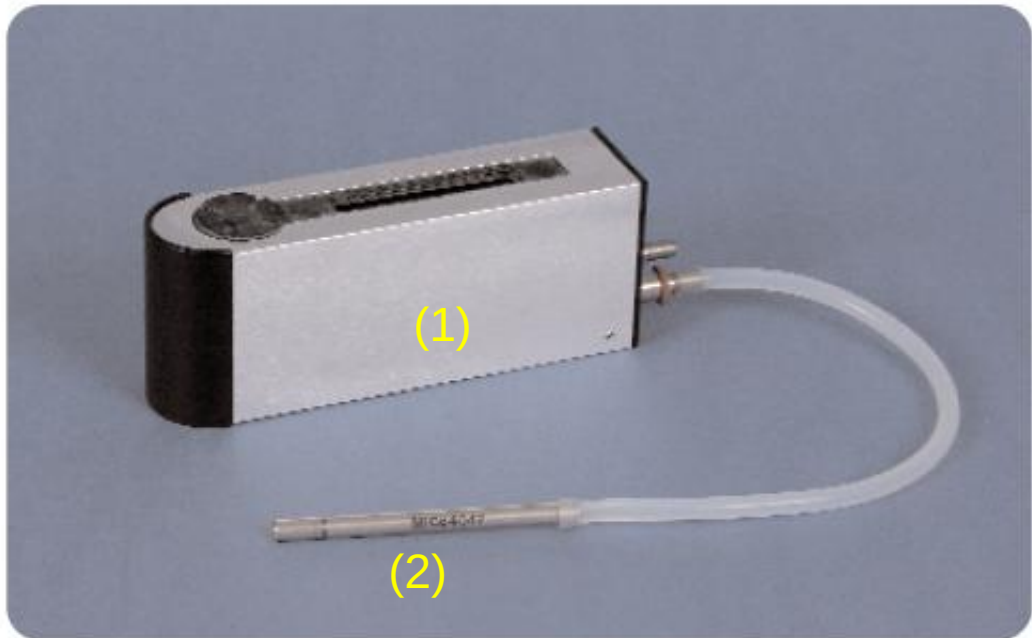
- Sample tube holds solid sample (1)
- Sample is heated for a specific time while an inert gas (He) is blown through the sample
- Thermodesorbed analytes are transferred via heated capillary (2) to the cryo-trap (3)
- There analytes are trapped at low temperature
- The cryo-trap is heated quickly and the analytes are desorbed into the GC column (4)



TDS analysis of a foam-lined imitation leather for automotive industries

3.5.6 Thermodesorption analysis

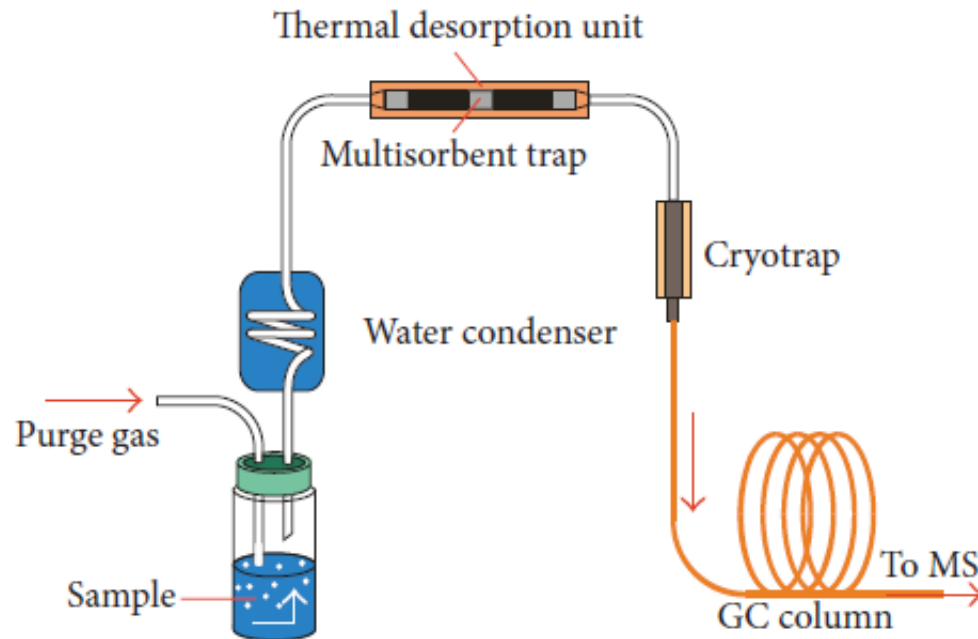
- Alternatively gas can be sucked via a pump (1)
- through a tube filled with an adsorbent (2)
- After a specified time (or volume of gas sucked through) the tube is introduced into the thermodesorption unit depicted on the previous slide and desorbed at high temperature
- Further steps of analysis see previous slide
- Important application:
 - Volatile organic compounds (VOCs) from (indoor) air



(2)

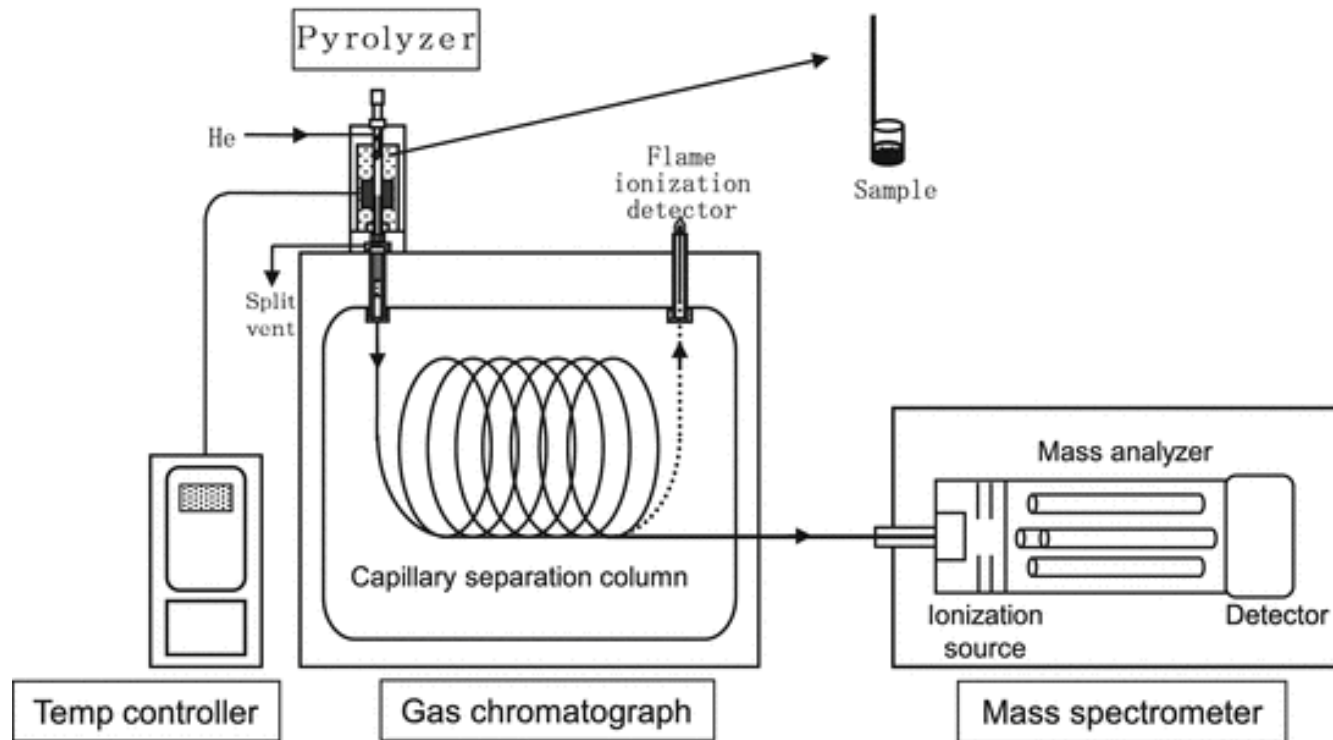


3.5.7 Purge and trap analysis



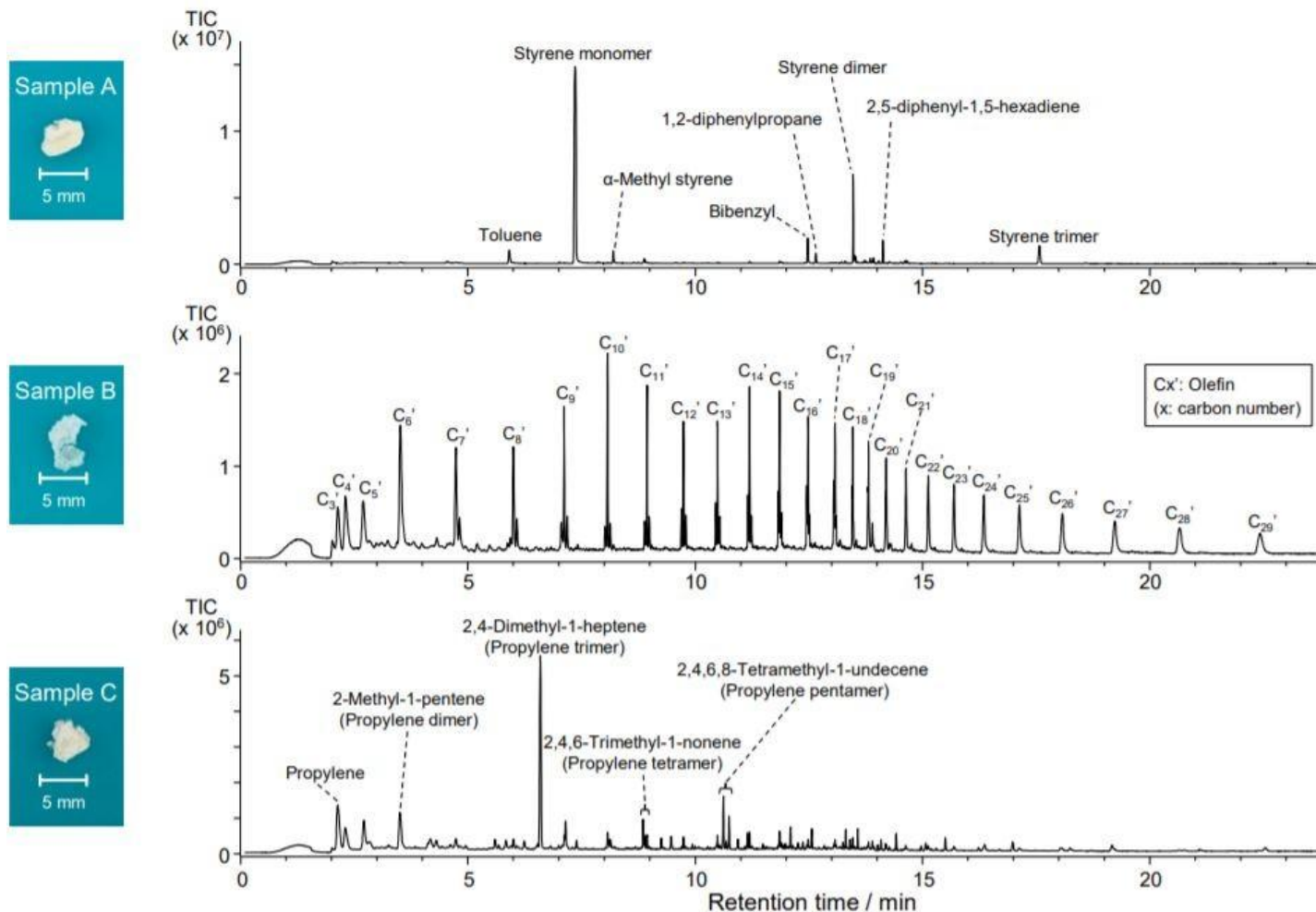
Purge and trap works similar as the approach described on the previous slide – but here a liquid is sampled by blowing an inert gas through the liquid. Vapours are trapped on a tube filled with an adsorbent – that is subsequently introduced into the thermodesorption system

3.5.8 Pyrolysis GC



Large molecules (polymers) are disintegrated into their building blocks by applying high temperature ($>1000^\circ$) in an inert gas atmosphere

3.5.8 Pyrolysis GC



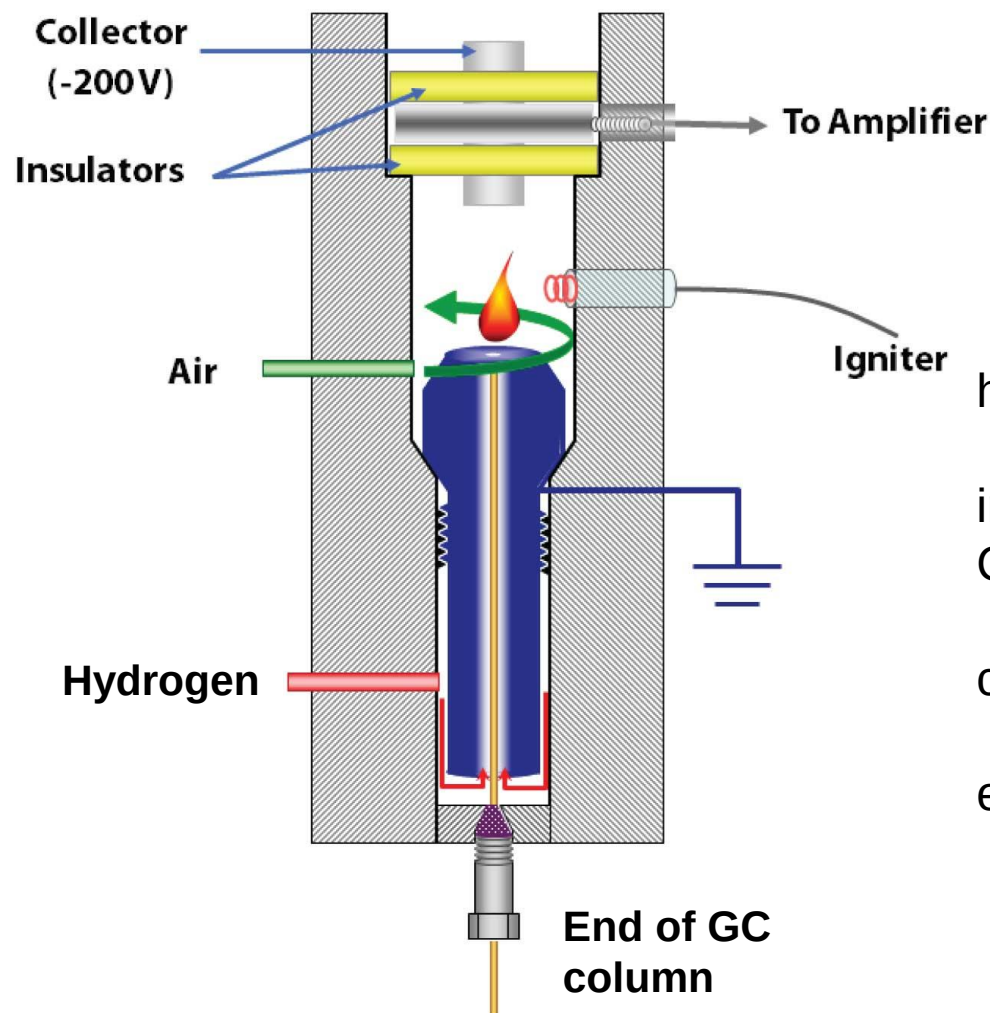
Pyrolysis-GC-(MS) of Microplastics

Source: Quantum Analytics

3.6 GC detectors

- **Flame ionization detector (FID)**
 - See next slides
- **Electron capture detector (ECD)**
 - See next slides
- **Thermal conductivity detector**
 - See next slides
- **Mass spectrometry (MS)**
 - See MS part

3.6.1 Flame Ionization Detector



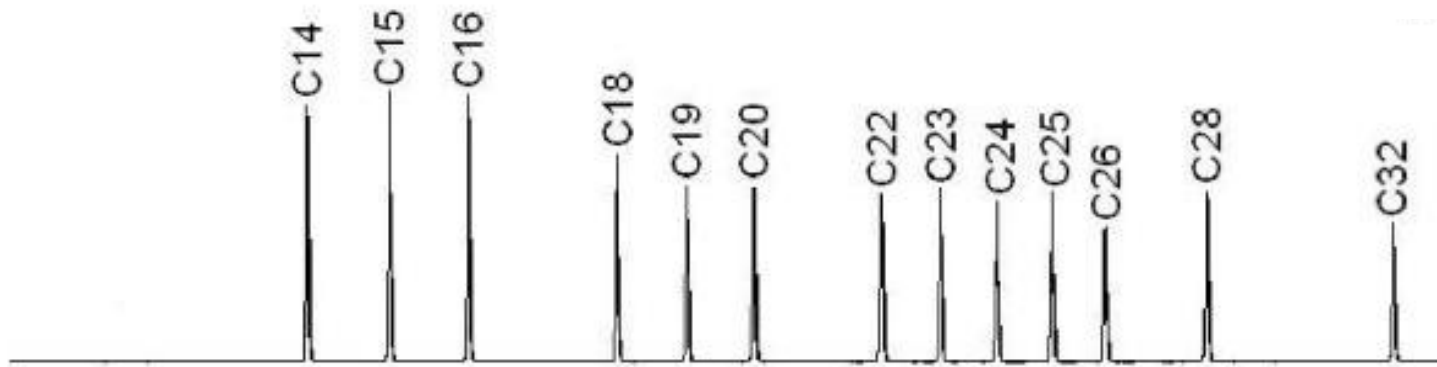
in the flame hydrocarbons react into CHO^+ ions

cathode collects CHO^+ ions

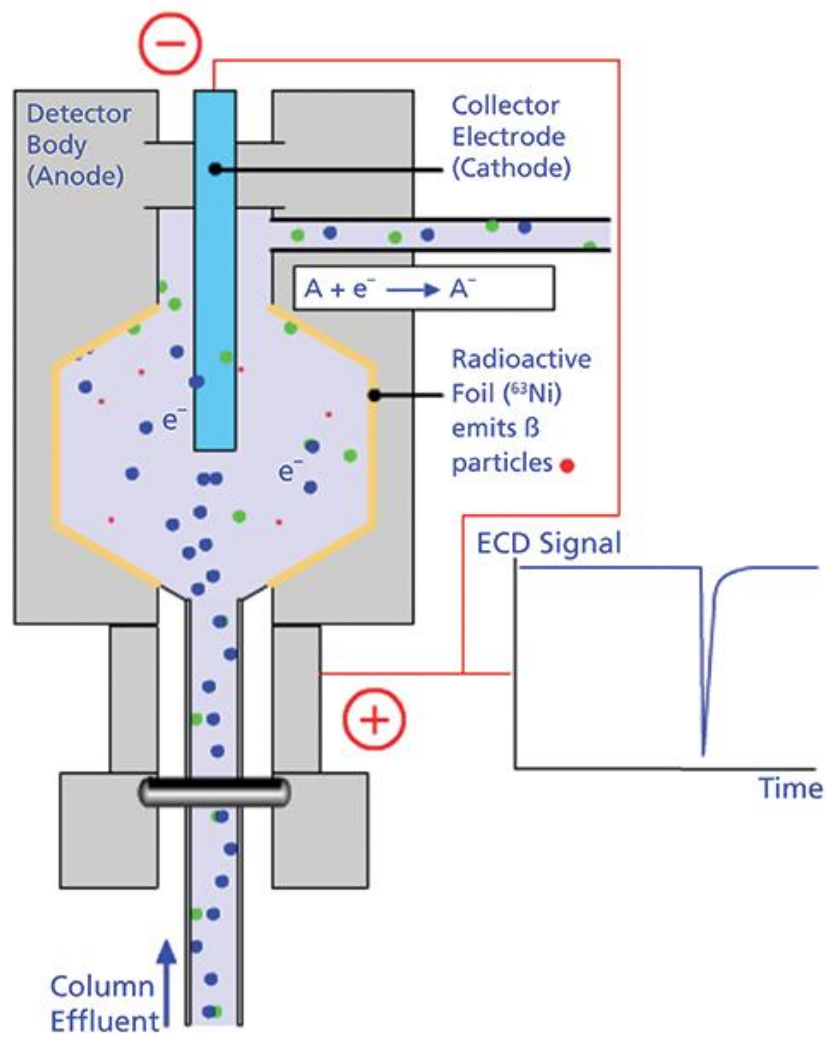
electric current is measured

3.6.1 Flame Ionization Detector

- Universal detector for compounds containing C (best for hydrocarbons)
- Destructive detector
- Wide linear range 10^7
- Detection limit (LOD) in the low pg range

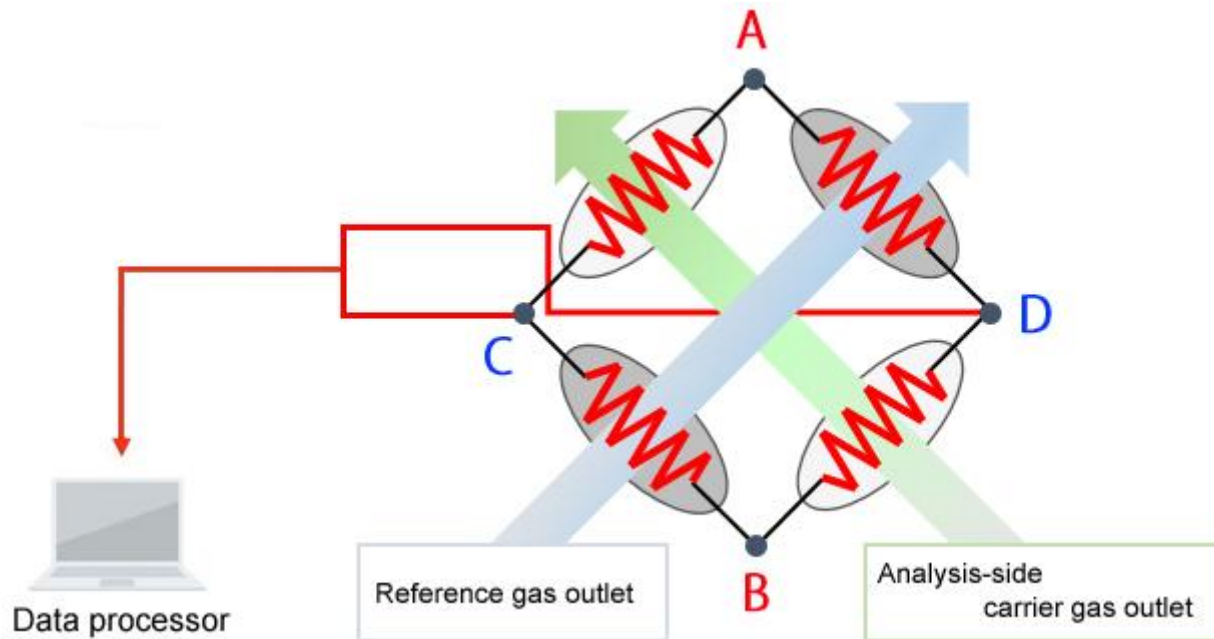


3.6.2 Electron Capture Detector



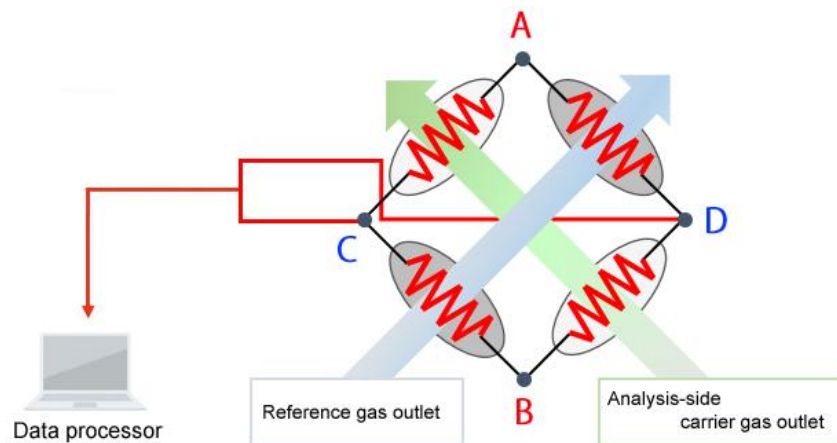
- The ECD is used for detecting electron-absorbing components such as halogenated compounds
- The ECD uses a radioactive electron emitter together with a makeup gas flowing through the detector chamber.
- The electron emitter typically consists of a foil (^{63}Ni), the make up gas is often N_2
- The electrons emitted from the electron emitter collide with the makeup gas, resulting in many more free electrons
- These are accelerated towards the anode, generating a current
- Electron-absorbing analyte molecules capture electrons and thereby reduce the current by forming ion-pairs
- In the pulsed mode the frequency of pulses is selected to obtain a constant current
- So the frequency rises when analyte molecules are detected

3.6.3 Thermal conductivity detector



- Voltage is applied between A and B.
- When only the carrier gas is flowing at a constant flow rate each filament maintains a constant temperature and a constant voltage is produced between C and D.
- Components eluted from the analysis-side column trigger a change in filament temperature which changes the resistance value and subsequently the voltage between C and D

3.6.3 Thermal conductivity detector



Analytical target component :
Methanol

Thermal conductivity: 52

Carrier gas : He

Thermal conductivity: 408

Methanol reaches
the detector.

Filament temperature
increases.

Resistance value increases.
Potential difference
(voltage value) increases.

Detected as a signal

Analytical target component: H₂

Thermal conductivity: 547

Carrier gas : N₂

Thermal conductivity: 73

H₂ reaches the detector.

Filament temperature
decreases.

Resistance value decreases.
Potential difference
(voltage value) increases.

Detected as a signal

Shimadzu

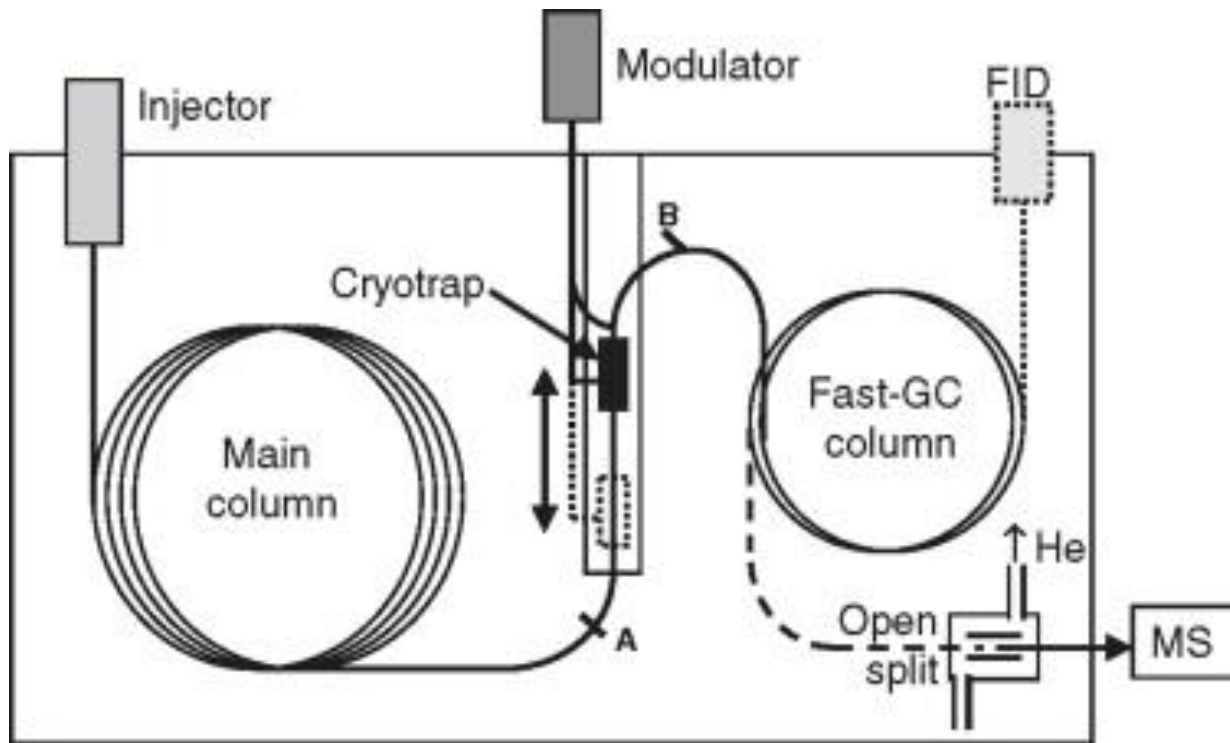
Primarily used for gas samples!

3.6.4 Detectors based on spectroscopy

- flame photometric detector (FPD)
 - for Sulfur and Phosphorous
- chemiluminescence detector (only few applications)
- atomic emission detector (AED)
 - element selective detection; excitation by microwave induced plasma
- infrared absorption detector
 - for example used in combination with pyrolysis GC for identifying the building blocks of polymer resins

Only rarely used nowadays – therefore not further discussed in this lecture!

3.7 Two Dimensional GC (GCxGC)

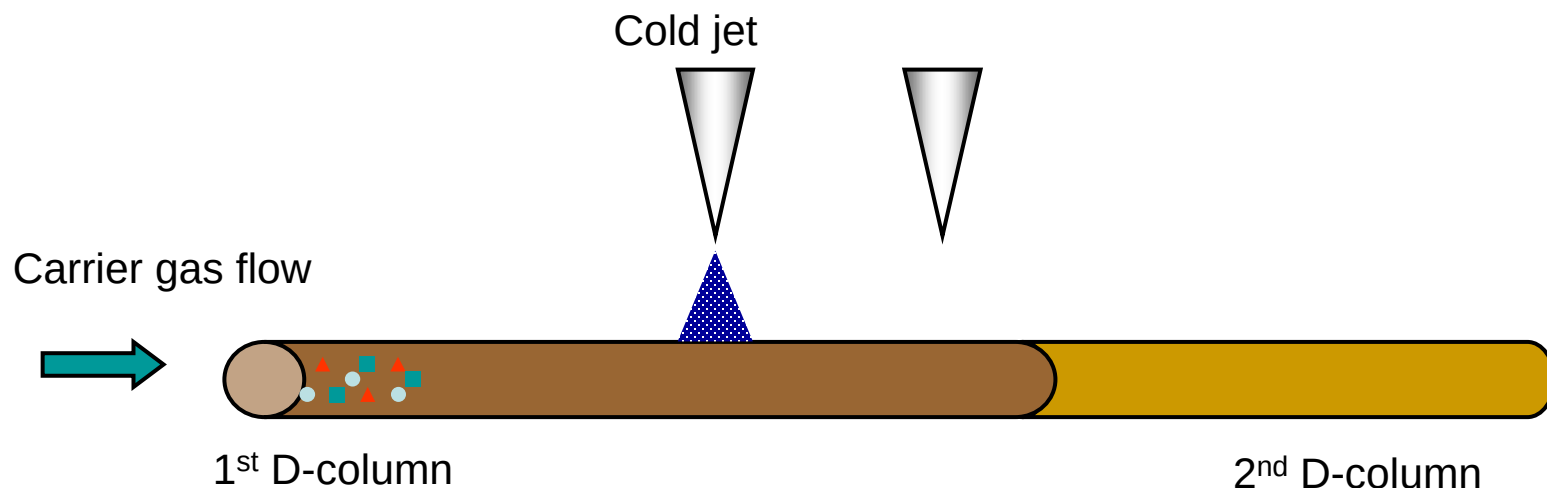


- Either „heart-cutting“ or comprehensive 2D GC
- Need for two columns with orthogonal separation (in GC mostly one non-polar and one polar column)
- The first column commonly is much longer than the second one
- Separation in the second dimension must be much faster than in the first one
- Need for a modulator – here only cryogenic modulation discussed

3.7 Two Dimensional GC (GCxGC)

Cryogenic Modulation for Comprehensive 2D-GC

Operation principle of a thermal modulator for 2D-GC based on two cold/hot jets:

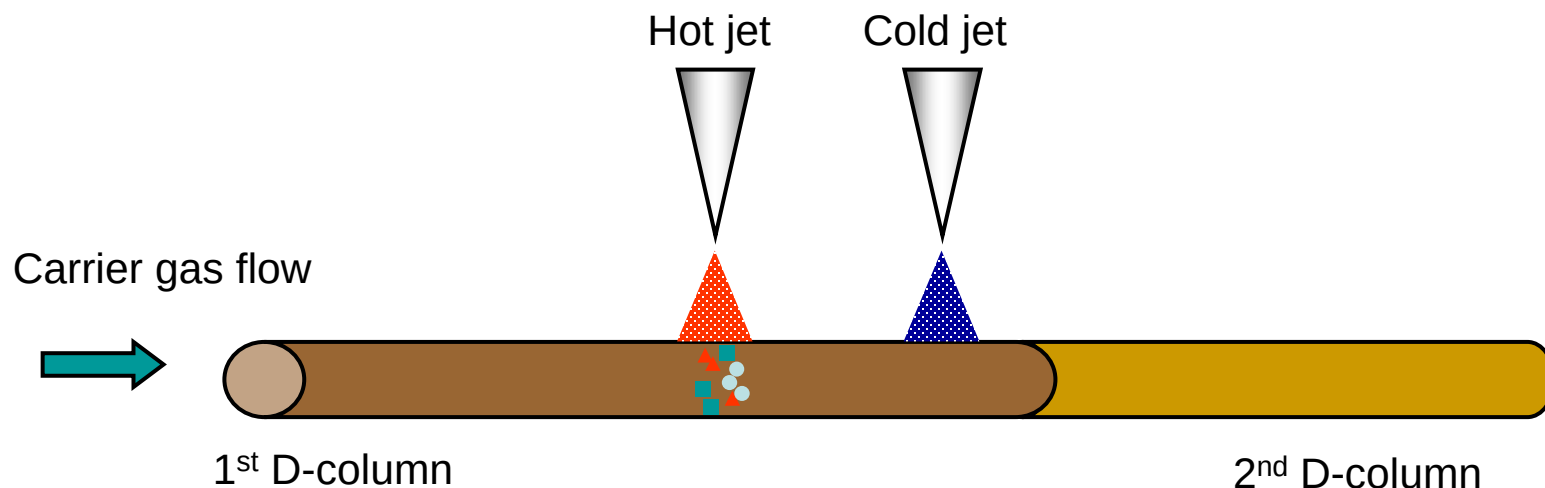


Phase 1: Focusing of analytes after 1st D column

3.7 Two Dimensional GC (GCxGC)

Cryogenic Modulation for Comprehensive 2D-GC

Operation principle of a thermal modulator for 2D-GC based on two cold/hot jets:

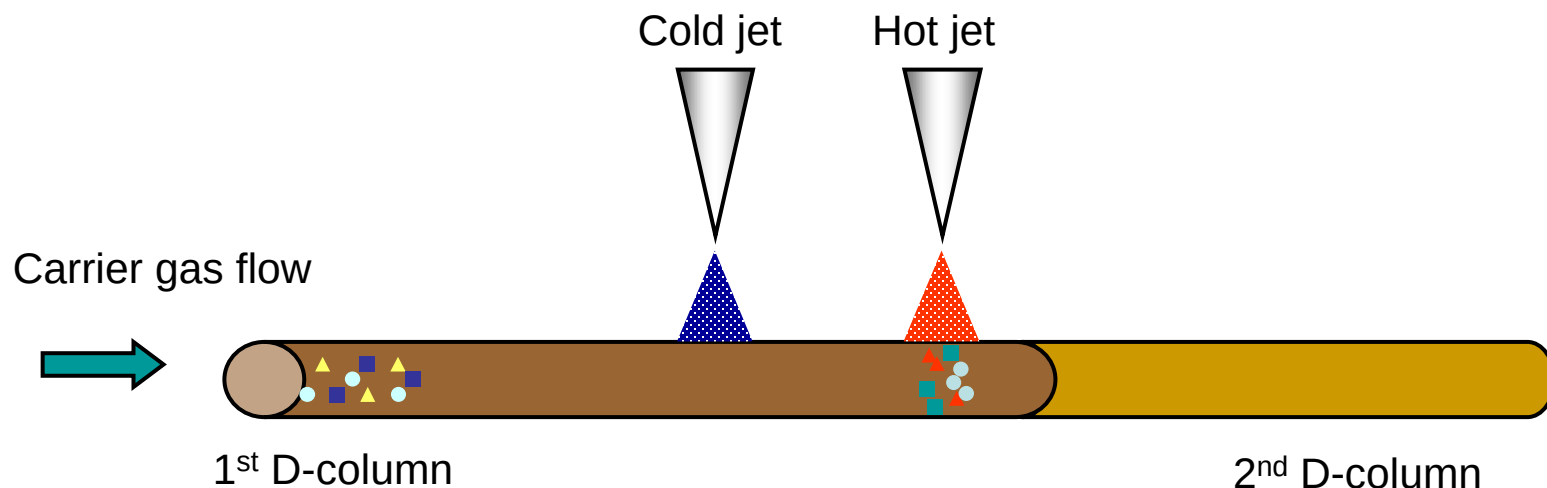


Phase 2: Transfer of analytes to 2nd stage of interface

3.7 Two Dimensional GC (GCxGC)

Cryogenic Modulation for Comprehensive 2D-GC

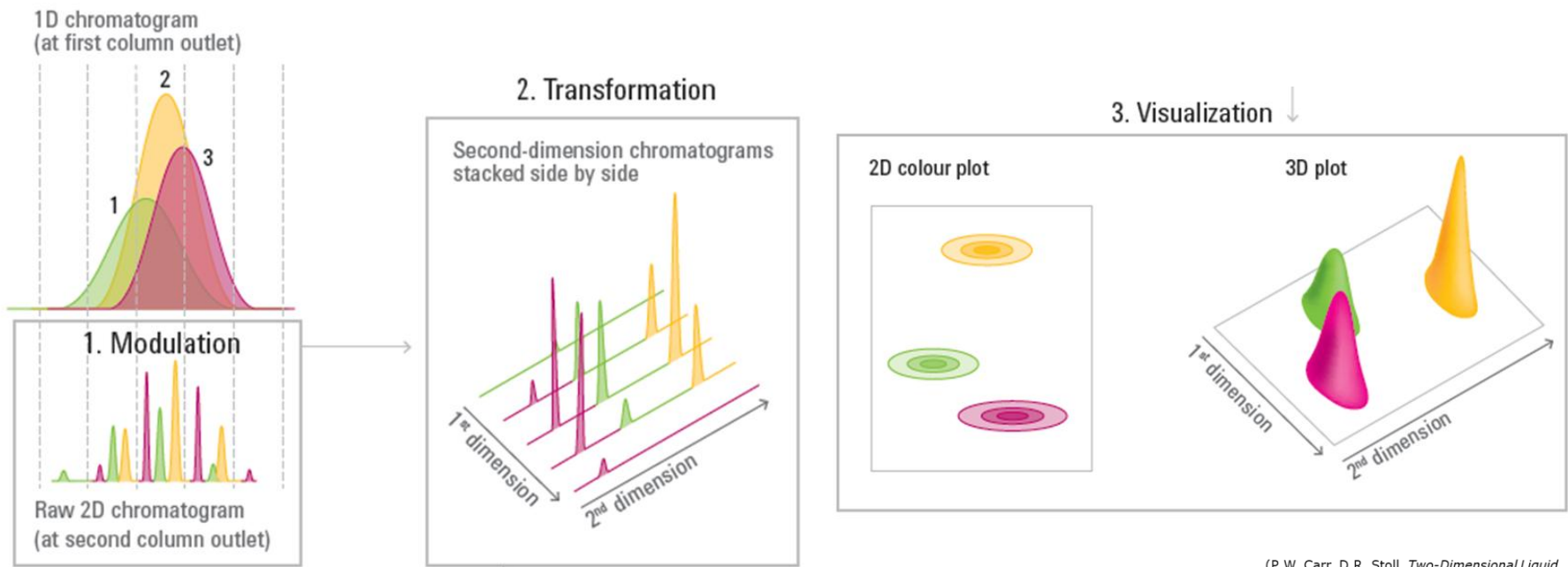
Operation principle of a thermal modulator for 2D-GC based on two cold/hot jets:



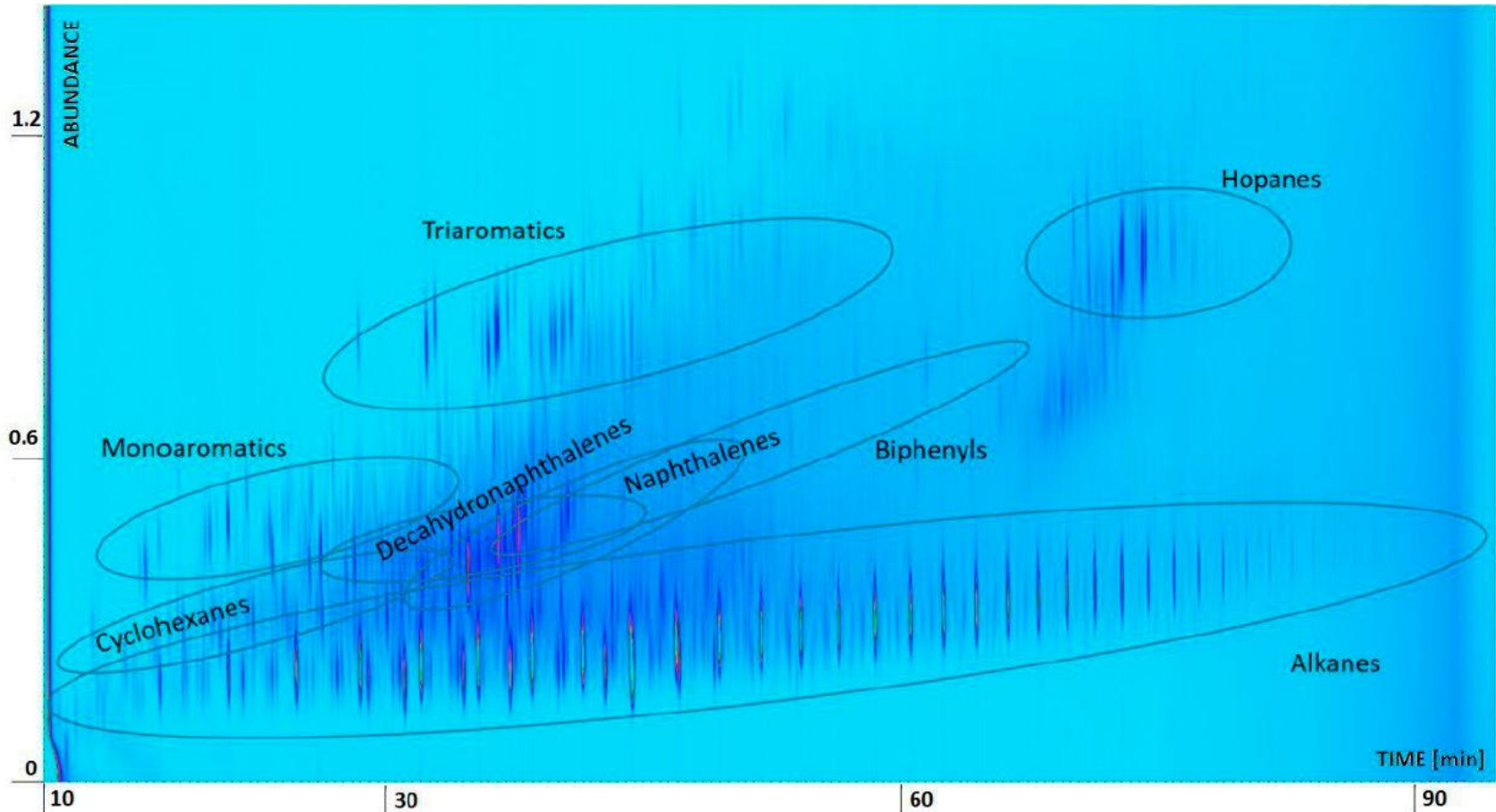
Phase 3: Transfer of analytes to 2nd D-column

3.7 Two Dimensional GC (GCxGC)

- Separation in the second dimension must be completed while peaks eluting from the first dimension are cryofocused
- Separation from the first dimension should be maintained
- Substances eluting from the first dimension may appear in several fractions
- The chromatogram is re-constructed after the second dimension



3.7 Two Dimensional GC (GCxGC)



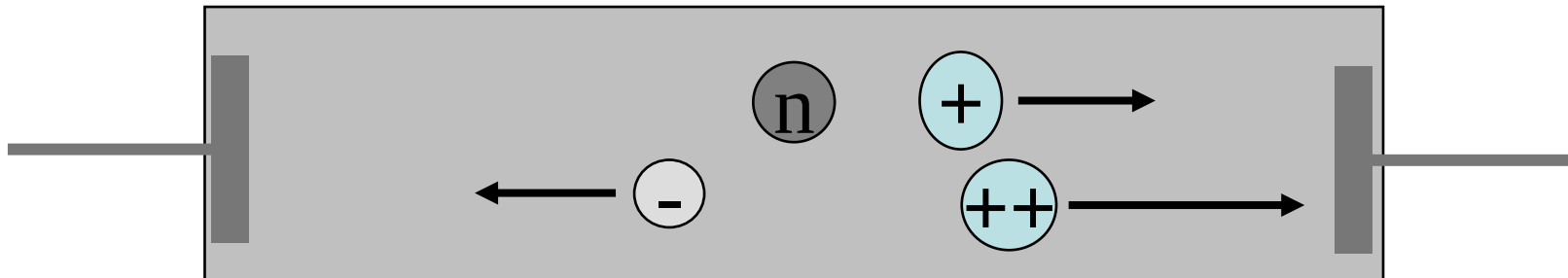
Comprehensive 2D GC of an oil spill

4. Electrophoresis

Electrophoretic separations are based on differences in migration of charged analytes (ions) in an electric field in solution

(+) anode

(-) cathode



4. Electrophoresis

Electrophoretic separations can be carried out:

- in free buffered solution  Capillary Electrophoresis
- in buffered solution immobilized in a gel or a similar support
 Capillary Gel Electrophoresis

4. Electrophoresis

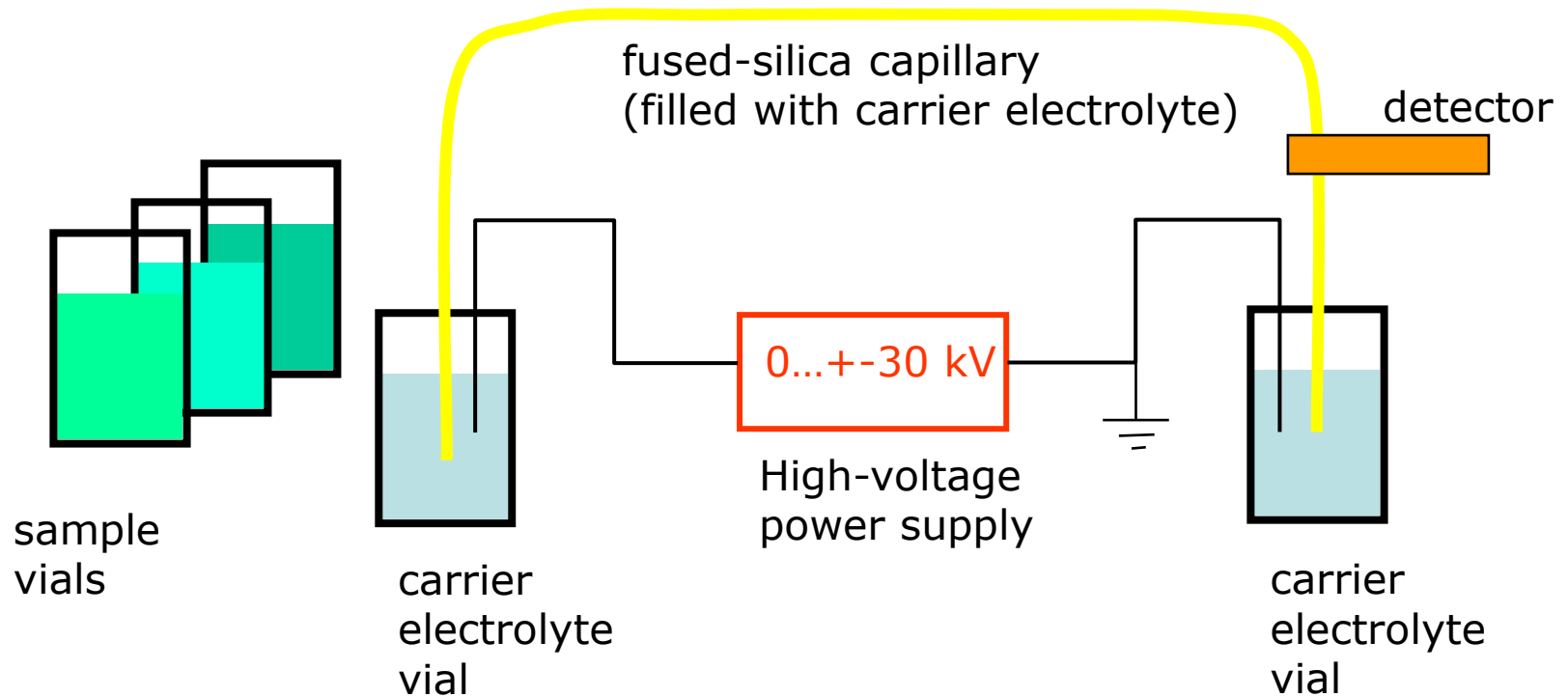
Modes of Capillary Electrophoresis

- Capillary zone electrophoresis (CZE)
- Micellar electrokinetic chromatography (MEKC)
- Micro emulsion electrokinetic chromatography (MEEKC)
- Capillary electrochromatography (CEC)
- Capillary gel electrophoresis (CGE)
- Isoelectric focussing (IEF)
- Isotachopheresis (ITP)

* no longer relevant - will not be discussed in this lecture

4.1 Capillary Zone Electrophoresis

- CZE-Instrumentation



4.1 Capillary Zone Electrophoresis



Casette for
separation
capillary-
see next
slide

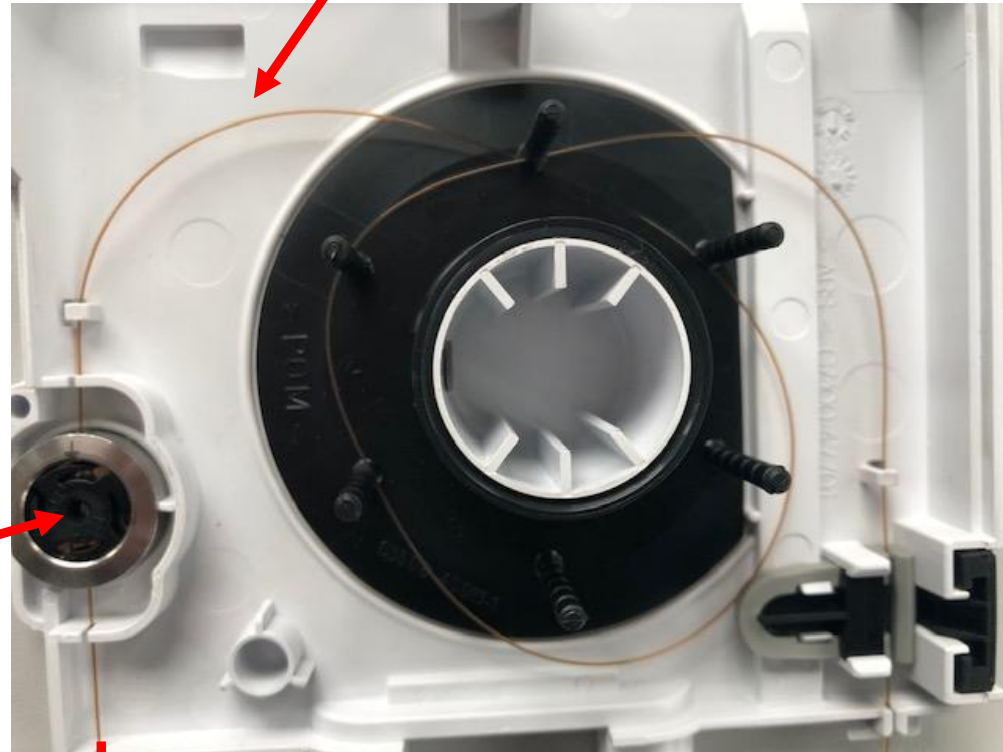


Tray for vials
filled with
sample,
electrolyte or
cleaning
/conditioning
solutions

4.1 Capillary Zone Electrophoresis



Separation capillary



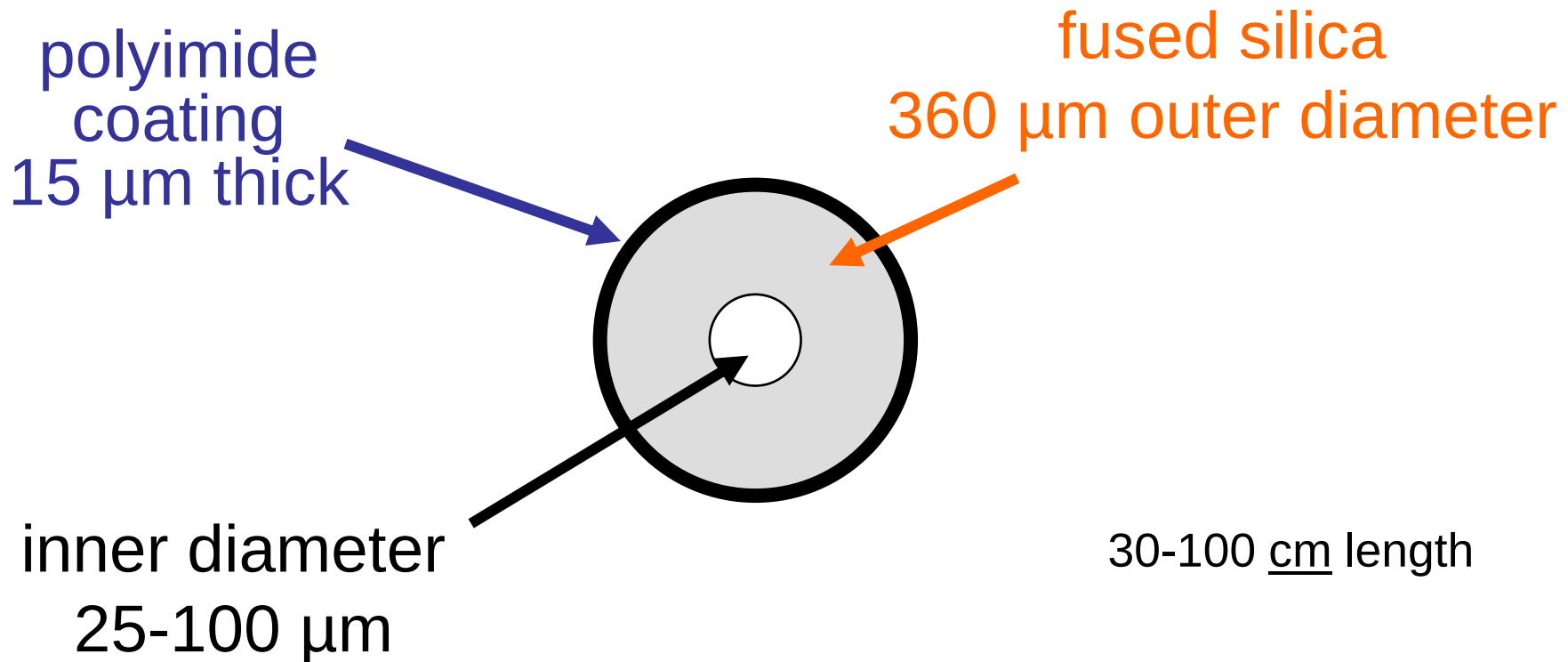
Interface
for
detection

Detection
side end

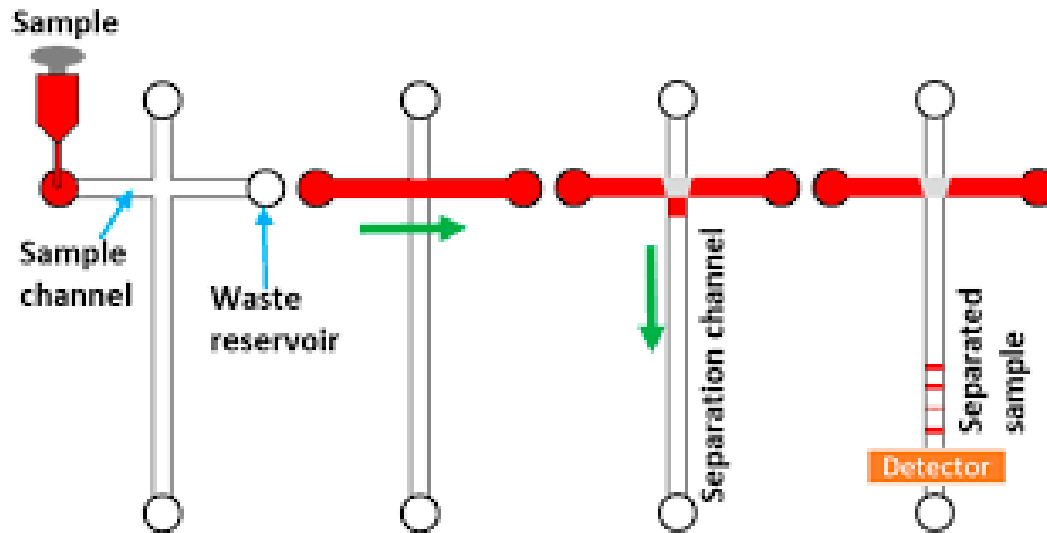
Injection
side end

4.1 Capillary Zone Electrophoresis

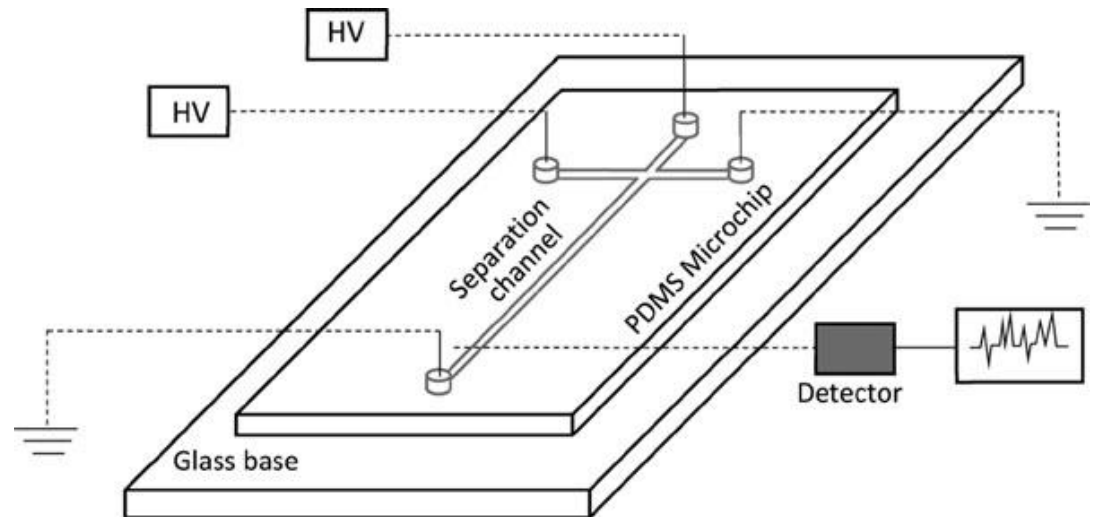
- Fused silica capillary:



4.1 (Chip) Capillary Zone Electrophoresis



CZE can also be performed using miniaturized devices i.e. so called chip electrophoresis



4.1.1 Electrophoretic mobility/velocity

Acceleration:

$$F_{ac} = z e E$$

z charge number, e charge unit
E electric field strength (E = voltage / length)

Friction (Stoke's Law):

$$F_{fr} = 6 \pi \eta r v_{ep}$$

v_{ep} electrophoretic velocity (cm / s)
r radius of ion η viscosity

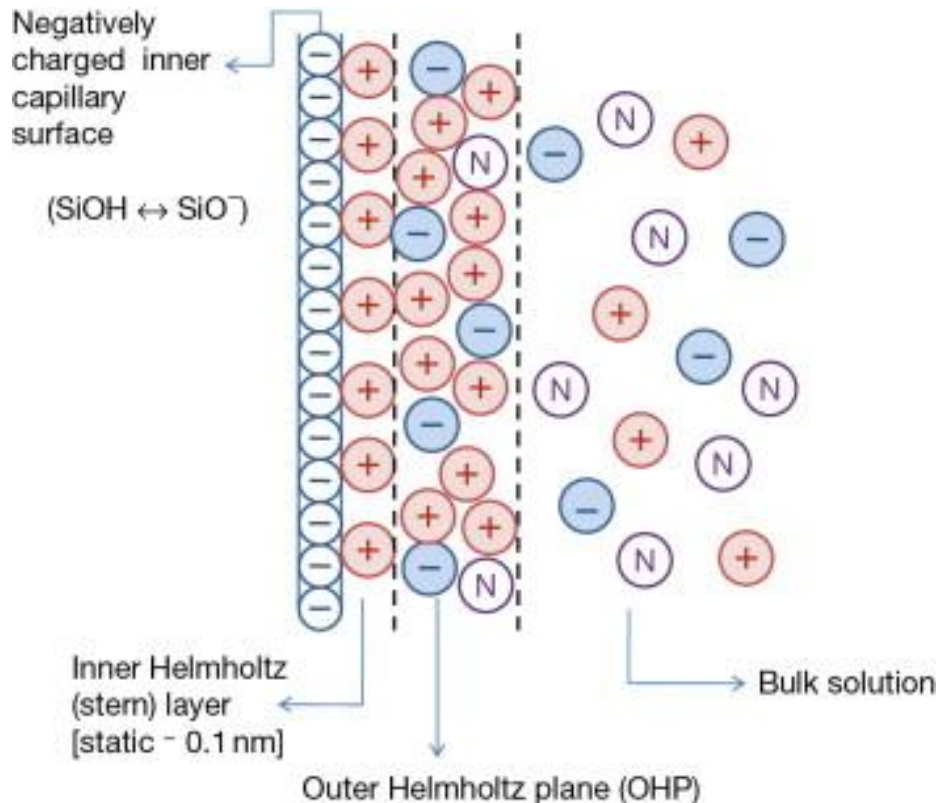
Steady-state: $F_{ac} = F_{fr}$

$$v_{ep}/E = z e / 6 \pi \eta r$$

v_{ep}/E electrophoretic mobility μ_{ep}

$$\mu_{ep} = \lambda / F \quad !! \quad (F \text{ Faraday constant, } \lambda \text{ ionic equivalence conductance})$$

4.1.2 Electroosmotic flow

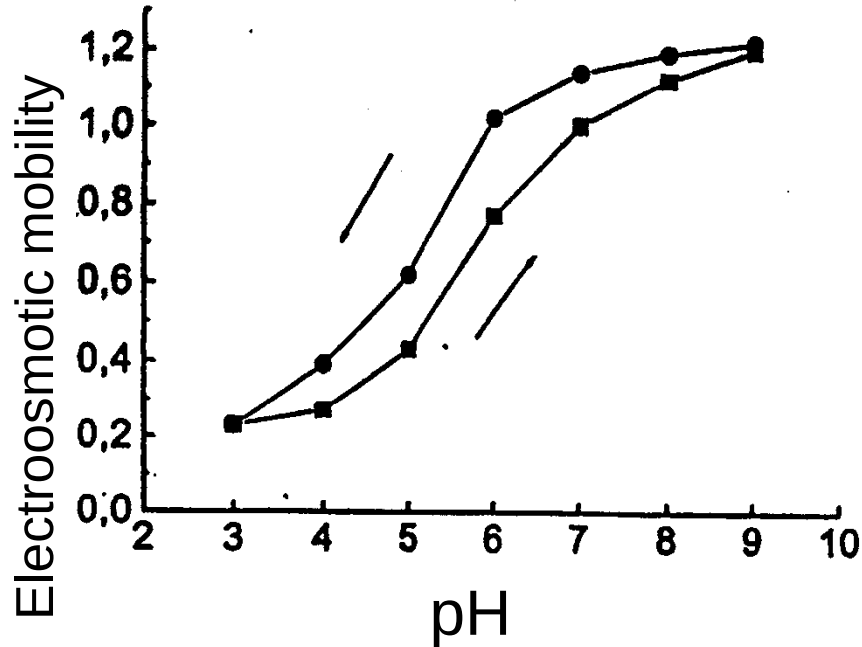


v_{eo} electroosmotic velocity
 μ_{eo} electroosmotic mobility
 $v_{eo} = \mu_{eo} E$

- fused silica has exposed silanol groups (Si-OH) on the inner surface, which become deprotonated at $\text{pH} \geq 3$
- This causes the buildup of a fixed layer of cations (inner Helmholtz layer) and a layer of mobile cations from the carrier electrolyte - upon application of voltage this outer Helmholtz layer (that contains more positive than negative charges) moves to the cathode - leading to a flow of the whole electrolyte to the cathode

4.1.2 Electroosmotic flow

Dependence of EOF on pH

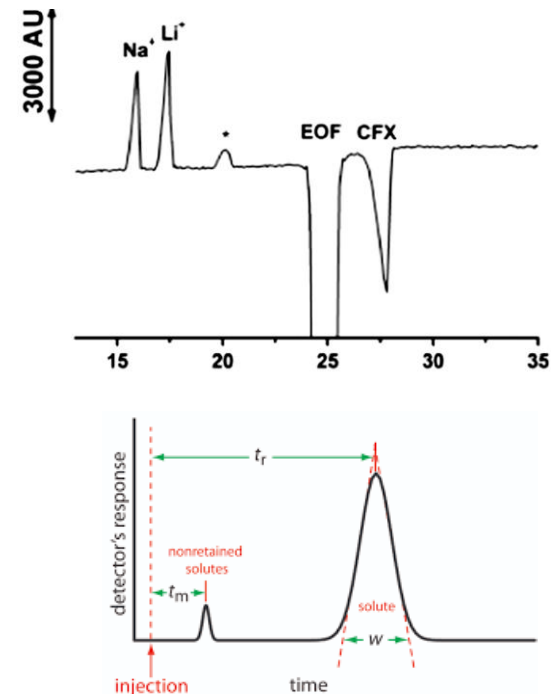
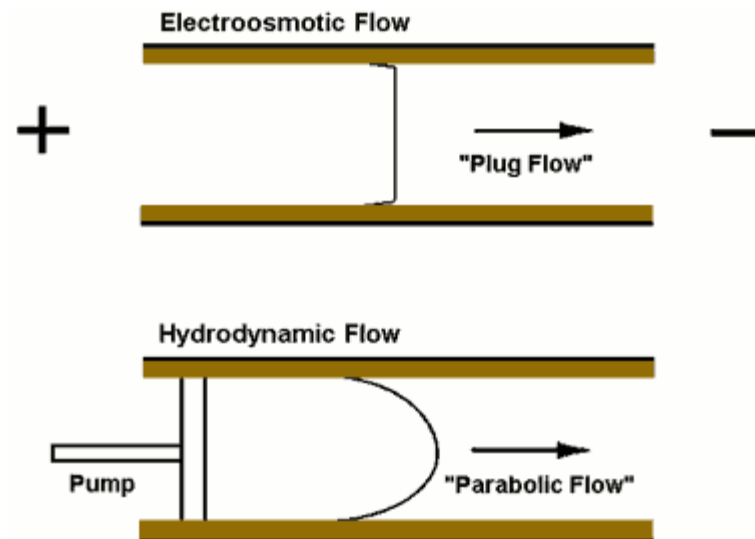


μ_{eo} electroosmotic mobility; depends on the zeta potential ζ (charge of the inner surface), ε (relative permittivity) and η (viscosity)

$$\mu_{eo} = \text{prop } \varepsilon \zeta / \eta$$

4.1.2 Electroosmotic flow

Flow profiles



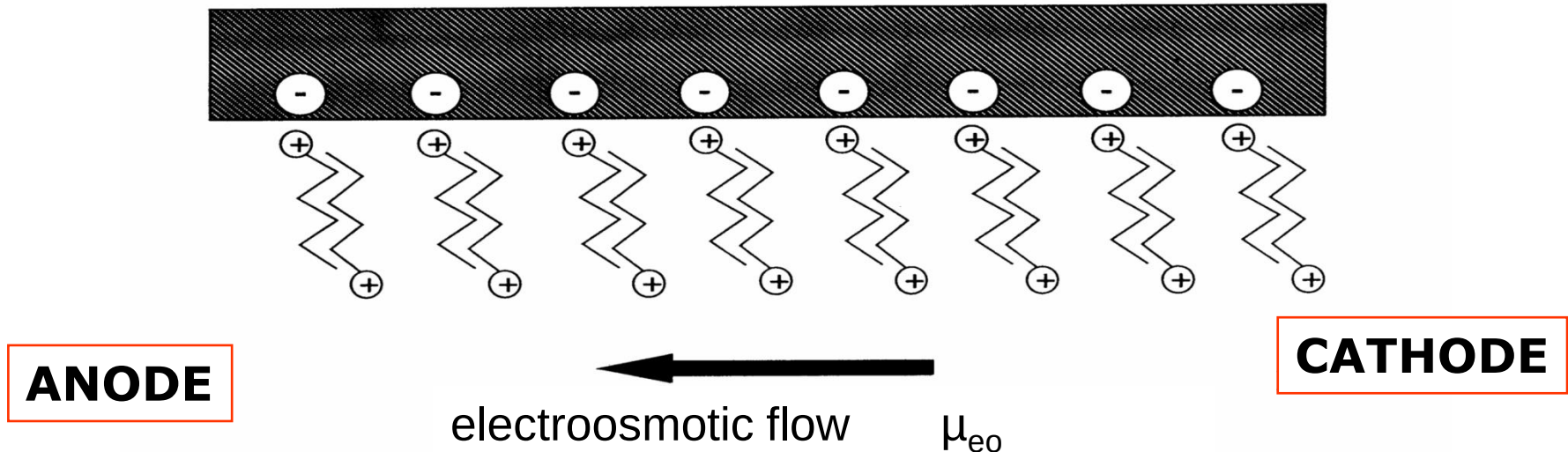
Electroosmotic flow gives less peak broadening than hydrodynamic laminar flow and leads to triangular peak shapes (instead of Gaussian as in chromatography)

4.1.2 Electroosmotic flow

How to manipulate magnitude and direction of EOF:

- Increase of ionic strength of carrier electrolyte decreases EOF
- Organic solvents in carrier electrolyte generally decrease EOF
- Addition of EOF-modifiers can reduce or even reverse EOF

Example for EOF-modifier: Tetradecyltrimethylammonium



4.1.3 Mobilities

- apparent mobility of an analyte ion:
- apparent velocity of an analyte ion:

$$\mu_{app} = \mu_{ep} + \mu_{eo}$$

$$v_{app} = v_{ep} + v_{eo}$$

$$\mu_{app} = \frac{v_{app}}{E} = \frac{L_d / t}{V / L_t} = \frac{L_d L_t}{V t}$$

L_d = length of capillary from injector to detector

t = migration time

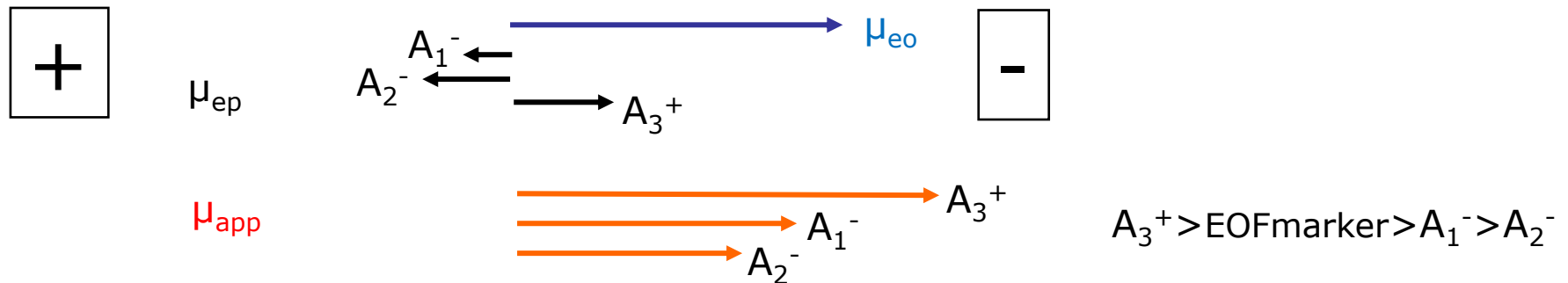
L_t = total length of capillary

V = voltage applied to capillary

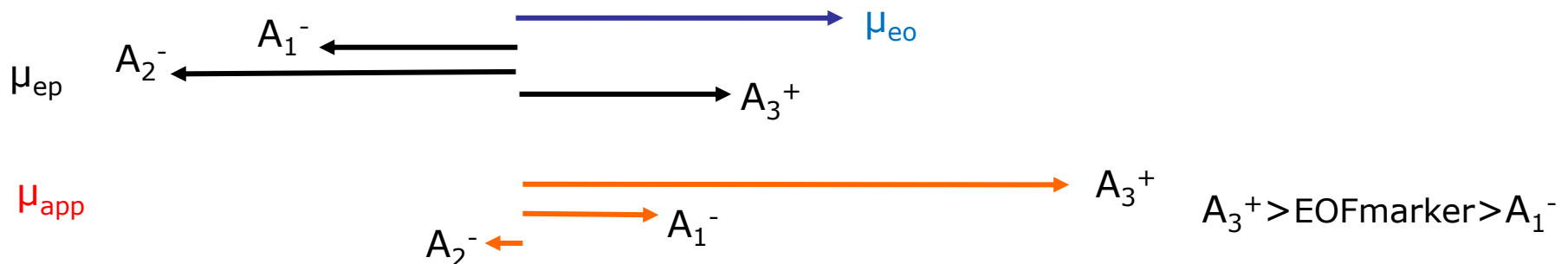
4.1.3 Mobilities

- μ_{ep} / μ_{eo} / μ_{app}

high- and medium-molecular-mass (size) analyte ions



low-molecular-mass (size) analyte ions

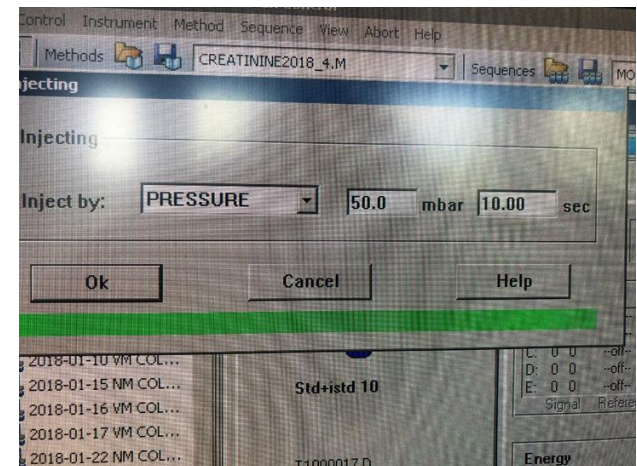
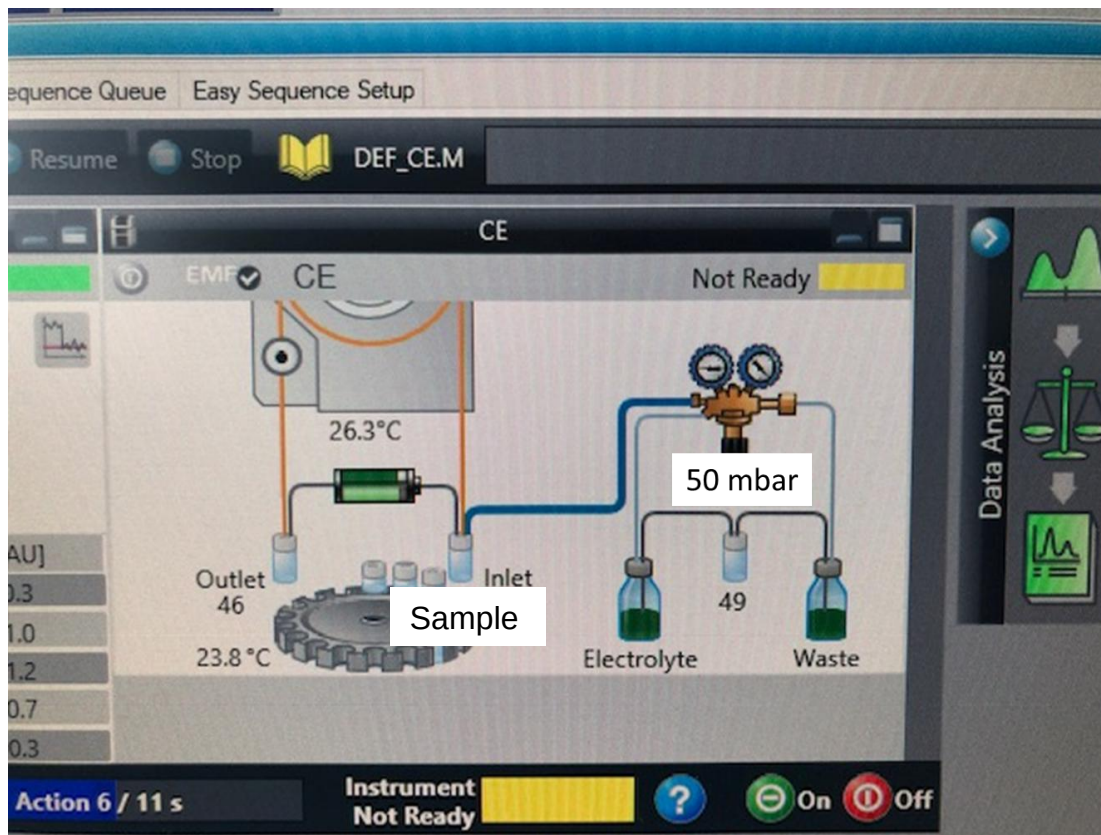


4.1.4 Sample injection

- by pressure
- electrokinetic



very small sample sizes - 10^{-9} liters

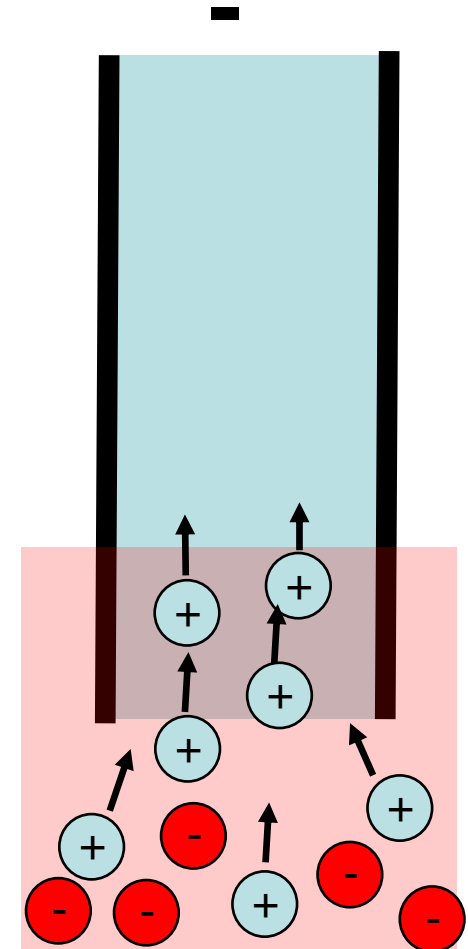


4.1.4 Sample injection

For electrokinetic injection a voltage (+ or -) is applied for a certain time while placing the sample vial at the inlet end of the separation capillary. This leads to:

- Ionic compounds with the respective charge (+ or -) migrate into the capillary
- Additionally the induced EOF transports liquid from the sample vial into the capillary or in the case of reversed polarity liquid from the capillary is brought back into the vial

Electrokinetic injection is highly discriminating - a fact that can be employed to inject a much larger number of the analyte ions of interest („enrichment“) into the capillary than in simple pressure injection.

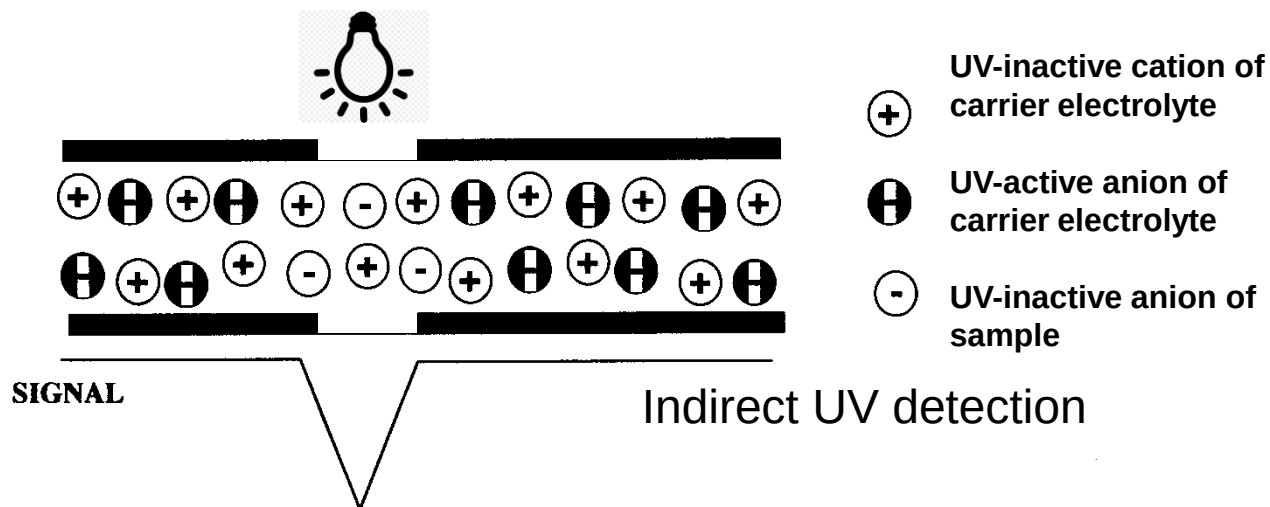
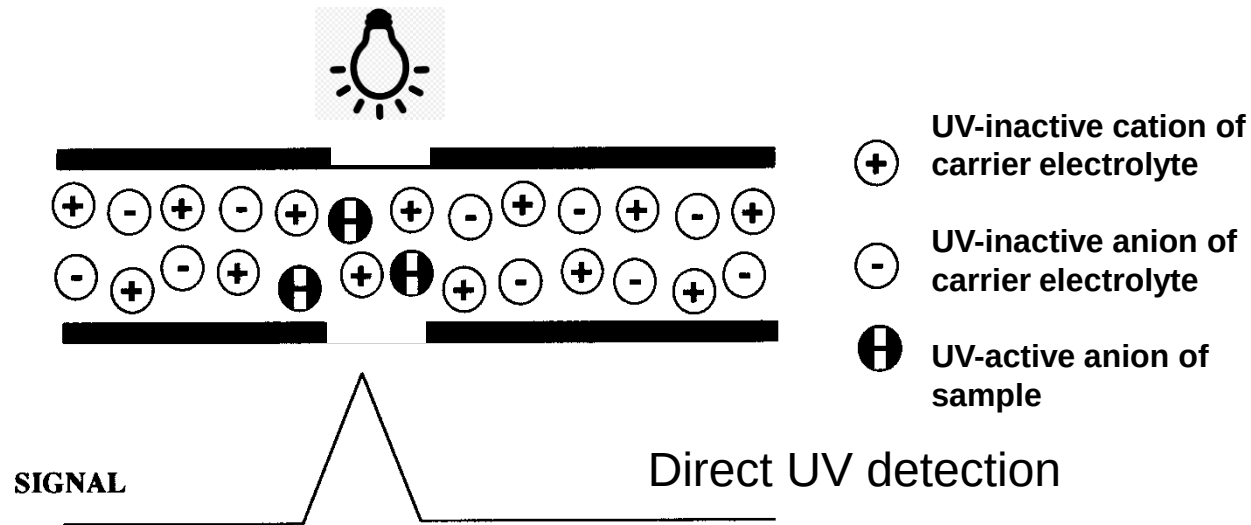


4.1.5 Detectors

- UV absorbance (direct or indirect mode)
(in-capillary detection)
- Fluorescence
(in-capillary detection)
- Conductivity Detection
(in-capillary detection)
- Mass Spectrometry (see MS part)

4.1.5.1 UV- detection

- direct and indirect UV detection

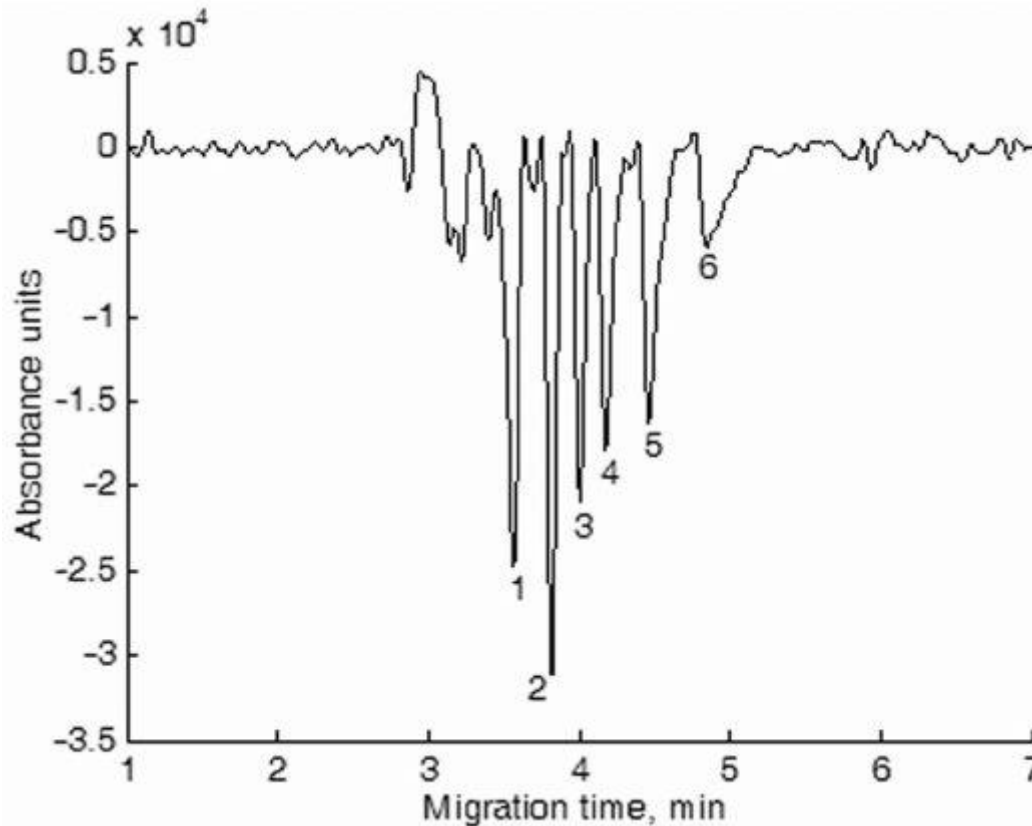


BGE

Phosphate
Borate
(both UV-transparent)

Chromate
Naphthalenedi-
carboxylic acid
(for anions)
Imidazole
(for cations)

4.1.5.1 UV- detection



Separation of methylated phosphonic acids by capillary zone electrophoresis with indirect UV-detection. For improved clarity, peaks are often flipped.

4.1.6 CZE Mechanism

Summarizing the mechanisms of capillary zone electrophoresis (CZE):

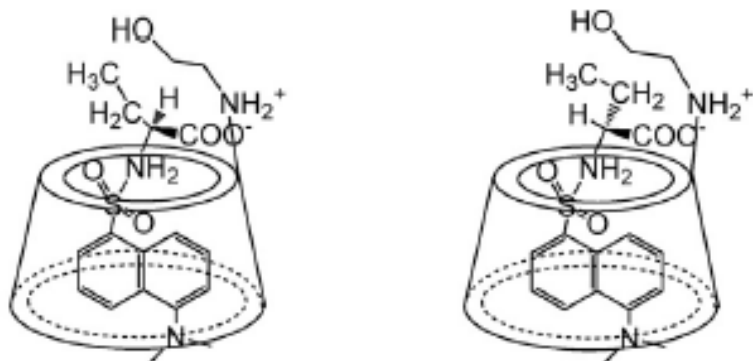
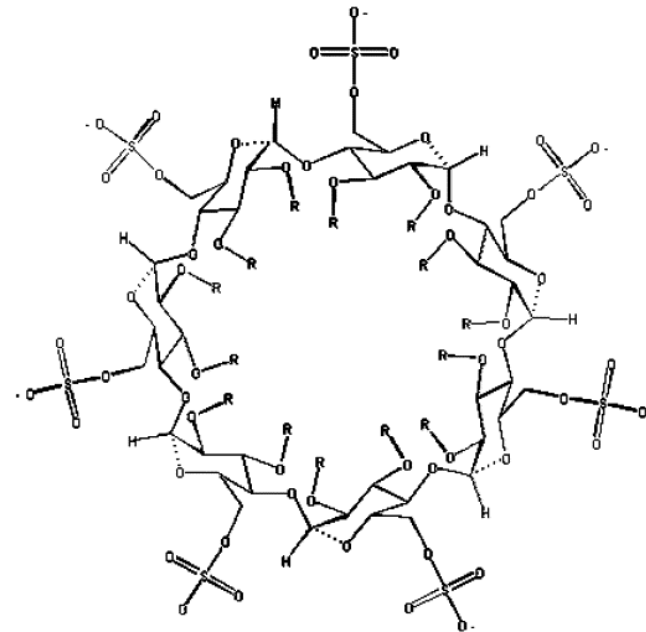
- Capillary filled with (buffered) carrier electrolyte
- Separation according to electrophoretic mobilities (therefore according to radius/charge ratio of analyte)
- Optimization of separation can be done by:
 - changing the pH (changes the degree of dissoziation of the analytes or even reverses the charge)
 - addition of complex-forming agents (changes in charge and/or size) to the carrier electrolyte

4.1.7 Chiral separations by CZE

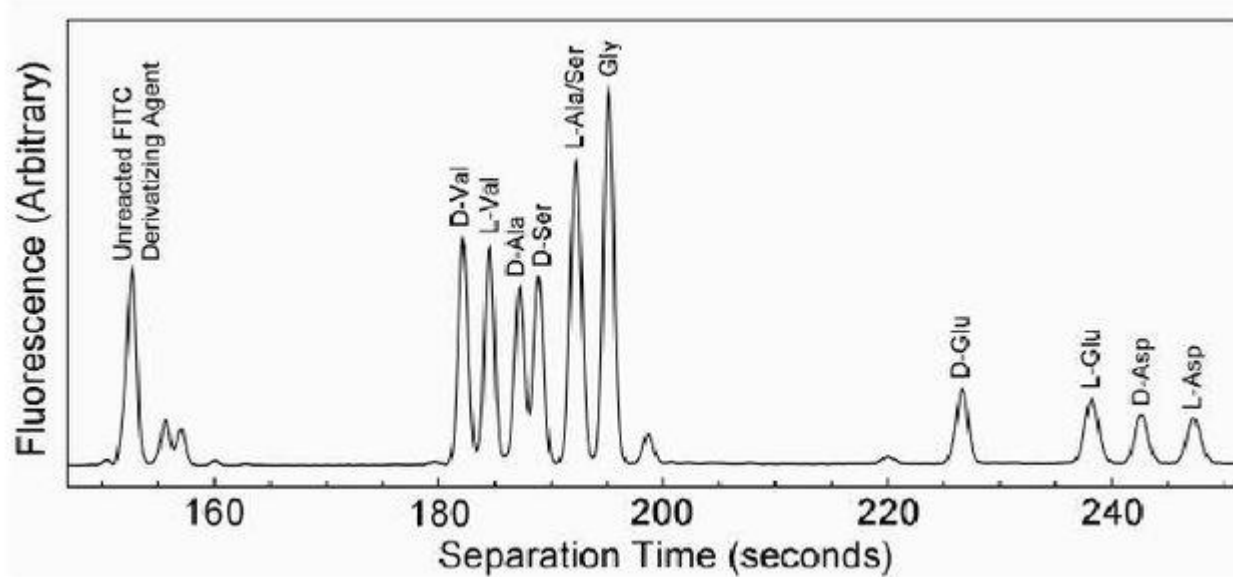
Addition of cyclodextrins (chiral selectors) to the carrier electrolyte.
Formation of diastereomeric complexes between the enantiomeric analytes and the chiral selectors

α -, β - or γ -cyclodextrin
methyl derivatives of β -CD
2-hydroxypropyl- β -CD
carboxymethyl- β -CD
sulfated β -CD
sulphobutylether- β -CD
2-hydroxypropyltrimethyl-
ammonium- β -CD

....



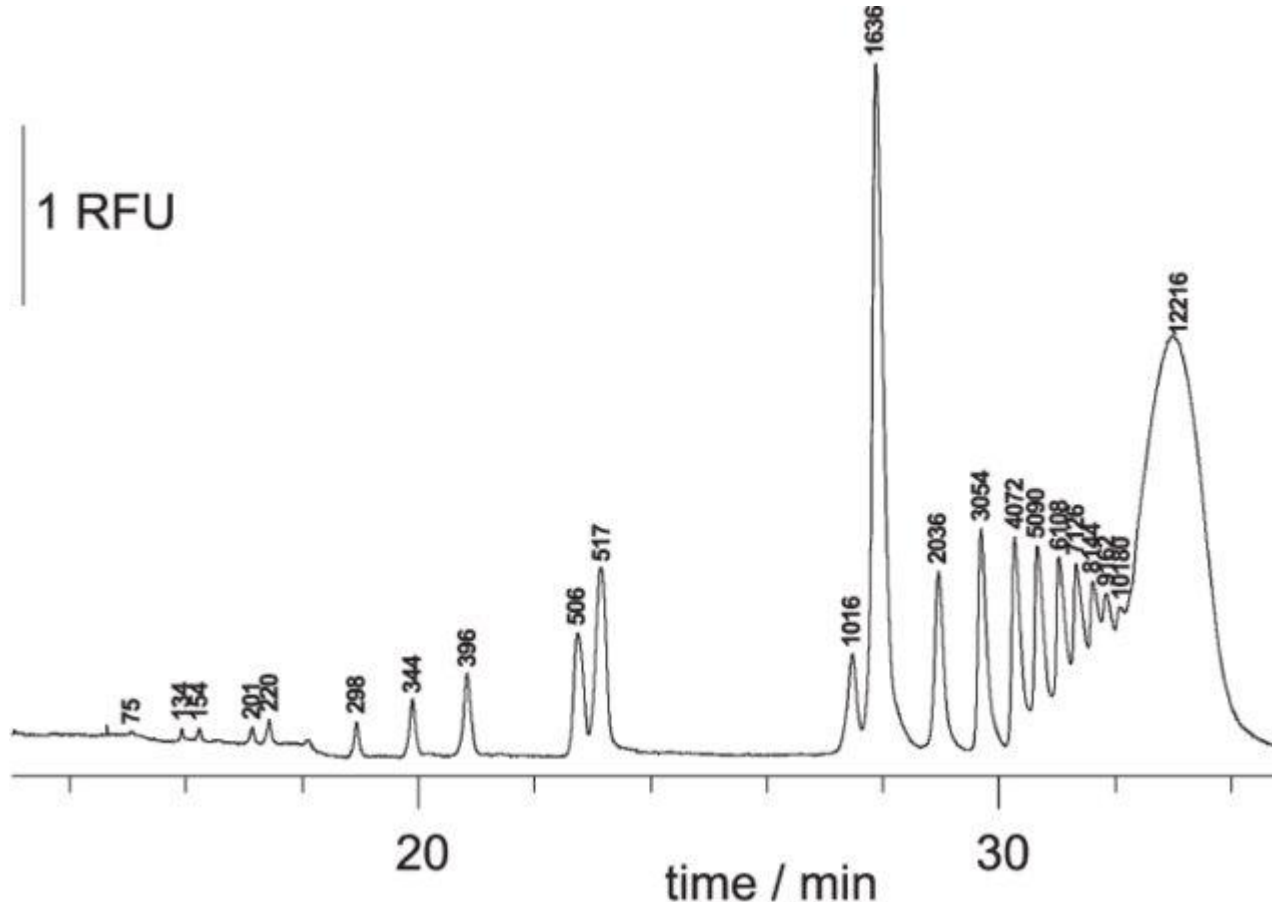
4.1.7 Chiral separations by CZE



Separation of amino acids by chip CZE

Run conditions: sodium dodecyl sulfate/ γ -cyclodextrin, pH 10.0, carbonate electrophoresis buffer, separation voltage of 550 V/cm at 10° C.

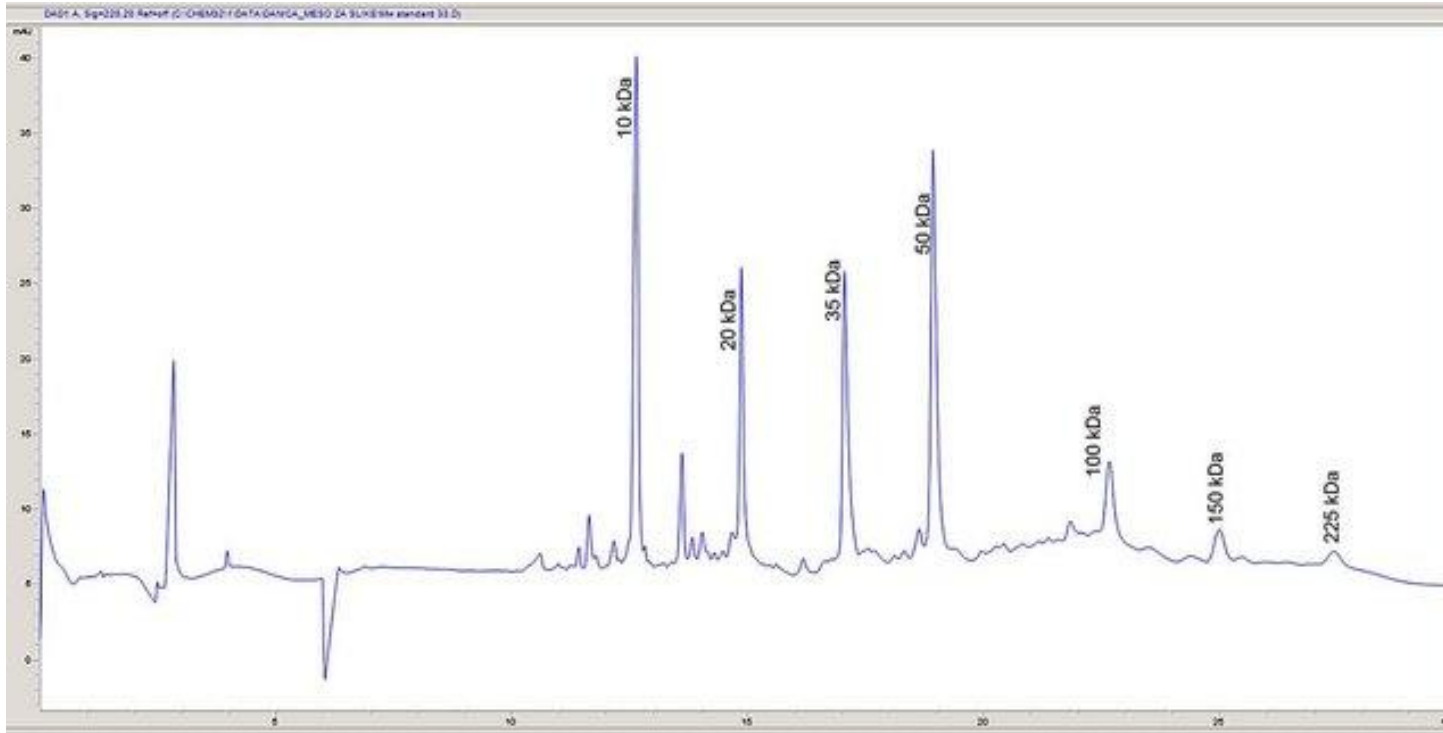
4.2 Capillary gel electrophoresis



Model separation of 1-kbp DNA ladder in capillary electrophoresis using hydroxyethylcellulose as sieving matrix. A 75 μm id PVA coated capillary column with 47 cm long (40 cm effective length) was filled with 0.5 % HEC solution in 100 mmol L⁻¹ TTE buffer and the separation was carried out with 200 V cm⁻¹ electric field.

Separation of DNA; number refers to number of base pairs

4.2 Capillary gel electrophoresis



Capillary SDS gel electrophoresis separation of a standard protein test mixture with a molecular weight range of 10 kDa to 225 kDa, on bare-fused silica capillaries filled with SDS-MW gel buffer, SDS sample buffer-100 mM Tris-HCl with pH 9.0 and 1% SDS.

SDS CGE of proteins

SDS is used to form negatively charged complexes with proteins. Thereby, a constant ratio of charge/mass is achieved for all proteins, allowing a separation in the gel due to differences in size.

4.3 Isoelectric focusing

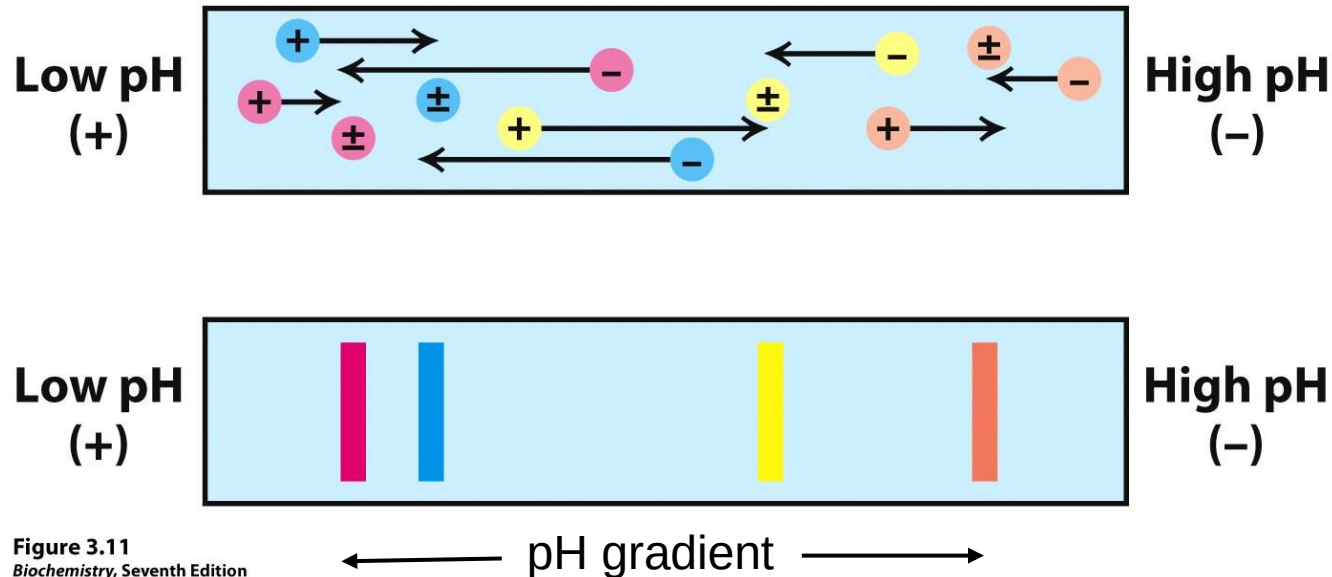
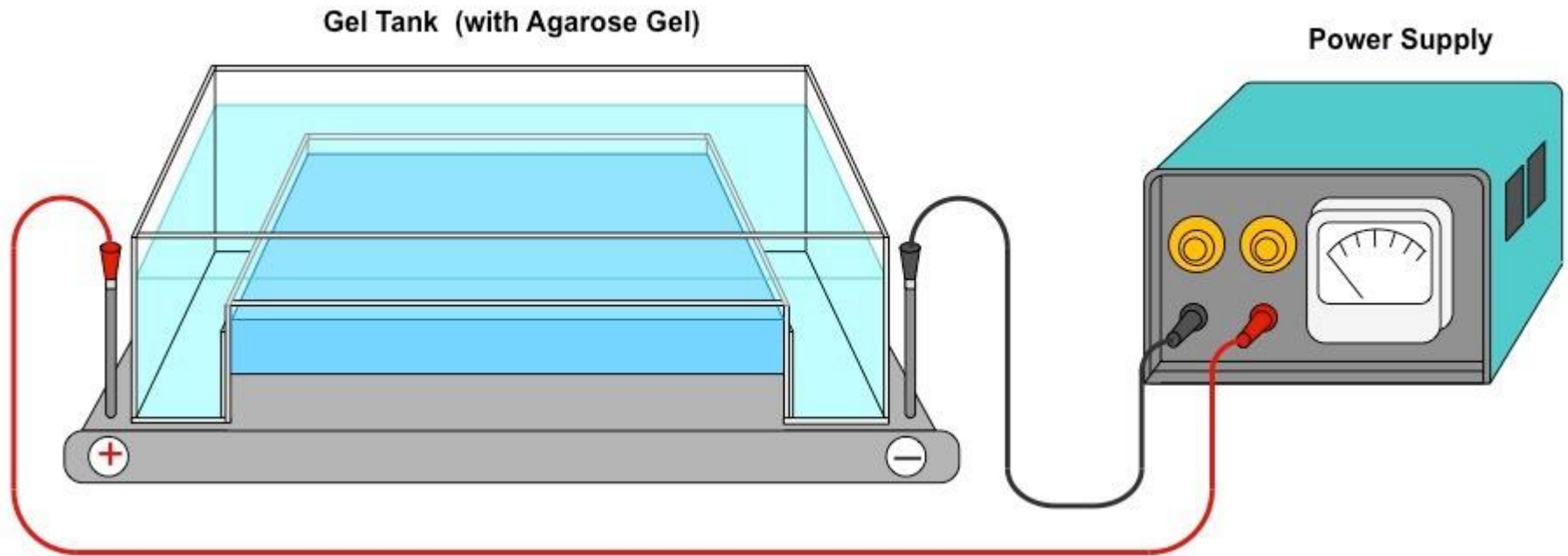


Figure 3.11
Biochemistry, Seventh Edition
© 2012 W. H. Freeman and Company

Capillary filled with mixture of ampholytes (aminocarboxylic acids of different pH) leading to a pH gradient after applying voltage.

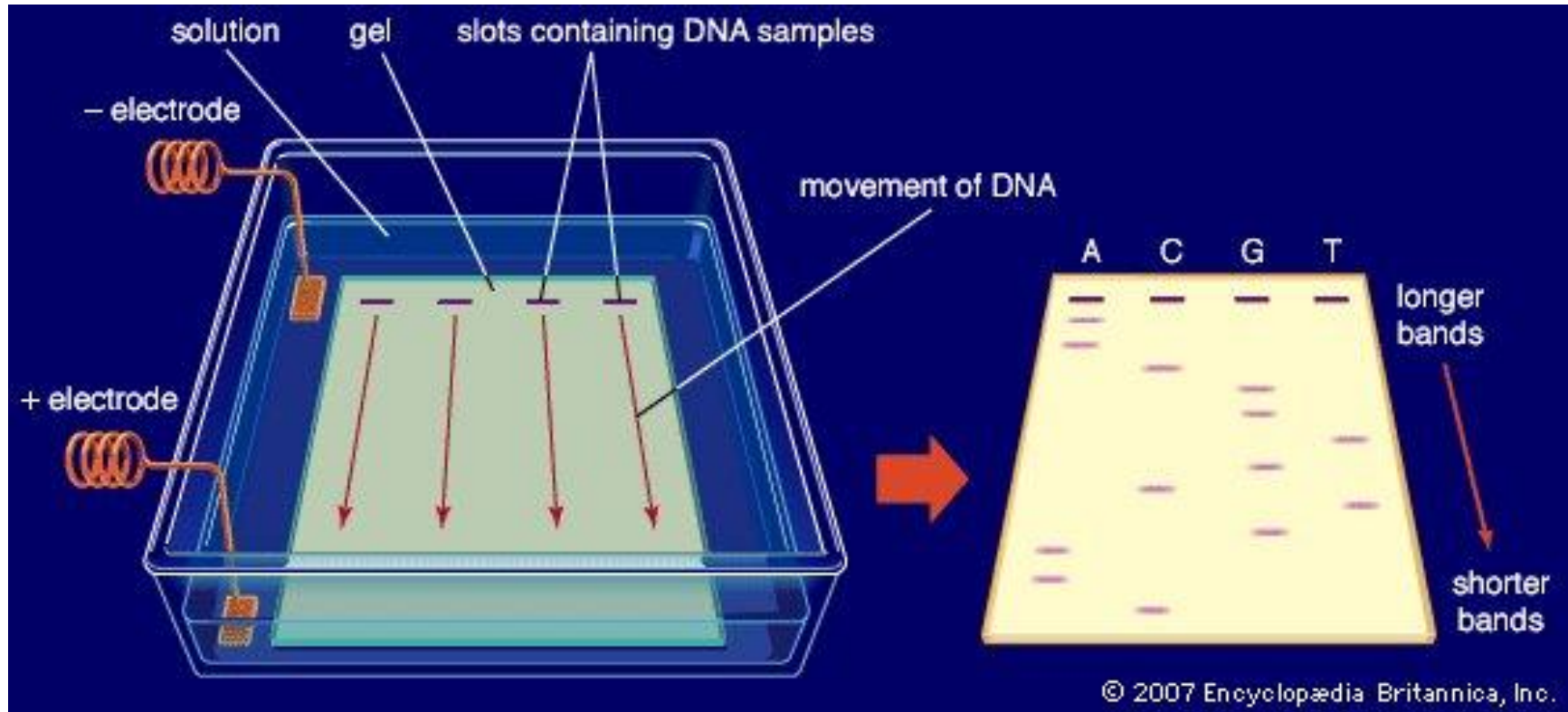
Ampholytic analytes (proteins, ...) are separated according to their pI (isoelectric point) values. Analytes migrate in the pH gradient until they reach the pH zone corresponding to their pI value and become neutral.

4.4 Gel electrophoresis



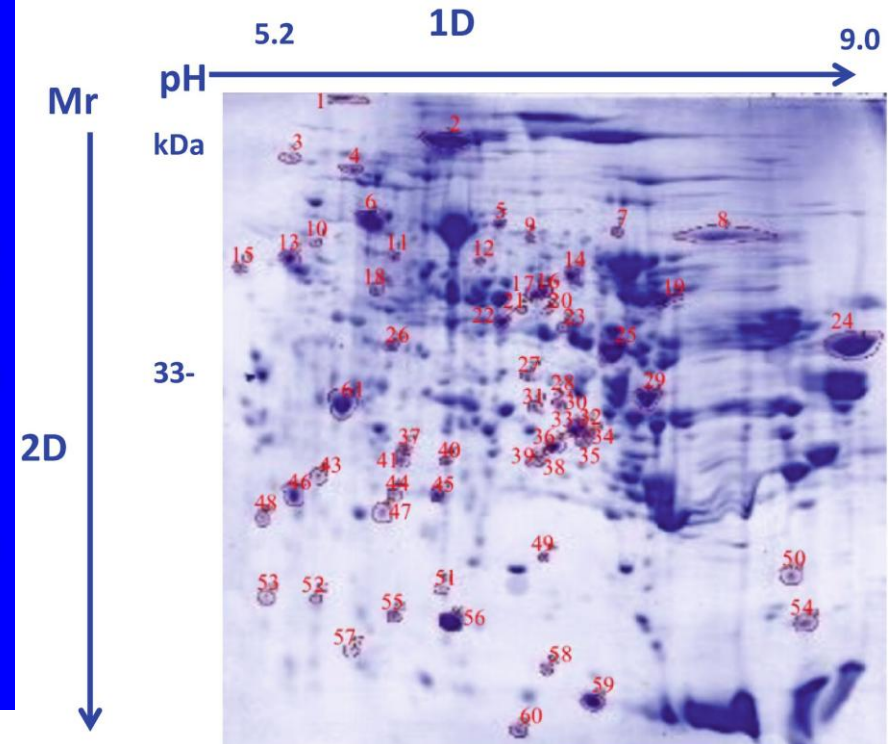
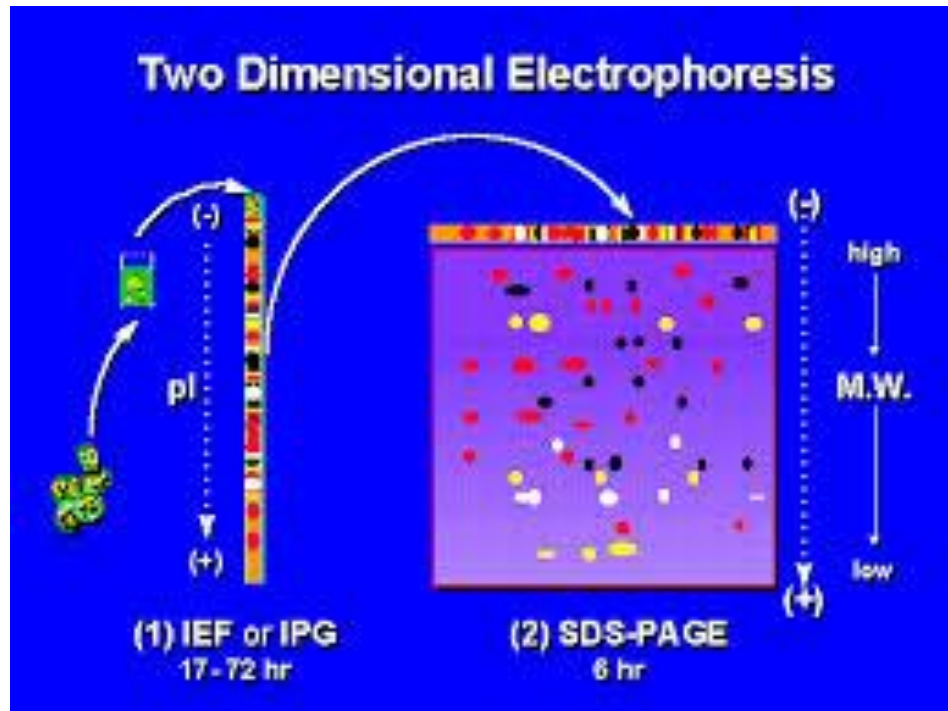
In gel electrophoresis a layer of an anticonvective support (polyacrylamide, agarose or cellulose acetate) is coated on a glass plate. Upon application of high voltage analytes (e.g. proteins) are separated

4.4 Gel electrophoresis



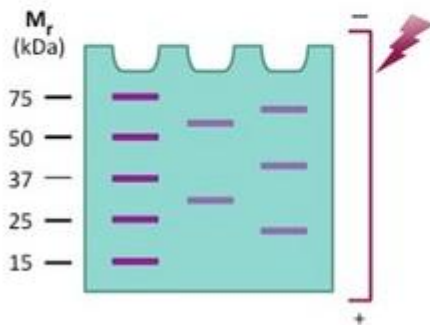
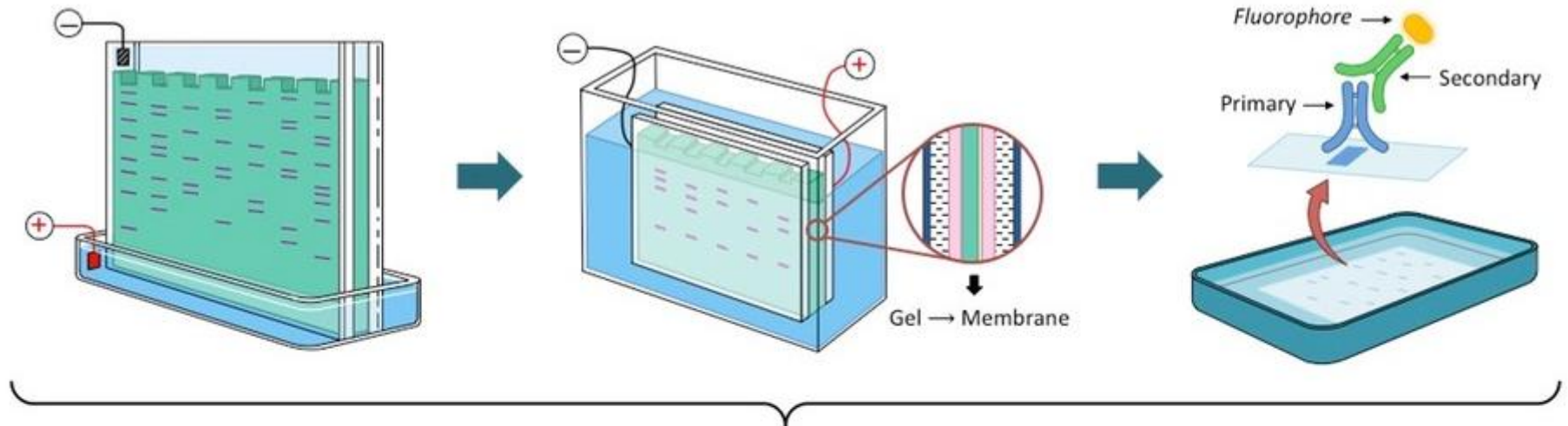
In gel electrophoresis a layer of an anticonvective support (polyacrylamide, agarose or cellulose acetate) is coated on a glass plate. Upon application of high voltage analytes (e.g. proteins) are separated

4.4 Gel electrophoresis

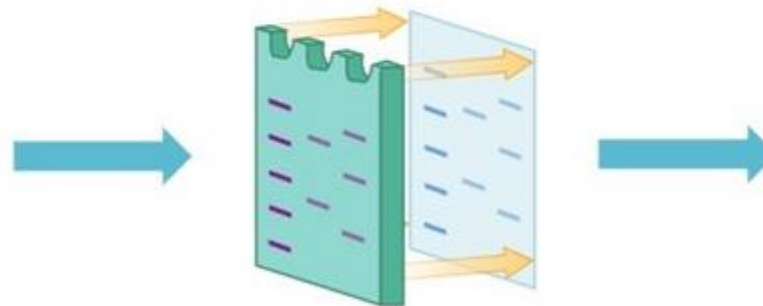


In SDS – polyacrylamide gel electrophoresis (SDS-PAGE) samples are separated in two dimensions. First according to pI (IEF) and subsequently according to size (molecular weight) - SDS gel electrophoresis

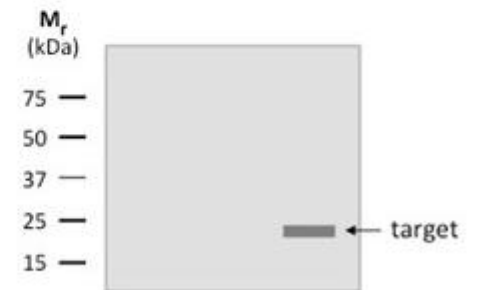
4.4 Gel electrophoresis



Electric charge separates proteins according to size



Proteins transferred to a solid support membrane



Proteins stained with specific antibodies for visualisation

For details see next slide

Spectroscopy and Mass Spectrometry

5. Infrared Spectroscopy

5.1. Instrumentation

5.1.1. Radiation Source for IR

5.1.2. Monochromators for IR

5.1.3. Interferometers for FT-IR

5.1.4. Detectors for IR

5.2. Principles of Vibrations in IR

5.2.1. The Harmonic Oscillator

5.2.2. The Non-Harmonic Oscillator

5.3 Principles of Rotations in IR

5.4. Specific Vibrations in IR

5.5 Sampling Techniques

5.6 IR Spectra

6. Raman Spectroscopy

7. Mass Spectrometry

7.1. Ionization Sources

7.1.1. Electron (Impact) Ionization

7.1.2. Chemical Ionization

7.1.3. Atmospheric Pressure Ionization

7.1.3.1. ESI

7.1.3.2 APCI

7.1.4. MALDI

7.2. Mass Analyzers

- 7.2.1. Quadrupol Mass Analyzer
- 7.2.2. Triple Quadrupol Mass Analyzer
- 7.2.3. Ion Trap Mass Analyzer
- 7.2.4. Electrostatic Trap (Orbitrap)
- 7.2.5. Magnetic Field Mass Analyzer
- 7.2.6. Time of Flight Mass Analyzer
- 7.2.7. Hybrid Instruments

7.3. Detectors

- 7.3.1. SEV and CEM
- 7.3.2. Micro Channel Plate
- 7.3.3. FT-based Detectors

7.4 Hyphenation

- 7.4.1. GC-MS
- 7.4.2. HPLC-MS
- 7.4.3. CE-MS

7.5. Mass Spectra

- 7.5.1. Isotopic Pattern
- 7.5.2. High Resolution Mass Spectrometry-Exact Mass
- 7.5.3. Multiple Charging-Deconvolution
- 7.5.4. Protein Identification

8. Inductively Coupled Plasma Spectroscopy

8.1 ICP-AES

8.2 ICP-MS

8.3 Comparison of Techniques

8.4 Laser Ablation ICP-MS

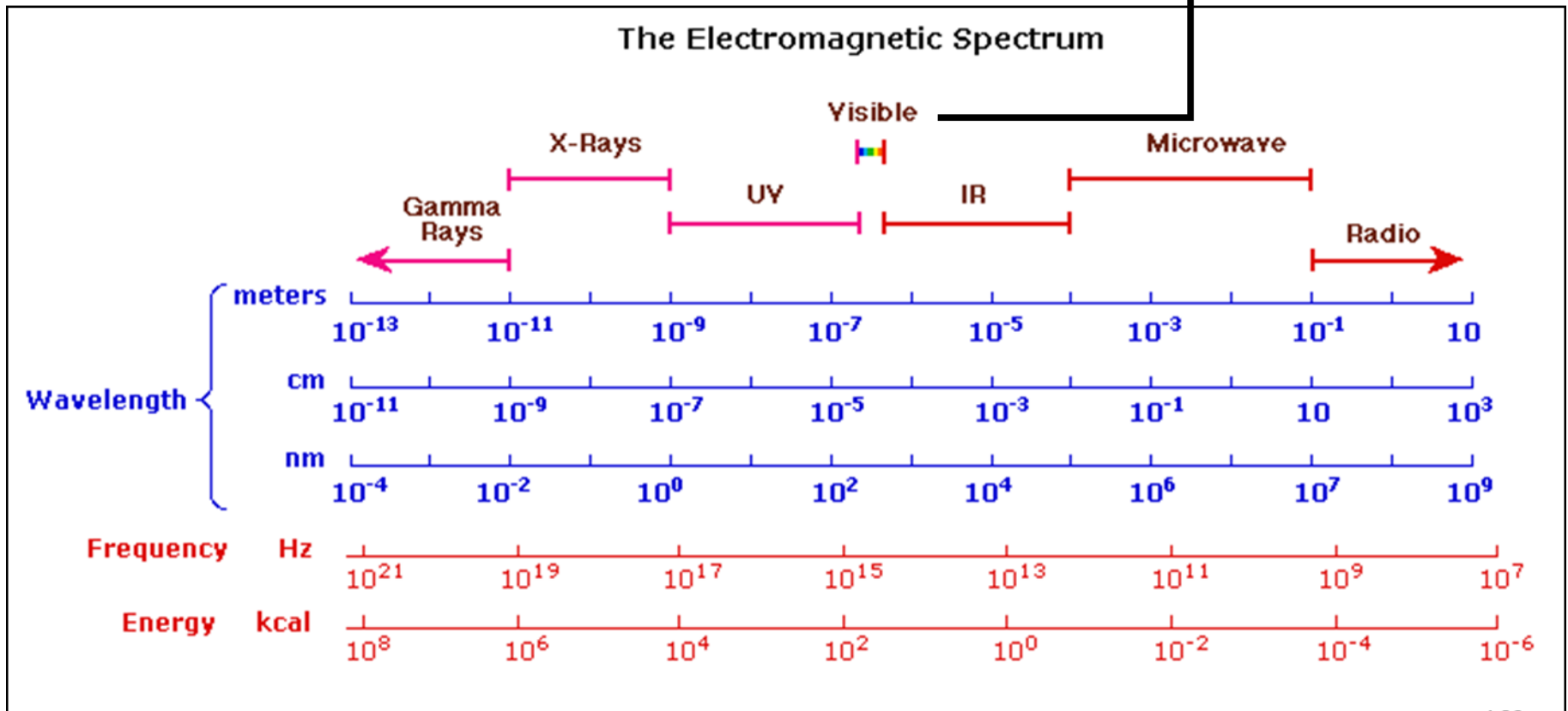
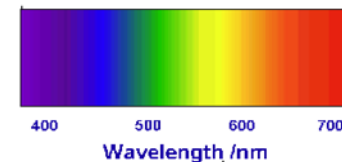
The following techniques will be discussed in detail:

- Infrared-Spectroscopy
- Raman Spectroscopy
- Mass-Spectrometry
- Inductively Coupled Plasma Spectroscopy (coupled to MS)

5. Spectroscopy

A short Introduction to Spectroscopy

5. Spectroscopy



5. Spectroscopy

Remember:

$$E = h\nu = hc/\lambda = hc\bar{\nu}$$

E.....Energy

h.....Planck's constant ($6,6 \cdot 10^{-34} \text{ Js}^{-1}$)

νFrequency [Hz]

λWavelength [cm]

c.....Speed of light ($2,99 \cdot 10^{10} \text{ cms}^{-1}$)

$\bar{\nu}$Wavenumber [cm^{-1}]

5. Spectroscopy

The energy of the radiation employed determines the possible changes induced in the molecule and thereby also determines the effects that can be observed:

- E n e r g y +		- NMR	flipping nuclear spin
		- ESR	flipping spin of unpaired electrons
		- Microwaves	rotation of molecules
		- IR/Raman	vibration of molecules
		- UV/vis	excitation of valence electrons
		- X-Rays	excitation of electrons of the inner shells
		- Mößbauer-spectroscopy	excitation of the nucleus

5. Spectroscopy

Radiation can interact with molecules or atoms and thereby induce changes (as described on the previous slide). Spectroscopical methods can measure parameters related to these changes. In principle there are different groups determined by the way of measurement. **Absorption-methods** are based on the absorption of part of the radiation by the molecule. Thereby the differences between the radiation before and after interaction with the molecule/atom are determined. Typical absorption-methods are:

- NMR (Nuclear Magnetic Resonance)
- ESR (Electron Spin Resonance)
- Microwave-spectroscopy
- IR spectroscopy
- UV-vis spectroscopy, Atomabsorption-spectroscopy
- X-ray spectroscopy
- Mößbauer-spectroscopy

5. Spectroscopy

Radiation can interact with molecules or atoms and thereby induce changes (as described on the previous slide). Spectroscopical methods can measure parameters related to these changes. In principle there are different groups determined by the way of measurement. **Emission-methods** are based on emission of radiation by the molecule/atom after absorption of radiation leading to an excited state. The emission process is related to the relaxation of the molecule/atom. Typical emission-methods are:

- Atomic emission spectroscopy (AES, ICP-AES)
- Fluorescence spectroscopy
- Auger-Electron spectroscopy
- ESCA (Electron Spectroscopy for Chemical Analysis)

5. Spectroscopy

Besides the two different principles of measurement described on the previous two slides, other possible ways of measurement exist; some other methods based on the interaction of radiation (light) with molecules and atoms are:

- diffraction of light (Nephelometry, Turbidimetry)
- diffraction of X-rays
- reflectance of radiation
- refraction of radiation (light) (refractive-index)
- methods based on changes in the polarization of light

5. IR-Spectroscopy

Infrared-Spectroscopy

5. IR-Spectroscopy

IR-spectroscopy is based on radiation interfering with molecules and inducing molecular vibrations. Depending on the wavelength range employed, 3 different ranges of IR-spectroscopy can be distinguished:

N(ear)IR 14.000 cm^{-1} – 4.000 cm^{-1} or 720 nm – 2,5 mm

M(id)IR 4.000 cm^{-1} – 200 cm^{-1} or 2,5 mm – 50 mm

F(ar)IR 200 cm^{-1} – 20 cm^{-1} or 50 mm - 500 mm

In particular MIR and NIR play an important role in Analytical Chemistry.

Remember:

$$E = h\nu = hc/\lambda = hc\bar{\nu}$$

E.....Energy

h.....Planck's constant ($6,6 \cdot 10^{-34} \text{ Js}^{-1}$)

νFrequency [Hz]

λWavelength [cm]

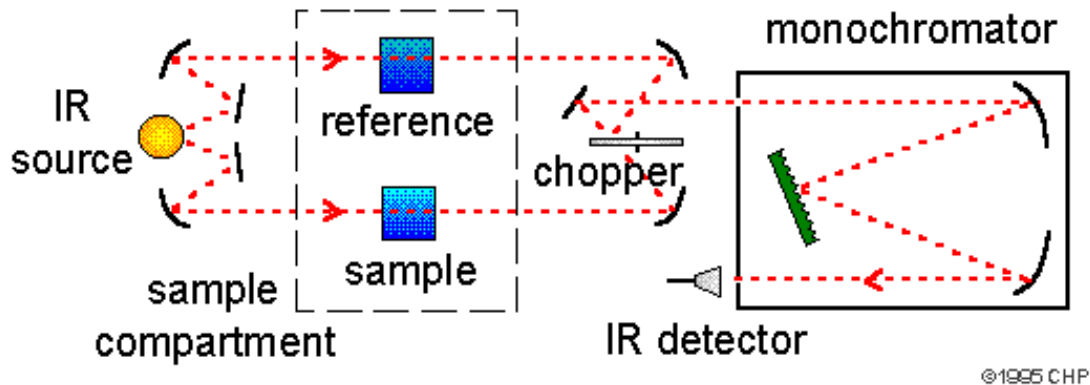
c.....Speed of light ($2,99 \cdot 10^{10} \text{ cms}^{-1}$)

$\bar{\nu}$Wavenumber [cm^{-1}]

5.1 IR-Spectroscopy-Instrumentation

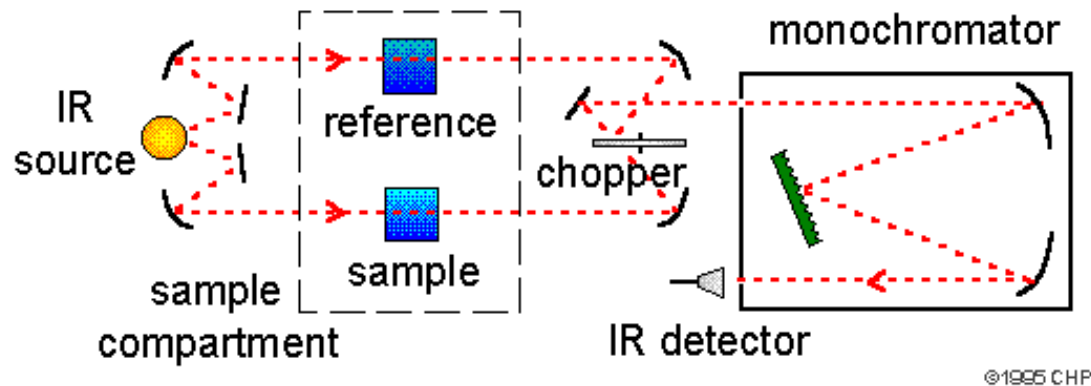
Instrumentation for MIR:

For a long time, standard or **dispersive** MIR instruments were dual-beam instruments using gratings for wavelength selection (monochromator). A typical layout for such an instrument is depicted below. The dual-beam design allows the simultaneous measurement of an IR-spectrum of the background (or solvent) and the sample. The continuous background subtraction provides a background-corrected IR-spectrum for the sample.



The more recently developed F(ourier)T(ransform)-IR instruments are mostly based on a single beam design. This means two separate spectra (background and sample) have to be recorded.

5.1 IR-Spectroscopy-Instrumentation



- Dual beam instrument allows on-line compensation of solvent spectrum
- Disadvantages of **dispersive** (instruments with monochromators) IR-instruments (layout see above):

- long time needed for acquisition-no accumulation of spectra possible $\overline{S/N} \cong \sqrt{n}$
- low yield of radiation intensity due to slit aperture
- no constant spectral resolution over the wave-length range

5.1.1 Radiation Source for IR

Radiation sources employed:

For MIR

Globar: glower made from siliciumcarbide

- ☹ relatively low intensity of radiation
- ☺ mechanically robust; primarily important for out-door

instruments

Nernst glower: glower made from zirconiumoxide with oxides from rare earth elements

- ☺ high radiation intensity
- ☹ mechanically less robust than globar; mainly used for lab-instruments.

For NIR

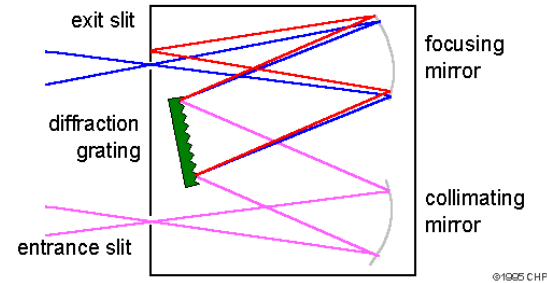
Quartz-halogen-lamp

Tungsten-lamp

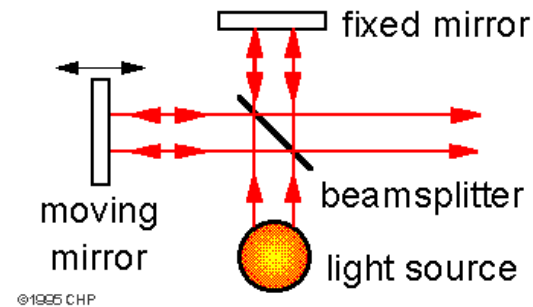
5.1.2 Monochromators for IR

Monochromators: (wave-selection)

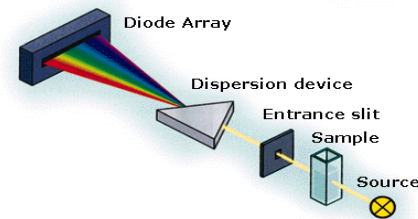
- gratings



- interferometers

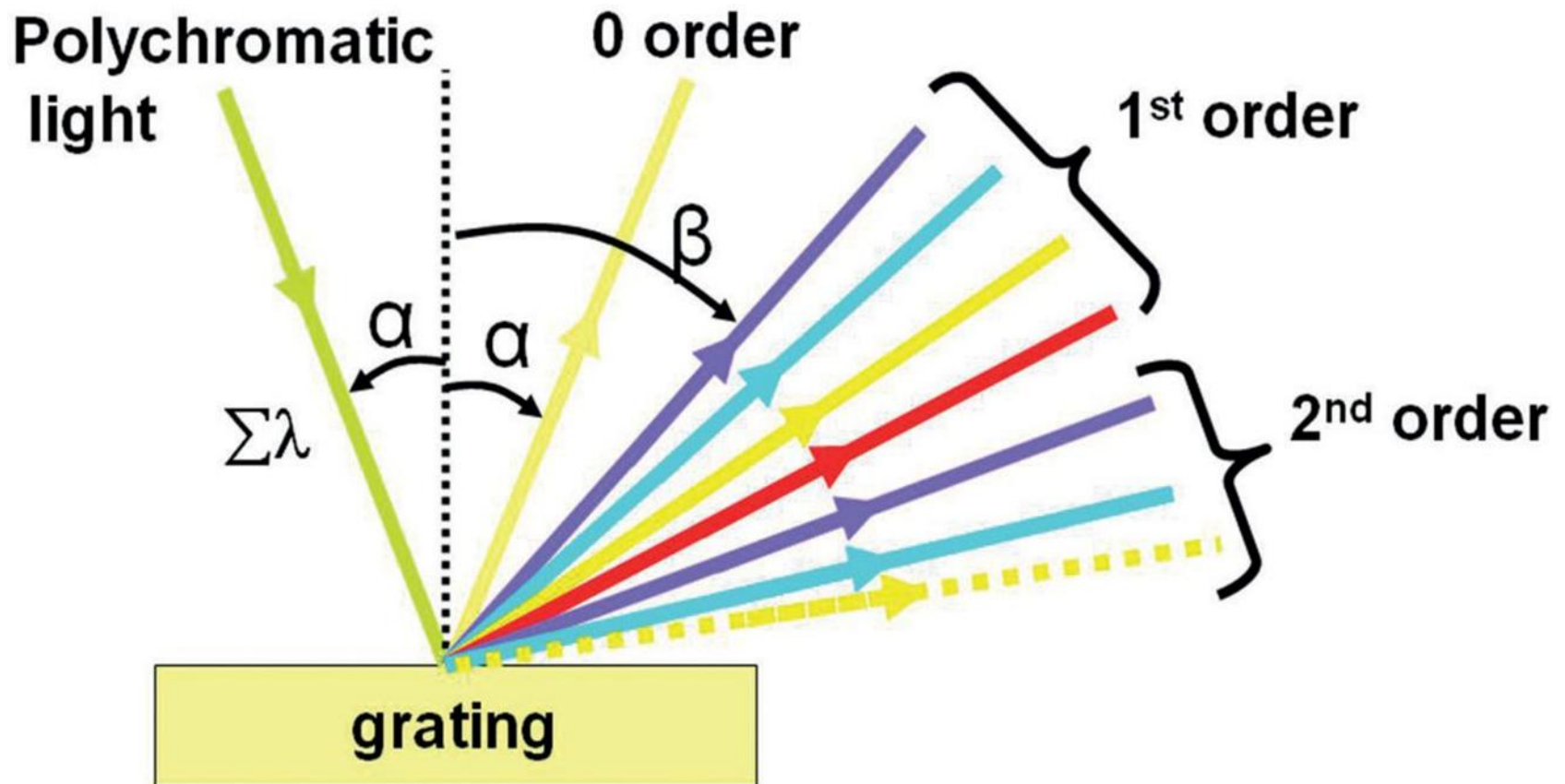


- diode array (NIR only)



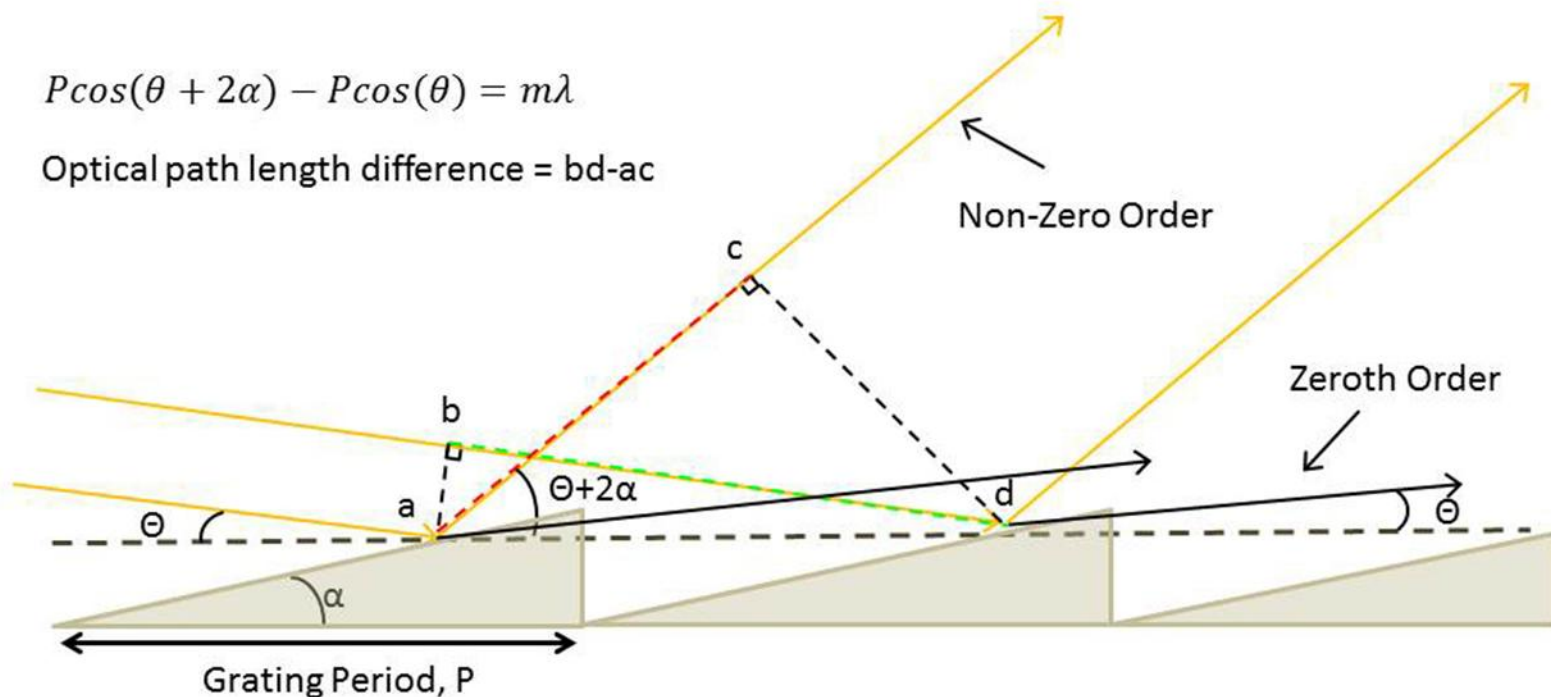
5.1.2 Monochromators for IR

Gratings are used as dispersive element and are a very common tool for wavelength selection.



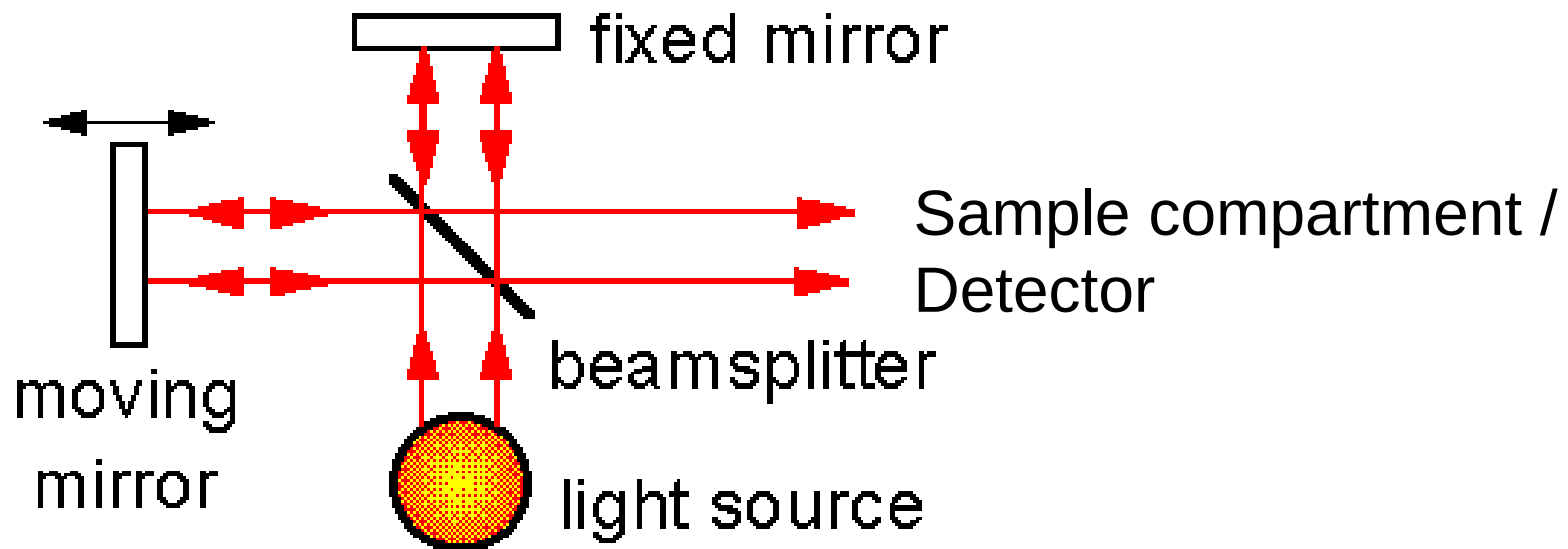
5.1.2 Monochromators for IR

Gratings are used as dispersive element and are a very common tool for wavelength selection.



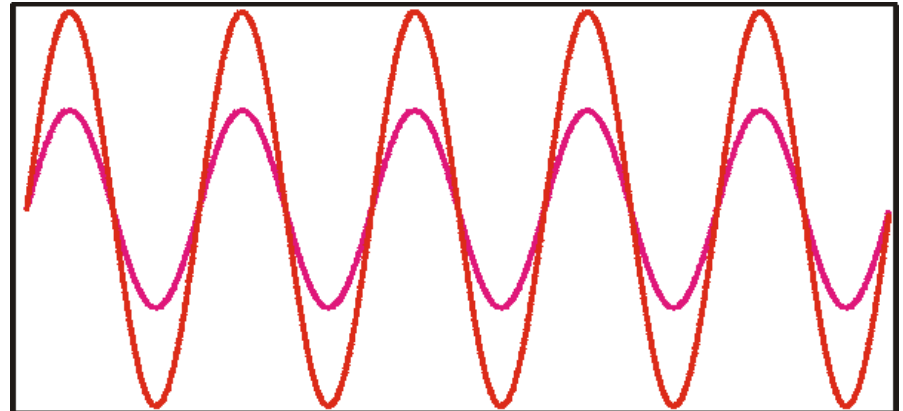
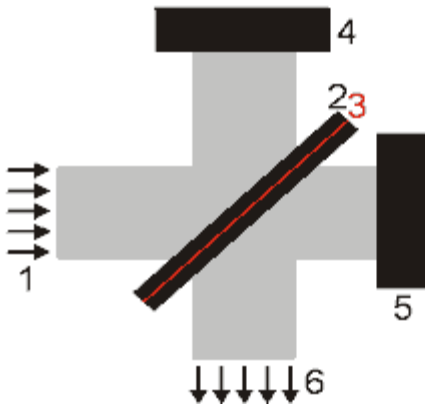
5.1.3 Interferometers for FT-IR

Michelson Interferometer:



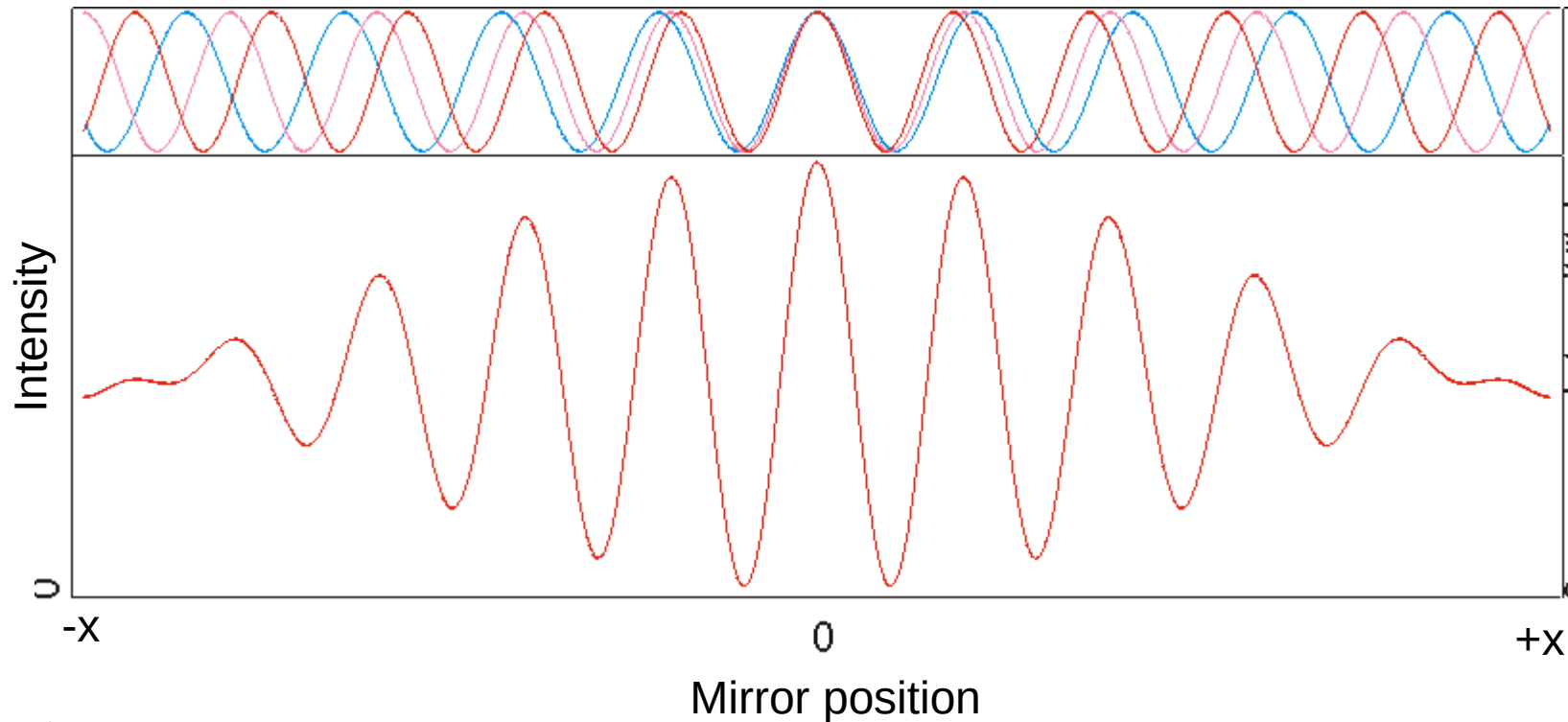
5.1.3 Interferometers for FT-IR

Michelson Interferometer:

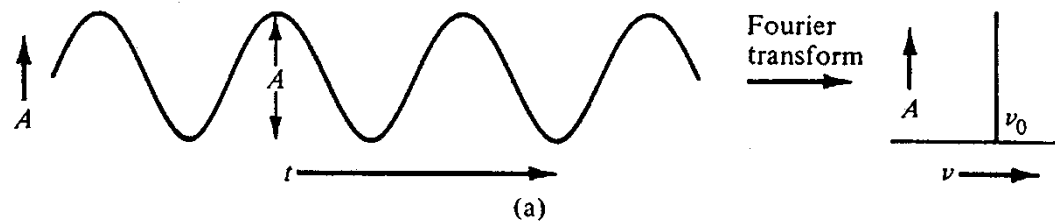


<http://www.ir-spektroskopie.de/spec/ftir-prinzip/michelson.gif>

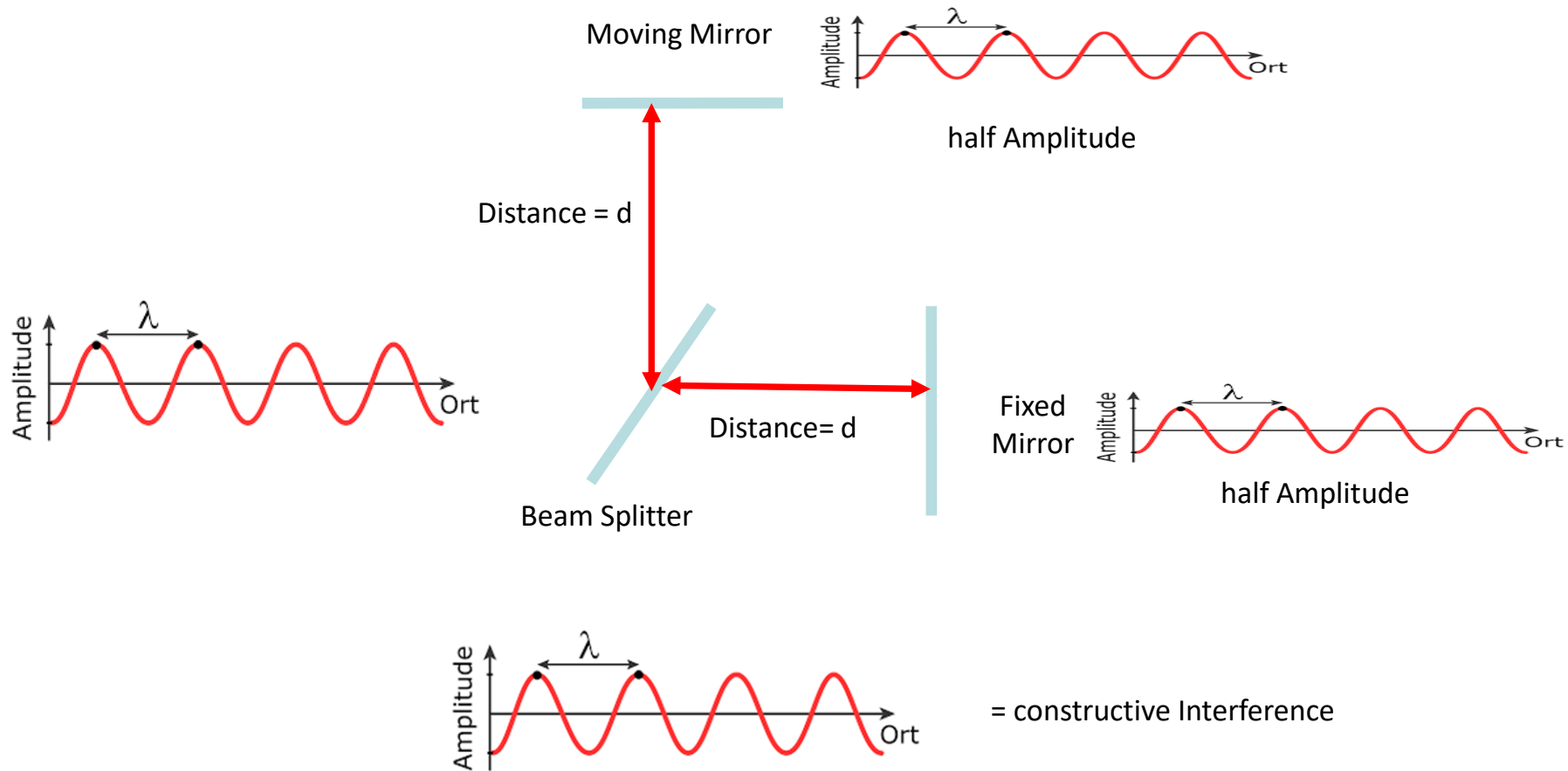
5.1.3 Interferometers for FT-IR



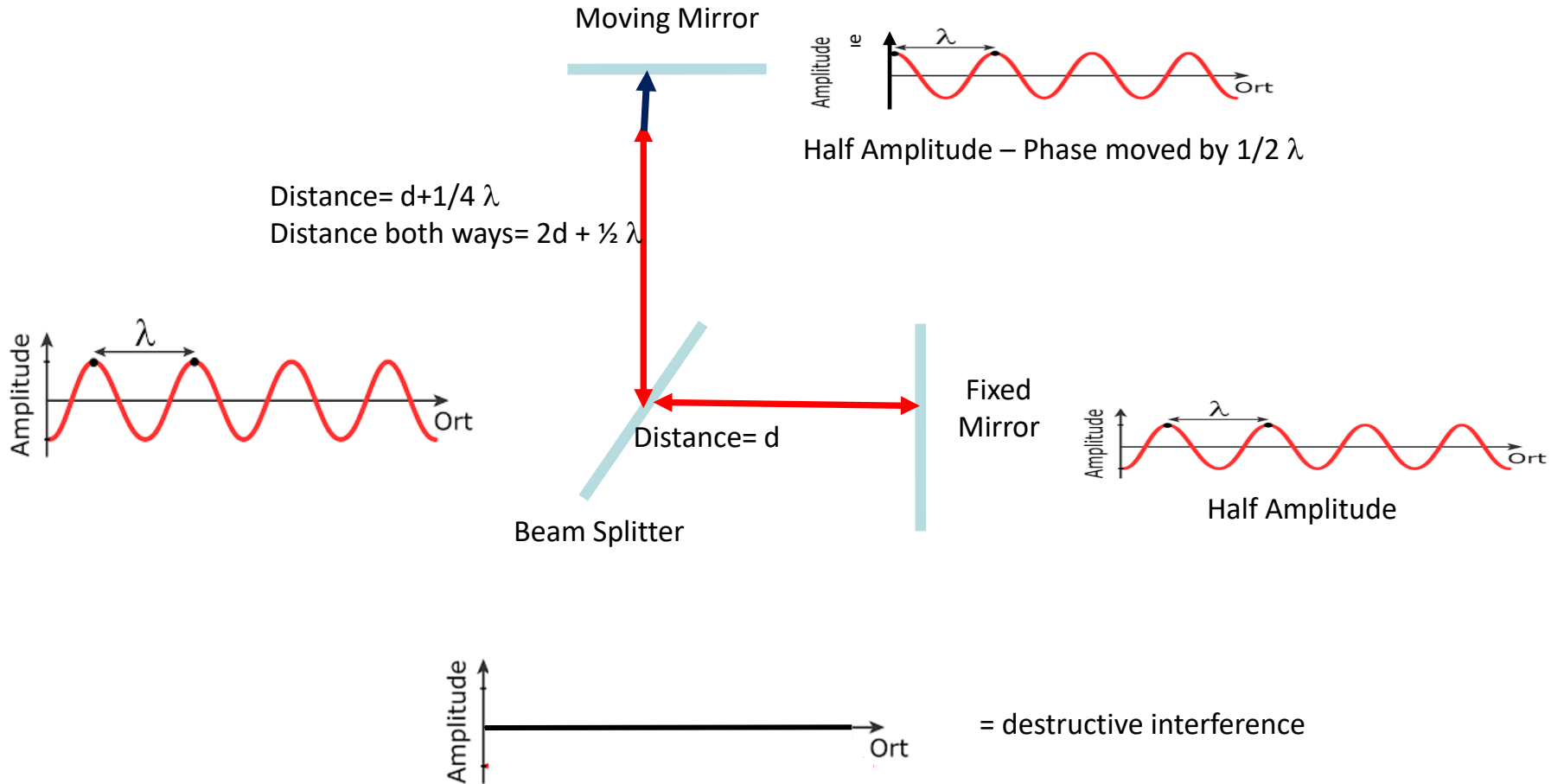
Simple Fourier transform pairs: (a) cosine function



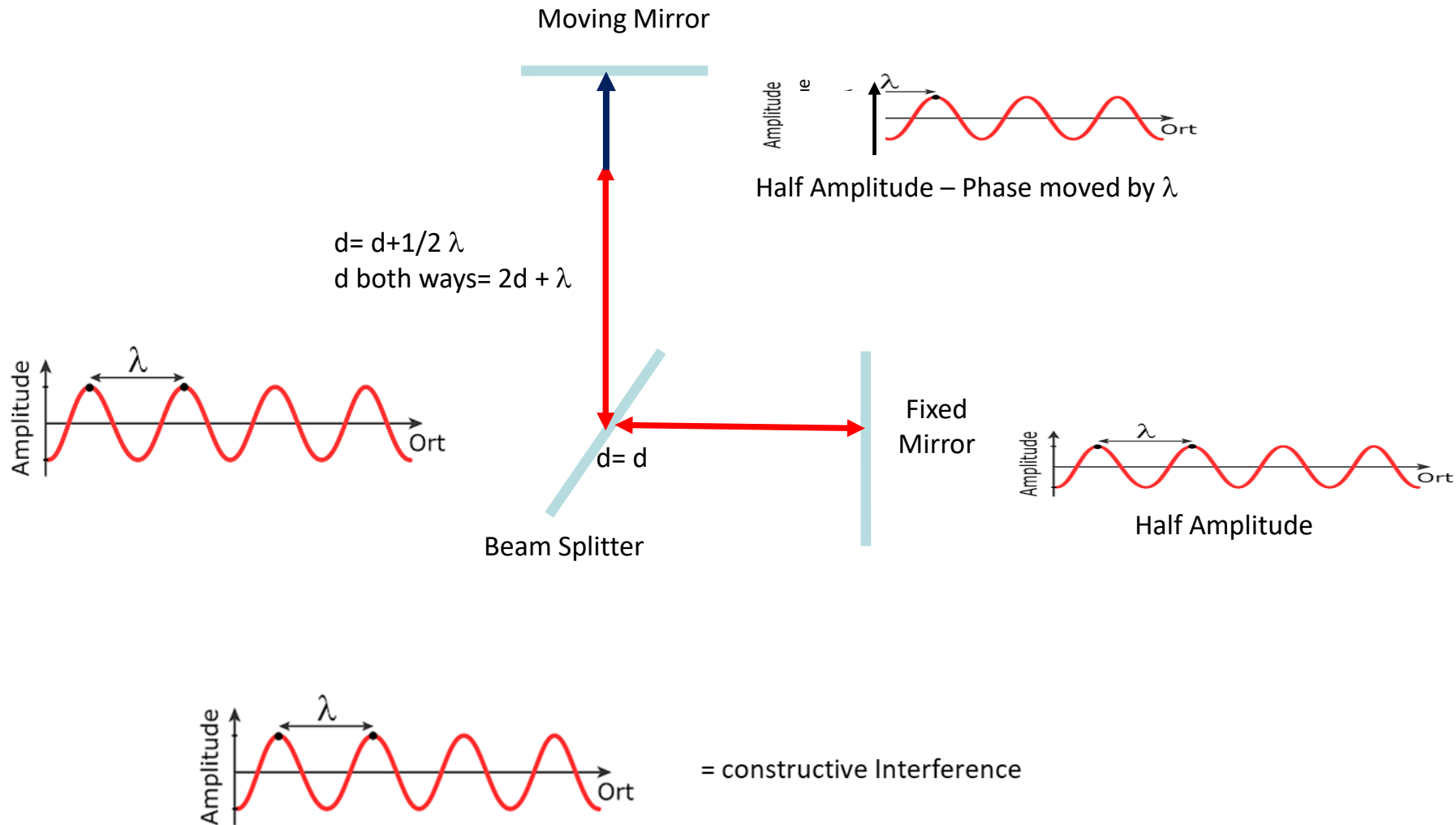
5.1.3 Interferometers for FT-IR



5.1.3 Interferometers for FT-IR



5.1.3 Interferometers for FT-IR

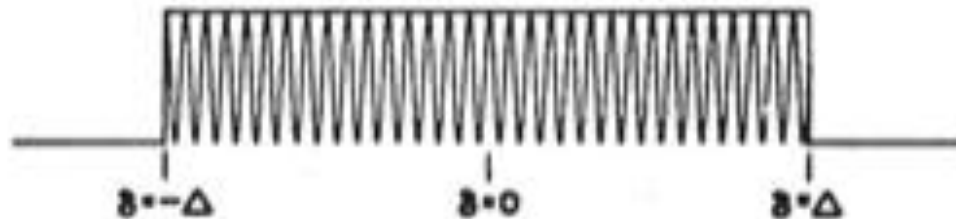


5.1.3 Interferometers for FT-IR

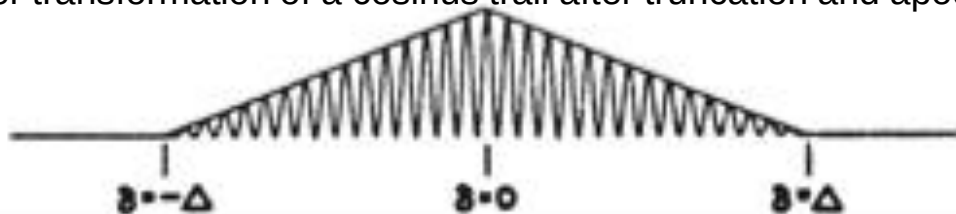
Fourier transformation of an infinitely long cosine trail



Fourier transformation of a cosine trail after truncation



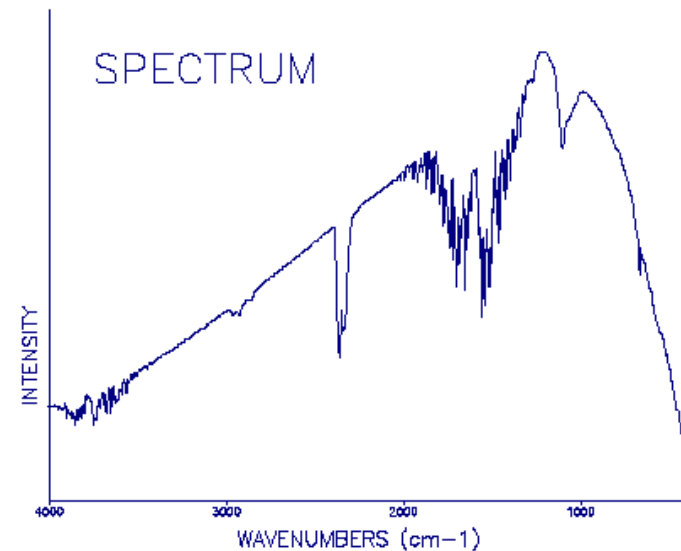
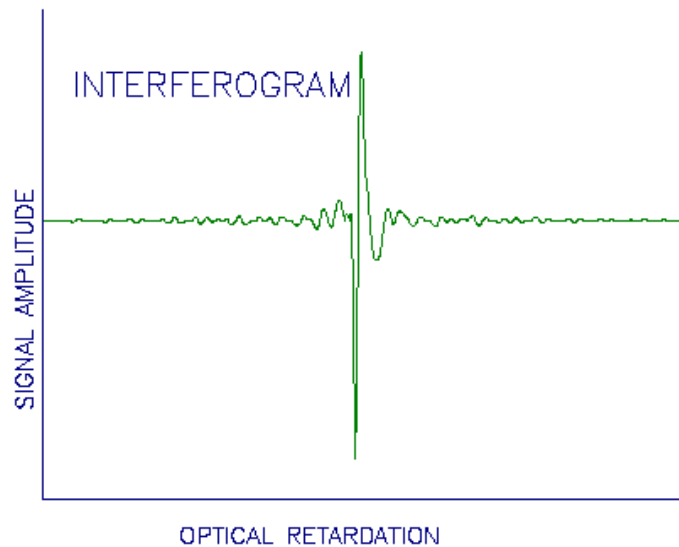
Fourier transformation of a cosine trail after truncation and apodization



5.1.3 Interferometers for FT-IR

Fourier Transformation:

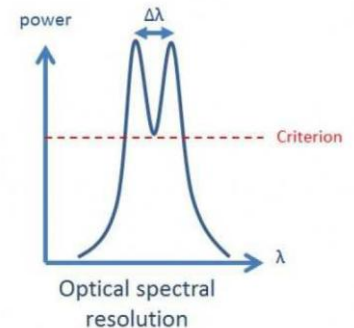
===== FOURIER TRANSFORM =====>>



5.1.3 Interferometers for FT-IR

Fourier Transform Infrared Spectroscopy:

- no monochromator – instead Interferometer
- each Interferogram contains information for all wavenumber/frequencies
- no sequential scanning of wavenumbers
- constant spectral resolution →
- obtained Interferograms require Fourier-Transformation to be converted into spectra
- much faster than dispersive instruments – allows accumulation of spectra – improvement of signal/noise ratio:
$$\overline{S/N} \cong \sqrt{n}$$

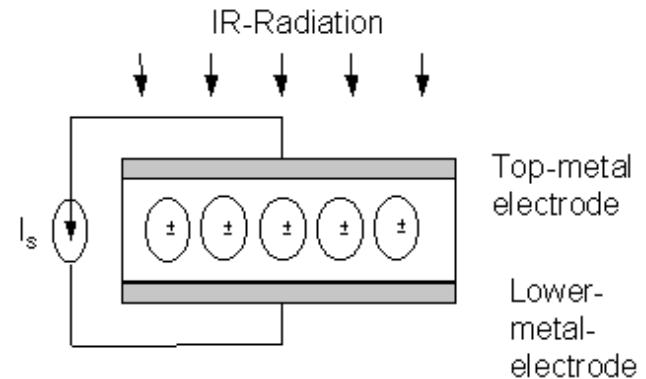


5.1.4 Detectors for IR

Pyroelectrical Detector:

Deuterated **T**ri **G**lycine **S**ulfate

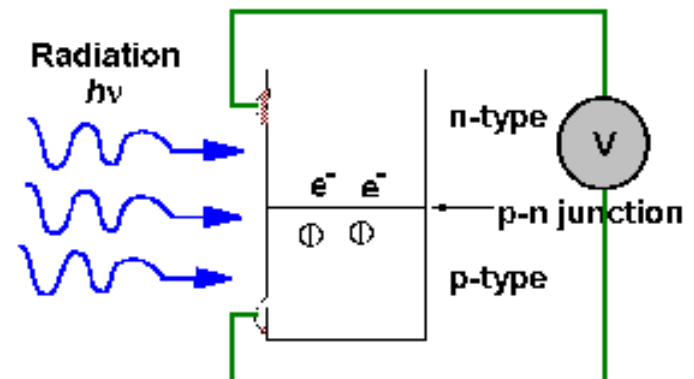
- ☹ less sensitive
- ☺ works @ room temperature



Photovoltaic Detector:

- ☺ high sensitivity
- ☹ has to be cooled with N_2 (l)

Detector type	(μm)
Si	0.2 - 1.1
Ge	0.4 - 1.8
InAs	1.0 - 3.8
InSb	1.0 - 7.0
InSb (77K)	1.0 - 5.6
HgCdTe (77K)	1.0 - 25.0



5.2 Principles of Vibrations in IR

IR Spectroscopy: Vibrations + Rotations

The vibration of a molecule consisting of two atoms can be modelled using two masses m_1 and m_2 connected via a spring (=chemical bond) with a distance of r° . Pulling back the two masses (atoms) and releasing them leads to a vibration of the system.

To simplify the system the two masses m_1 and m_2 can be replaced by a single mass μ (reduced mass) which is oscillatingly suspended on the spring. So we can deduct:

$$\mu = \frac{m_1 \cdot m_2}{m_1 + m_2}$$

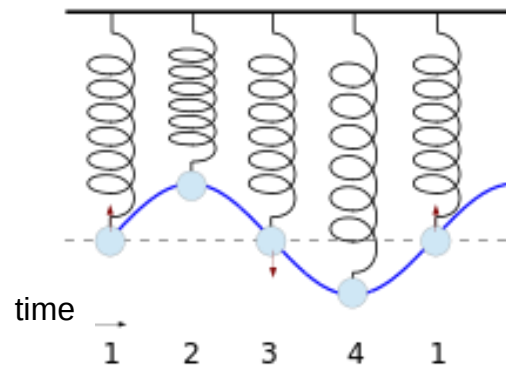
5.2.1 The Harmonic Oscillator

Harmonic Oscillator

If the reduced mass μ is pulled back from the equilibrium position by a distance of y , we can deduct the following according to the “harmonic oscillator” model: the restoring force F can be regarded as proportional to the distance y (Hookes law). The factor for this proportionality is called k and depends on the spring (=chemical bond): $F = -ky$

If the mass μ is moved from position y to $y + dy$, the necessary work equals the difference in E_{pot} of the system (equilibrium-excited state):

$$E = \int_0^y ky \, dy = \frac{1}{2}ky^2$$



When released the potential energy of the system is transformed into kinetic energy whereby upon reaching r° the kinetic energy reaches its maximum. (as $\Sigma E_{\text{pot}} + E_{\text{kin}} = \text{const.}$).

Plotting energy versus y leads to a parabolic curve

5.2.1 The Harmonic Oscillator

Harmonic Oscillator:

To calculate the frequency of the harmonic oscillator, one has to combine Hookes law and the relationship force = mass x acceleration:

$$\mu \frac{d^2 y}{dt^2} = -ky$$

Leading to:

$$v_{osz} = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

So the frequency is independent from the distance y.

5.2.1 The Harmonic Oscillator

Harmonic Oscillator:

The classical model of the harmonic oscillator would allow any energy (frequency) for the vibrating mass μ . In a molecular/atomistic system energy is a quantized parameter. Therefore only certain levels of energy are allowed:

$$E = h \nu_{osc} \left(v + \frac{1}{2} \right)$$

v vibration quantum number (0, 1, 2, 3,)

For a transition between two vibrational levels the following rule for selection has to be obeyed:

$$\Delta v = \pm 1$$

If by absorption of light the molecule is transferred from level $v=0$ to $v=1$, the molecule increases its energy by ΔE :

$$\Delta E = h \nu_{osc} \left(1 + \frac{1}{2} \right) - h \nu_{osc} \left(0 + \frac{1}{2} \right) = h \nu_{osc}$$

The energy of the light absorbed equals therefore: $h \nu_{osc}$

So the frequency of the absorbed light equals the frequency of oscillation of the molecule. As the latter depends on k and μ , information on the molecular structure can be deducted.

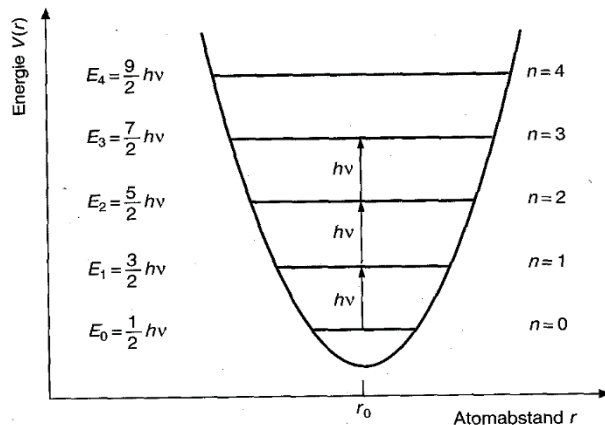
5.2.2 The Non-Harmonic Oscillator

(Non)Harmonic Oscillator:

The model of the harmonic oscillator only describes a single transition as all transitions show equal amounts of energy. Switching to the non-harmonic oscillator we add several new facts:

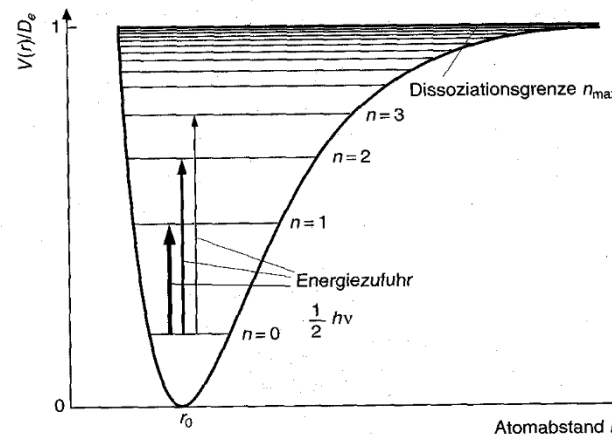
- at a certain energy level the bond will break
- the different levels of vibrational energy are no longer equidistant
- additional transitions such as $\Delta v = \pm 2$, $\Delta v = \pm 3$ are possible

For the gaseous HCl-Molecule we observe in addition to the basic $v=0 \rightarrow v=1$ (wavenumber 2890 cm^{-1}) overtones such as $v=0 \rightarrow v=2$ (wavenumber 5668 cm^{-1}) and $v=0 \rightarrow v=2$ (wavenumber 8347 cm^{-1})



Harmonic oscillator

$$E = h\nu_{\text{osc}} \left(n + \frac{1}{2} \right)$$



Anharmonic oscillator

$$E = h\nu_{\text{osc}} \left(n + \frac{1}{2} \right) - \chi_e h\nu \left(n + \frac{1}{2} \right)^2 + \gamma_e h\nu \left(n + \frac{1}{2} \right)^3 \dots$$

5.3 Principles of Rotations in IR

Rotationen of di-atomic molecules:

Excited levels of rotation need only about 1/1000 of the energy needed to move from one level of vibration to the next one. Transitions of outer shell electrons (UV-spectroscopy) require another 1000 times more energy.

As a model for rotation of diatomic molecules we can use the so called rigid rotator. Here we have two masses m_1 and m_2 , with a constant distance of r .

The moment of inertia I is described as:

$$I = \mu r^2 \qquad \mu = \frac{m_1 \cdot m_2}{m_1 + m_2}$$

The following levels of rotational energy E_{rot} exist for the rigid rotator:

$$E_{\text{rot}} = h c B J (J + 1)$$

Rotational quantum number $J = 0, 1, 2, 3, \dots$

$$B = \frac{h}{8\pi^2 c I}$$

For transition between the levels of rotational energy the following rule of selection is valid: $\Delta J = \pm 1$

B = rotational constant

5.3 Principles of Rotations in IR

The change in energy for a transition $J=0 \rightarrow J=1$ equals $2hcB$, for a transition $J=1 \rightarrow J=2$ $4hcB$, and for a transition $J=2 \rightarrow J=3$ $6hcB$, etc.

The following light wavenumbers are absorbed:

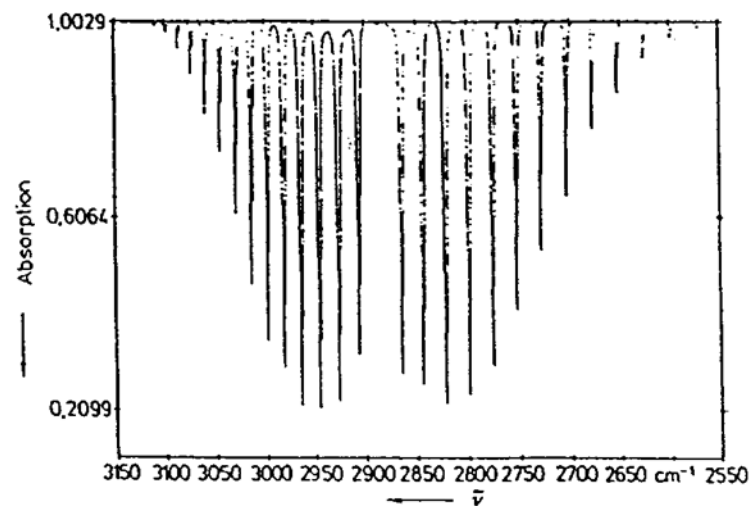
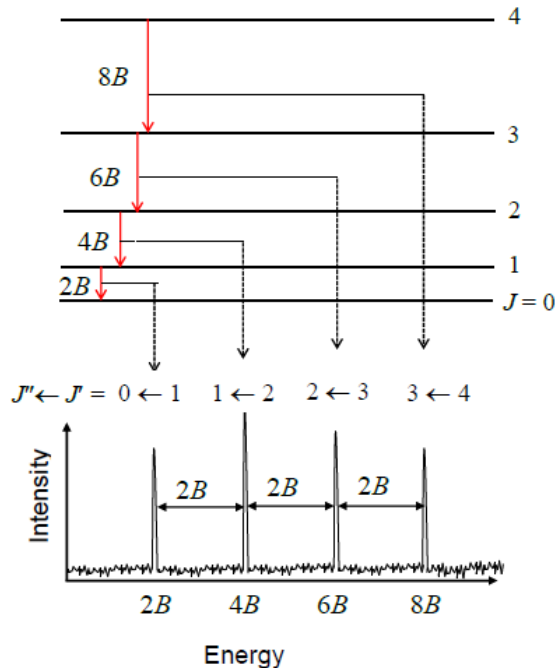
Transition $J=0 \rightarrow J=1$ wavenumber = $2B$

Transition $J=1 \rightarrow J=2$ wavenumber = $4B$

Transition $J=2 \rightarrow J=3$ wavenumber = $6B$

(So wavenumbers are equidistant!)

In IR spectroscopy (4000 bis 400 cm^{-1}) we observe a combination of vibrations and rotations. The latter can only be observed as discrete signals if we measure gaseous molecules; otherwise rotational levels merge into one broad band.

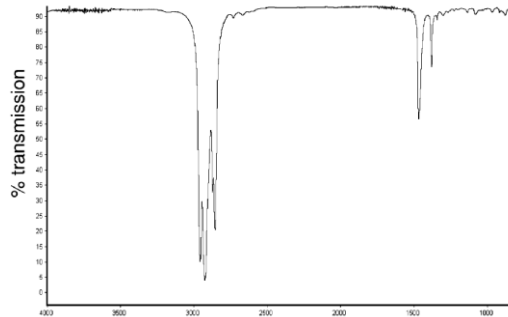


Rotational bands of the gaseous HCl molecule

5.4. Specific Vibrations in IR

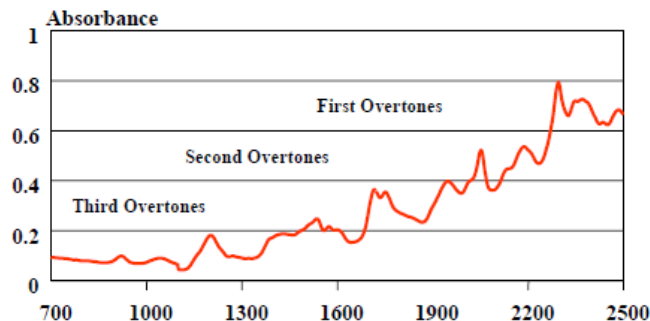
MIR:

- mainly basic vibrations



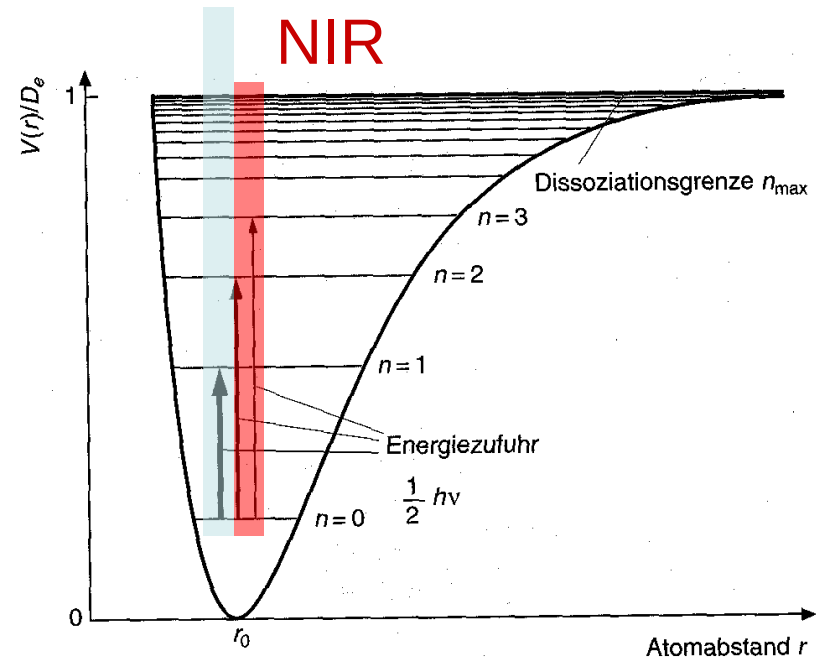
NIR:

- overtones
- combined vibrations



MIR

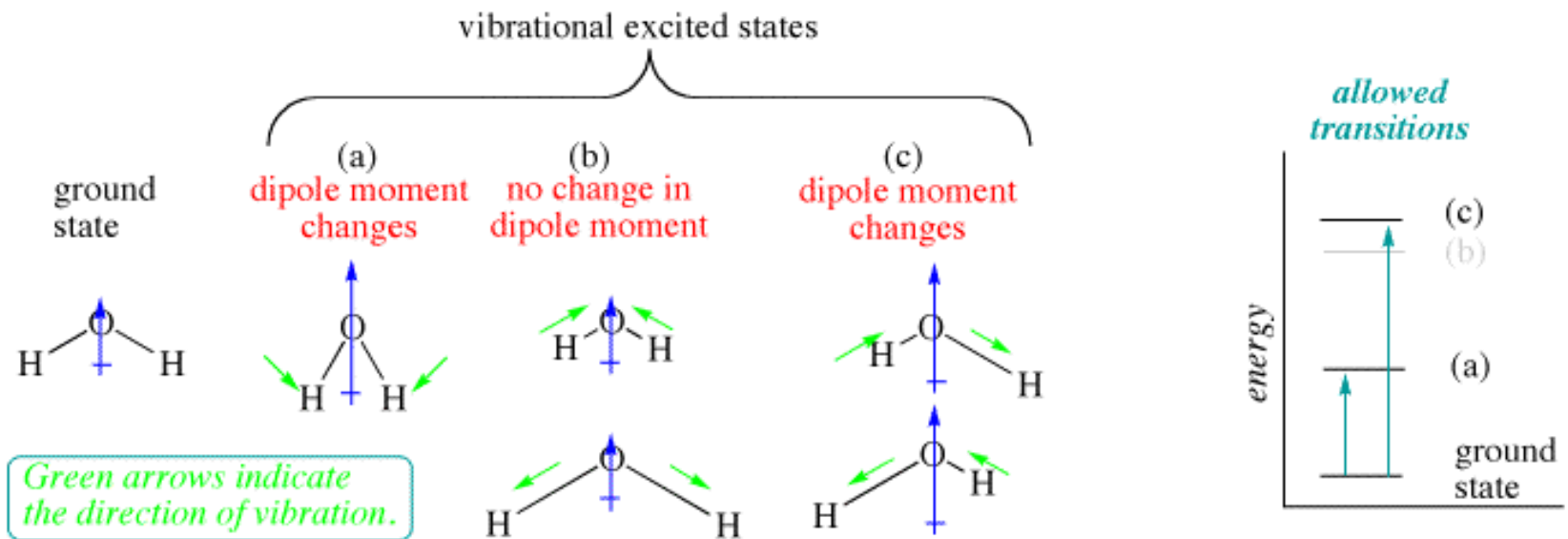
NIR



Quantized energy in an anharmonic oscillator

5.4. Specific Vibrations in IR

Selection rules for vibrations

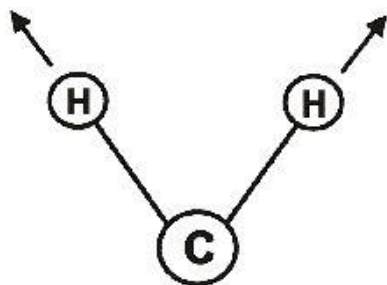


Only transitions involving the change of the dipole momentum are allowed!

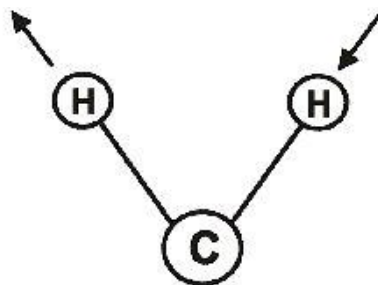
Case (b) will be discussed in the next chapter (Raman spectroscopy)

5.4. Specific Vibrations in IR

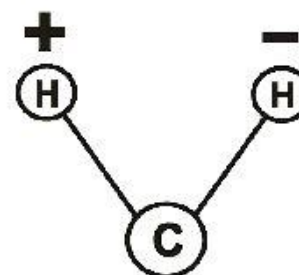
Vibrations



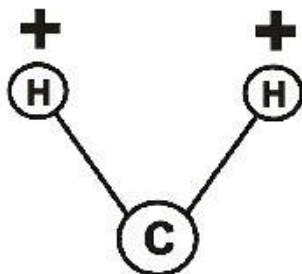
**Symmetrical
stretching**



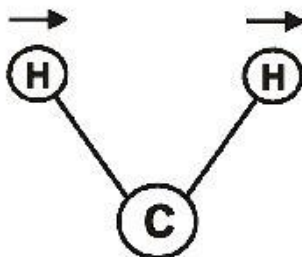
**Antisymmetrical
stretching**



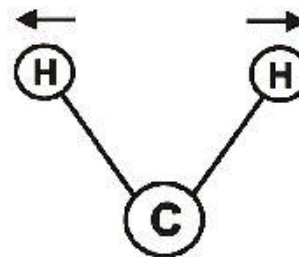
**Twisting
(deformation out-of-plane)**



**Wagging
(deformation out-of-plane)**



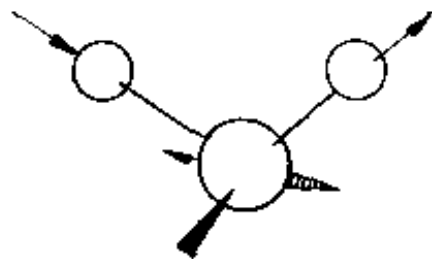
**Rocking
(deformation in-plane)**



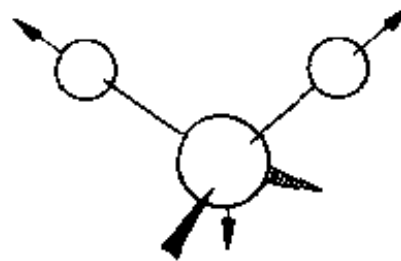
**Scissoring
(deformation in-plane)**

5.4. Specific Vibrations in IR

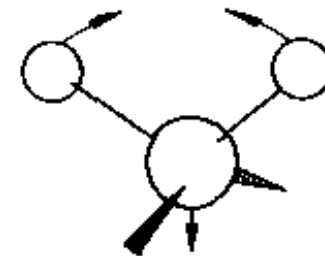
CH₂ vibrations



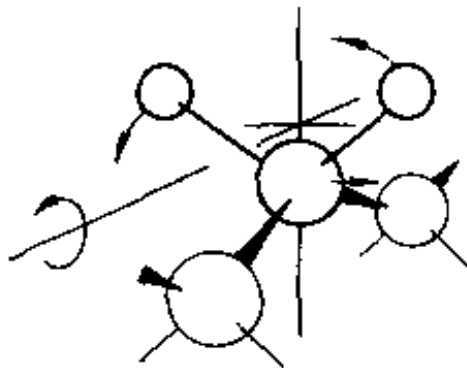
a) ν_{as} 2930 cm⁻¹



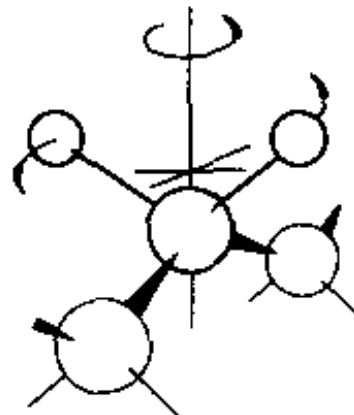
b) ν_s 2850 cm⁻¹



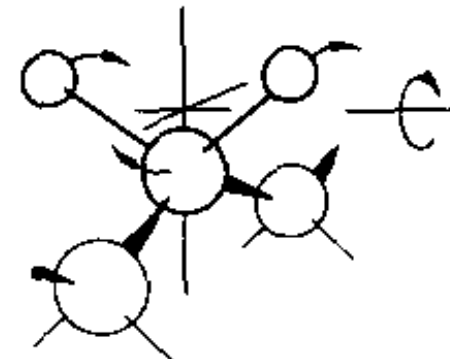
c) δ_s 1470 cm⁻¹



a) rocking: 720 cm⁻¹



b) twisting: ~ 1300 cm⁻¹



c) wagging: 1305 cm⁻¹

5.4. Specific Vibrations in IR

energy/wave-numbers of bands:

type of bond:

C—C ca. 1000 cm⁻¹

C=C ca. 1650 cm⁻¹

C≡C ca. 2200 cm⁻¹

Atoms involved in vibration: small μ leads to an increase in ν

C—H ca. 3000 cm⁻¹

C—D ca. 2100 cm⁻¹

C—C ca. 1000 cm⁻¹

C—Cl ca. 700 cm⁻¹

e.g. $m_H = 1/2 m_D$; $\nu_{X-D} = 0.7 \nu_{X-H}$

$$\nu_{osz} = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

5.5. Sampling Techniques

how to get spectra of:

solids:

KBr pellets

Diffuse or regular Reflectance

Attenuated Total Reflectance

liquids:

Solution in suitable solvent

(e.g. CHCl_3)

liquid film

Attenuated Total Reflectance

gaseous:

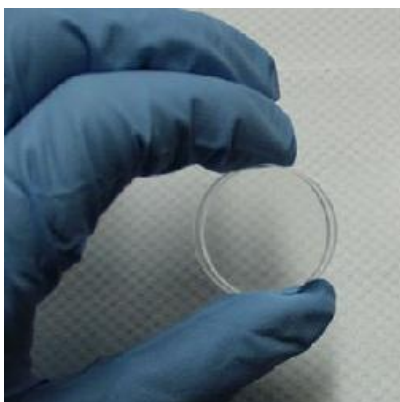
Gas cell

(often with heating for non-permanent gases)

5.5. Sampling Techniques



Liquid in-solution



Liquid film



KBr-pellet



Gas-cuvette

5.5. Sampling Techniques

ATR:

ZnSe crystal or diamond with evanescent waves on the interface between the two surfaces; radiation intrudes into the sample thereby gaining information about its composition

- ☺ a variety of samples with different shape can be analyzed
- ☺ de facto no sample preparation

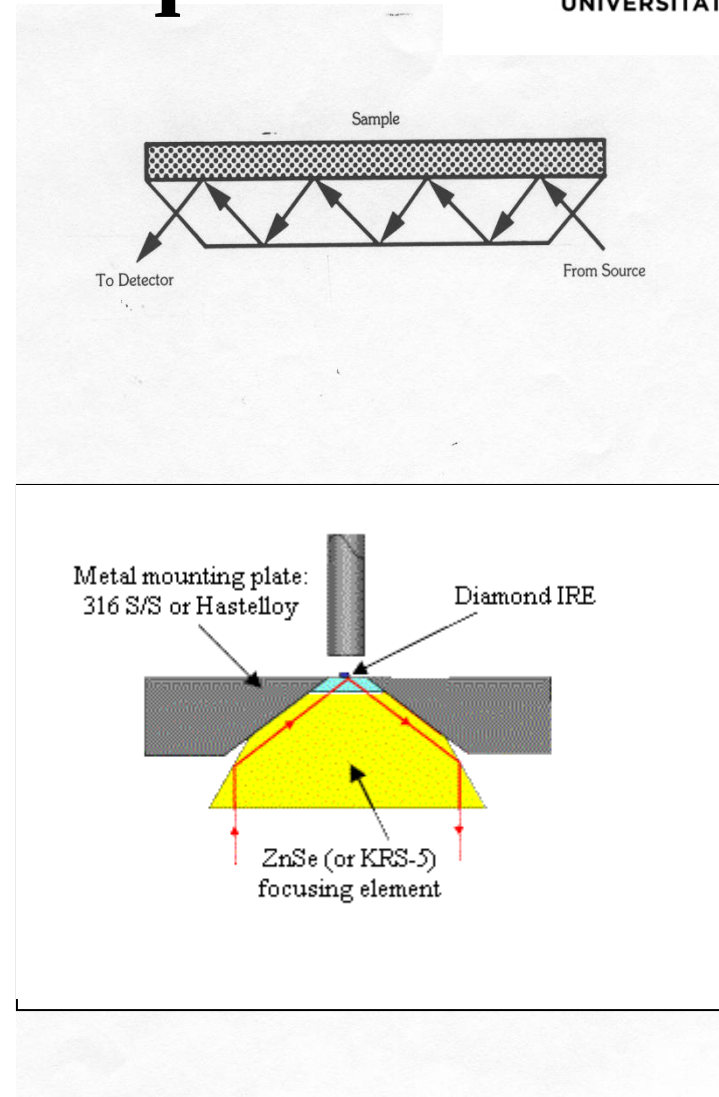
n_1, n_2 : refraction index of crystal and sample

α : incident angle

l_{ATR} : penetration

λ : wavelength

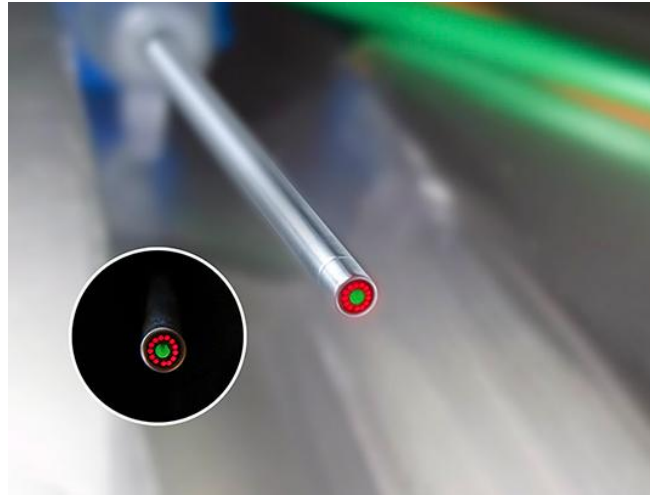
$$l_{ATR} = \frac{\lambda_{IR}}{2\pi n_1 \sqrt{\sin^2 \alpha_{ATR} - \left(\frac{n_2}{n_1}\right)^2}},$$



5.5. Sampling Techniques

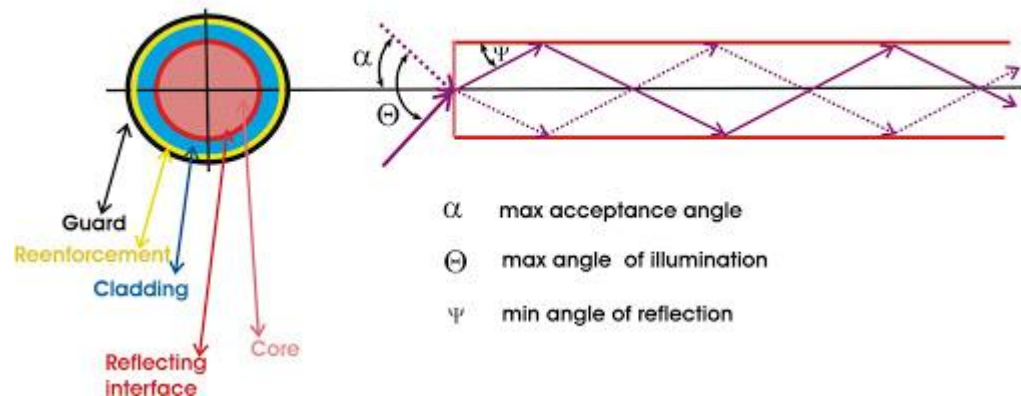


NIR-sensor for food analysis



Principle of optical fibers (based on ATR)

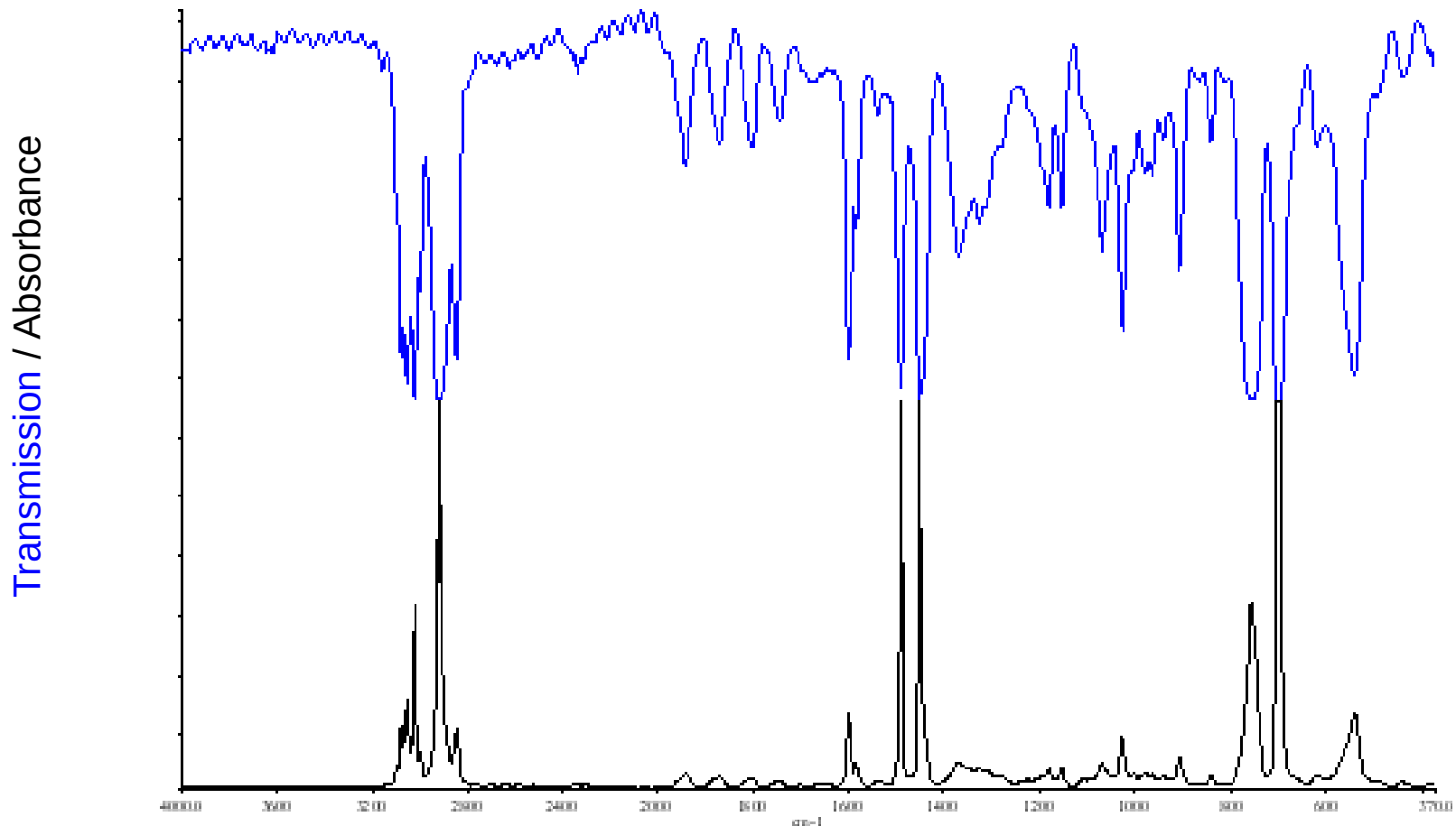
The Principal of Fibre Optics



5.6. IR-Spectra

IR-Spectrum of Polystyrene: Transmission-Modus (blue) -Absorbance-Mode (black)

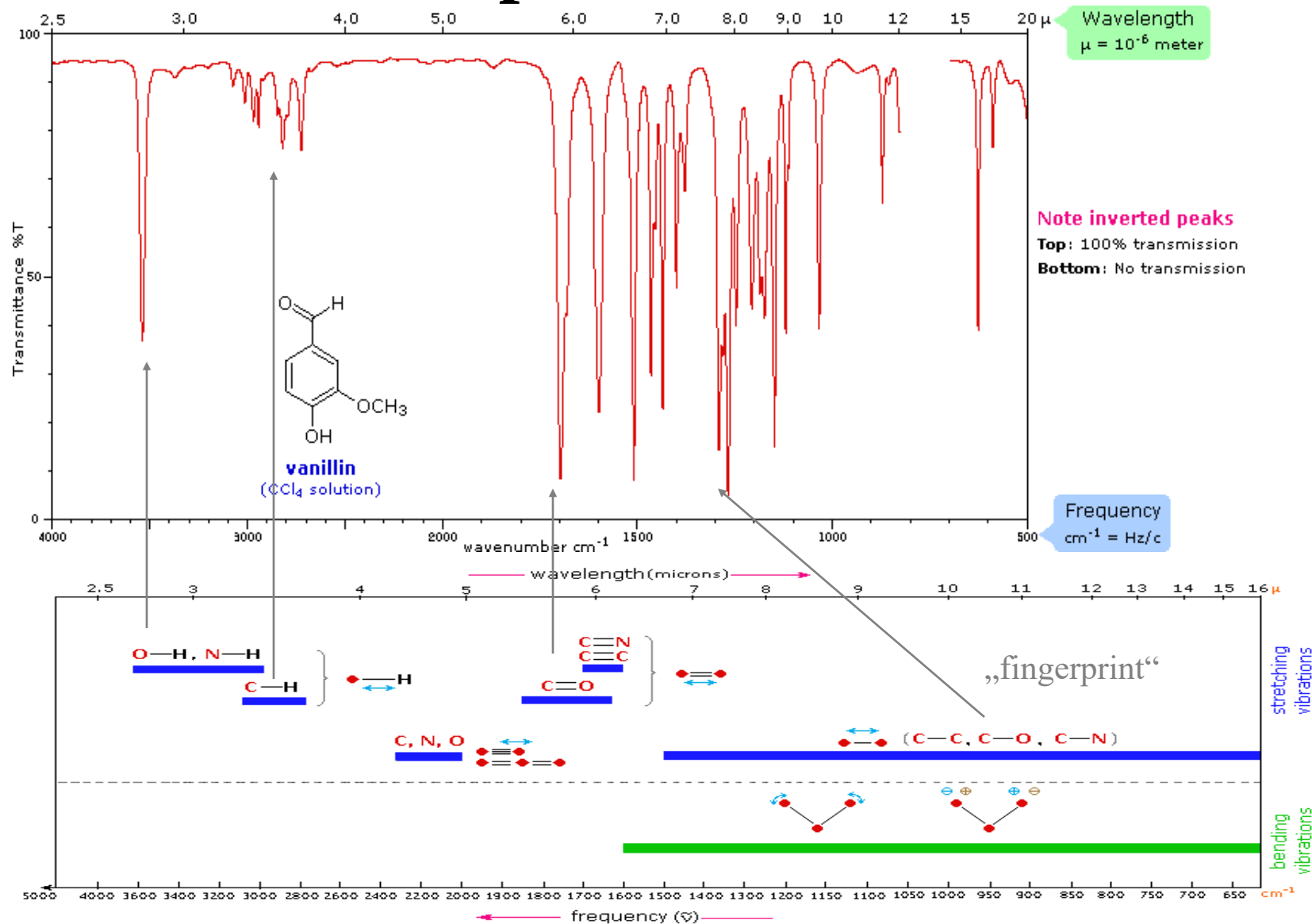
Transmission $T = I/I_0$ rel. Transmission = $T \times 100$



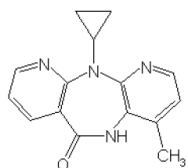
Absorbance = $-\log T = \log I_0/I$

wavenumber / cm⁻¹

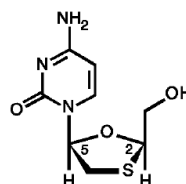
5.6. IR-Spectra



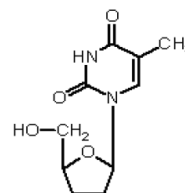
5.6. IR-Spectra



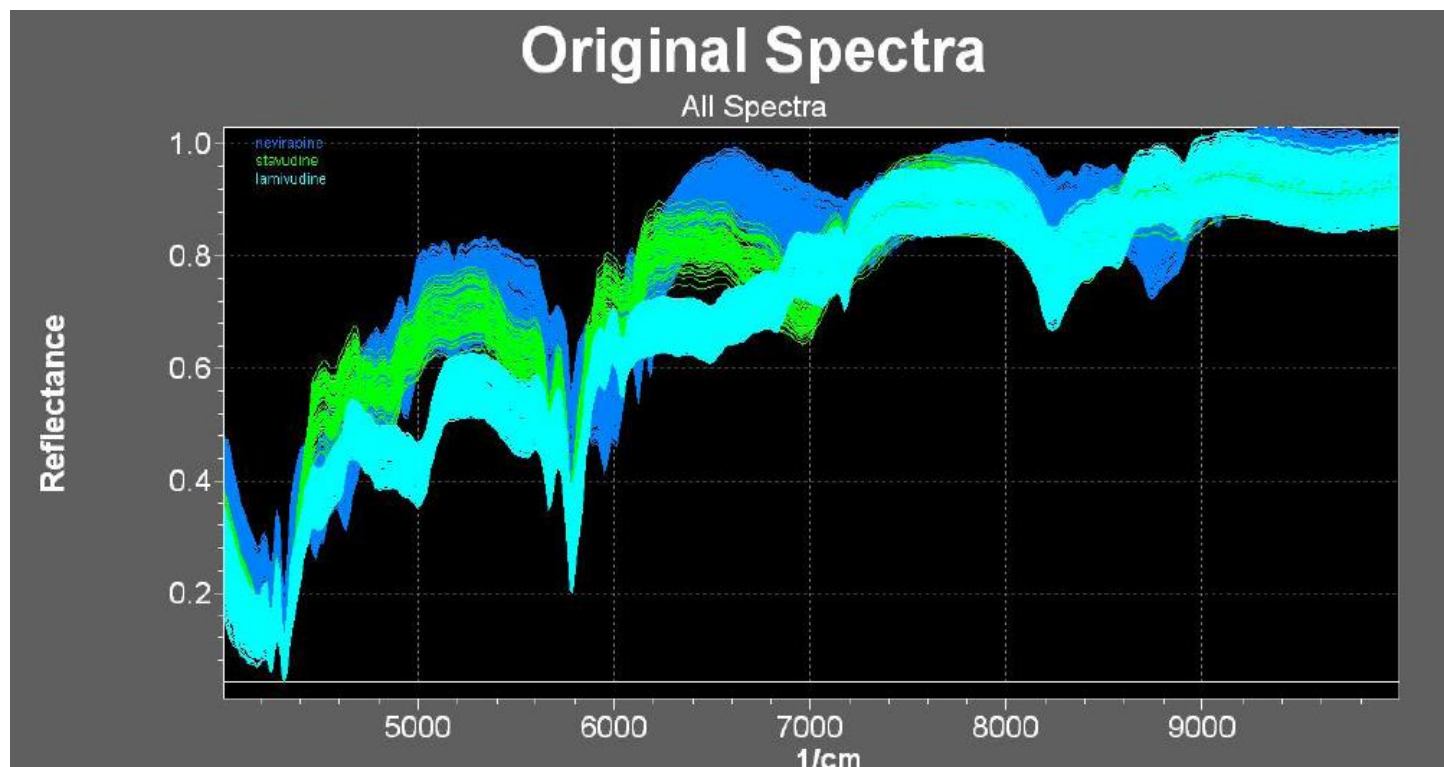
NEVIRAPINE



LAMIVUDINE

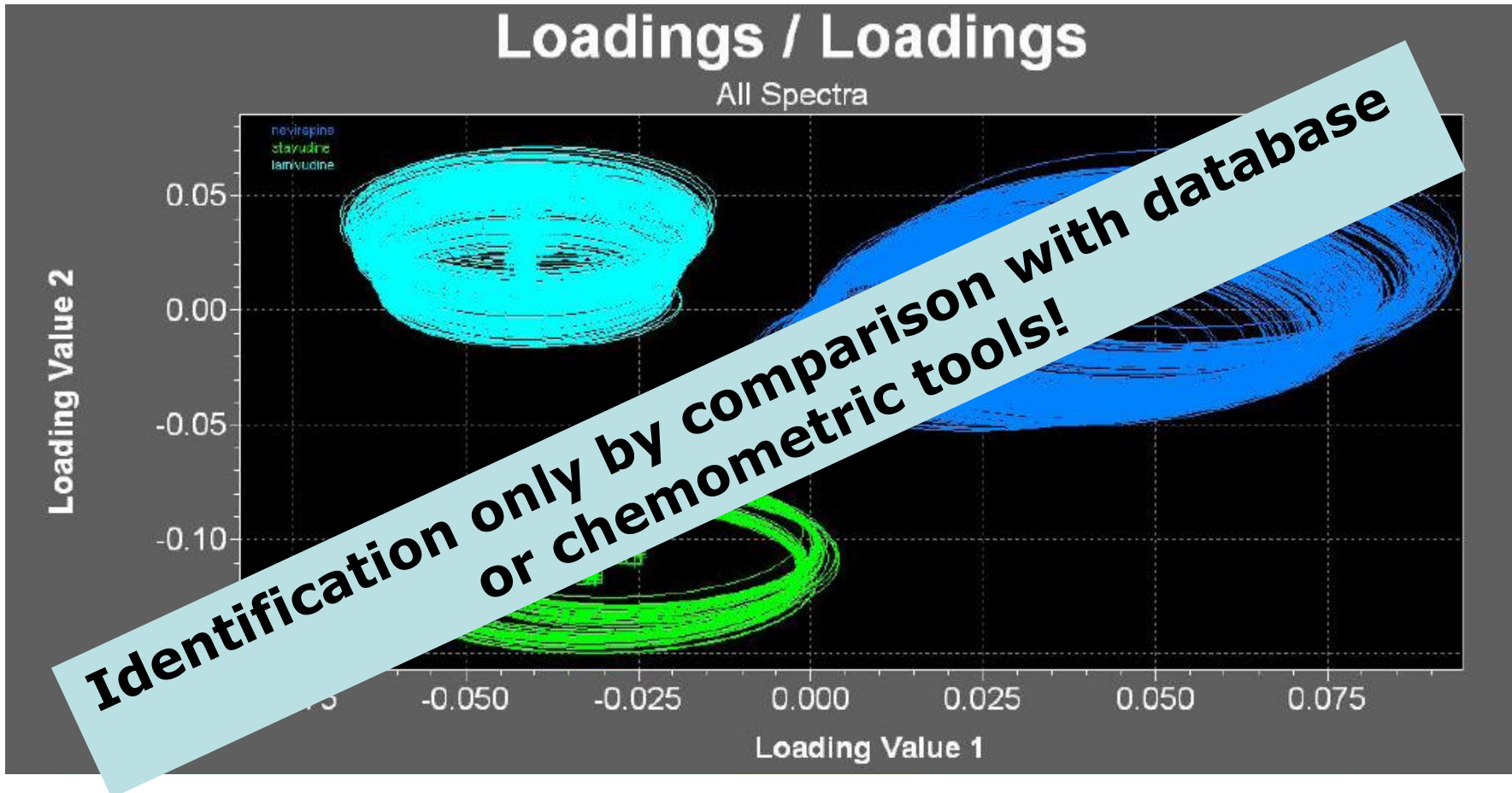


STAVUDINE

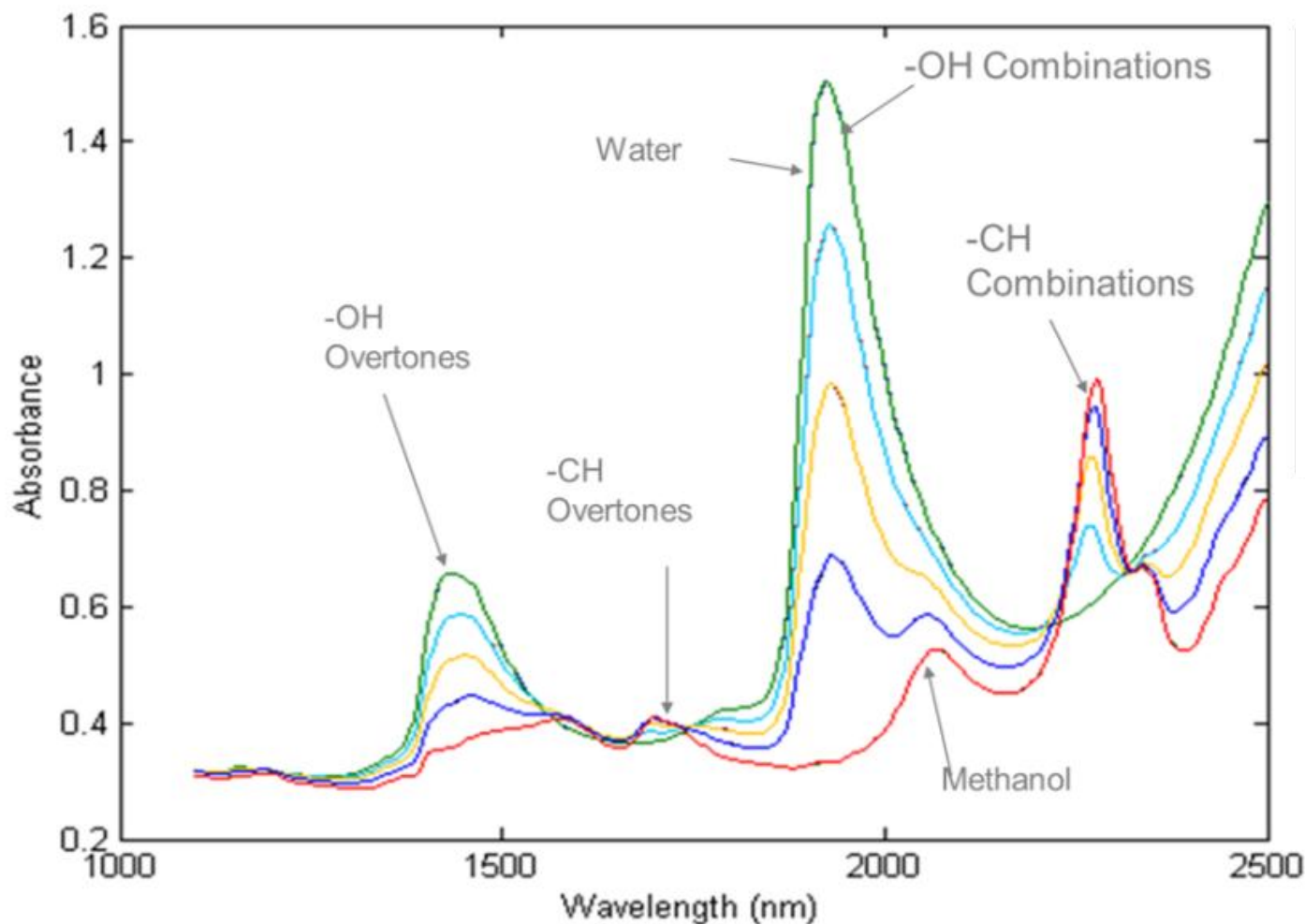


NIR-Spectra

5.6. IR-Spectra



5.6. IR-Spectra



MeOH/H₂O

0/100

25/75

50/50

75/25

100/0

NIR - Quantitation

5.6. IR-Spectra

MIR:

- Structural elucidation
- Direct assigning of signals to functional groups is possible
- Mainly for qualitative analysis
- Less frequent in quantitative analysis
- Possible use as detector in chromatography (e.g. in GC) – but rarely employed

NIR:

- Identification via comparison with database
- Widely used in quantitative analysis
- On-line use for controlling chemical plants
- Spectra evaluation needs chemometric methods

6. Raman-Spectroscopy

Raman-Spectroscopy

6. Raman-Spectroscopy

A New Type of Secondary Radiation

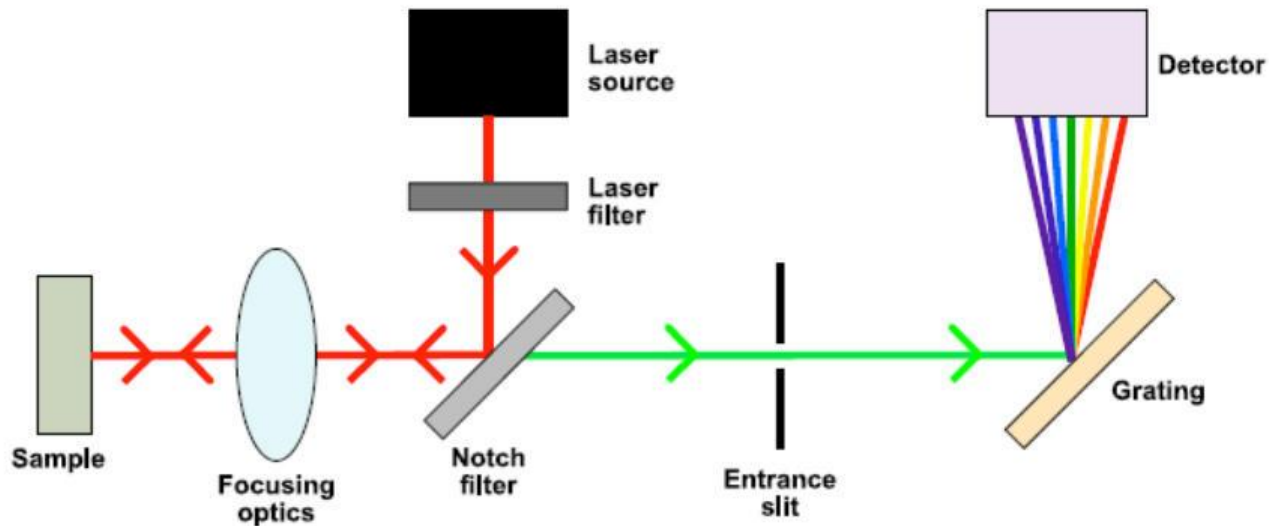
C. V. Raman and K. S. Krishnan, Nature, 121(3048), 501, March 31, 1928

If we assume that the X-ray scattering of the 'unmodified' type observed by Prof. Compton corresponds to the normal or average state of the atoms and molecules, while the 'modified' scattering of altered wave-length corresponds to their fluctuations from that state, it would follow that we should expect also in the case of ordinary light two types of scattering, one determined by the normal optical properties of the atoms or molecules, and another representing the effect of their fluctuations from their normal state. It accordingly becomes necessary to test whether this is actually the case. The experiments we have made have confirmed this anticipation, and shown that in every case in which light is scattered by the molecules in dust-free liquids or gases, the diffuse radiation of the ordinary kind, having the same wave-length as the incident beam, is accompanied by a modified scattered radiation of degraded frequency. The new type of light scattering discovered by us naturally requires very powerful illumination for its observation. In our experiments, a beam of sunlight was converged successively by a telescope objective of 18 cm. aperture and 230 cm. focal length, and by a second lens was placed the scattering material, which is either a liquid (carefully purified by repeated distillation *in vacuo*) or its dust-free vapour. To detect the presence of a modified scattered radiation, the method of complementary light-filters was used. A blue-violet filter, when coupled with a yellow-green filter and placed in the incident light, completely extinguished the track of the light through the liquid or vapour. The reappearance of the track when the yellow filter is transferred to a place between it and the observer's eye is proof of the existence of a modified scattered radiation. Spectroscopic confirmation is also available.

Some sixty different common liquids have been examined in this way, and every one of them showed the effect in greater or less degree. That the effect is a true scattering, and secondly by its polarisation, which is in many cases quire strong and comparable with the polarisation of the ordinary scattering. The investigation is naturally much more difficult in the case of gases and vapours, owing to the excessive feebleness of the effect. Nevertheless, when the vapour is of sufficient density, for example with ether or amylene, the modified scattering is readily demonstrable

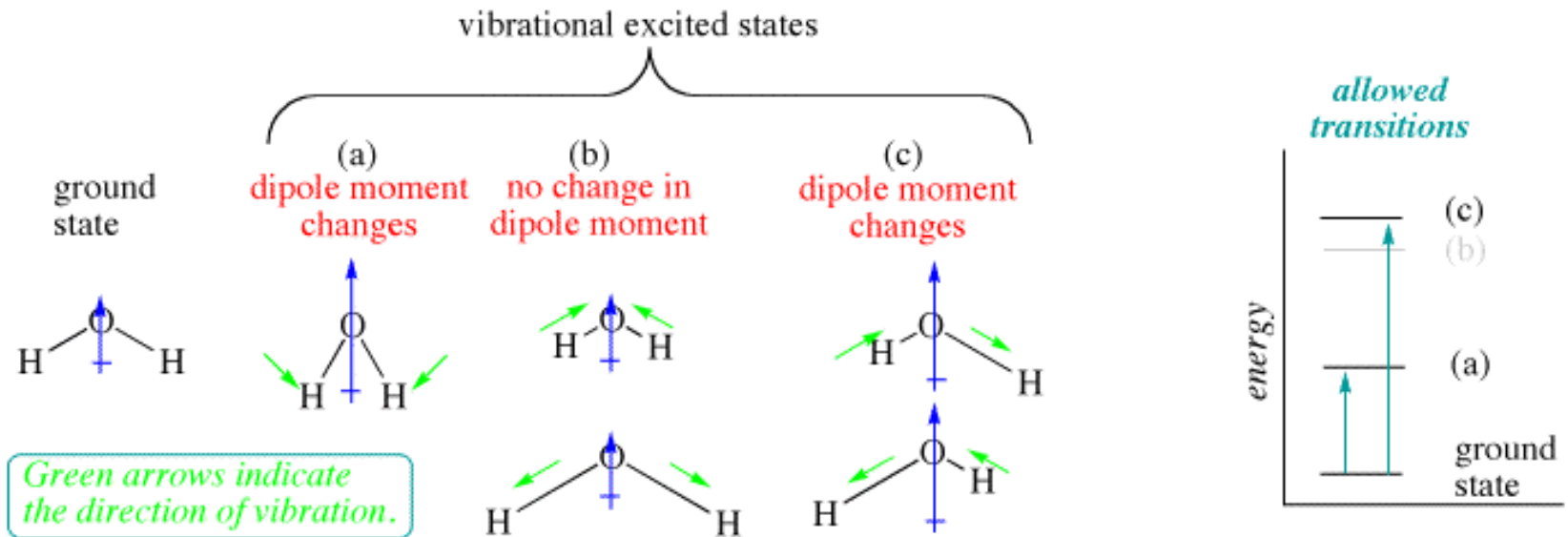
6. Raman-Spectroscopy

Raman Spectrometer



Source: Hg vapor (435,8 nm or 22.946 cm^{-1})
Nd YAG Laser (1064 nm or 9.398 cm^{-1})
Ar Laser (488 nm or 20.492 cm^{-1})
Detector: Si-Diode
In-Ga-As-Diode

6. Raman-Spectroscopy



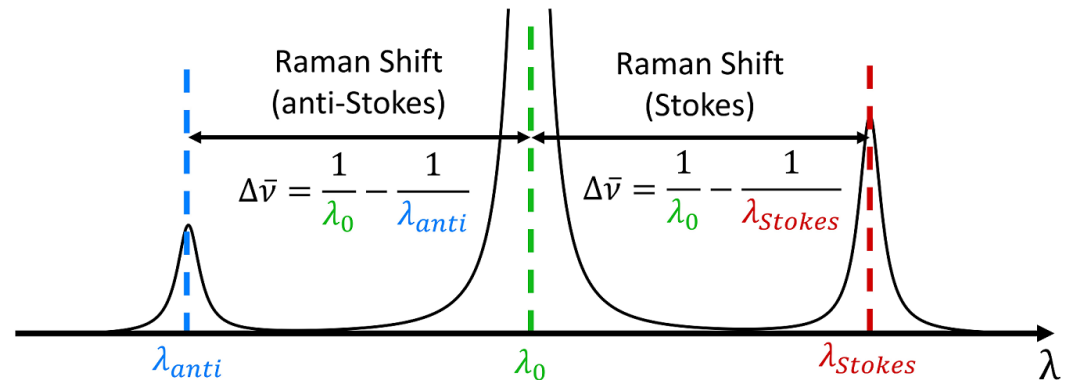
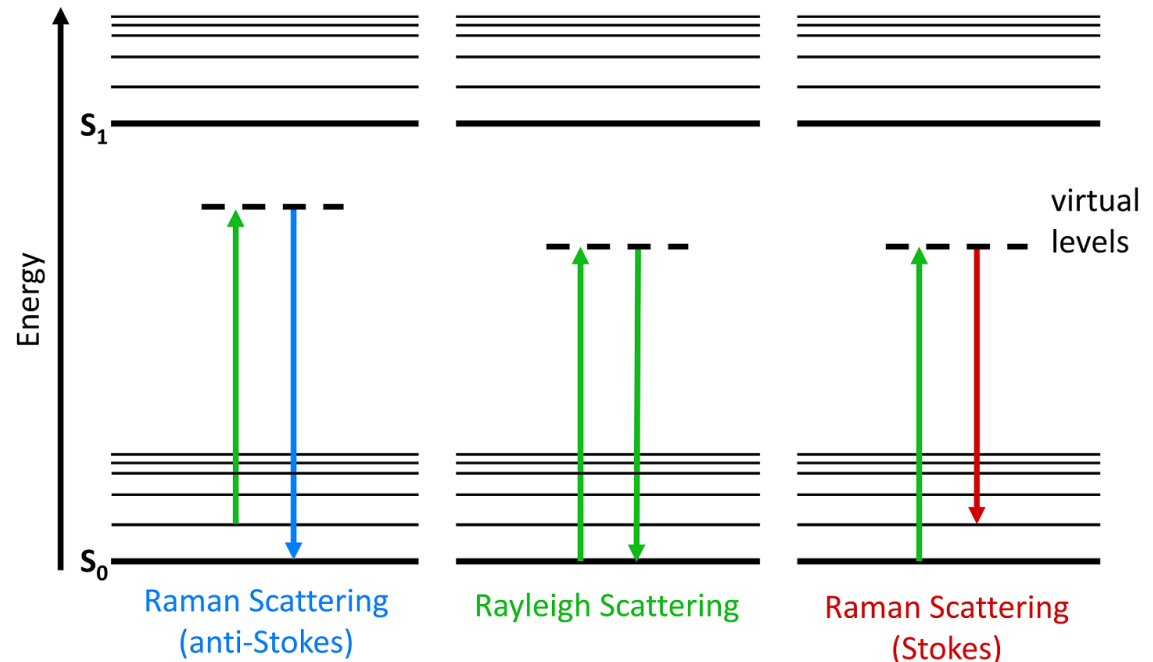
In contrast to IR spectroscopy where transitions to (a) and (c) were observed, in Raman spectroscopy transition to (b) with no change of the dipole moment is monitored

6. Raman-Spectroscopy

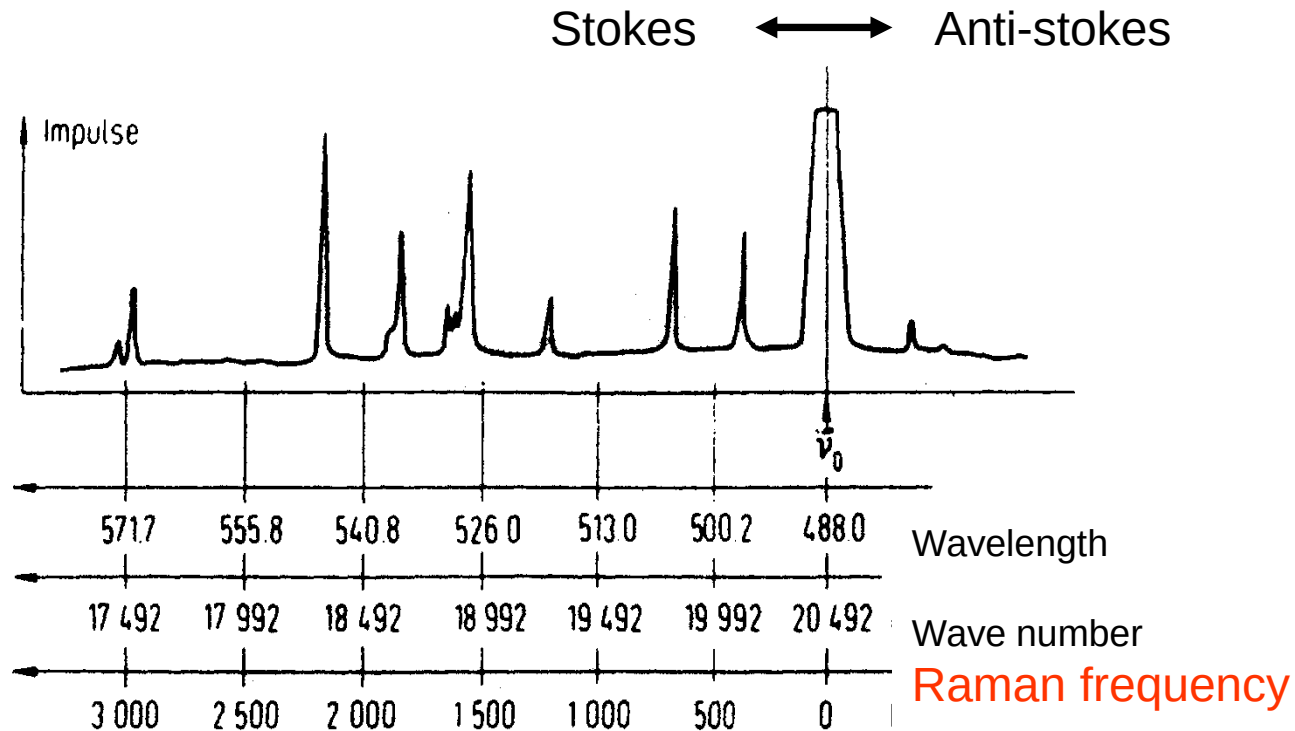
Raman scattering:

Laser-light hits the molecule:

- mostly light just passes through (elastic)
- 10^{-4} Rayleigh scattering
- 10^{-8} Raman scattering (Stokes and Anti-Stokes)



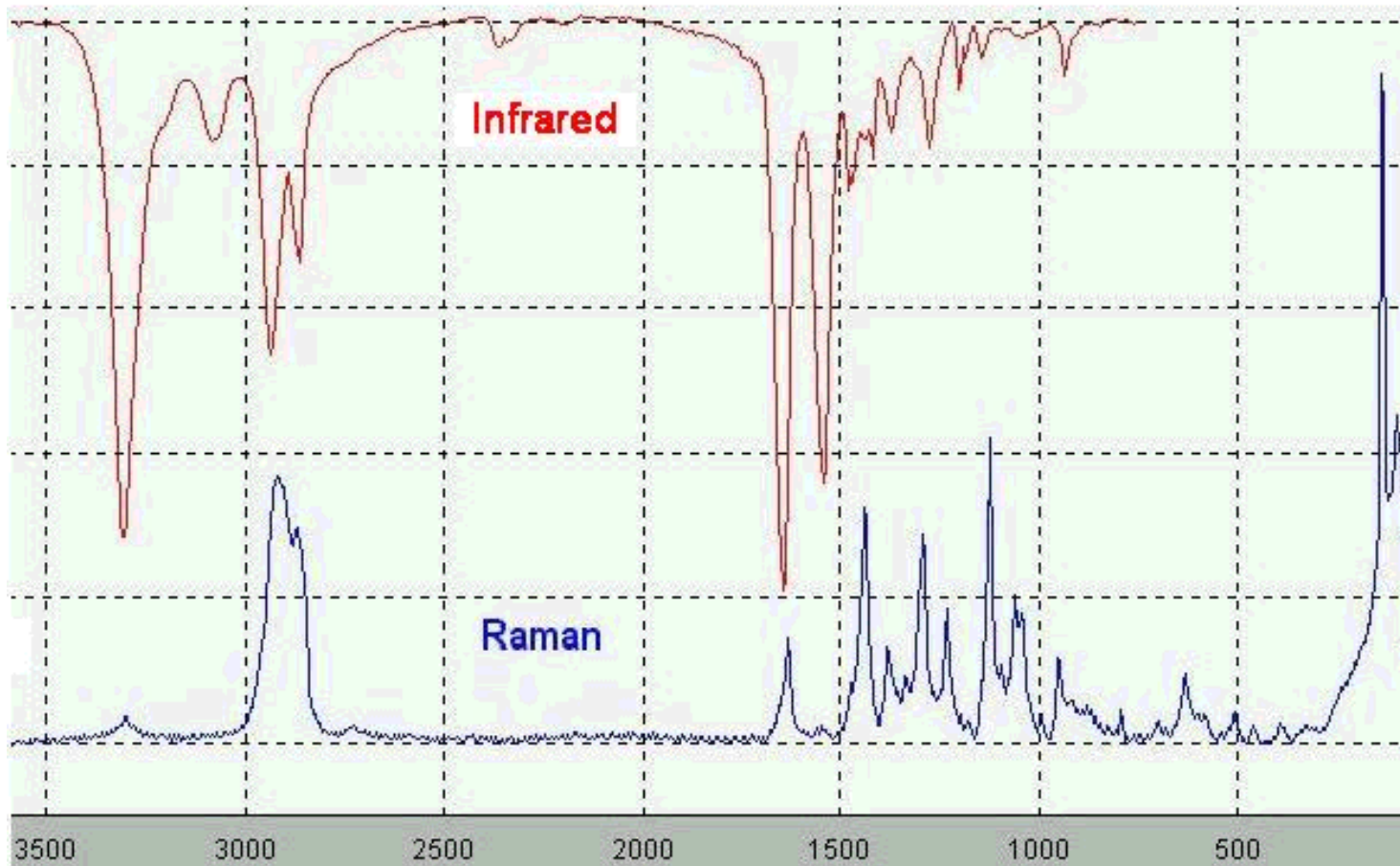
6. Raman-Spectroscopy



$$\nu_{\text{laser}} - \nu_{\text{scattered}} = \Delta\nu_{\text{Raman}}$$

6. Raman-Spectroscopy

Infrared and Raman Spectra of Polyamide (Nylon 66)



ν for IR and $\Delta\nu$ for Raman respectively

7. Mass Spectrometry

Mass-Spectrometry

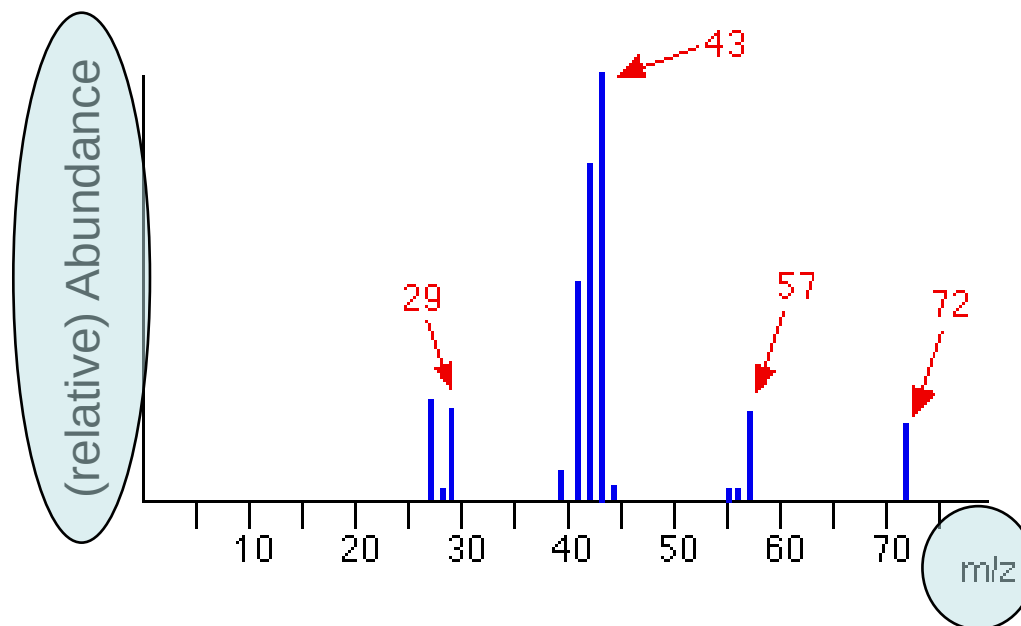
7. Mass Spectrometry

- Structural Elucidation
- Detector for Separation Techniques (GC, HPLC, Electrophoresis.....)
- Quantitation of mixtures

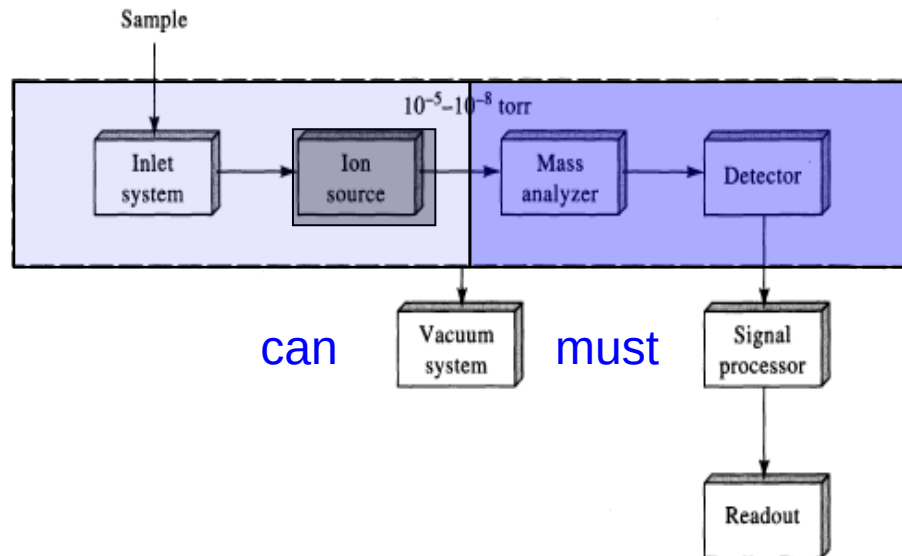
Mass in u = $\frac{1}{12}$ mass of $^{12}_6\text{C} = 1.0000000$ actually measured m/z i.e. mass/charge

simplified mass spectrum of pentane - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

Mass spectrum:



7.1. Ionization Sources



The first major component of a MS is the ion-source. Its major aim is the production of analyte-ions that can be transferred to the connected mass analyzer (see next chapter). According to the analytical problem (type of analytes investigated) and the type of mass analyzer used (see next chapter) different types of ion-sources providing different ionization mechanisms can be employed. In the following slides some important ionization techniques are listed and information with respect to important characteristics of the individual ionization technique are provided.

7.1. Ionization Sources

Ionization method	Typical Analytes	Sample Introduction	Mass Range	Method Highlights
Electron ^(Impact) Ionization (EI)	Relatively small volatile	GC or liquid/solid probe	to 1,000 Daltons	Hard method versatile provides structure info
Chemical Ionization (CI)	Relatively small volatile	GC or liquid/solid probe	to 1,000 Daltons	Soft method molecular ion peak $[M+H]^+$

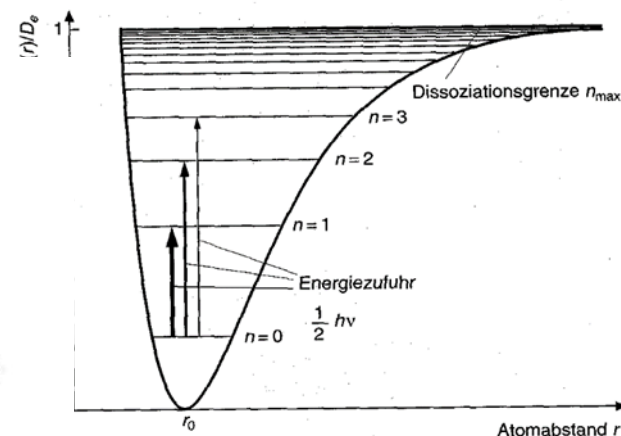
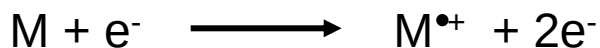
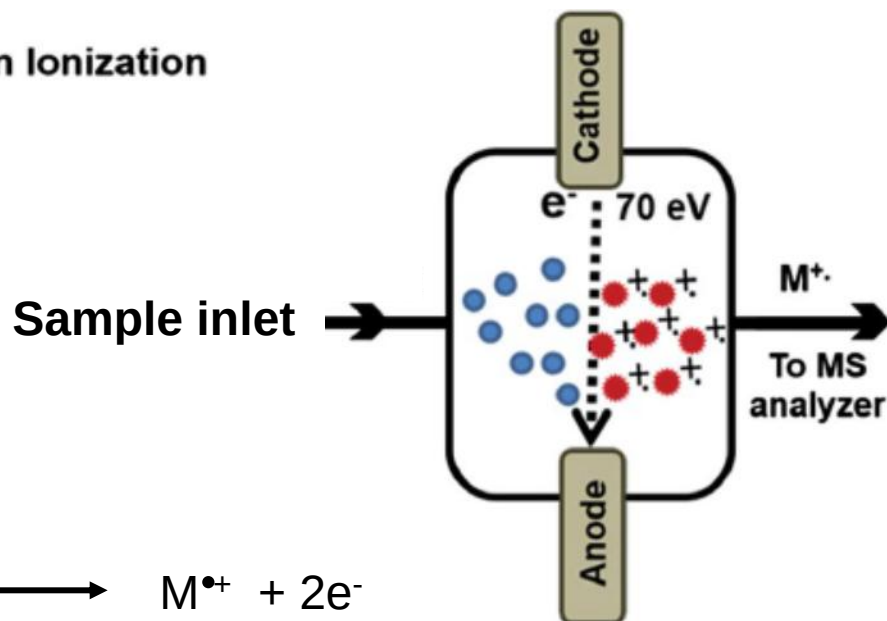
Methods requiring vacuum conditions for ionization
Atmospheric pressure ionization techniques

Hard method: strong fragmentation
Soft method: almost no fragmentation

7.1.1. Electron (Impact) Ionization

- Only suitable for small analytes (< 1000u)
- Gas-phase ionization - so analytes have to be in the gas phase
- Vaporization at high temperature (200-300°C) and low pressure (vacuum)
- Analytes must be stable under these conditions
- Primarily a radical cation is formed – but much energy is introduced into the molecule
- Subsequently fragmentation („hard ionization technique“) occurs

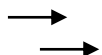
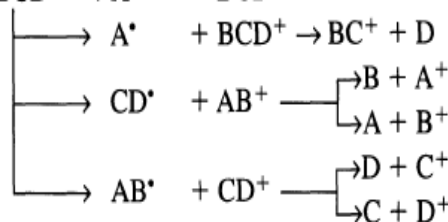
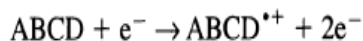
Electron Ionization



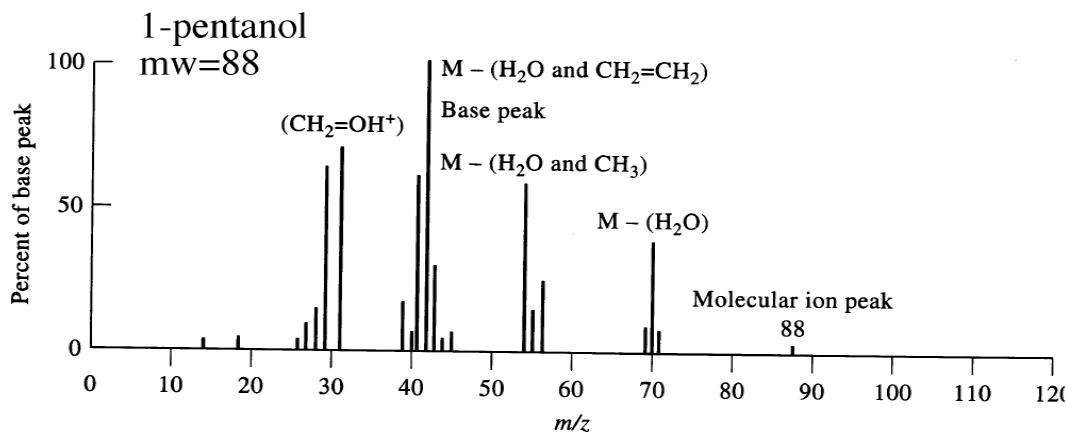
7.1.1. Electron (Impact) Ionization

Molecular ion formation

Fragmentation

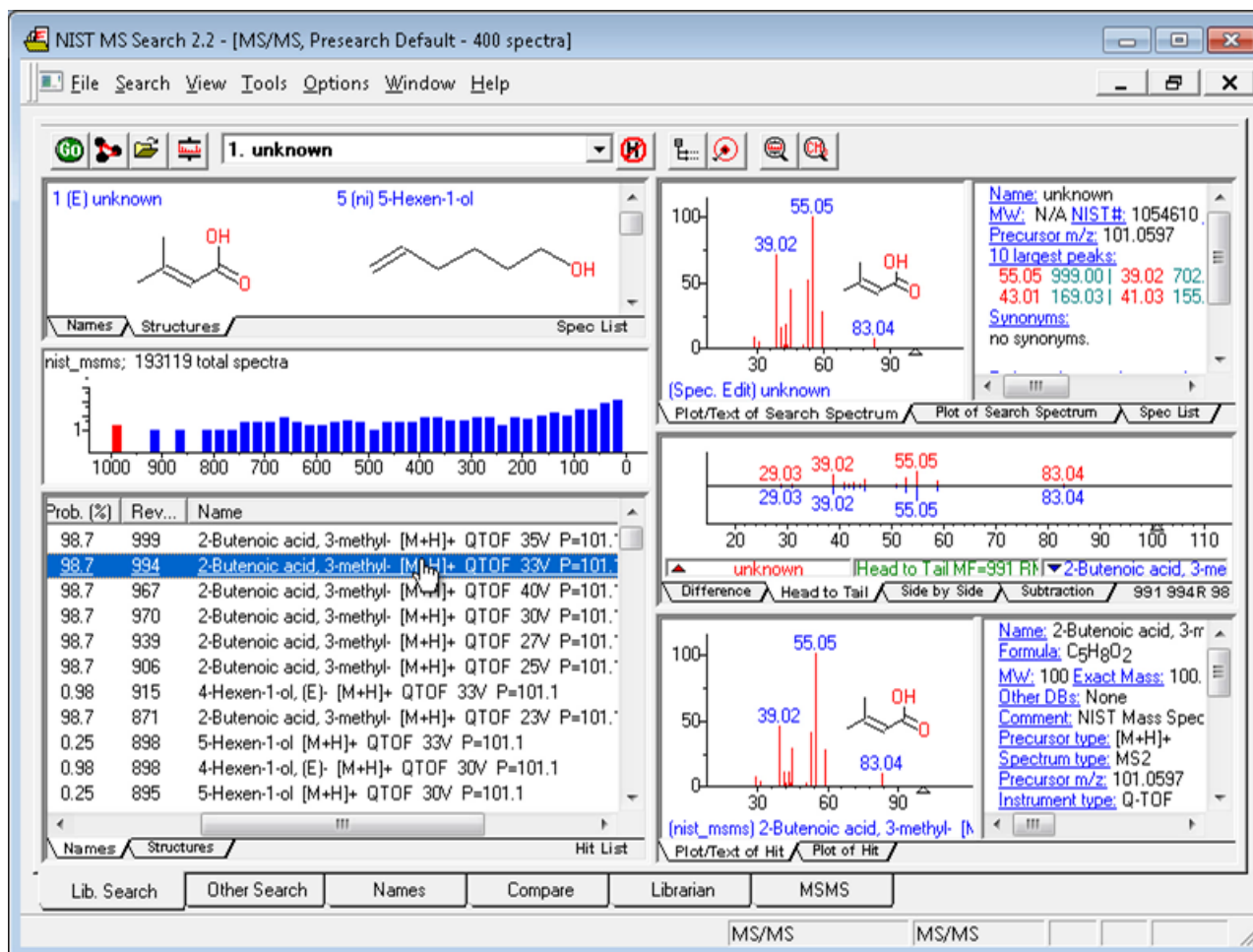


This fragmentation can be regarded as beneficial, as typical fragmentation patterns facilitate the unambiguous identification of molecules. Fragmentation also provides information on molecule – structure! If primarily the molecular mass of the analyte is of interest– other methods (e.g. CI – see following slides) may be used.



7.1.1. Electron (Impact) Ionization

Almost all
databases
based on
ionization
with electrons
@ 70 eV

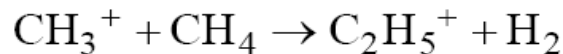
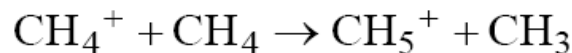


As fragmentation is to a large degree independent from the instrument used (i.e. which manufacturer) – large databases (e.g. NIST, Wiley...) exist for substance identification.

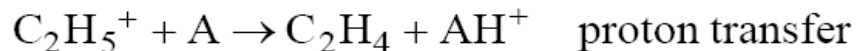
7.1.2. Chemical Ionization

Chemical ionization (CI)

CI ionization is performed using a common EI source. Major difference to EI is that electron impact does not ionize the actual analyte molecules but primarily an excess reactant gas (e.g. methane, isobutane or ammonia). So focusing on the example below, e^- are hitting CH_4 , ejecting one e^- and producing a CH_4^+ radical cation.

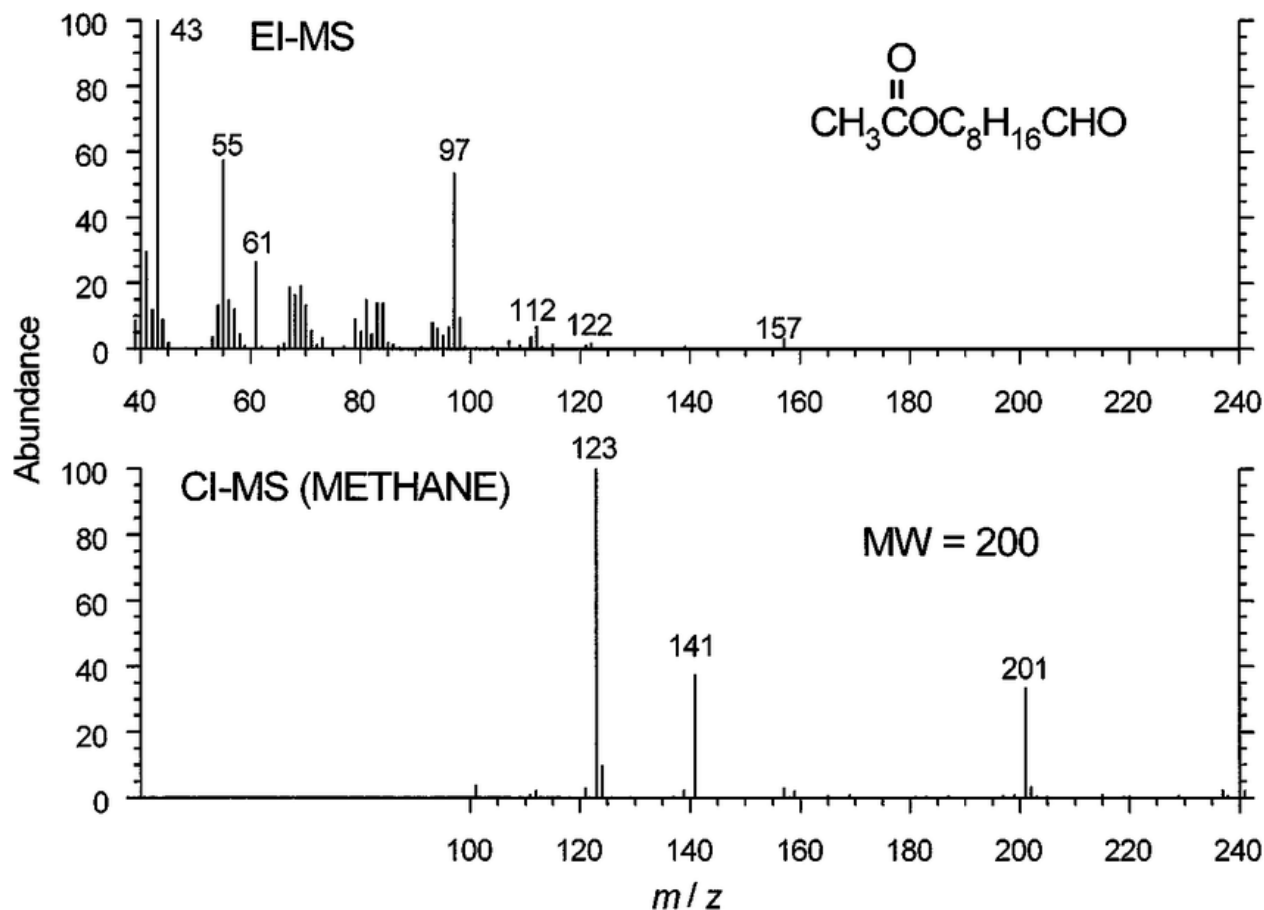


Further reaction of these ions from the reactant gas with the analyte (A) leads to:



CI can be seen as a „soft“ ionization method leading to reduced fragment formation and allowing to obtain signals for intact $[\text{M}+\text{H}]^+$ molecular ions. Anyhow restrictions with respect to molecules suitable for CI are the same as for EI (sample molecules need to be volatile as they have to be vaporized first)

7.1.2. Chemical Ionization



Comparison of EI (70 eV) and CI ionization of 9-acetyloxynonanal

7.1.3. Atmospheric Pressure Ionization

Ionization method	Typical Analytes	Sample Introduction	Mass Range	Method Highlights
Electrospray (ESI)	Peptides Proteins nonvolatile	Liquid Chromatography or syringe	to 200,000 Daltons	Soft method ions often multiply charged
Atmospheric Pressure Chemical Ionization (APCI)	Relatively small volatile	Liquid Chromatography or syringe	to ~1,200 Daltons	Soft method for less polar analytes
Atmospheric Pressure Photo Ionization (APPI)	Relatively small volatile	Liquid Chromatography or syringe	to ~1,200 Daltons	Soft method for less polar analytes

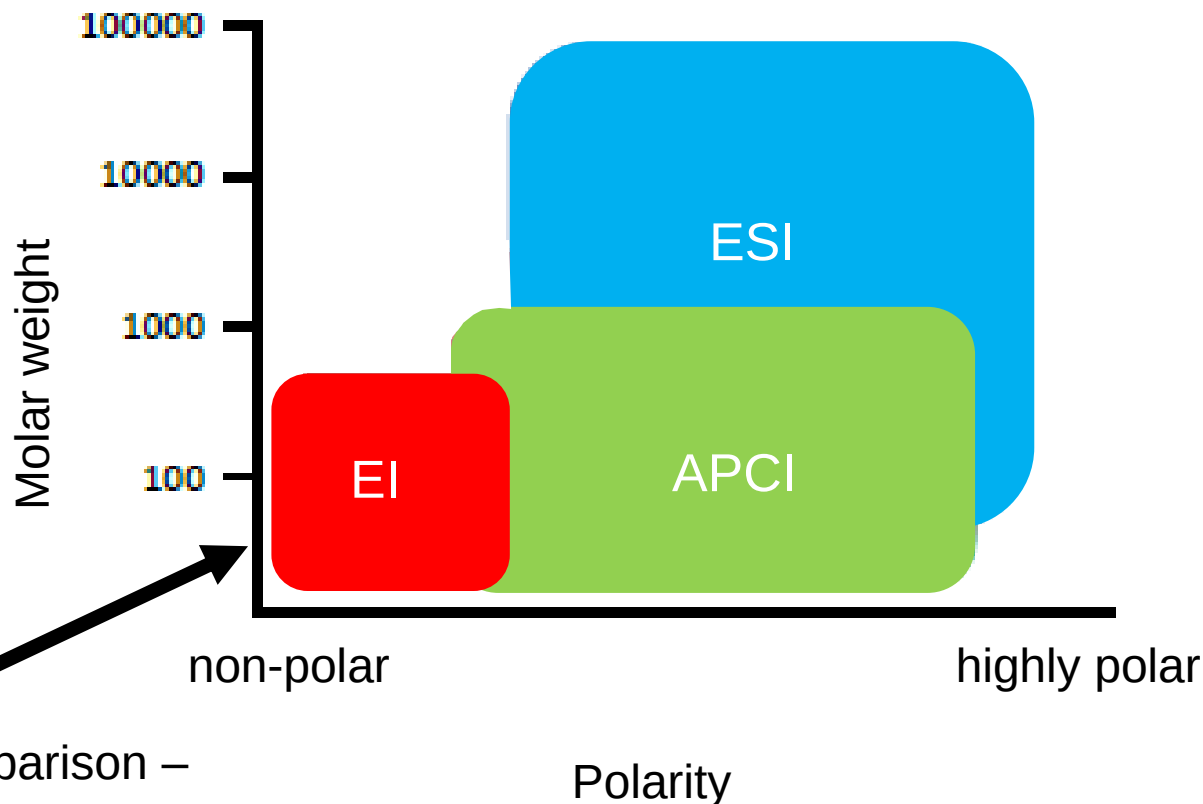
Methods requiring vacuum conditions for ionization

Atmospheric pressure ionization techniques

Hard method: strong fragmentation

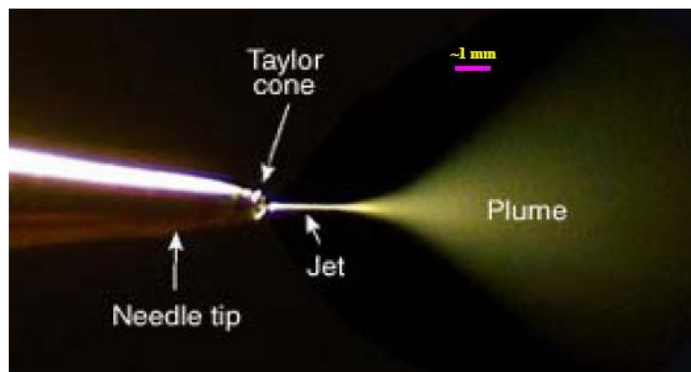
Soft method: almost no fragmentation

7.1.3. Atmospheric Pressure Ionization

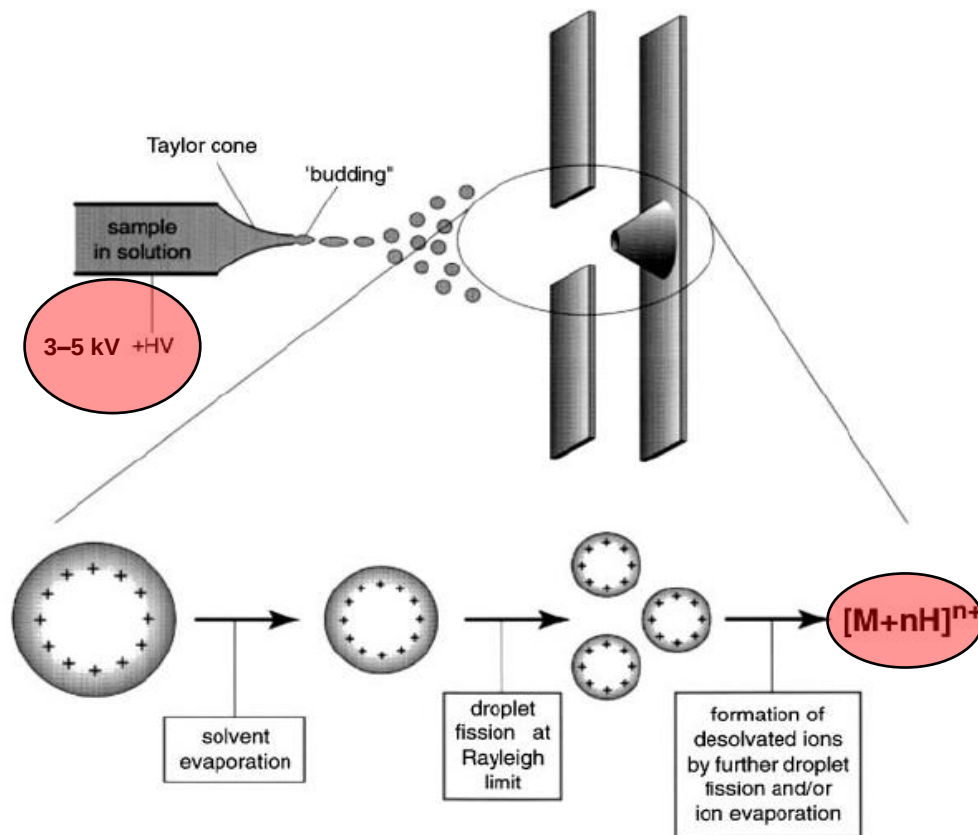


EI just for comparison –
no AP-ionization!

7.1.3.1. ElectroSprayIonization



Electrospray Ionization

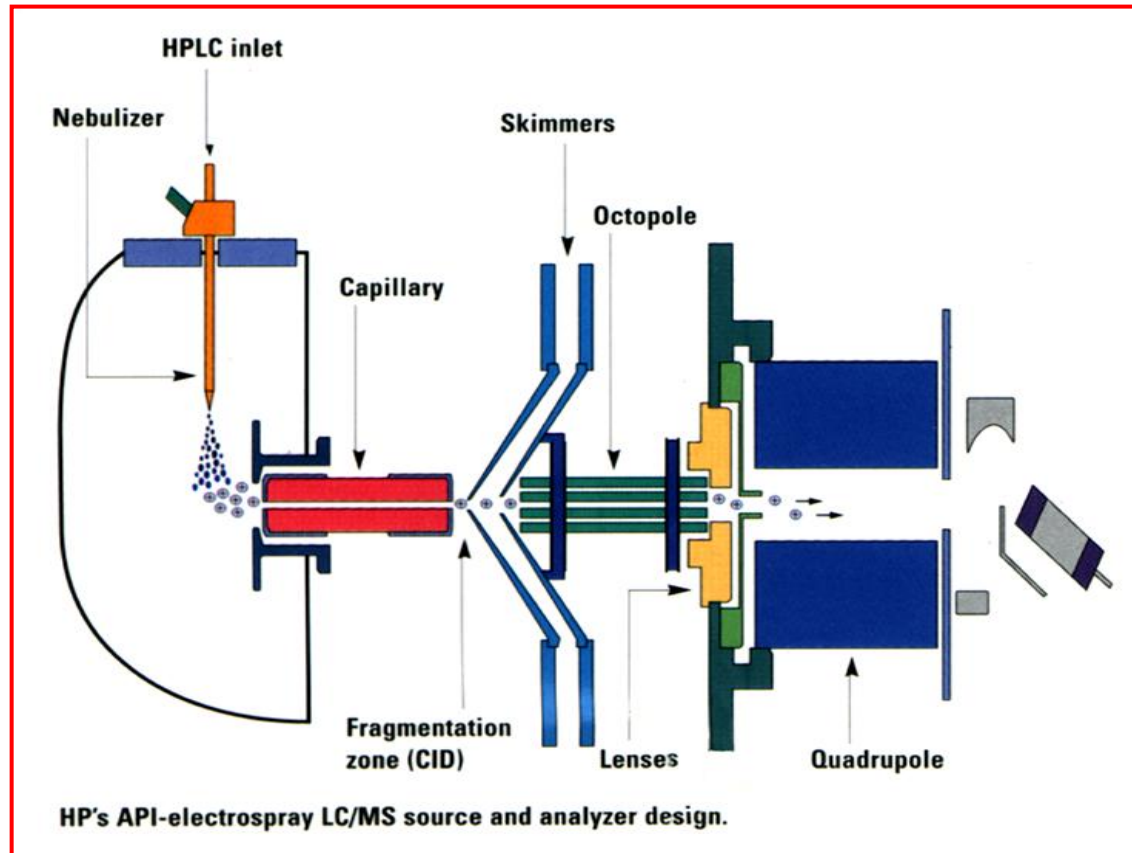


Actual ESI process: only
@ very low flow rates =
nano-spray

At higher flow rates (as in
HPLC) – additional
nebulization by a gas
(commonly N₂) is
required – **pressure
assisted ESI**

Particularly for larger bio-molecules multiply charged
ions are obtained by addition of more than one H⁺.

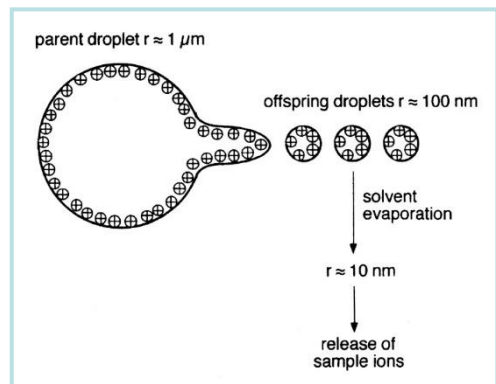
7.1.3.1. ESI



Electro Spray Ionization for HPLC coupling; e.g. orthogonal Sprayer from Agilent Technologies

7.1.3.1. ESI

Between the capillary outlet and the MS orifice a high voltage (several kV) is applied. This leads to separation of charges at the capillary outlet. On their way from the capillary outlet to the MS orifice (inlet) in an environment with high electrical field strength, the radius of the droplets decreases due to evaporation of the solvent. When the Rayleigh Limit is reached, the droplets disintegrate into several smaller droplets. The reason for this effect is the fact that the decreasing radius of the droplets increases the density of equal charges in the droplet. The Rayleigh Limit defines the maximum number of equal charges that can be stabilized inside the droplet. Equal charges repel each other due to Coulomb forces. This repulsion can to some degree be counterbalanced by the surface tension of the liquid. When the Rayleigh Limit is exceeded, the Coulomb forces are stronger than the surface tension and so called „Coulomb explosions“ occur, leading to the formation of smaller droplets.



Rayleigh Limit

$$q_r = \left(\frac{8\pi}{e} \right) (\gamma \epsilon_0)^{\frac{1}{2}} r^{\frac{3}{2}}$$

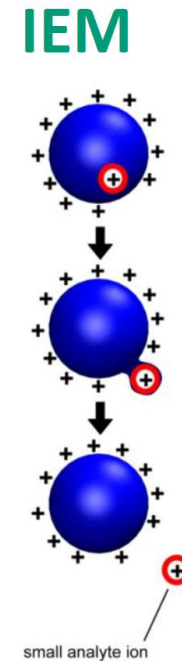
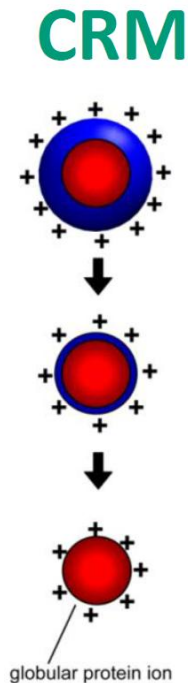
q_r
 r
 e
 γ
 ϵ_0

maximum radius
radius of droplet
elemental charge
surface tension
permittivity of vacuum

7.1.3.1. ESI

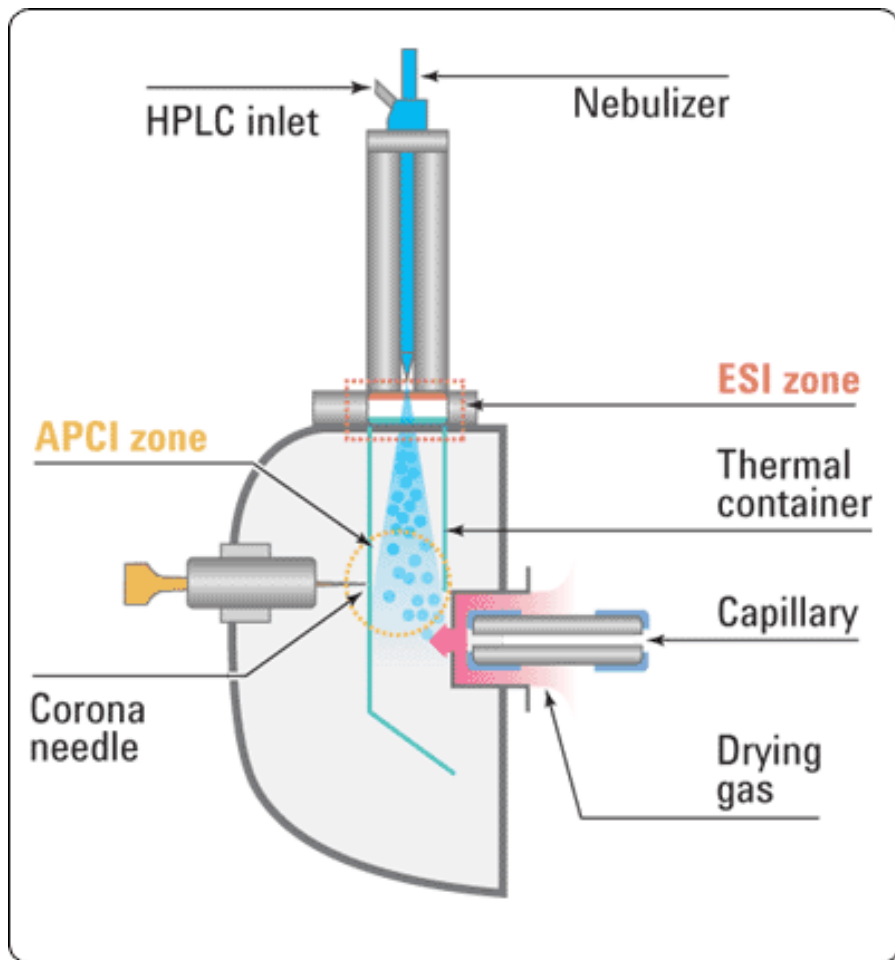
ESI – how are the ions set free finally? There is more than one model trying to explain this process!

Large molecules: **C**harged **R**esidue **M**odel
(„drying“ by evaporation of solvent)

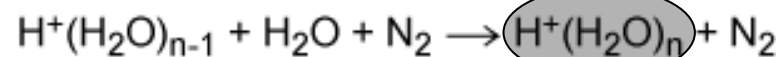
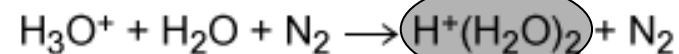
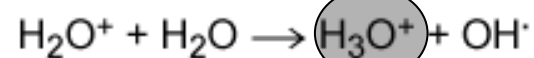
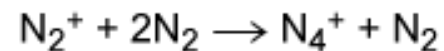


Small molecules: **I**on **E**vaporation **M**odel („ejection“)

7.1.3.2. Atmospheric Pressure Chemical Ionization



Reactions in Plasma Zone:



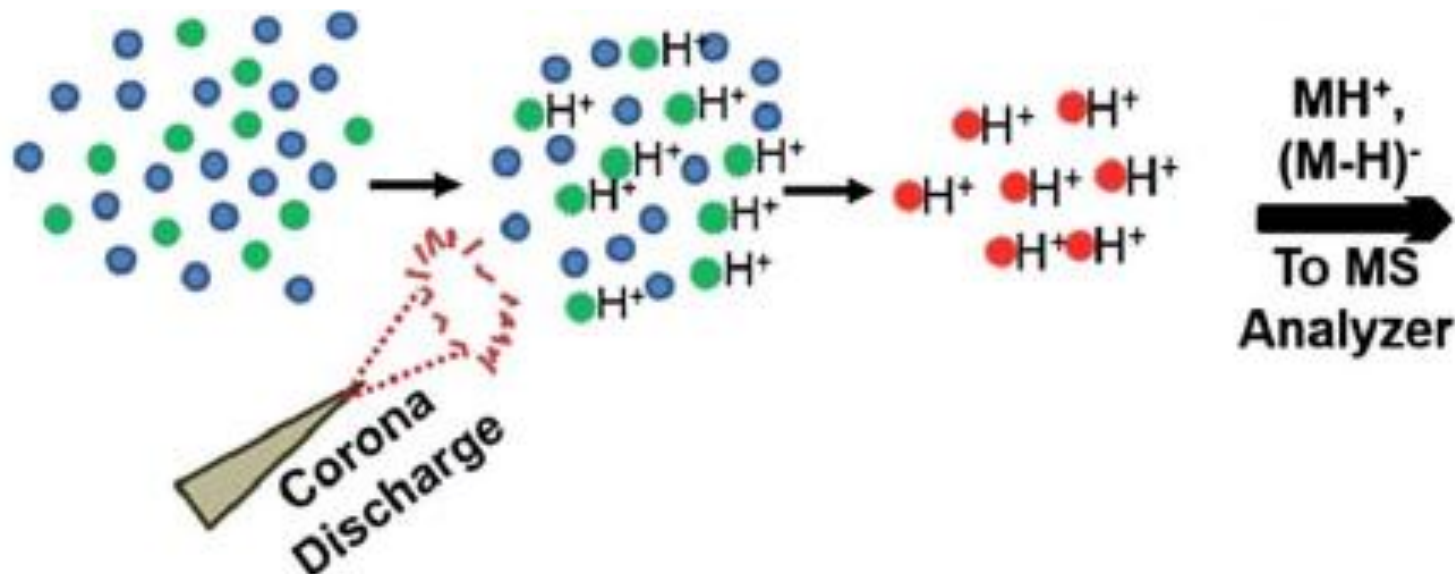
Finally analyte molecules (A) are protonated leading to $[\text{A}+\text{H}]^+$. In APCI (in contrary to ESI) multiple charging is rare.

7.1.3.2. Atmospheric Pressure Chemical Ionization

- Sample Neutral Vapors/Droplets
- Solvent Neutral Vapors/Droplets
- H^+ Protonated Solvent Molecule

• H^+ Protonated Analyte Ion

- Analytes need to be vaporized
- Ionization in the gas phase
- Primarily solvent molecules are ionized
- These transfer charge to the analyte



7.1.4. Matrix Assisted Laser Desorption Ionization

Ionization method	Typical Analytes	Sample Introduction	Mass Range	Method Highlights
Matrix Assisted Laser Desorption (MALDI)	Peptides Proteins Nucleotides	Sample mixed in solid matrix	to 500,000 Daltons	Soft method very high mass

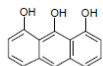
Methods requiring vacuum conditions for ionization
Atmospheric pressure ionization techniques

Hard method: strong fragmentation
Soft method: almost no fragmentation

7.1.4. Matrix Assisted Laser Desorption Ionization

Matrix

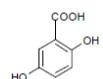
1,8,9-Trihydroxyanthracen
(Dithranol)



Suitable for

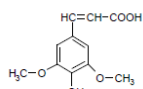
Polymere

2,5-Dihydroxybenzoessäure
(DHB)



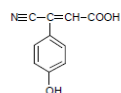
Proteine, Peptide,
Polymere

3,5-Dimethoxy-4-hydroxyzimtsäure
(Sinapinsäure)



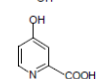
Proteine, Polymere

α -Cyano-4-hydroxyzimtsäure



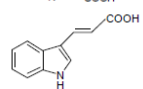
Peptide, (Polymere)

4-Hydroxypicolinsäure



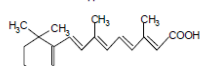
Oligonucleotide

Trans-Indol-3-acrylsäure
(IAA)



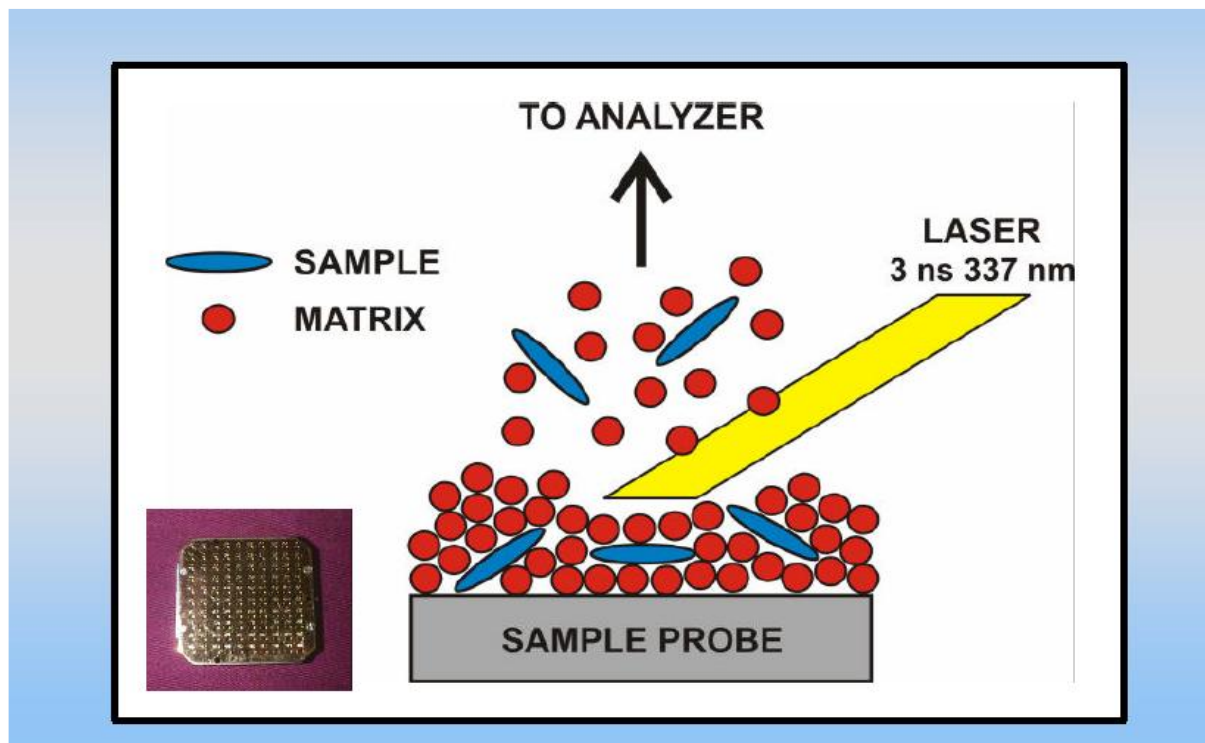
Polymere

Vitamin A-Säure

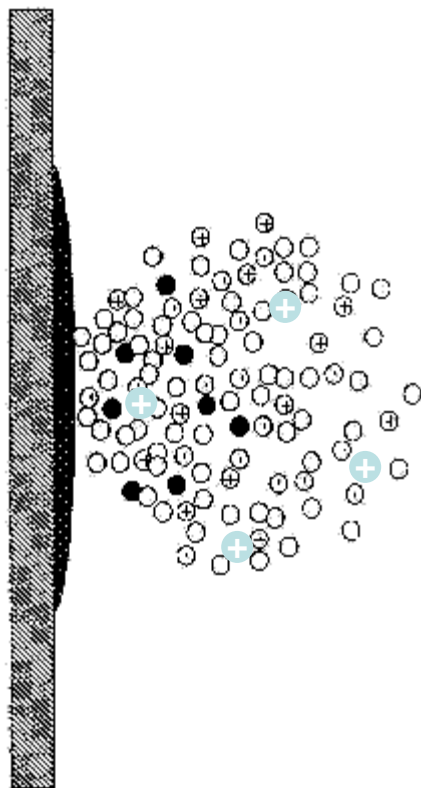


Polymere

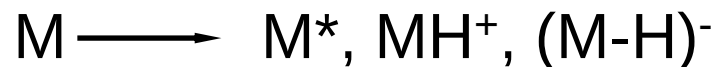
MALDI



7.1.4. MatrixAssistedLaserDesorptionIonization



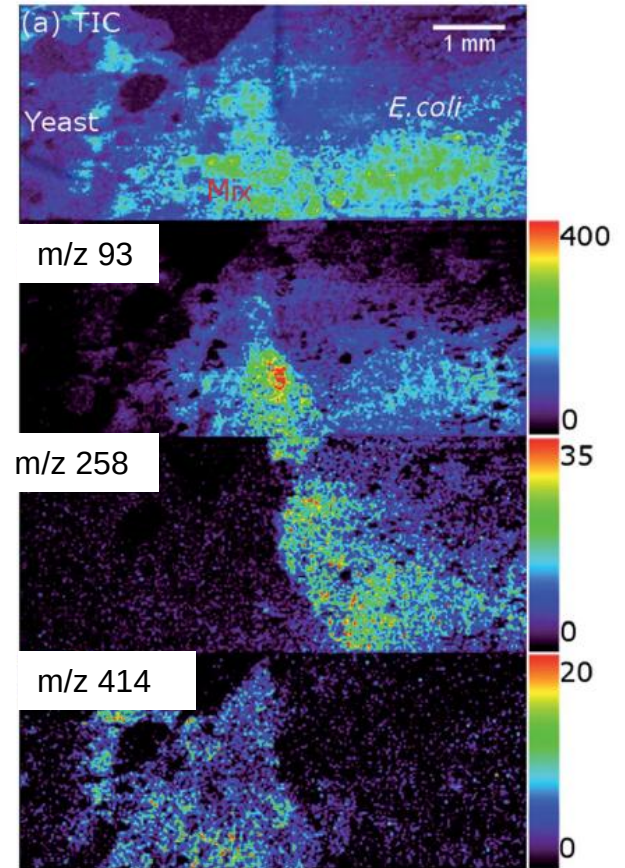
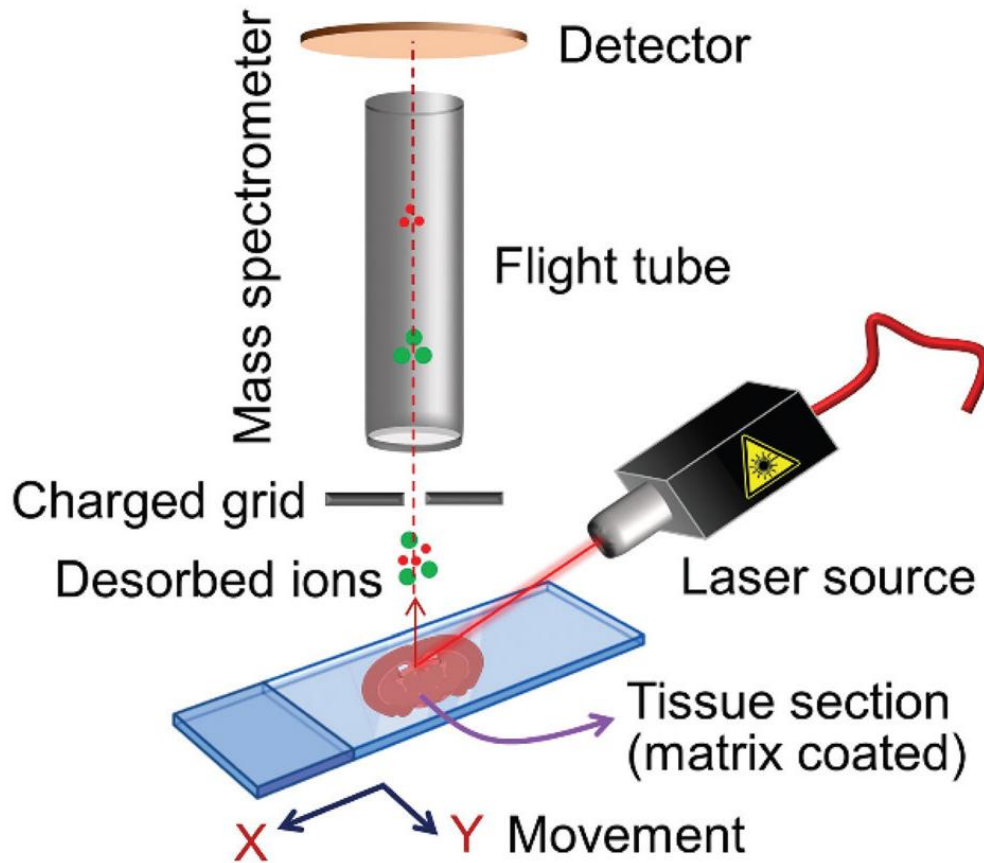
Laser flash produces matrix (M) neutrals, positive, negative ions and sample neutrals.



Sample molecules (A) are ionized by gas phase proton transfer



7.1.4. MatrixAssistedLaserDesorptionIonization



- MALDI can also be used for „Imaging“ by scanning a sample surface
- Thereby one laser shot is used for each „pixel“
- Finally the image is set together providing information on the lateral distribution of an analyte

7.1.5. SUMMARY

Ionization method	Typical Analytes	Sample Introduction	Mass Range	Method Highlights
Electron ^(Impact) Ionization (EI)	Relatively small volatile	GC or liquid/solid probe	to 1,000 Daltons	Hard method versatile provides structure info
Chemical Ionization (CI)	Relatively small volatile	GC or liquid/solid probe	to 1,000 Daltons	Soft method molecular ion peak $[M+H]^+$

Ionization method	Typical Analytes	Sample Introduction	Mass Range	Method Highlights
Matrix Assisted Laser Desorption (MALDI)	Peptides Proteins Nucleotides	Sample mixed in solid matrix	to 500,000 Daltons	Soft method only few charges

7.1.5. SUMMARY

Ionization method	Typical Analytes	Sample Introduction	Mass Range	Method Highlights
Electrospray (ESI)	Peptides Proteins nonvolatile	Liquid Chromatography or syringe	to 200,000 Daltons	Soft method ions often multiply charged
Atmospheric Pressure Chemical Ionization (APCI)	Relatively small volatile	Liquid Chromatography or syringe	to ~1,200 Daltons	Soft method for less polar analytes
Atmospheric Pressure Photo Ionization (APPI)	Relatively small volatile	Liquid Chromatography or syringe	to ~1,200 Daltons	Soft method for less polar analytes

7.2. Mass Analyzers

Mass analyzers / Mass separation units

Quadrupole Mass Analyzer (single und triple Quadrupole)

Ion-Trap (Paul Trap)

Electrostatic Trap (Orbitrap®)

Magnetic Field/Electrostatic Field Mass Analyzer

Time of Flight Mass Spectrometer (TOF-MS)

Hybride Mass Spectrometers

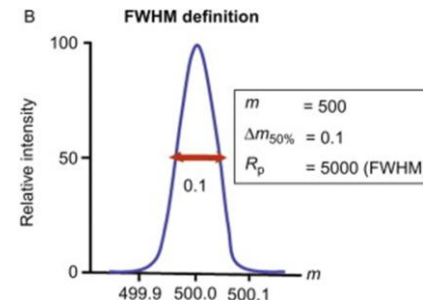
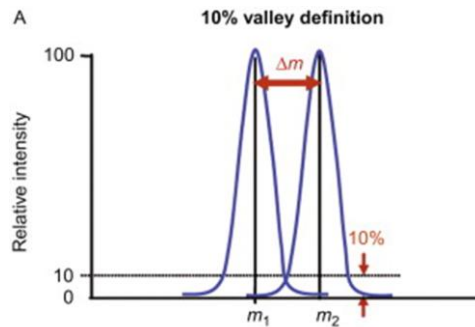
7.2. Mass Analyzers

Important Parameters for Characterization of Mass Separation units:

-Mass- range:

- small: for „small molecules“ up to 2.000 m/z
- high: for large molecules > 10.000 m/z

-Resolution:



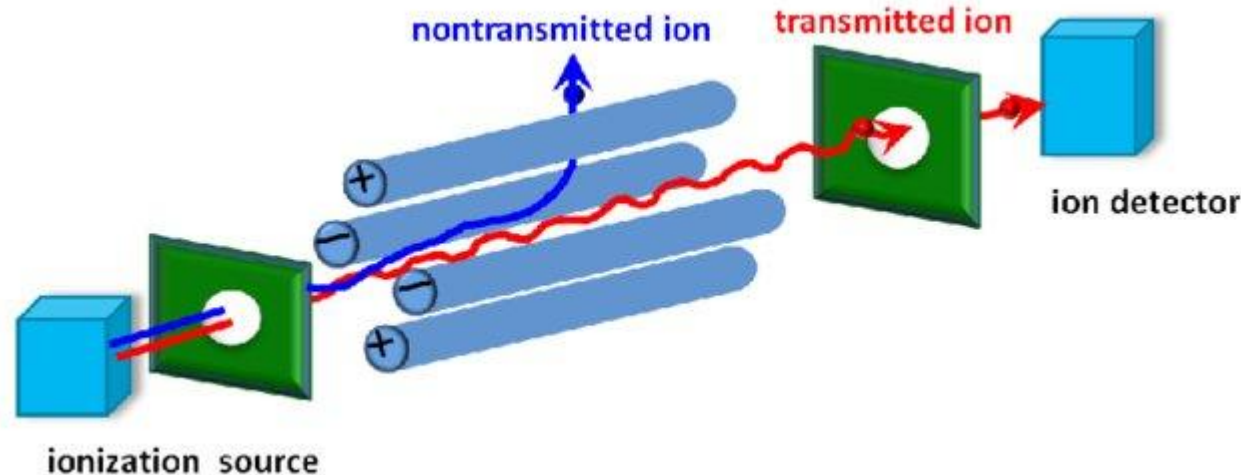
Up to more than 10⁶ !

Deviation from the theoretical monoisotopic mass in ppm

-Mass accuracy:

$$Error_{ppm} = \frac{m/z_{exp} - m/z_{theor}}{m/z_{theor}} * 10^6$$

7.2.1. Quadrupol Mass Analyzer

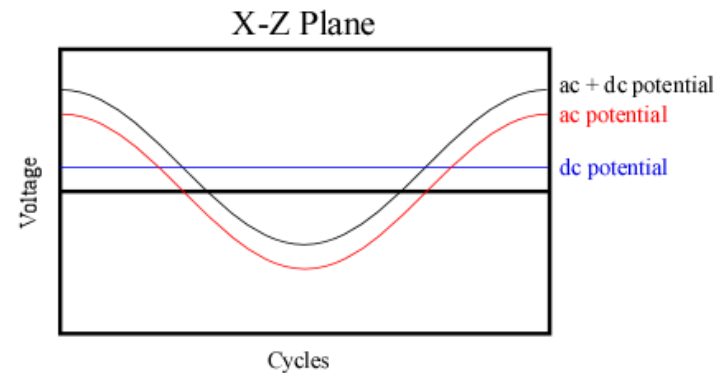
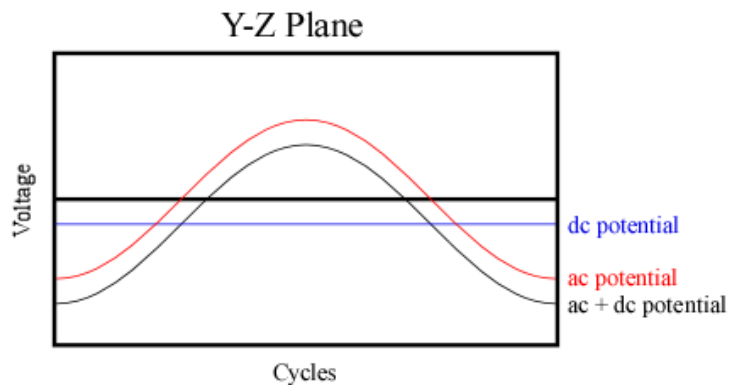
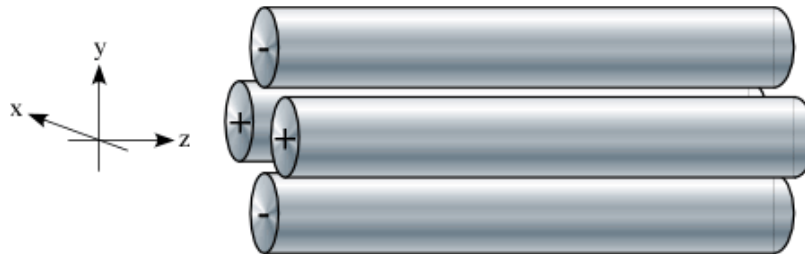
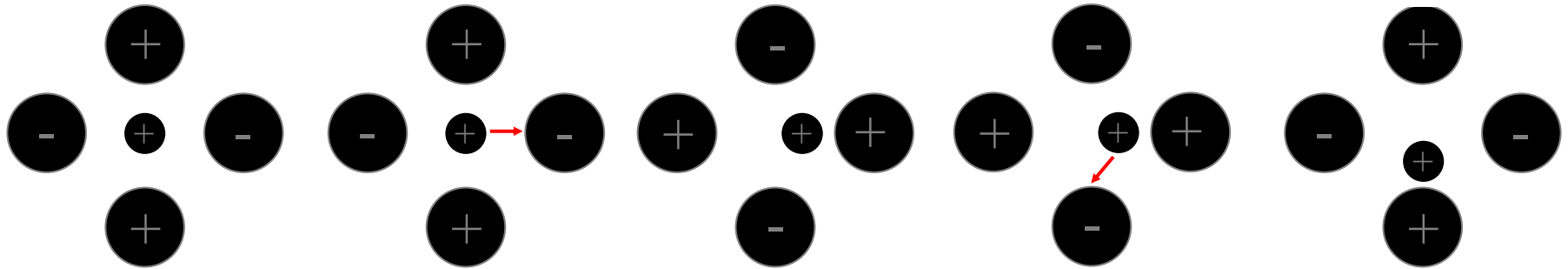


- ions travel parallel to four rods
- opposite pairs of rods have rapidly alternating potentials (AC)
- ions try to follow alternating field in helical trajectories
- stable path only for one m/z value for each field frequency

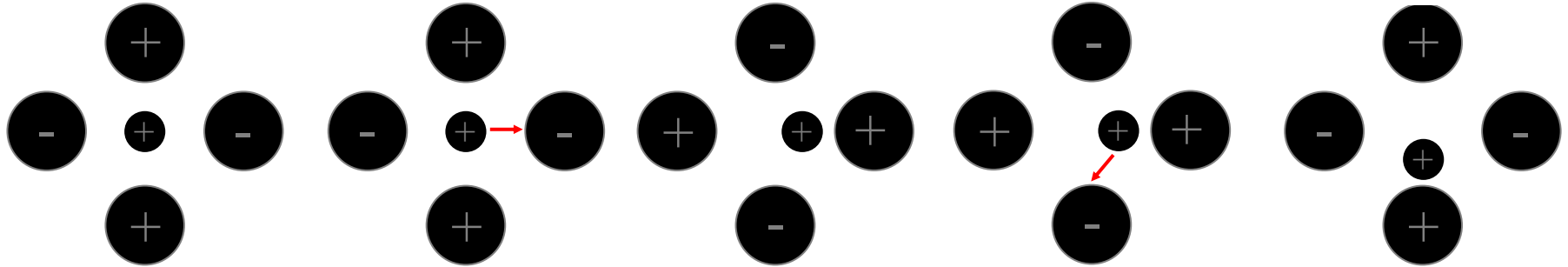
Quadrupol analyzers are easy to operate, inexpensive and very compact (bench-top instruments). They provide relatively low resolution (around 1000) a restricted mass range (up to m/z 2000) and low mass accuracy. They can be operated in the SCAN mode - potentials are varied with time so sequentially ions with different m/z ratios can travel through the quadrupol on a stable flightpath or the selected ion mode (SIM) with a fixed potential allowing only selected m/z ratios to pass.

7.2.1. Quadrupol Mass Analyzer

7.2.1. Quadrupol Mass Analyzer

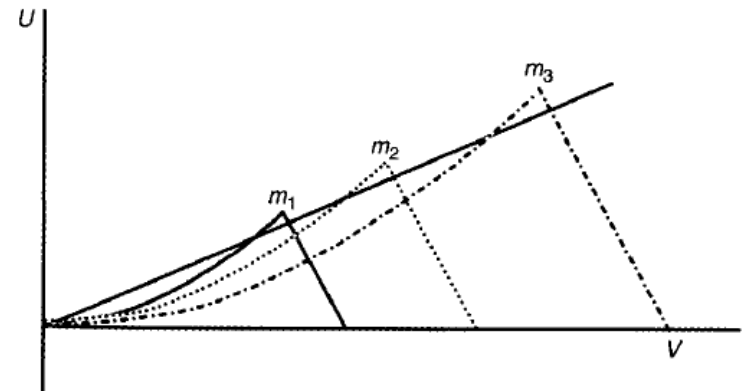


7.2.1. Quadrupol Mass Analyzer



$$a = \frac{8 z e U}{m r_0^2 \omega^2} \Rightarrow U = a \left(\frac{m}{z} \right) \frac{r_0^2 \omega^2}{8 e}$$

$$q = \frac{4 z e V_0}{m r_0^2 \omega^2} \Rightarrow V = q \left(\frac{m}{z} \right) \frac{r_0^2 \omega^2}{4 e}$$



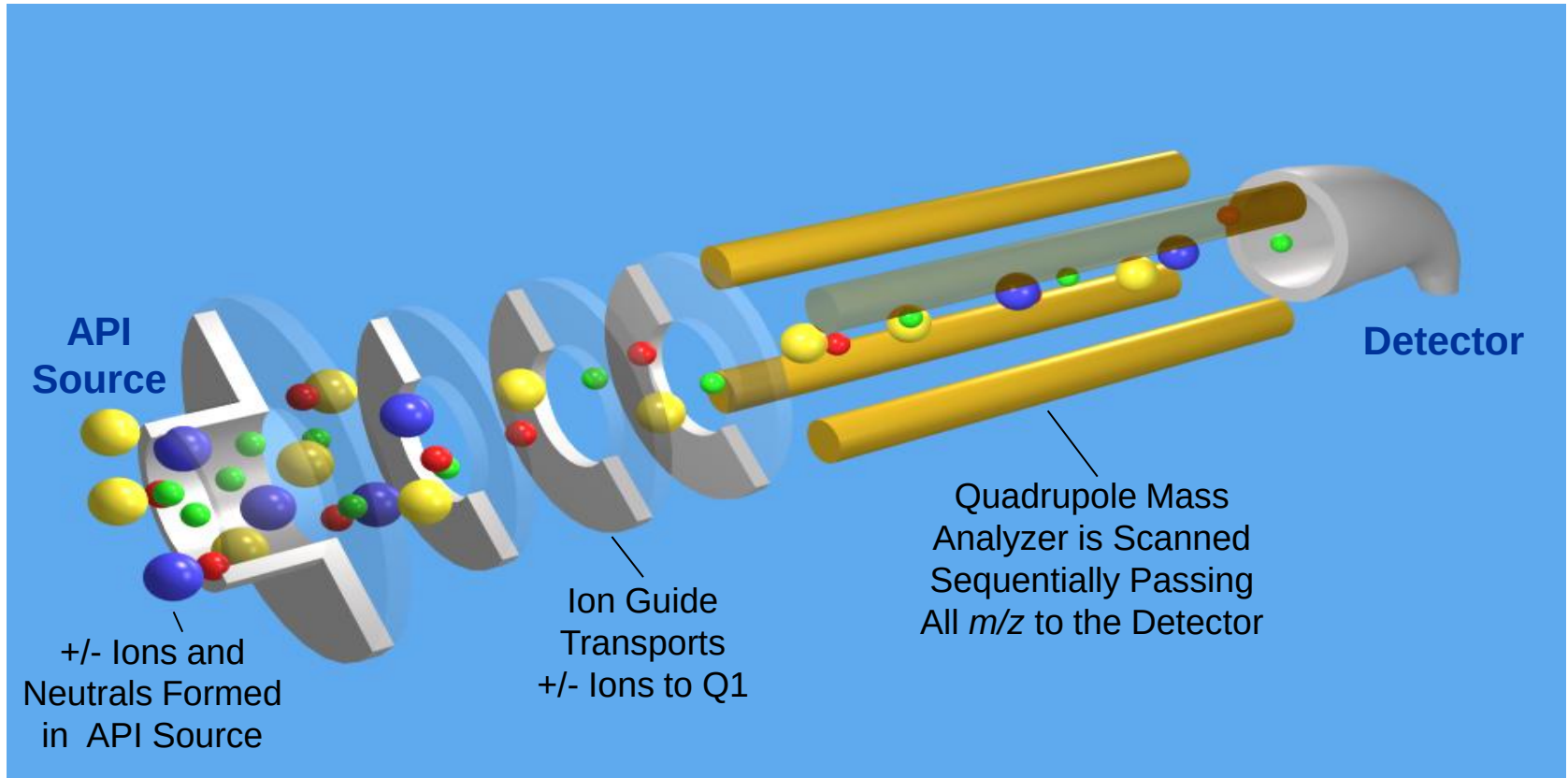
Matthieu Diagram

Scanning: Variation of U and V @ $U/V = \text{constant}$

Variation of m/z window: variation of U/V

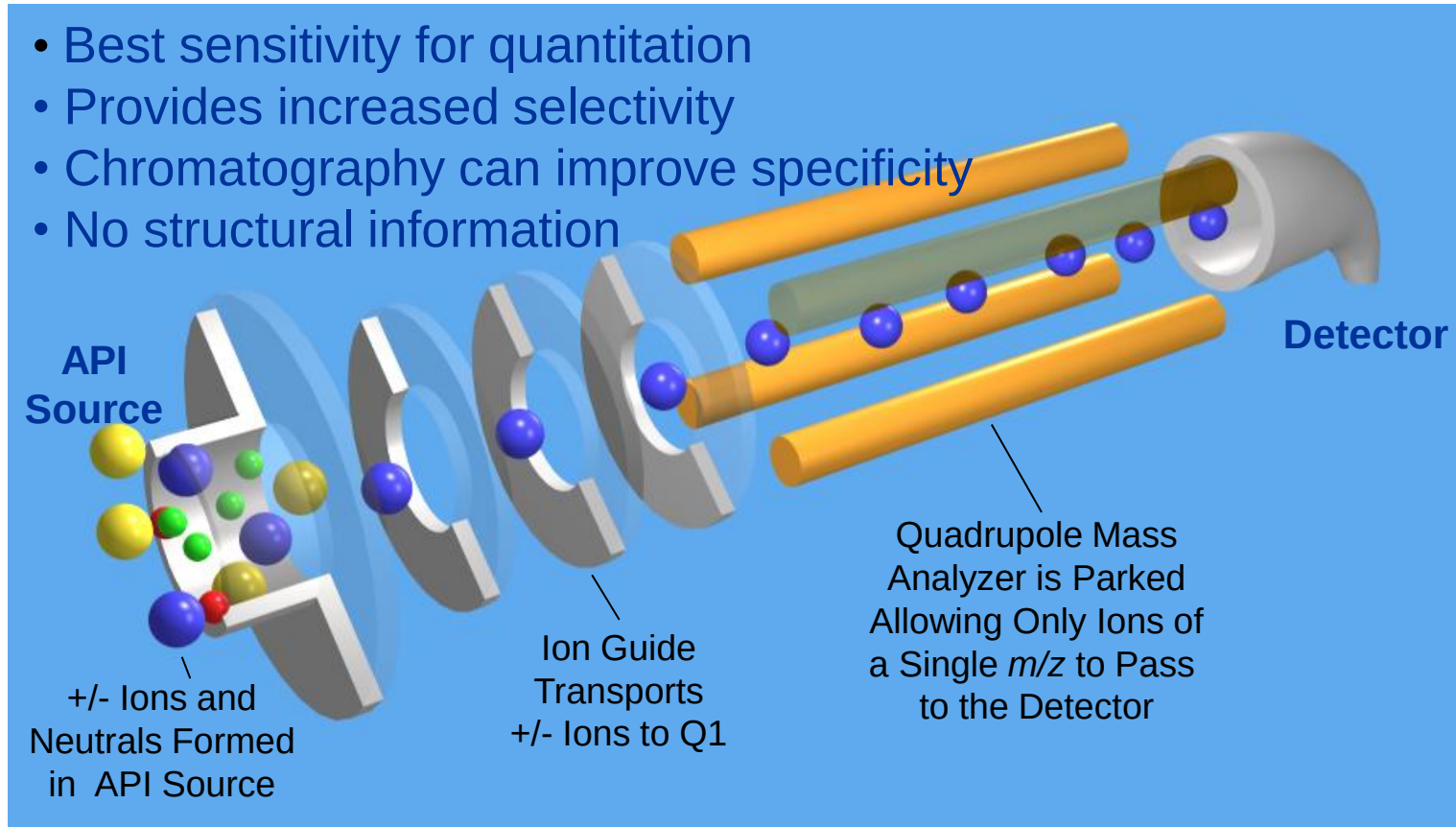
Transmission of all ions = longuide: @ $U=0$

7.2.1. Quadrupole Mass Analyzer



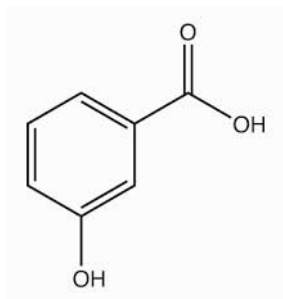
MS Single Quadrupole **SCAN** Mode)

7.2.1. Quadrupole Mass Analyzer

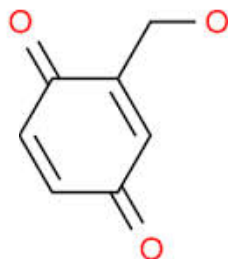
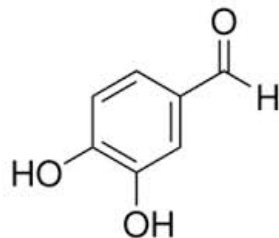


MS Single Quadrupole **S**(elected) **I**(on) **M**(onitoring) Mode)

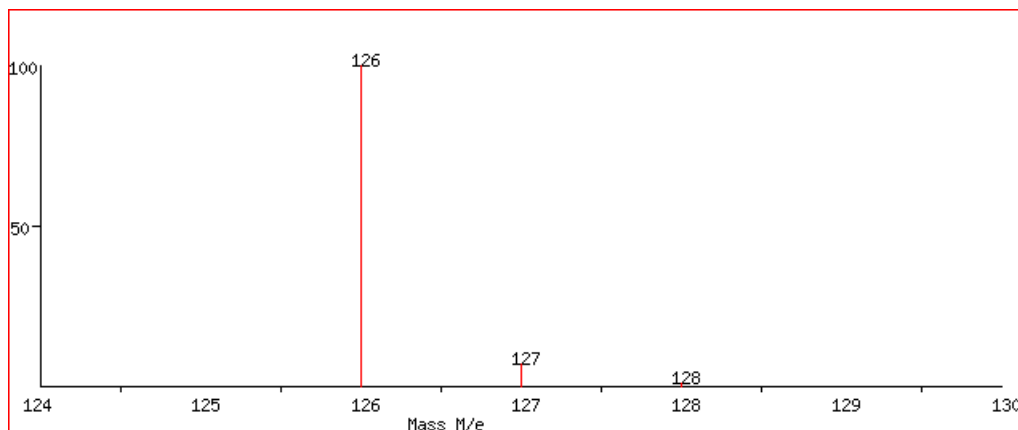
7.2.1. Quadrupol Mass Analyzer



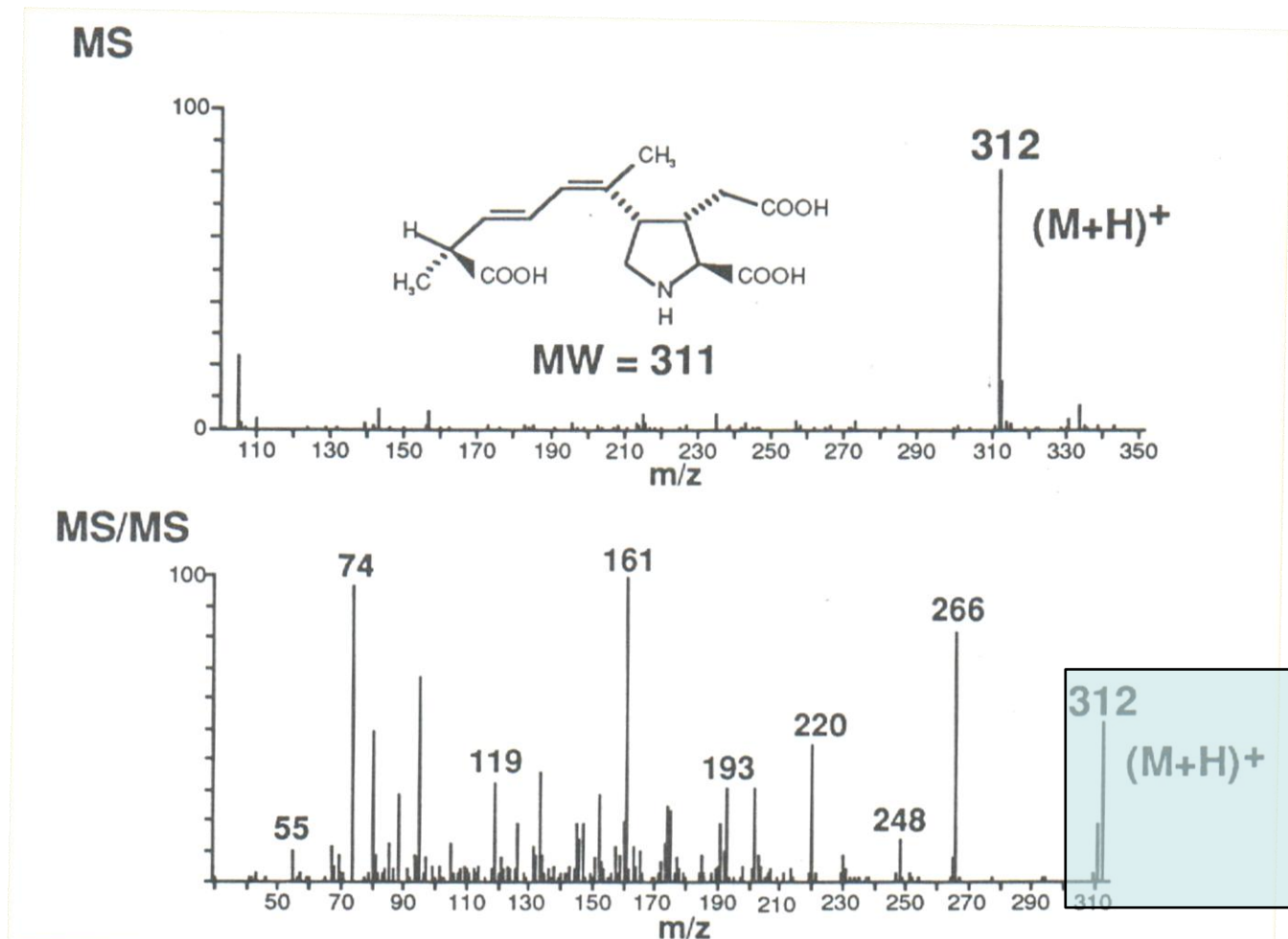
All $C_6H_6O_3$



- In the case of soft ionization (e.g. CI or ESI) – isobaric molecules cannot be distinguished as only the (protonated) molecular ion is displayed (no fragmentation pattern)



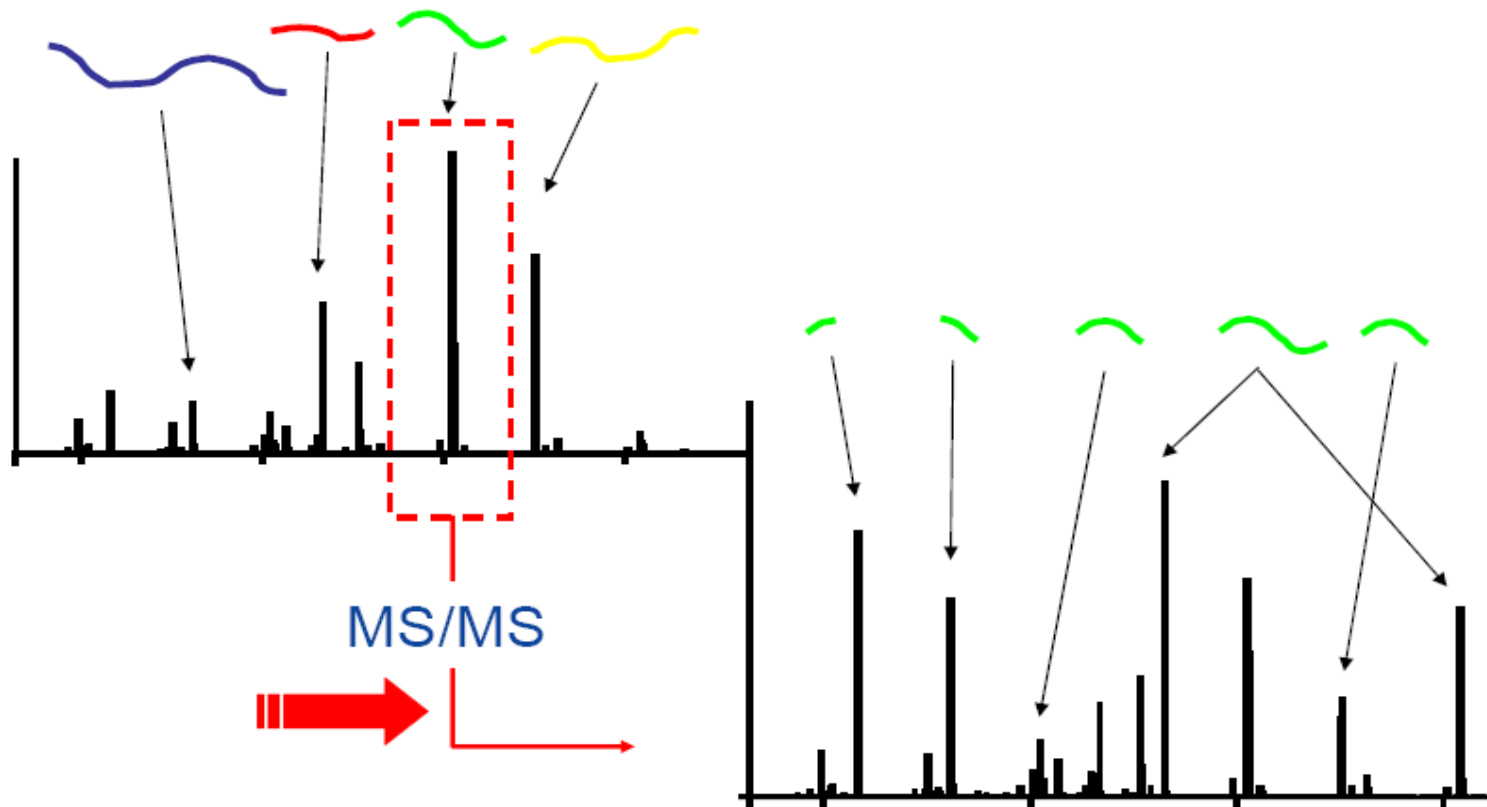
7.2.1. Quadrupol Mass Analyzer



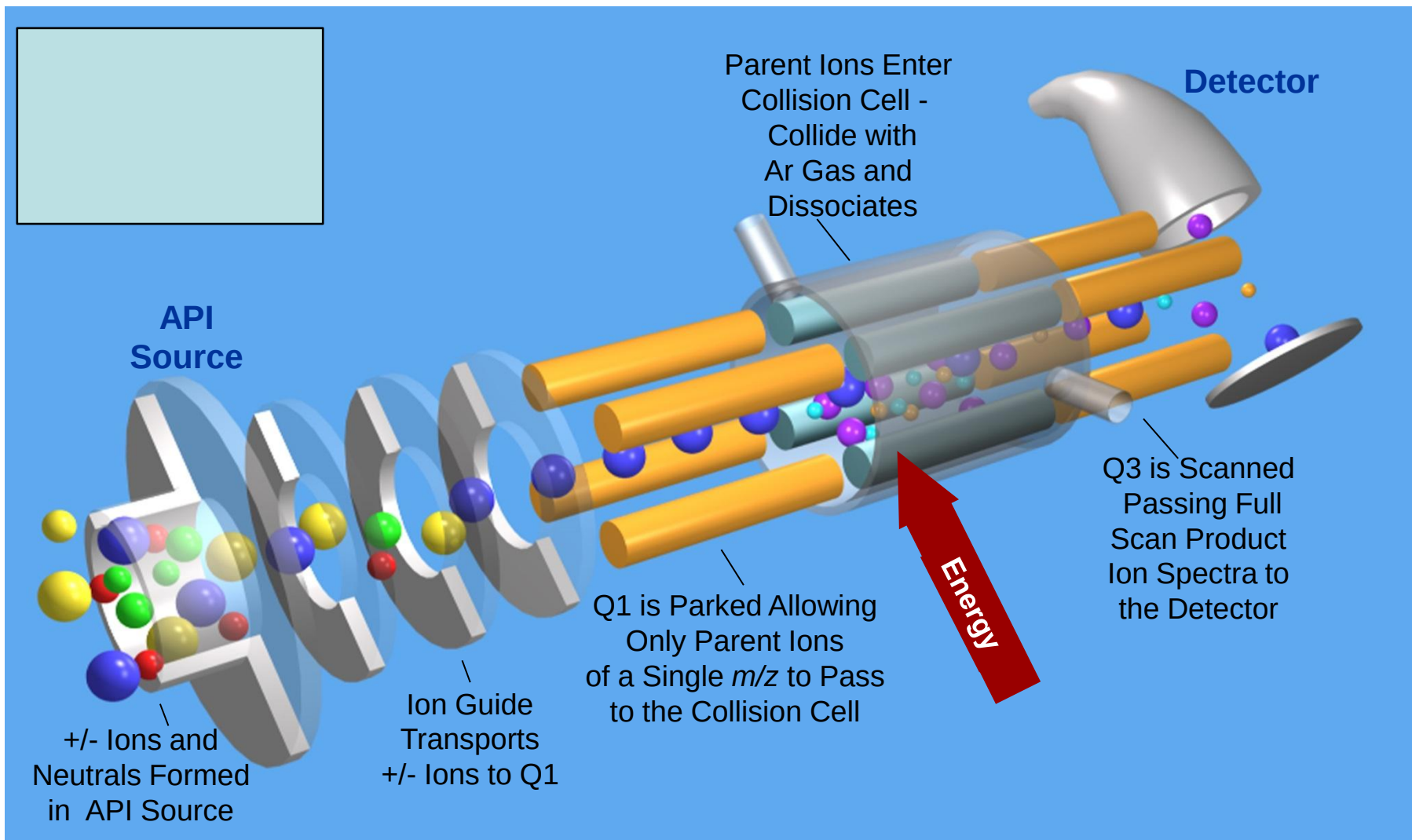
Selected precursor

7.2.2. Triple Quadrupol Mass Analyzer

Tandem Mass Spectrometry (MS/MS)

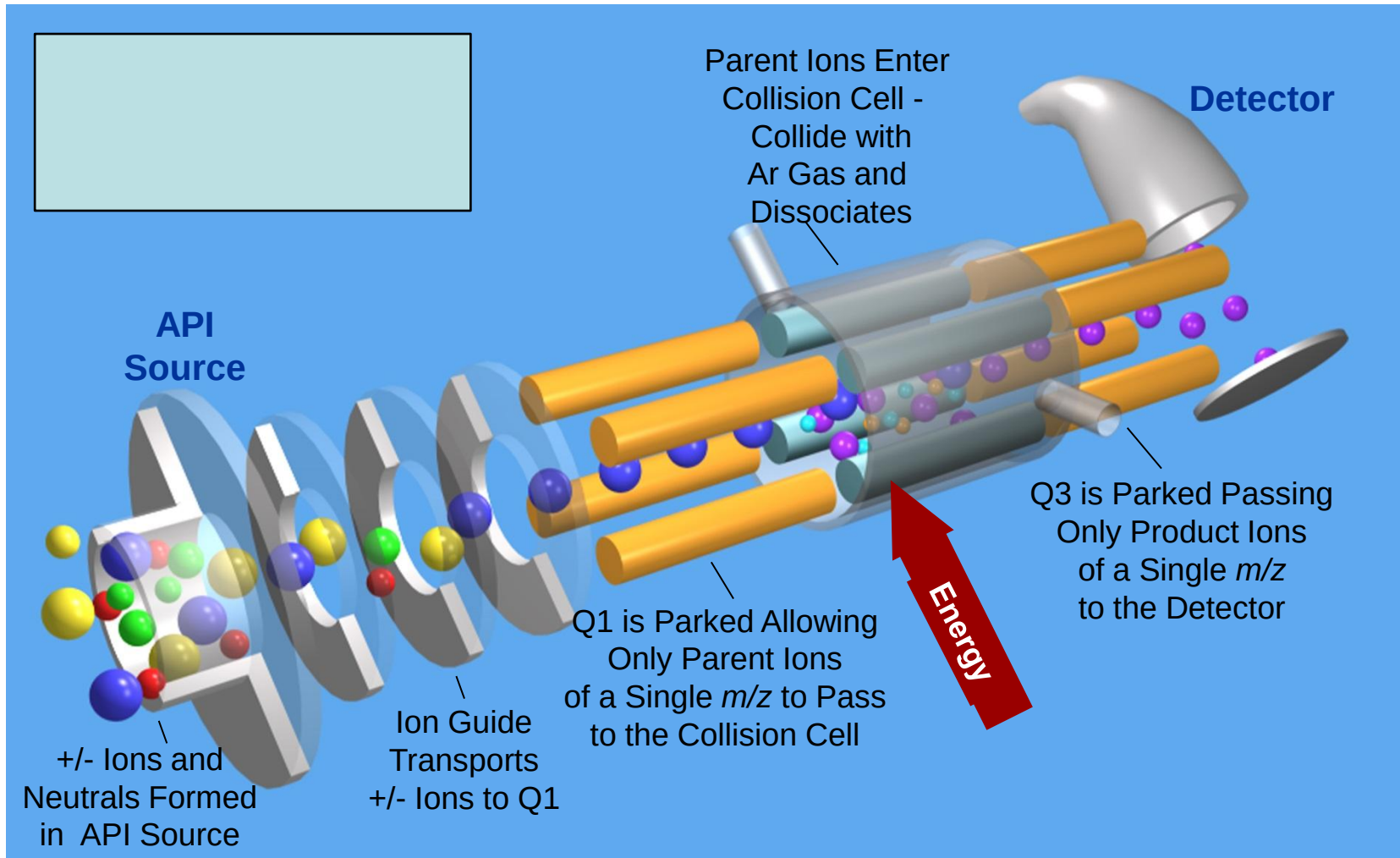


7.2.2. Triple Quadrupol Mass Analyzer



MS Triple Quadrupol Product Ion Scan MS/MS

7.2.2. Triple Quadrupol Mass Analyzer



MS Triple Quadrupol – MRM Mode

7.2.2. Triple Quadrupol Mass Analyzer

Quadrupol:

Single Quadrupol:

- SCAN: provides spectrum
- SIM: only recording selected m/z values e.g. for quantification

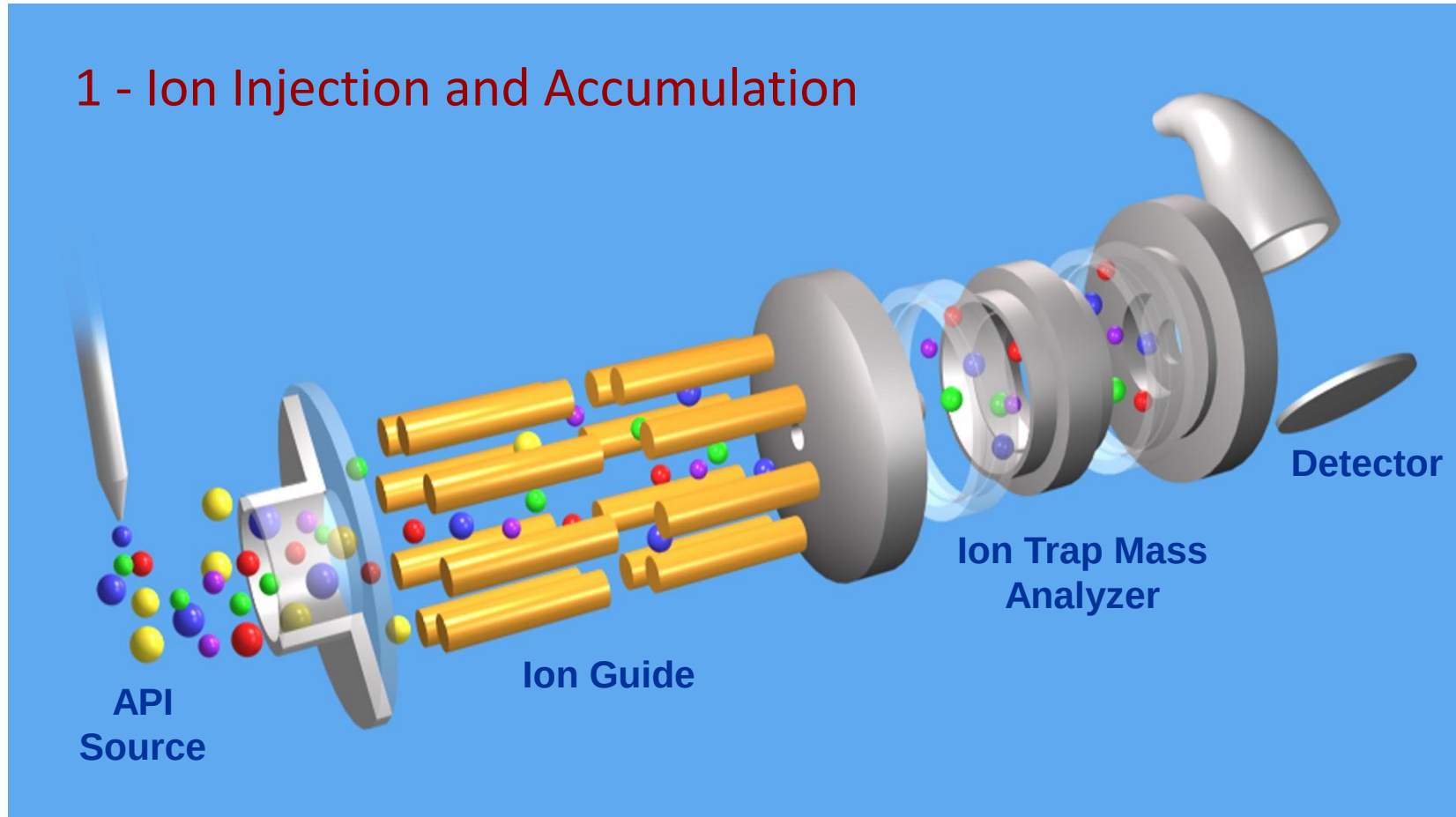
Triple Quadrupol:

- MS² (in space) possible
- Very sensitive for quantitative analysis
- MRM modus: Selection of pre-cursor and detection of specific fragments
- Large linear range for quantitative analysis

Resolution	~ 1000;	→	unit resolution
Mass-range	< 2000;		
Mass-Error	> 100ppm		

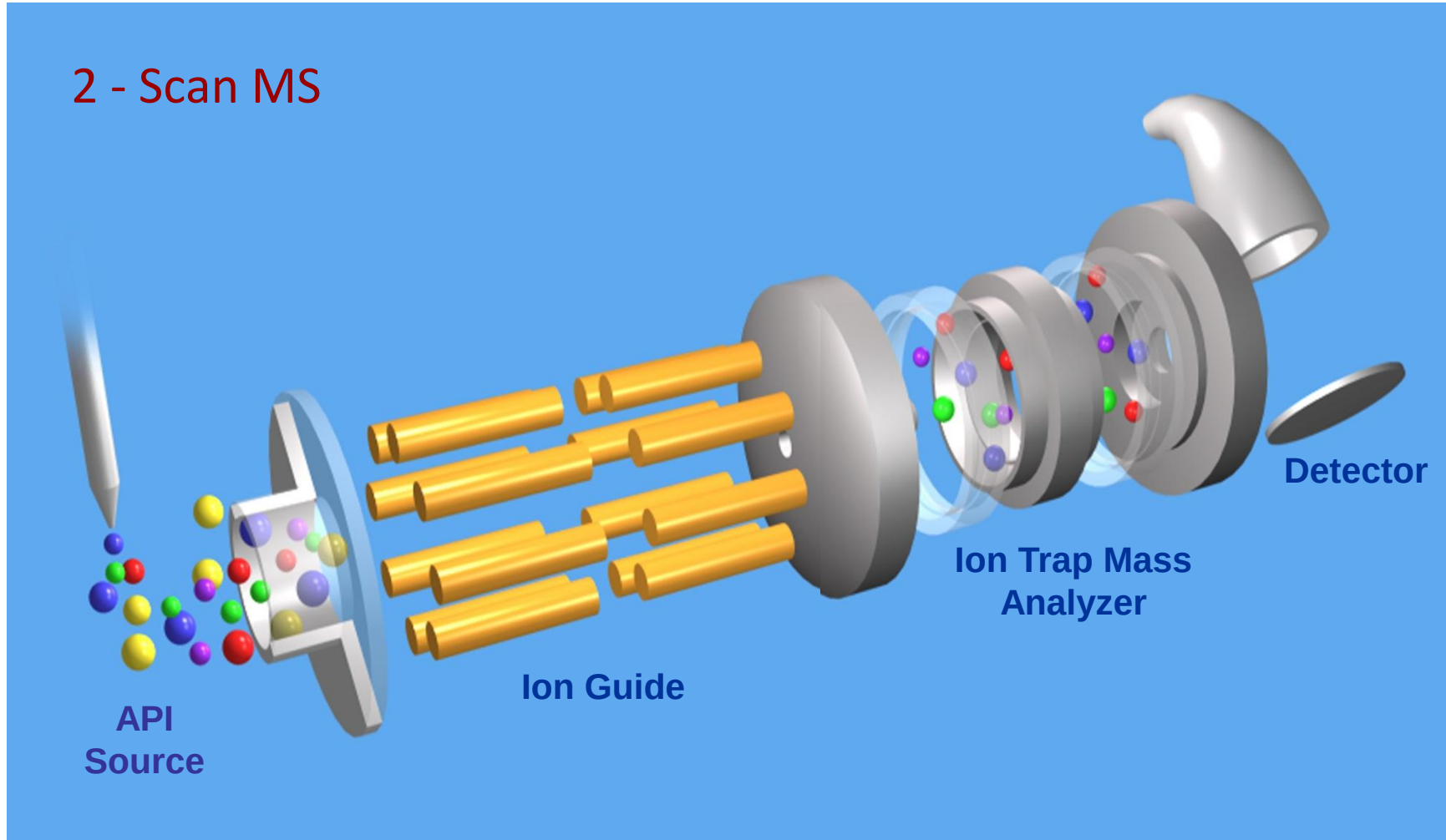
7.2.3. Ion Trap Mass Analyzer

1 - Ion Injection and Accumulation



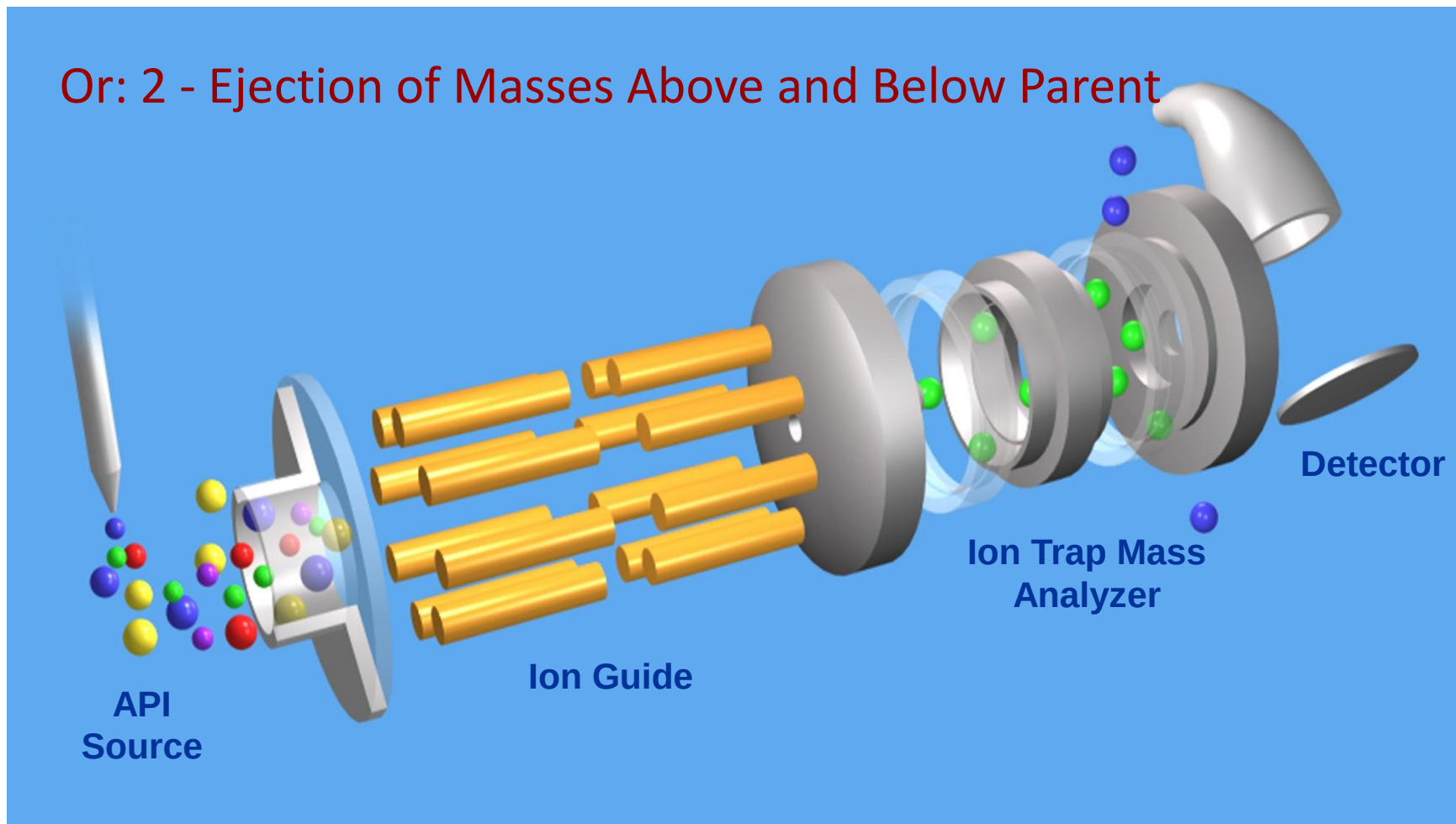
7.2.3. Ion Trap Mass Analyzer

2 - Scan MS



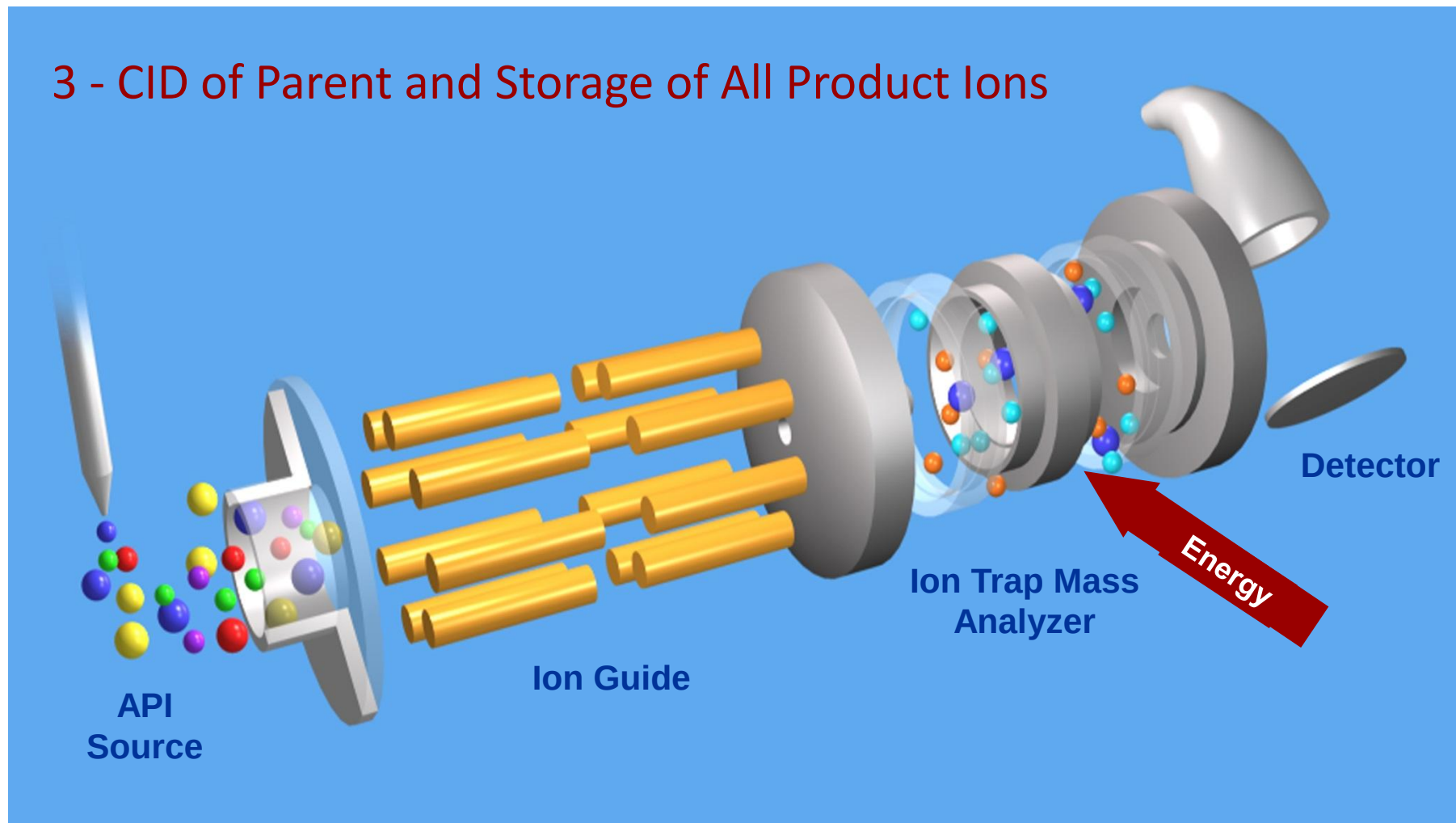
7.2.3. Ion Trap Mass Analyzer

Or: 2 - Ejection of Masses Above and Below Parent



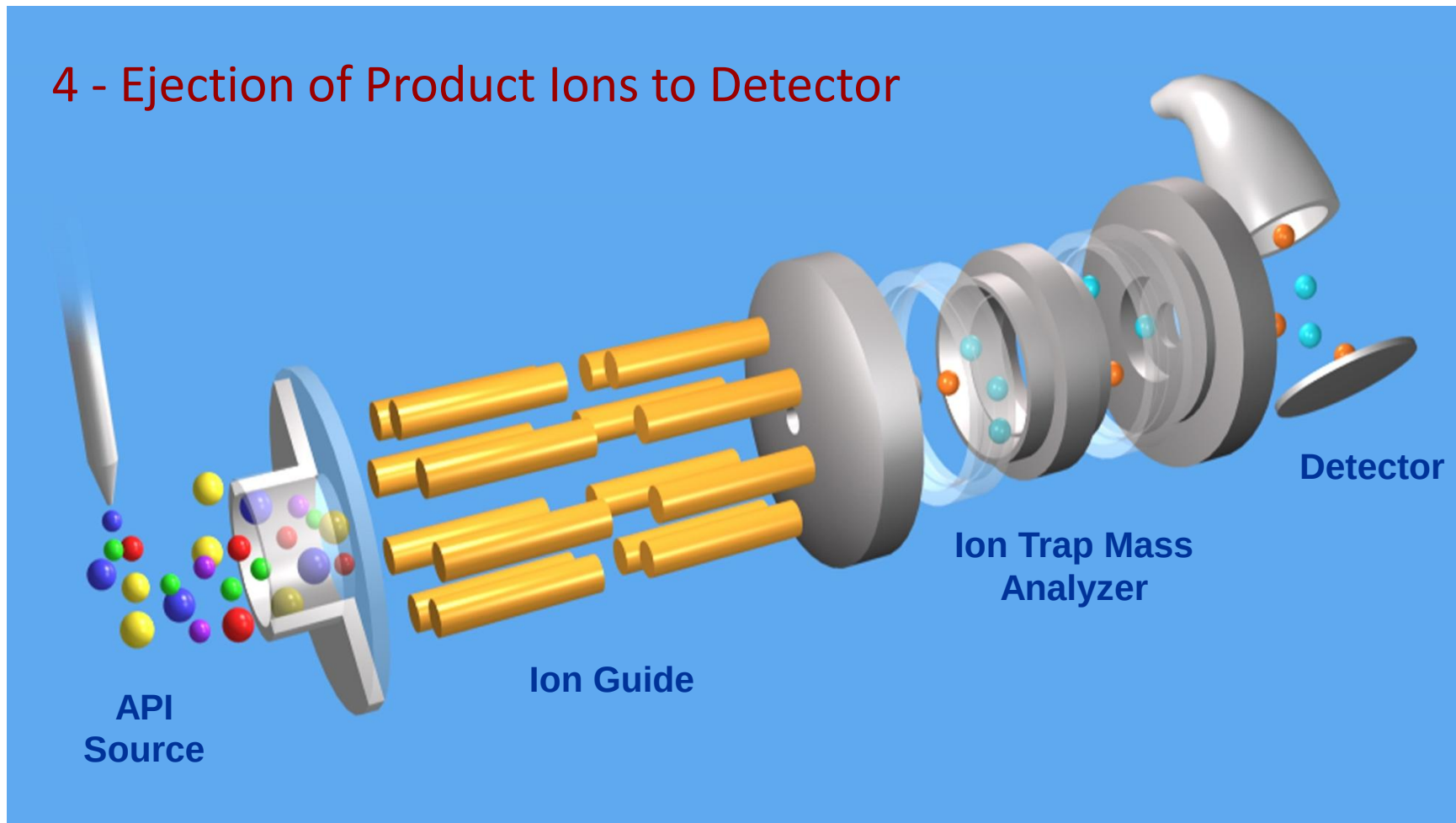
7.2.3. Ion Trap Mass Analyzer

3 - CID of Parent and Storage of All Product Ions

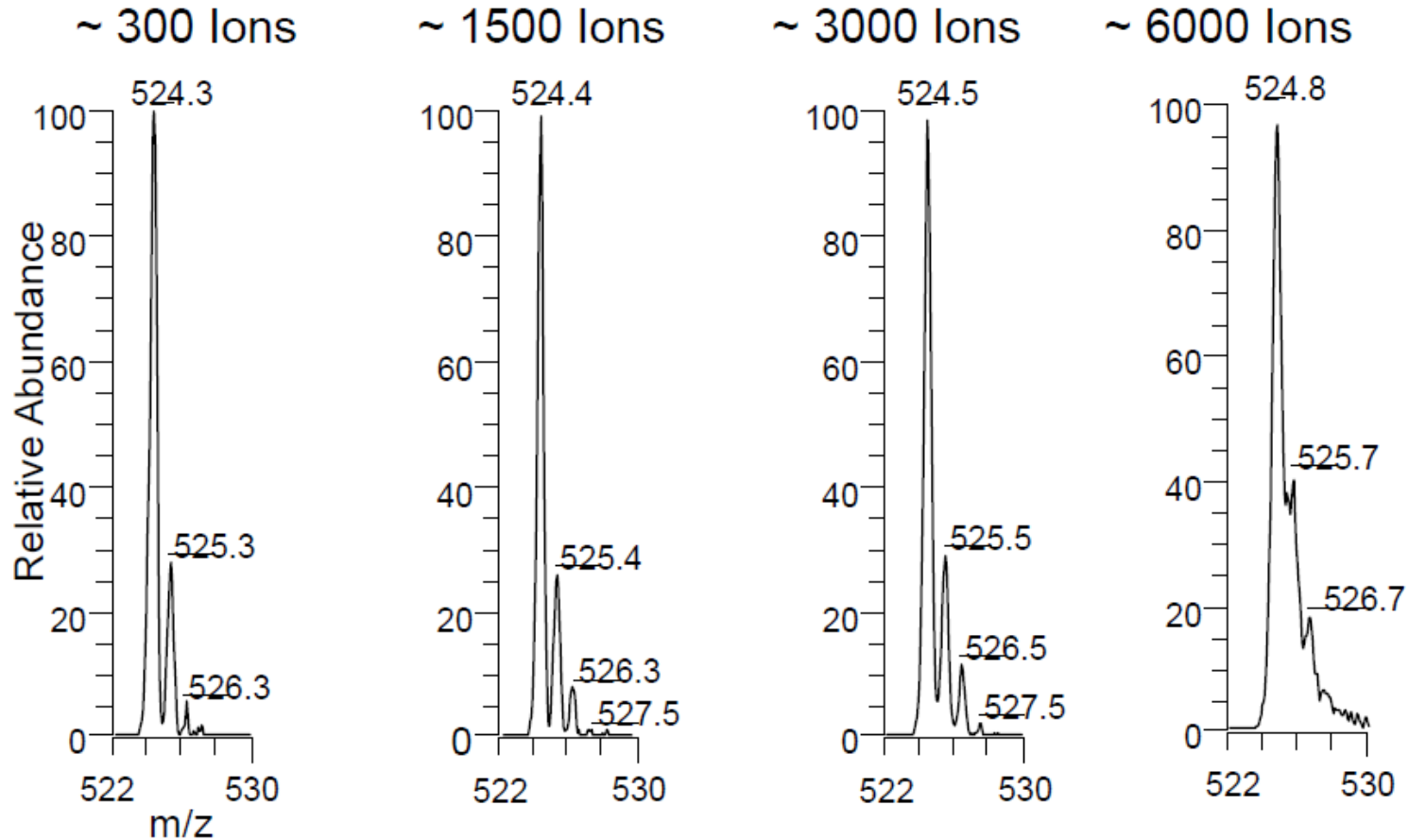


7.2.3. Ion Trap Mass Analyzer

4 - Ejection of Product Ions to Detector



7.2.3. Ion Trap Mass Analyzer



Resolution and Mass
accuracy acceptable

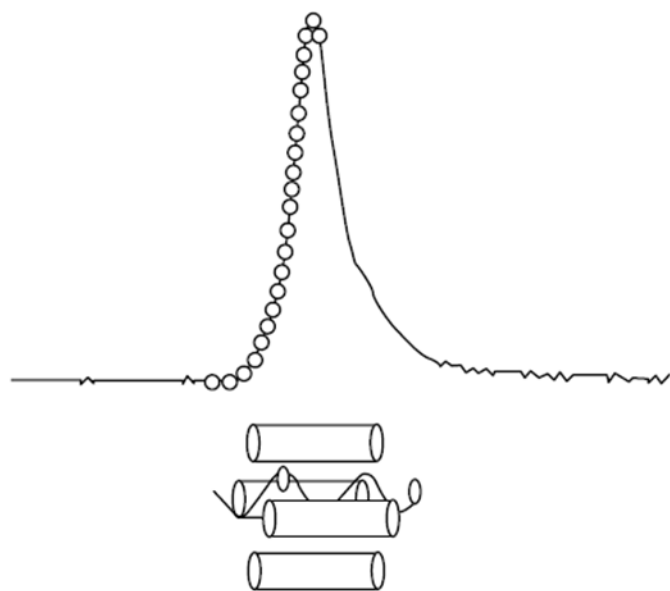


Resolution and Mass
accuracy deteriorating

7.2.3. Ion Trap Mass Analyzer

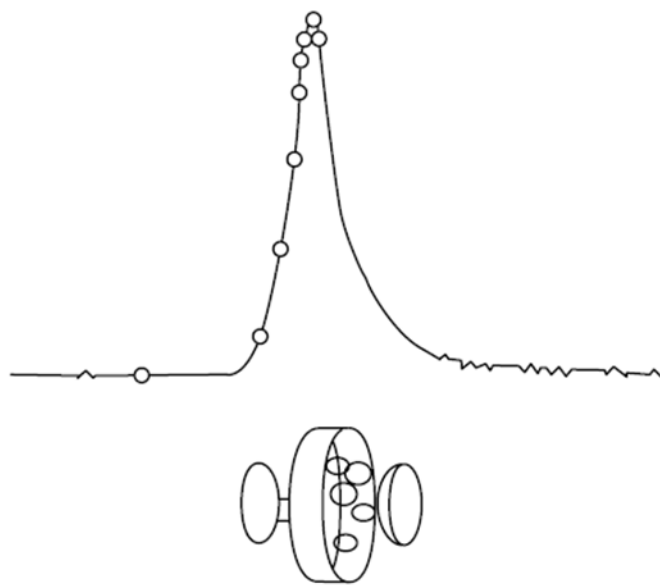
Triple Quadrupole

Data points are evenly spaced across the entire peak



Ion Trap

Data points are obtained most frequently at the top of the peak



7.2.3. Ion Trap Mass Analyzer

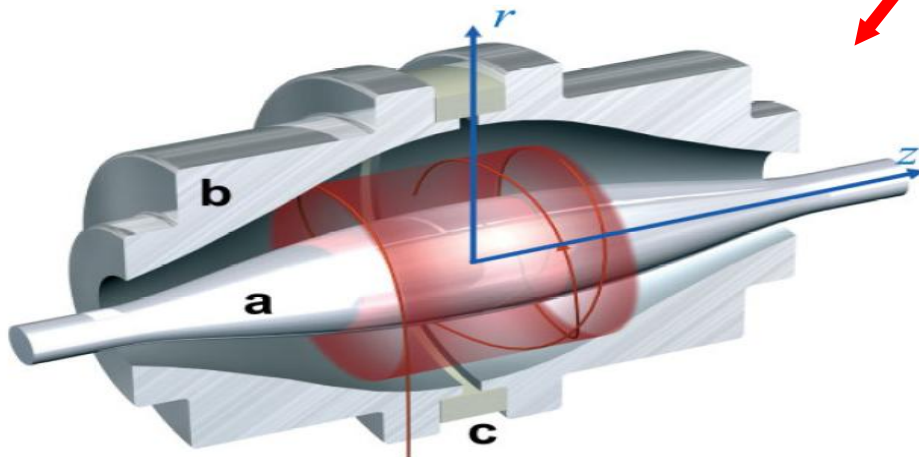
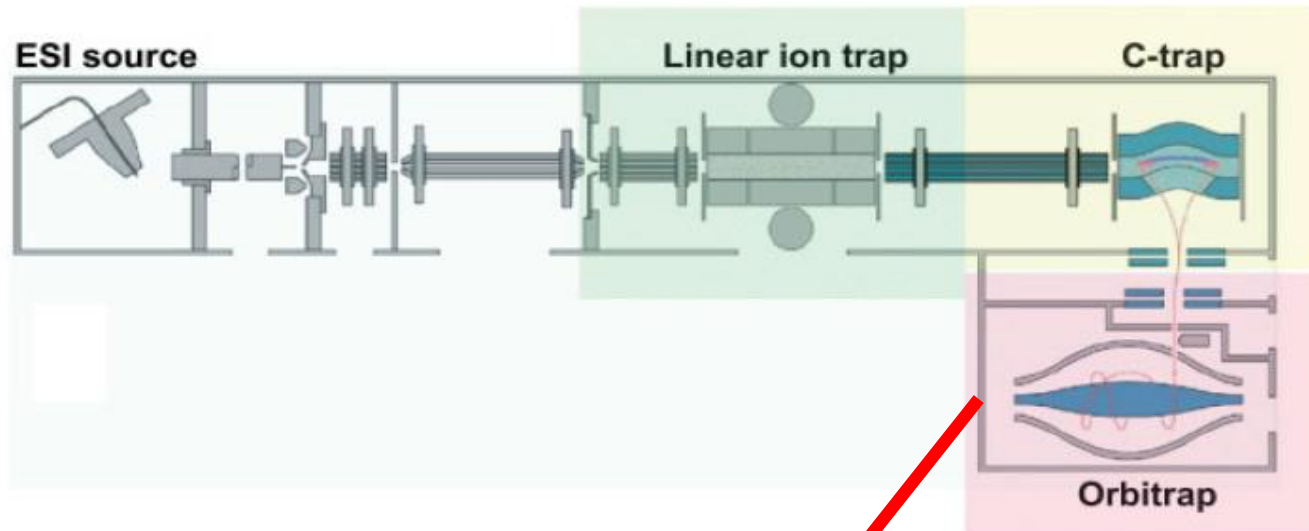
Ion Trap:

- Ions can be accumulated in the trap-
more sensitive than Quadrupole in SCAN mode
- MSⁿ Experiments (in time) possible – selected
Precursor can be fragmented and fragments can
again be defined as Precursor for further
fragmentation
- Due to space-charge effects less suitable for
quantitative analysis

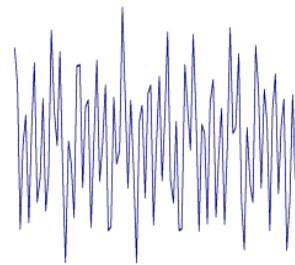
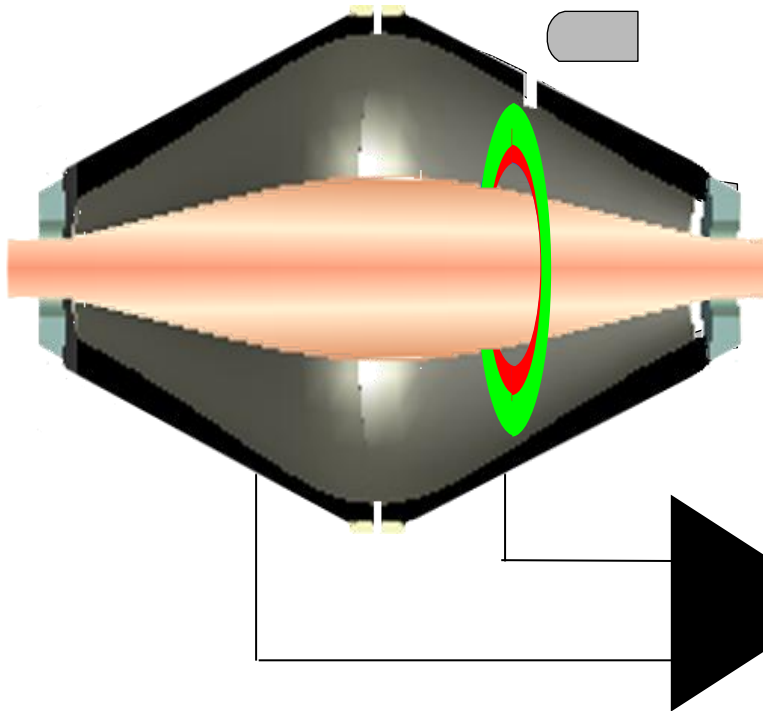
Resolution	< 2000;
Mass-range	< 4000;
Mass-Error	> 100ppm

7.2.4. Electrostatic Trap

Electrostatic trap = Orbitrap



7.2.4. Electrostatic Trap



ω
 k

$$\omega = \sqrt{\frac{k}{m/z}}$$

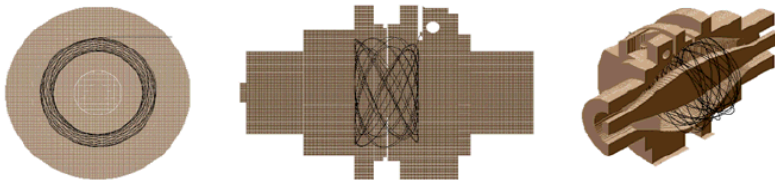
= Oscillation-Frequency
= Instrumental parameter

From: Alexander Makarov, Michaela Scigelova

7.2.4. Electrostatic Trap

Ion Motion in the Orbitrap

Simulation of ion trajectory with SIMION



Characteristic frequencies

– Frequency of rotation ω_ϕ

$$\omega_\phi = \frac{\omega_z}{\sqrt{2}} \sqrt{\left(\frac{R_m}{R}\right)^2 - 1}$$

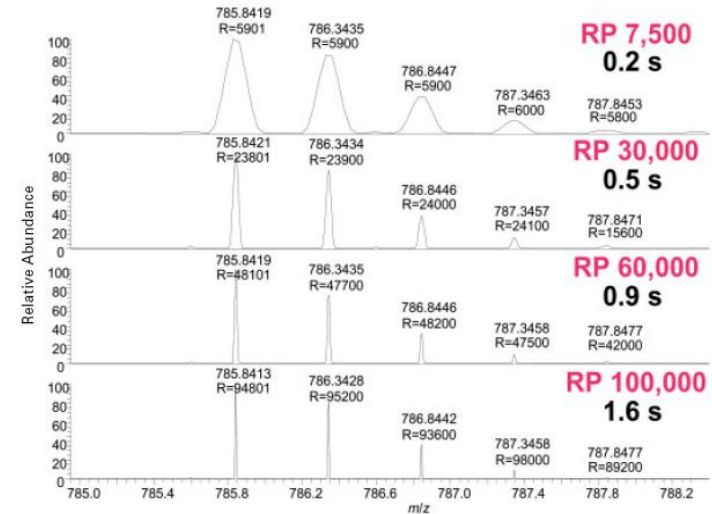
– Frequency of radial oscillations ω_r

$$\omega_r = \omega_z \sqrt{\left(\frac{R_m}{R}\right)^2 - 2}$$

– Frequency of axial oscillations ω_z

$$\omega = \sqrt{\frac{k}{m/z}}$$

Can be used to determine m/z due to independence from location and E_{kin} of the ions



Resolution		Transient
LQ Orbitrap	Orbitrap Elite	
–	15,000	48 ms
7500	30,000	96 ms
15,000	60,000	192 ms
30,000	120,000	384 ms
60,000	240,000	768 ms
120,000	480,000*	1536 ms

Resolution depends on measuring time and model!

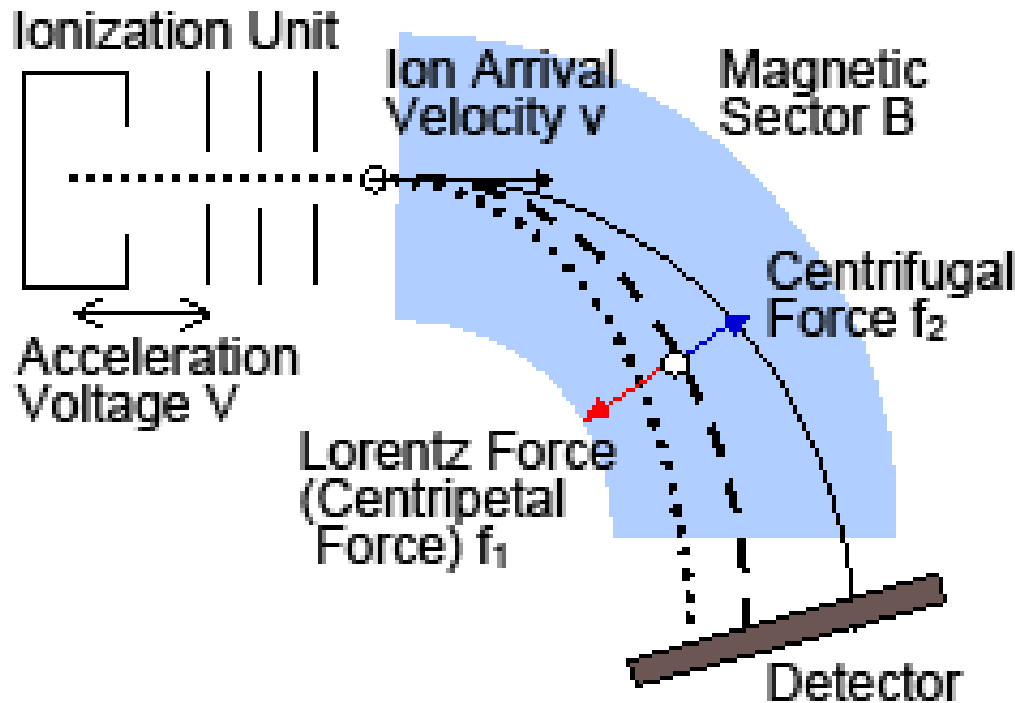
7.2.4. Electrostatic Trap

Electrostatic Trap:

- providing high resolution without the increased maintenance and operational effort of ICR-FT-MS instruments
- can be combined with Qq or Ion trap allowing MSⁿ⁽²⁾
- good sensitivity

Resolution	up to 500.000 (depending on the specific model);
Mass-range	< 4000;
Mass-Error	< 2ppm

7.2.5. Magnetic Field Mass Analyzer



$$f_1 = Bzev \dots (1)$$

$$f_2 = \frac{mv^2}{r} \dots (2)$$

$$\frac{mv^2}{r} = Bzev \dots (3)$$

Acceleration in electric field:

$$\frac{1}{2} mv^2 = zeV \dots (4)$$

(3) + (4) $\rightarrow$

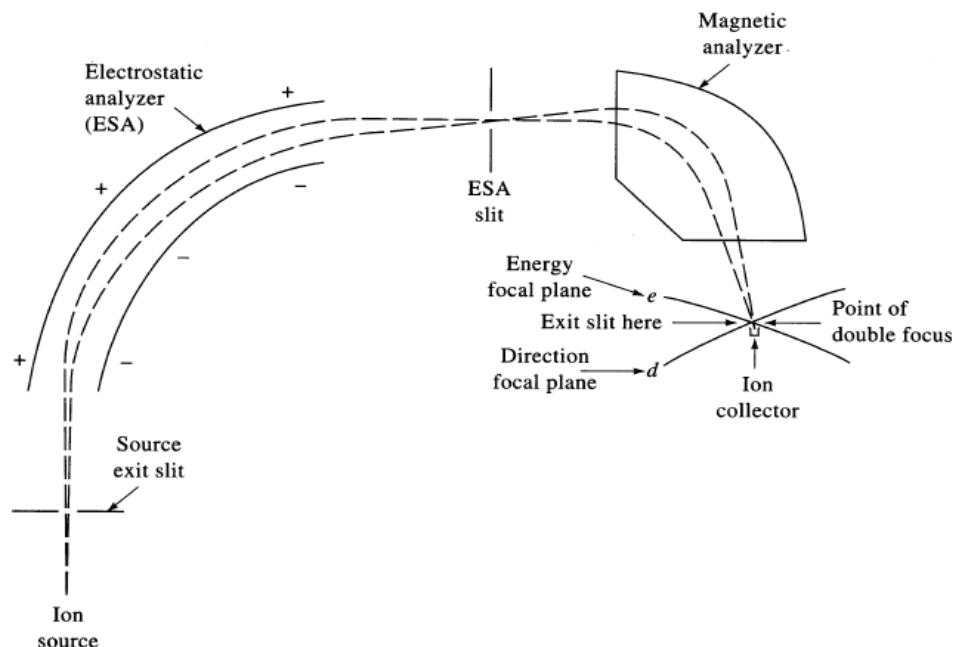
$$\frac{m}{z} = \frac{eB^2r^2}{2V} \dots (5)$$

m: mass of ion
z: charge of ion
v: velocity
V: acceleration voltage

B: strength of magnetic field
e: unit charge
r: radius

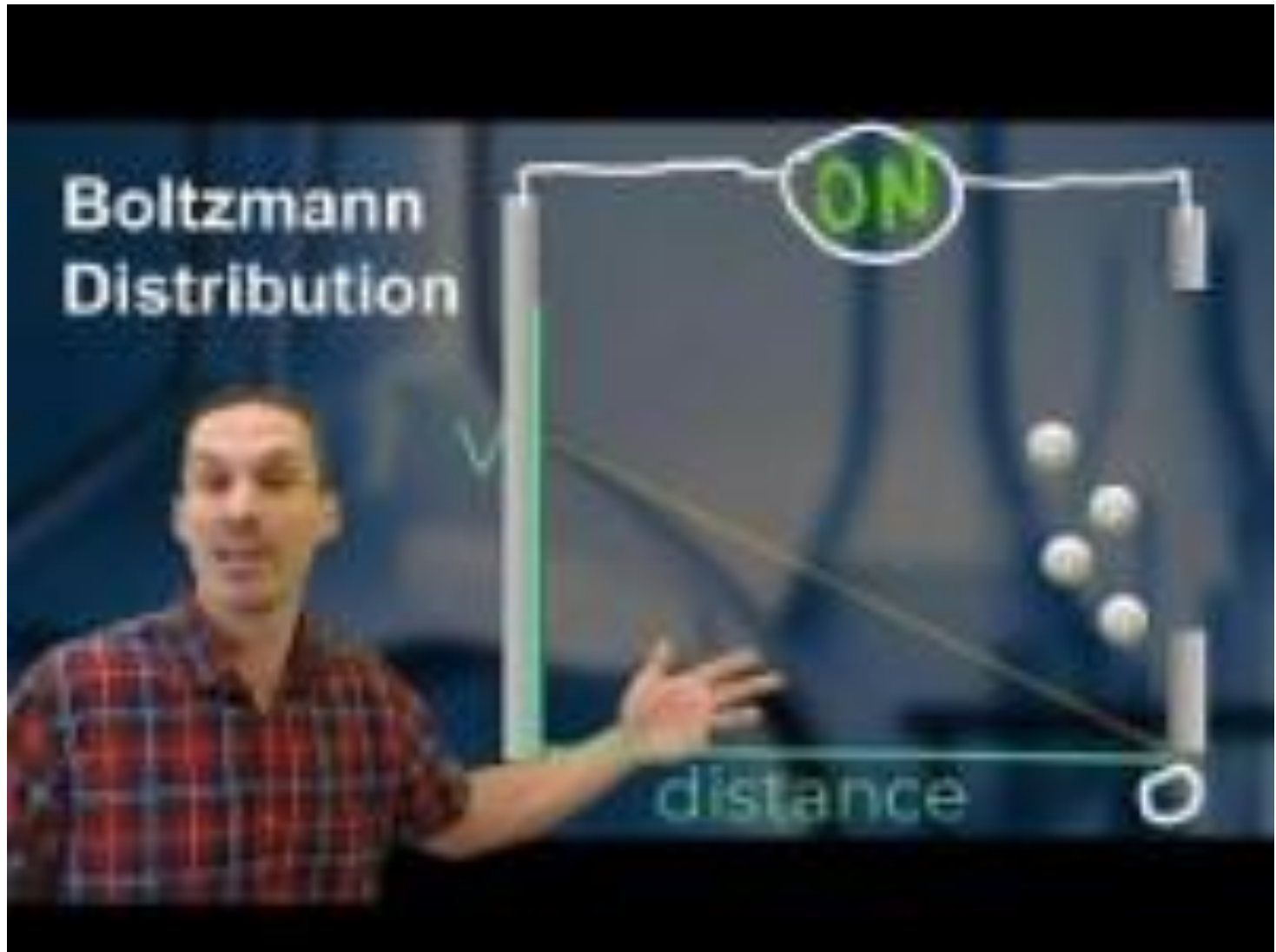
7.2.5. Magnetic Field Mass Analyzer

Simple magnetic sector instruments do not really outperform quadrupole or ion trap MS instruments. Therefore, to achieve high resolution ($R > 10.000$) double focusing instruments combining an electrostatic analyzer and a magnetic field analyzer are employed.

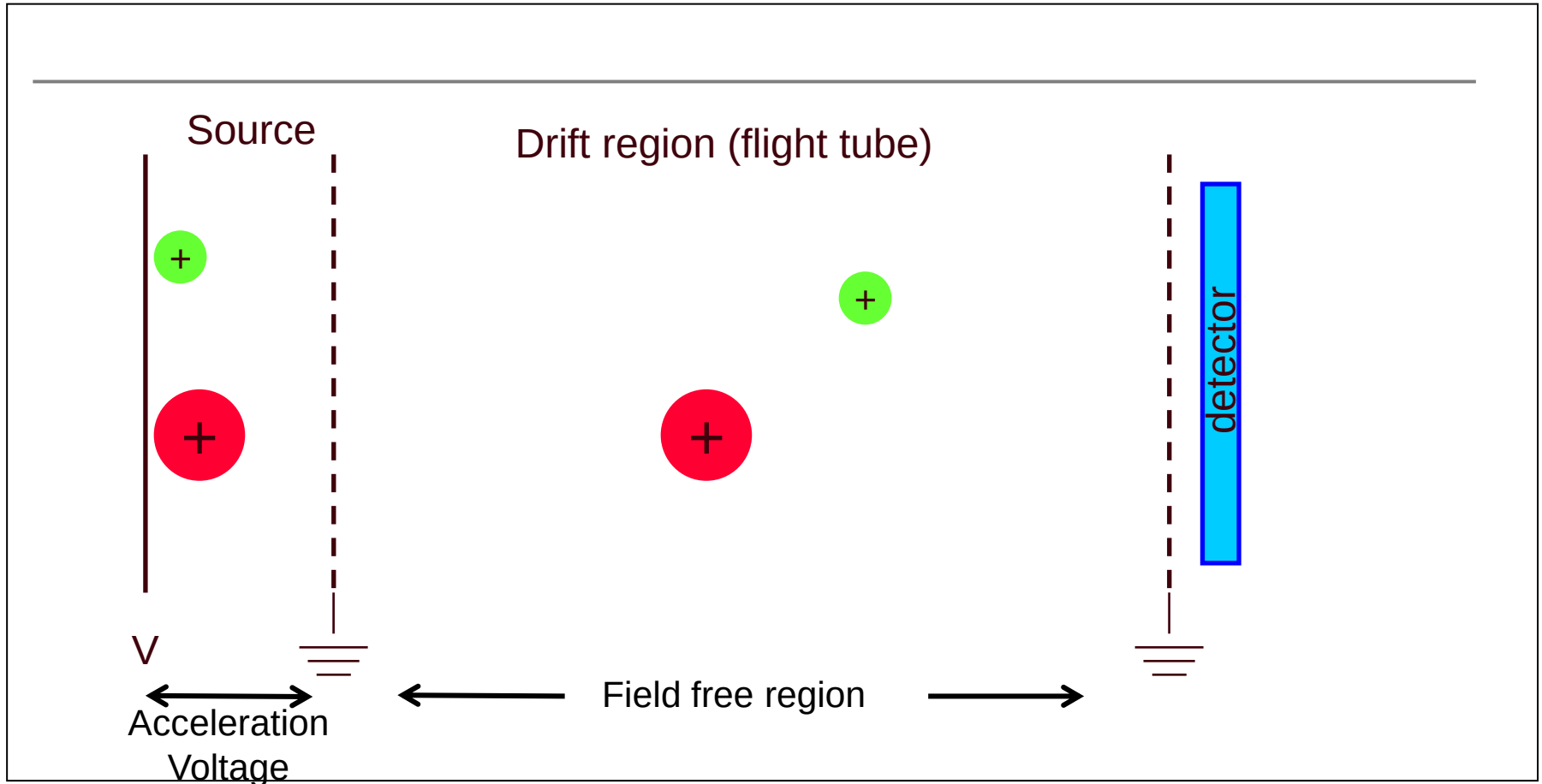


Double focusing instruments have an extended mass range (compared to quadrupole instruments), provide high resolution and improved mass accuracy. Commonly they do not allow multidimensional MS experiments. **Today only very few fields of application** - **isotope ratio (IR) MS and HR-ICP-MS.**

7.2.6. Time of Flight Mass Analyzer



7.2.6. Time of Flight Mass Analyzer



7.2.6. Time of Flight Mass Analyzer

Time of flight mass-spectrometer (TOF)

$$E = zV = \frac{m v^2}{2}$$

E = kinetic energy of the ion leaving the source
z = charge; V = acceleration voltage; m = mass;
v = velocity;

using $v = \frac{L}{t}$ L = length of the tube
t = time $\frac{m}{z} = \frac{2Vt^2}{L^2}$ and

$$t = L \sqrt{\frac{m}{z} \frac{1}{2V}}$$

e.g. L=100 cm, V=2kV:

H⁺ = 1.61 μs

N₂⁺ = 8.52 μs

7.2.6. Time of Flight Mass Analyzer

Time of Flight Mass Spectrometer (TOF)

$$t = L \sqrt{\frac{m}{z} \frac{1}{2V}}$$

eg.: a Molecule 721 Da and 1 charge:

$$m = 721 \text{ Da} \times 1.66053 \times 10^{-27} \text{ kgDa}^{-1}$$

$$z = 1.602 \times 10^{-19} \text{ C}$$

$$L = 2 \text{ m}$$

Acceleration Voltage: 19500 V

Flight velocity: $\sim 72\,000 \text{ m/s} = \sim 260\,000 \text{ km/h}$

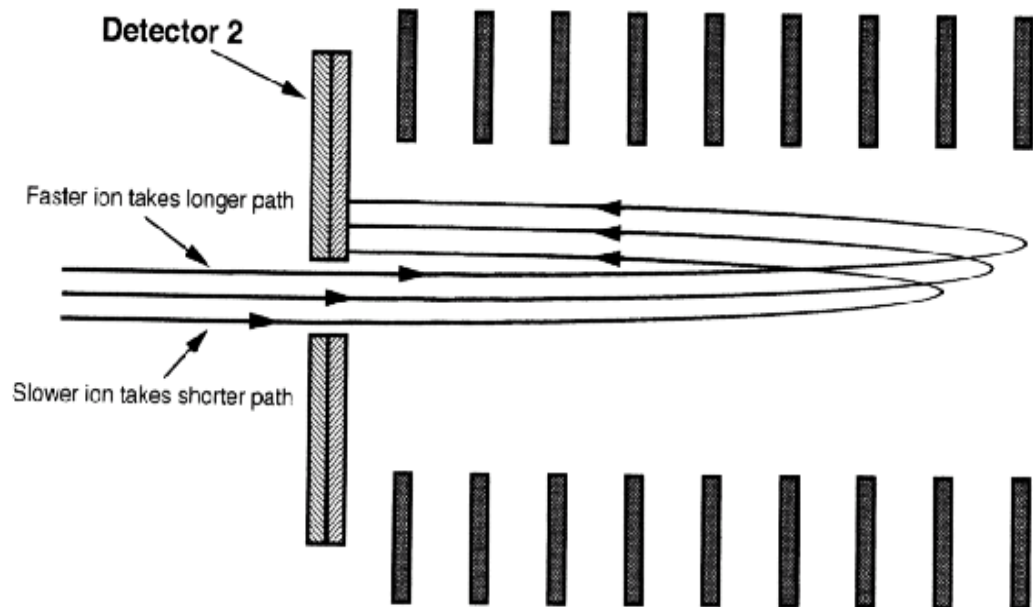
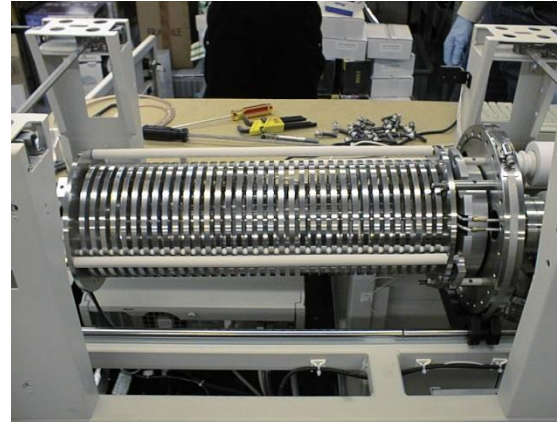
Time of Flight: approx. 27.68 μs

Simple TOF instruments are low-resolution MS!

For high-resolution a **reflectron** (see next slide) is needed!!

7.2.6. Time of Flight Mass Analyzer

TOF



The reflectron helps to compensate differences in kinetic energies of ions with identical m/z at. Those with higher kinetic energy intrude further into the reflectron and thereby take a longer way.

The reflectron is compulsory if increased resolution is required!
(actually high-resolution TOF instruments with multiple reflection exist!)

7.2.6. Time of Flight Mass Analyzer

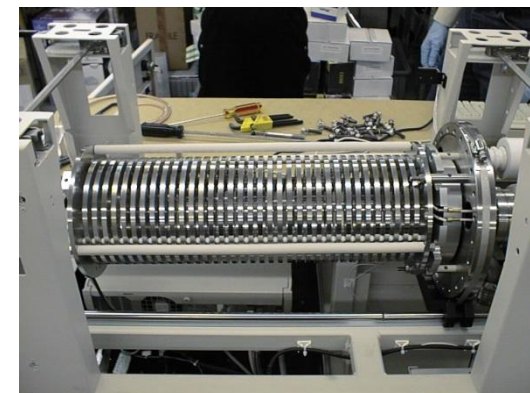
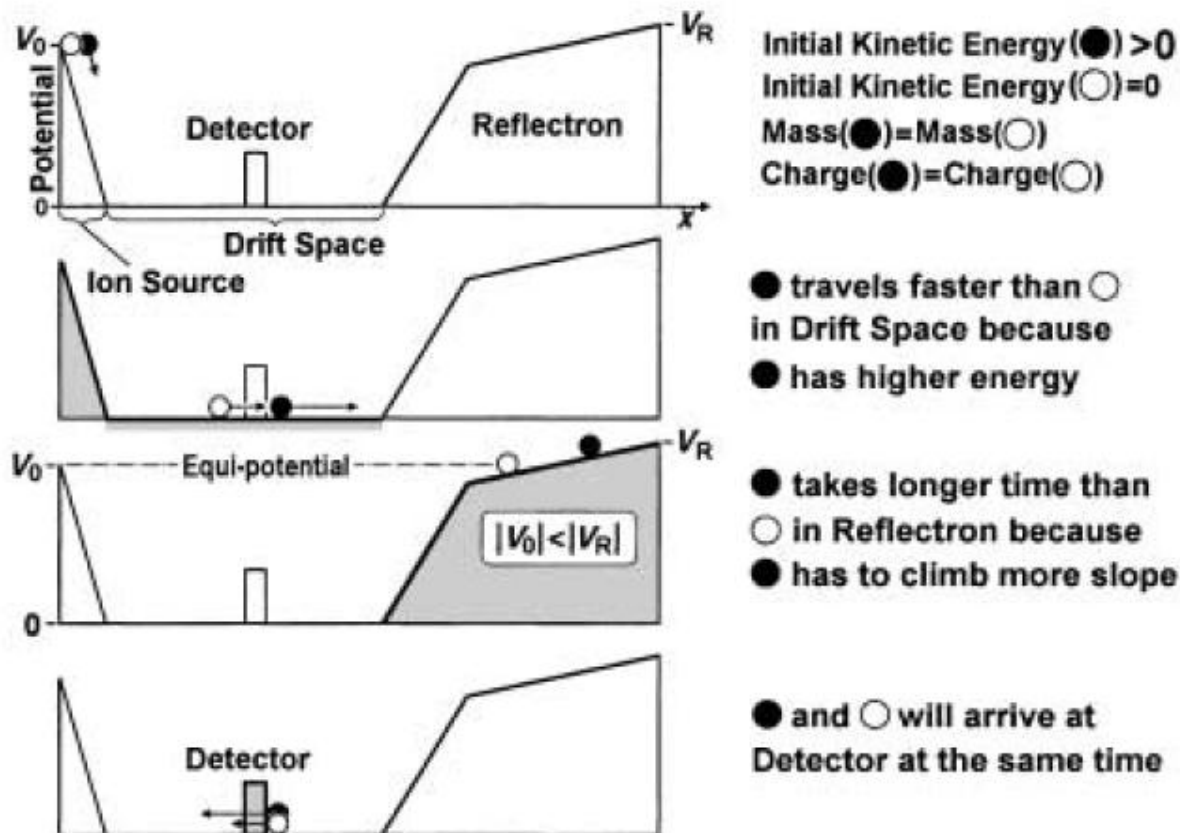


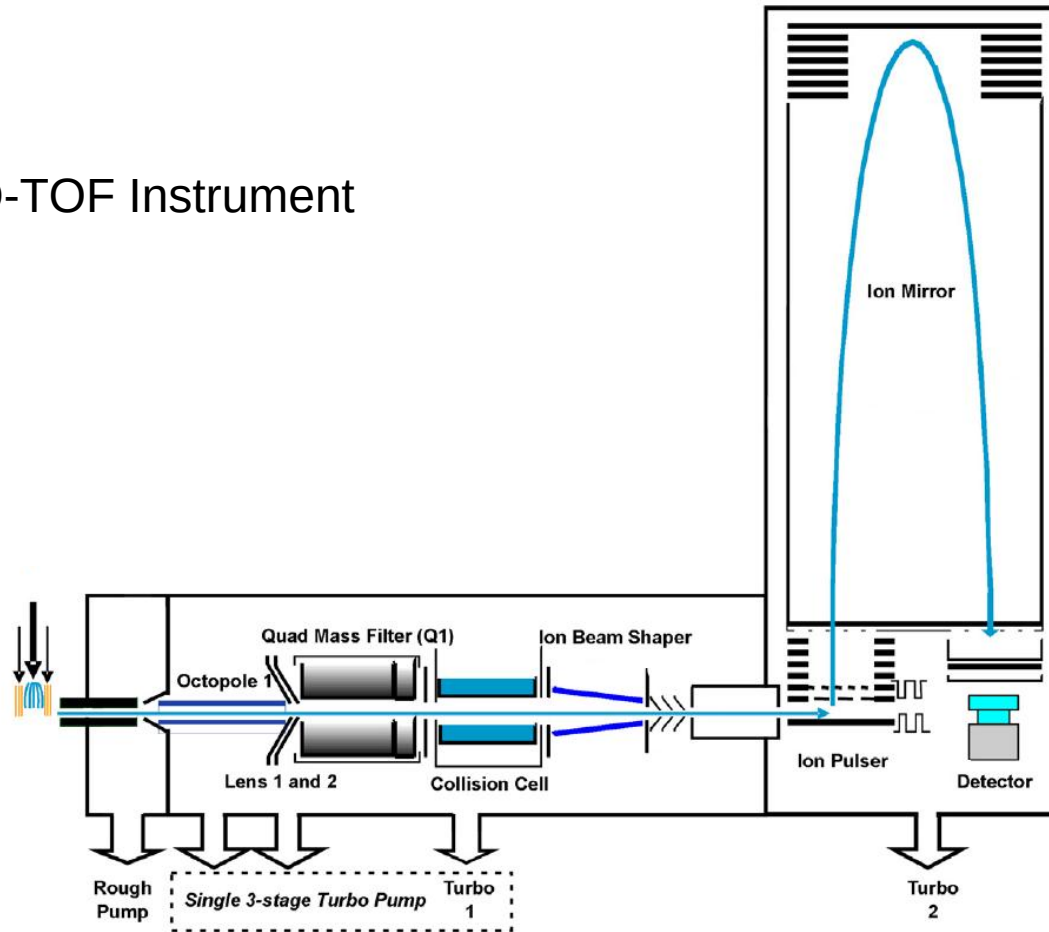
Figure 2. Improvement of mass resolution by reflectron.

Resolution (only with **REFLECTRON**)
Mass-range
Mass-Error

> 20.000 – typically ~40.000 (depending on m/z);
theoretically indefinite
< 2ppm

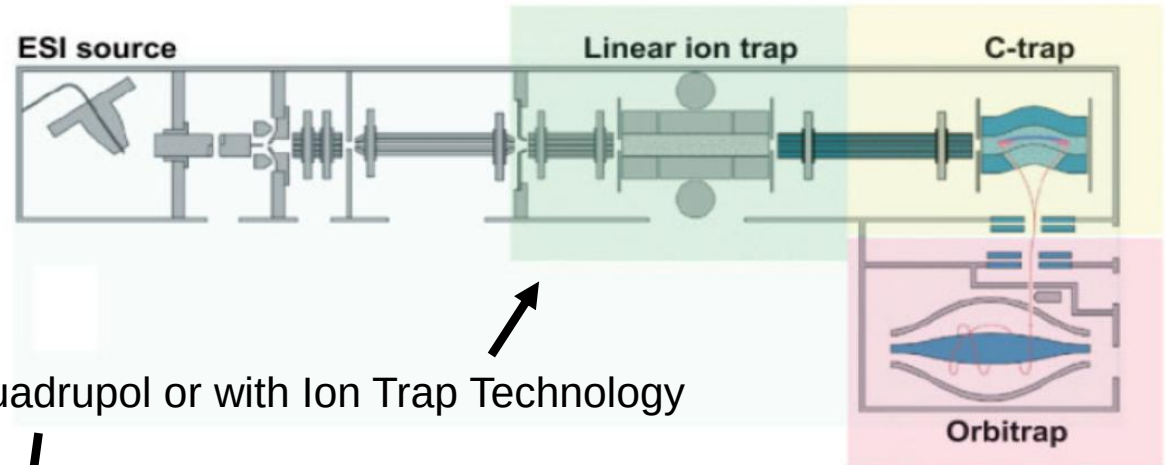
7.2.7 Hybrid MS Instruments

Q-TOF Instrument

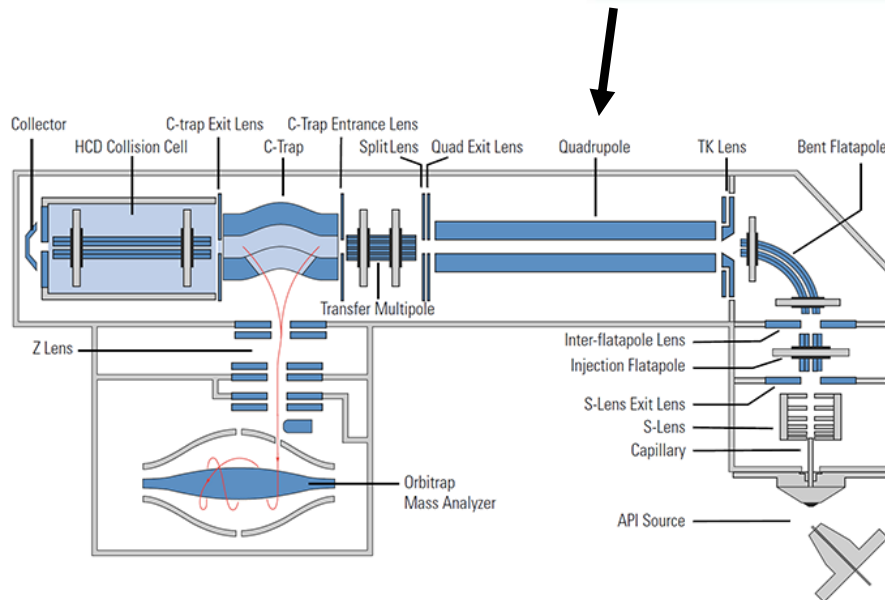


Hybrid mass spectrometers are combining mass separation-units with different working principles

7.2.7 Hybrid MS Instruments



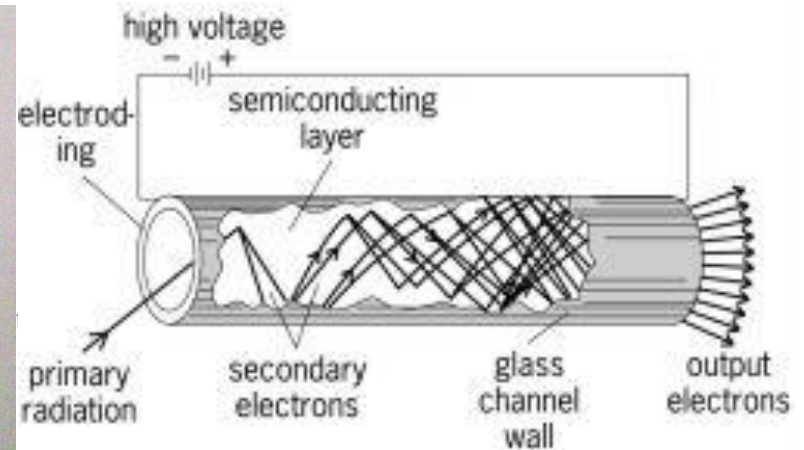
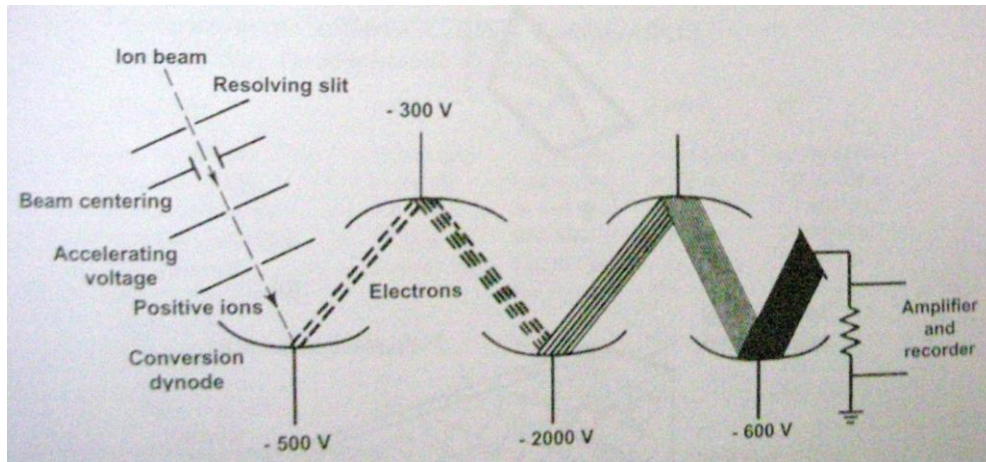
Combination of Orbitrap with Quadrupole or with Ion Trap Technology



Hybrid mass spectrometers are combining mass separation-units with different working principles

7.3 MS Detectors

7.3.1 SEV

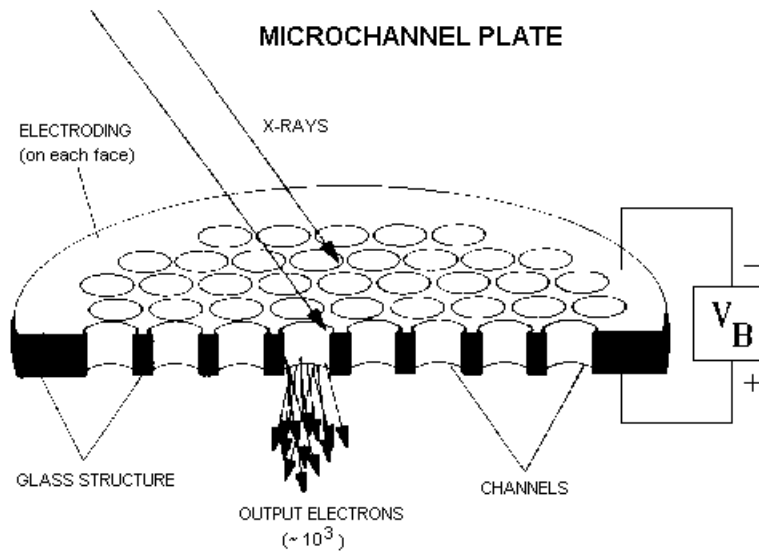


Secondary electron multiplier
- discrete dynodes

Channel electron multiplier (CEM)
- continuous dynode

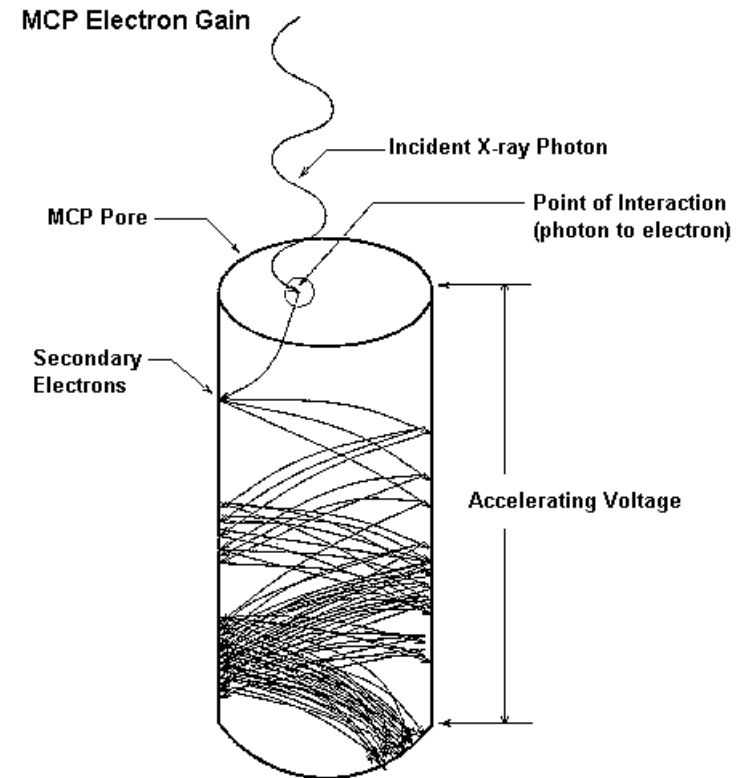
Detectors used for Quadrupole, Triple-Quadrupole, Magnetic Field/Electrostatic Field MS, and Ion Traps

7.3.2. Micro Channel Plate

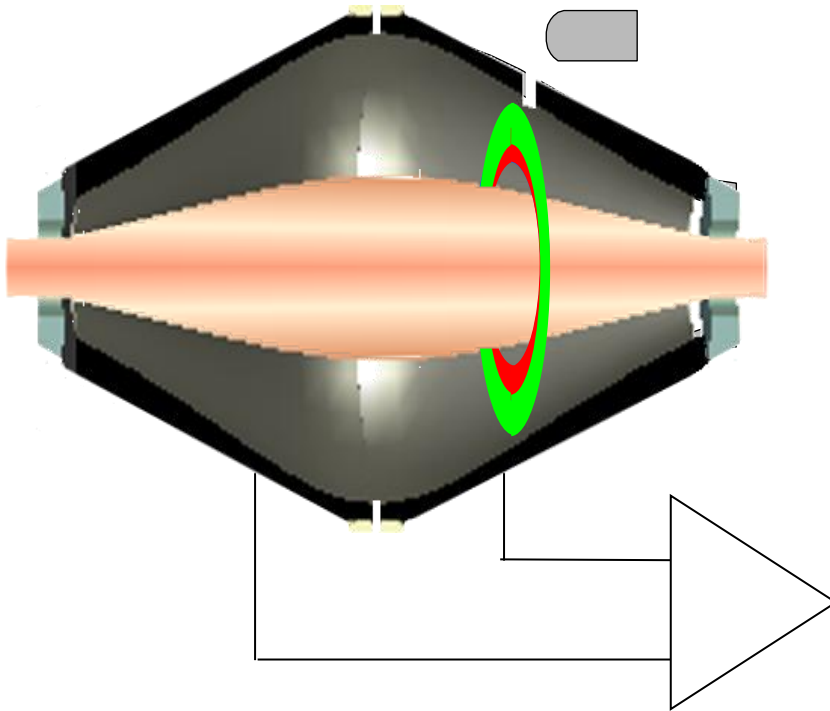


Used for TOF-MS

Array of CEM's

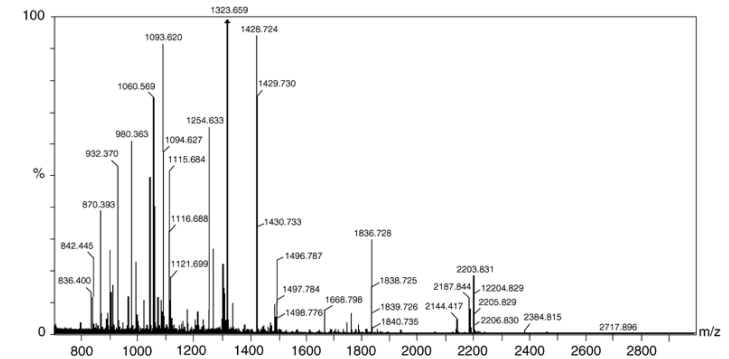
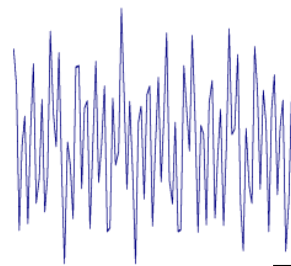


7.3.3. FT-based Detectors



Movement of Ions induces
current in outer Electrode

Used in Orbitraps

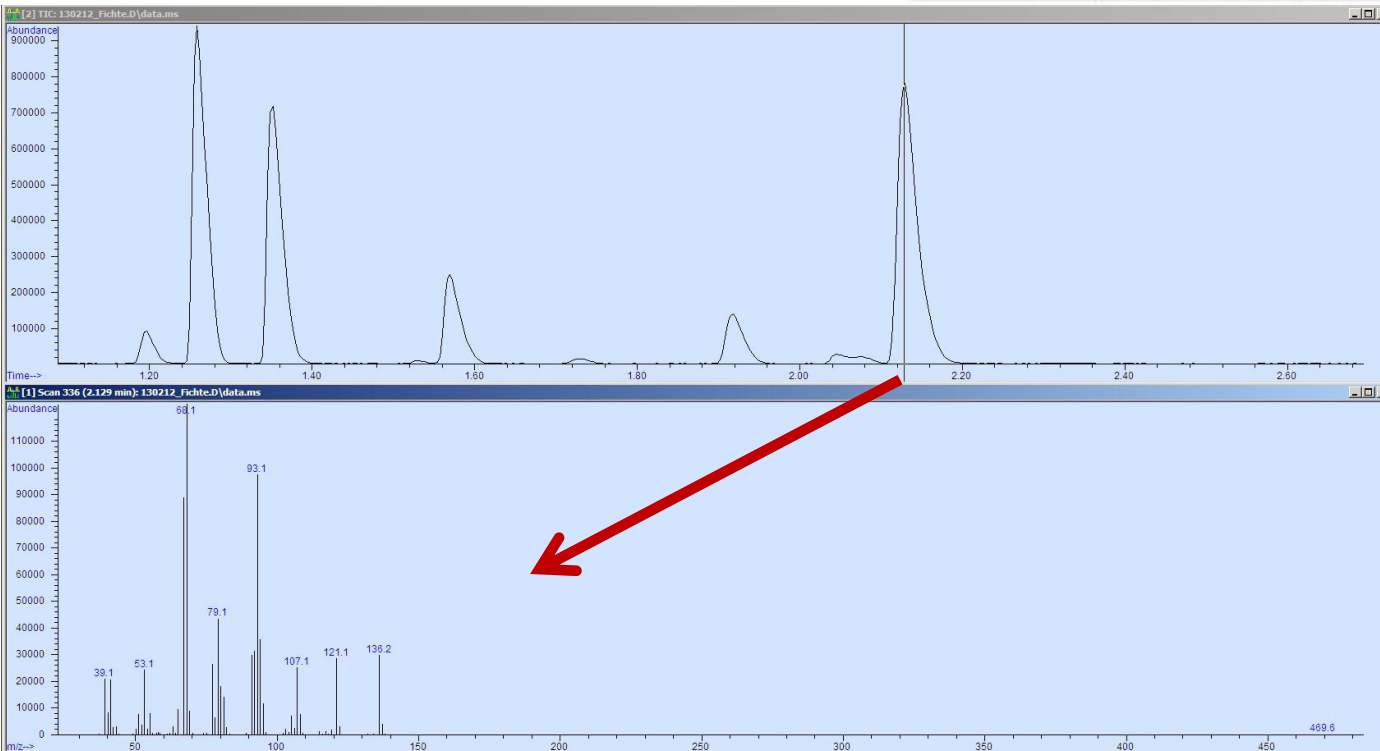
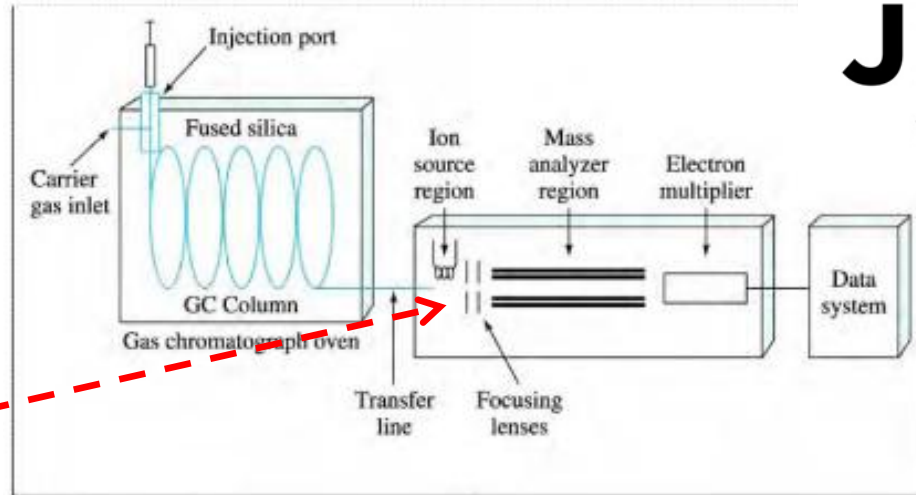
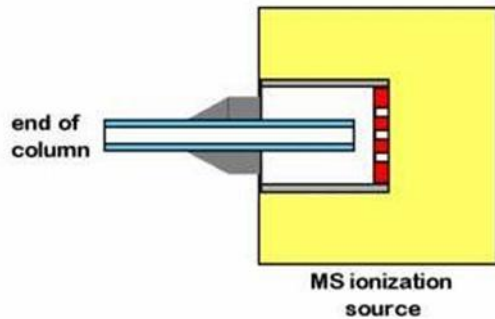


Fourier Transformation

7.4. Hyphenated Methods

Coupling of analytical
separation techniques
with mass spectrometry
(Hyphenated Techniques)

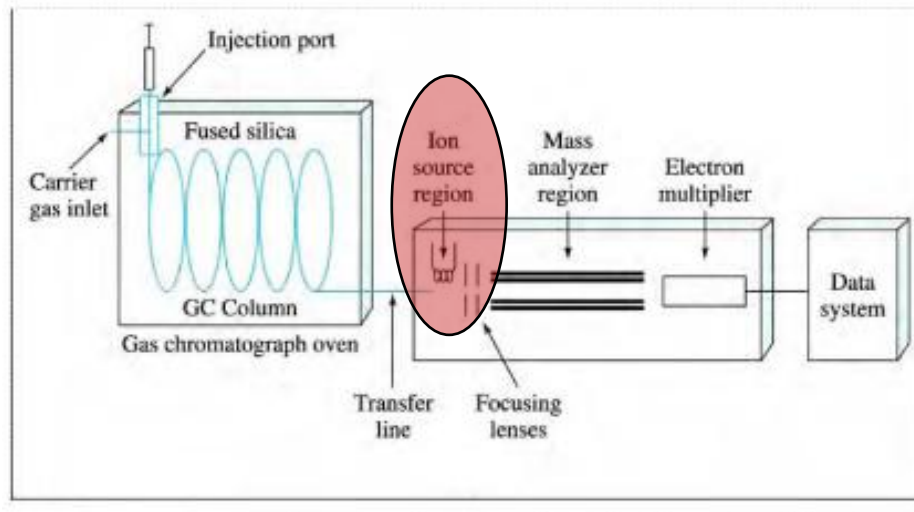
7.4.1. GC-MS



Chromatogram
(Total ion
chromatogram)

Mass spectrum
of peak at
retention time of
2,10 min

7.4.1. GC-MS

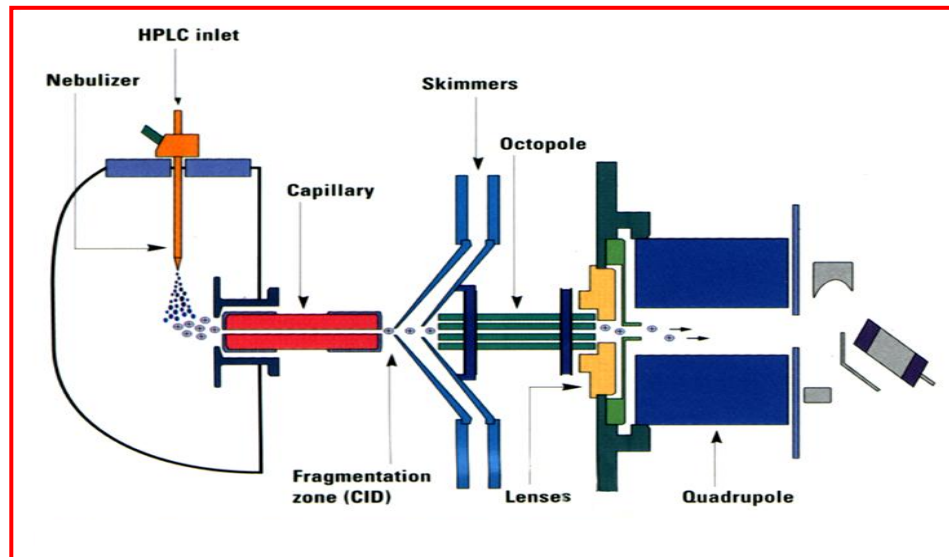


- Mostly in combination with Electron Ionization or Chemical Ionization as vacuum conditions can easily be achieved by simply pumping the gaseous mobile phase
- Majority of GC-MS instruments still with simple Quadrupole mass analyzers but QqQ-MS and QTOF-MS becoming more and more important

7.4.2. HPLC-MS

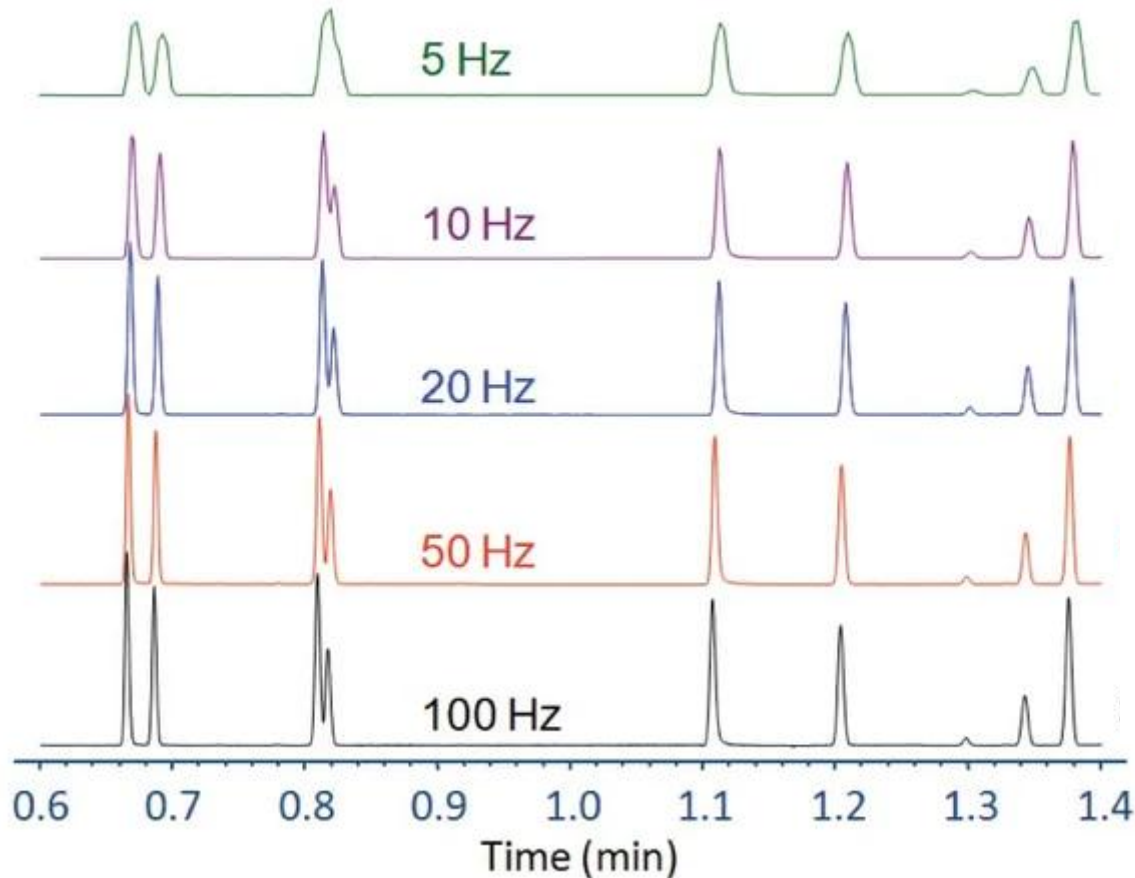
ESI (electrospray ionization)

APCI (atmospheric pressure chemical ionization)



- Solely in combination with Atmospheric Pressure Ionization techniques as vacuum conditions cannot be achieved easily as liquid mobile phase would have to be vaporized first and only then vacuum conditions could be achieved
- Combined with all sorts of Mass Analyzers
- Requires volatile mobile phase (no phosphate buffers!!)

7.4.2. HPLC-MS



© SepScience

To maintain chromatographic resolution, the sampling rate of the MS detector must be sufficiently high – **use fast instruments** (such as TOF-MS)!

7.5. Information from MS Spectra

- Isotopic Pattern – provides information on composition (types and number of elements present in a molecule)
- $m/z \longrightarrow$ mass $\longrightarrow$ particularly if $z \neq 1$
- Accurate mass – may (if accurate enough) allow suggesting an unambiguous molecular formula
- Fragmentation - with hard ionization techniques and with instruments offering MS^n capabilities (will be discussed in more detail in the course „Interpretation of MS and IR spectra“ in the Master Curriculum)

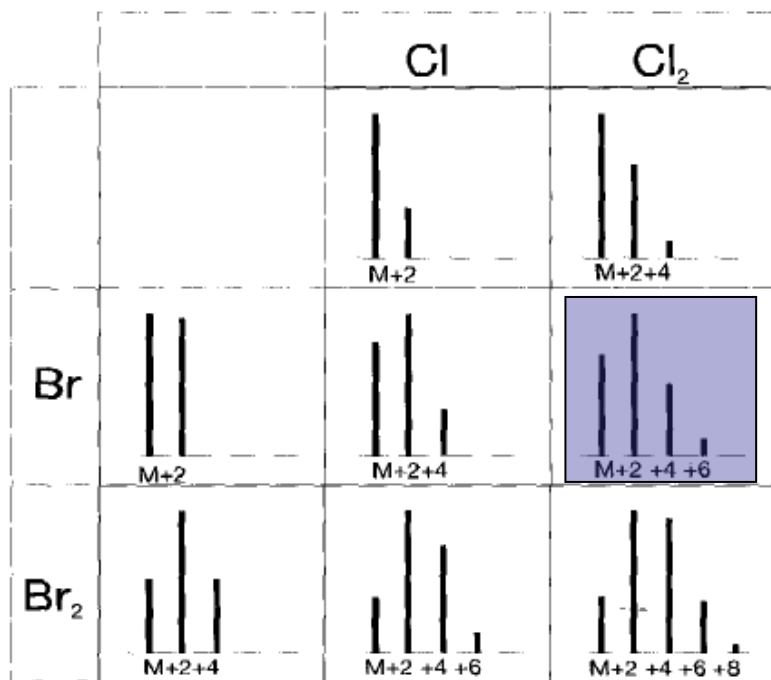
7.5.1. Isotopic Pattern

Table 5.2 Percent Abundance of Some Natural Isotopes

Symbol	Mass (amu)	Percent	Symbol	Mass (amu)	Percent
^1_1H	1.007825035	99.9885	$^{19}_9\text{F}$	18.99840322	100
^2_1H	2.014101779	0.0115	$^{32}_{16}\text{S}$	31.97207070	94.93
^3_2He	3.01602931	0.000137	$^{33}_{16}\text{S}$	32.97145843	0.76
^4_2He	4.00260324	99.999863	$^{34}_{16}\text{S}$	33.96786665	4.29
$^{12}_6\text{C}$	12 (exactly)	98.93	$^{36}_{16}\text{S}$	35.96708062	0.02
$^{13}_6\text{C}$	13.003354826	1.07	$^{35}_{17}\text{Cl}$	34.968852721	75.78
$^{14}_7\text{N}$	14.003074002	99.632	$^{37}_{17}\text{Cl}$	36.96590262	24.22
$^{15}_7\text{N}$	15.00010897	0.368	$^{39}_{19}\text{K}$	38.9637074	93.2581
$^{16}_8\text{O}$	15.99491463	99.757	$^{40}_{19}\text{K}$	39.9639992	0.0117
$^{17}_8\text{O}$	16.9991312	0.038	$^{41}_{19}\text{K}$	40.9618254	6.7302
$^{18}_8\text{O}$	17.9991603	0.205			

7.5.1. Isotopic Pattern

How to calculate an isotopic pattern:



Isotope	rel. Abundance
^{35}Cl	100
^{37}Cl	31.98
^{79}Br	100
^{81}Br	97.88

e.g. Cl_2Br

m/z 149 $1 * 1 * 1 = 1$

m/z 151 $1 * 0.32 * 1 = 0.32$

$0.32 * 1 * 1 = 0.32$ | 1.62

$1 * 1 * 0.98 = 0.98$

m/z 153 $0.32 * 0.32 * 1 = 0.10$

$0.32 * 1 * 0.98 = 0.31$ | 0.72

$1 * 0.32 * 0.98 = 0.31$

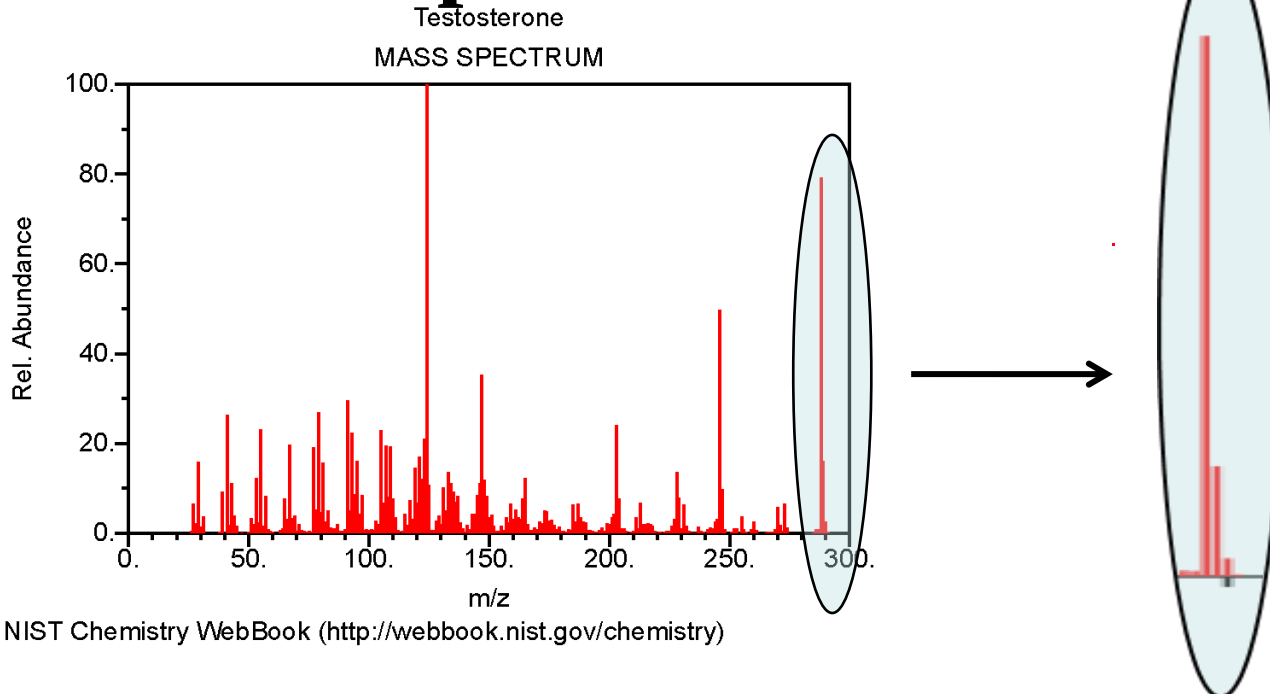
m/z 155 $0.32 * 0.32 * 0.98 = 0.10$

7.5.1. Isotopic Pattern

Table 5.2 Percent Abundance of Some Natural Isotopes

Symbol	Mass (amu)	Percent	Symbol	Mass (amu)	Percent
^1_1H	1.007825035	99.9885	$^{19}_9\text{F}$	18.99840322	100
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^4_2He	4.00260324	99.999863	$^{34}_{16}\text{S}$	33.96786665	4.29
$^{12}_6\text{C}$	12 (exactly)	98.93	$^{36}_{16}\text{S}$	35.96708062	0.02
$^{13}_6\text{C}$	13.003354826	1.07	$^{35}_{17}\text{Cl}$	34.968852721	75.78
$^{14}_7\text{N}$	14.003074002	99.632	$^{37}_{17}\text{Cl}$	36.96590262	24.22
$^{15}_7\text{N}$	15.00010897	0.368	$^{39}_{19}\text{K}$	38.9637074	93.2581
$^{16}_8\text{O}$	15.99491463	99.757	$^{40}_{19}\text{K}$	39.9639992	0.0117
$^{17}_8\text{O}$	16.9991312	0.038	$^{41}_{19}\text{K}$	40.9618254	6.7302
$^{18}_8\text{O}$	17.9991603	0.205			

7.5.1. Isotopic Pattern



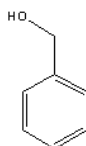
Acetone C_3H_6O

m/z	rel ab
58	100.0
59	3.2
60	0.2



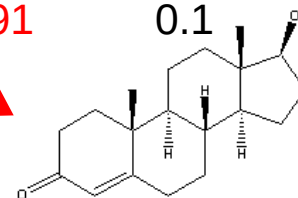
Benzylalcohol C_7H_8O

m/z	rel ab
108	100.0
109	7.7
110	0.4



Testosteron $C_{19}H_{28}O_2$

m/z	rel ab
288	100.0
289	21.0
290	2.3
291	0.1

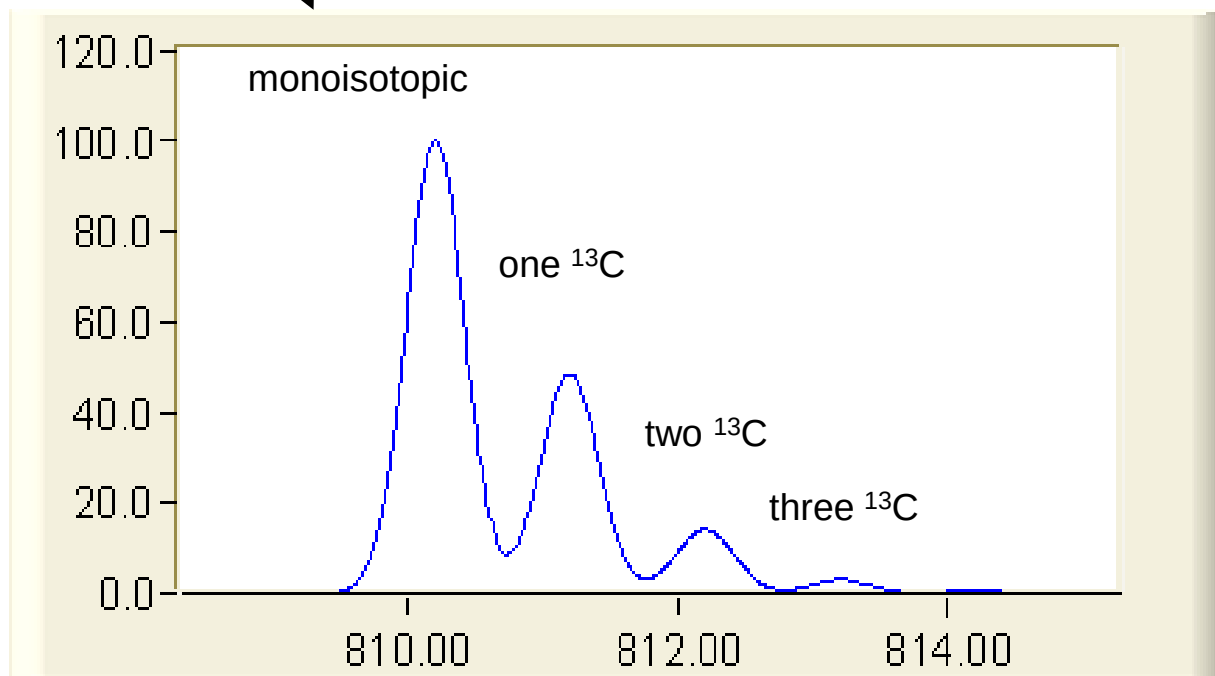


„Percentage“ of the first ^{13}C signal roughly correlates with the number of C-atoms

7.5.1. Isotopic Pattern

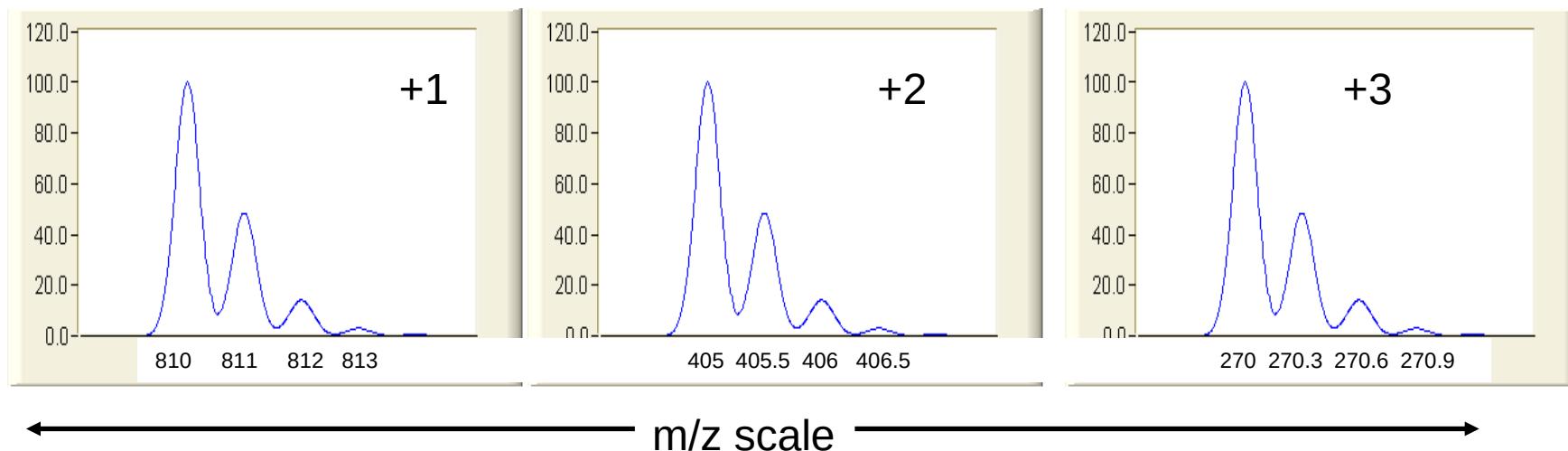
^{12}C only

Isotopic Pattern of C in Organic Compounds



7.5.1. Isotopic Pattern

Monoisotopic mass = 810



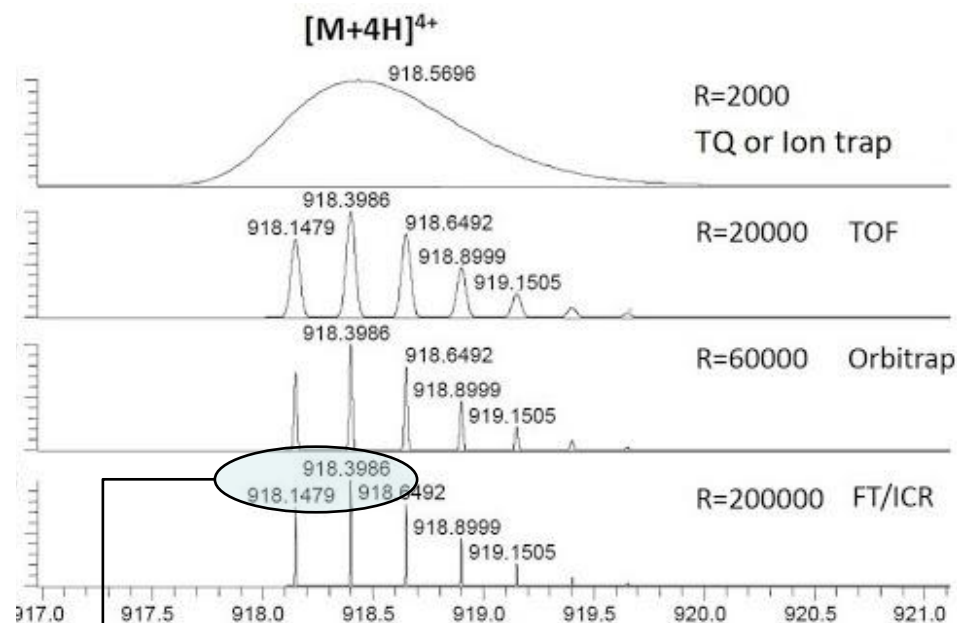
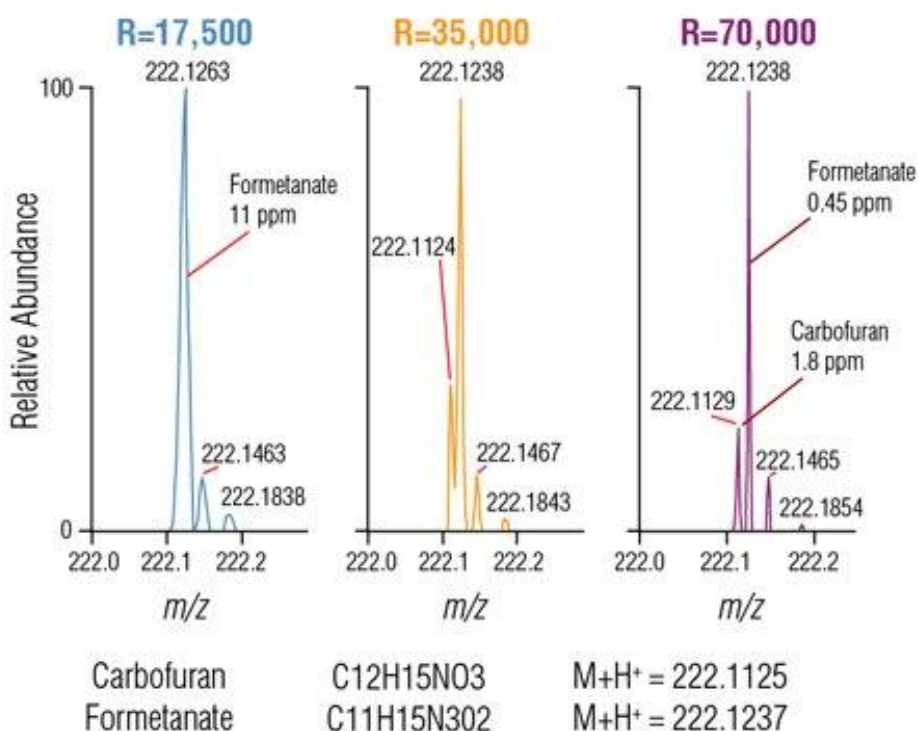
Difference (in m/z) between monoisotopic peak and first isotope peak (one ^{13}C):

- | | | |
|-----------------|--------|-------------------------|
| $\Delta = 1$ | —————→ | single charge z^{+1} |
| $\Delta = 0.5$ | —————→ | doubly charged z^{+2} |
| $\Delta = 0.33$ | —————→ | triply charged z^{+3} |

can be used to determine the
charge state in ESI ionization

7.5.2. High Resolution Mass Spectrometry

Importance of Resolution in Compound ID



Comparison of MS Resolution

see previous slide: $\Delta m \approx 0.25 \rightarrow z=4$

Importance for correct quantitative analysis

7.5.2. High Resolution Mass Spectrometry

- Nominal Mass

integer masses of the elements encountered are summed up, e.g. for CH_4 : $12 \text{ u} + 4 \times 1 \text{ u} = 16 \text{ u}$

- Mono-isotopic Mass

exact masses of the most abundant isotope of the elements encountered are summed up, e.g. for CH_4 : $12.00000 \text{ u} + 4 \times 1.00778 \text{ u} = 16.03112 \text{ u}$

- Molar Mass

exact atomic masses (i.e.= isotopic mass x abundance) of the elements encountered are summed up, e.g. for CH_4 :
 $12.011 \text{ u} + 4 \times 1.00794 \text{ u} = 16.04276 \text{ u}$

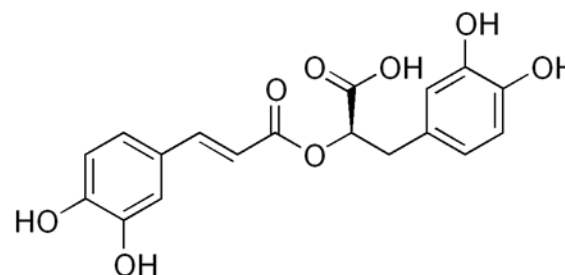
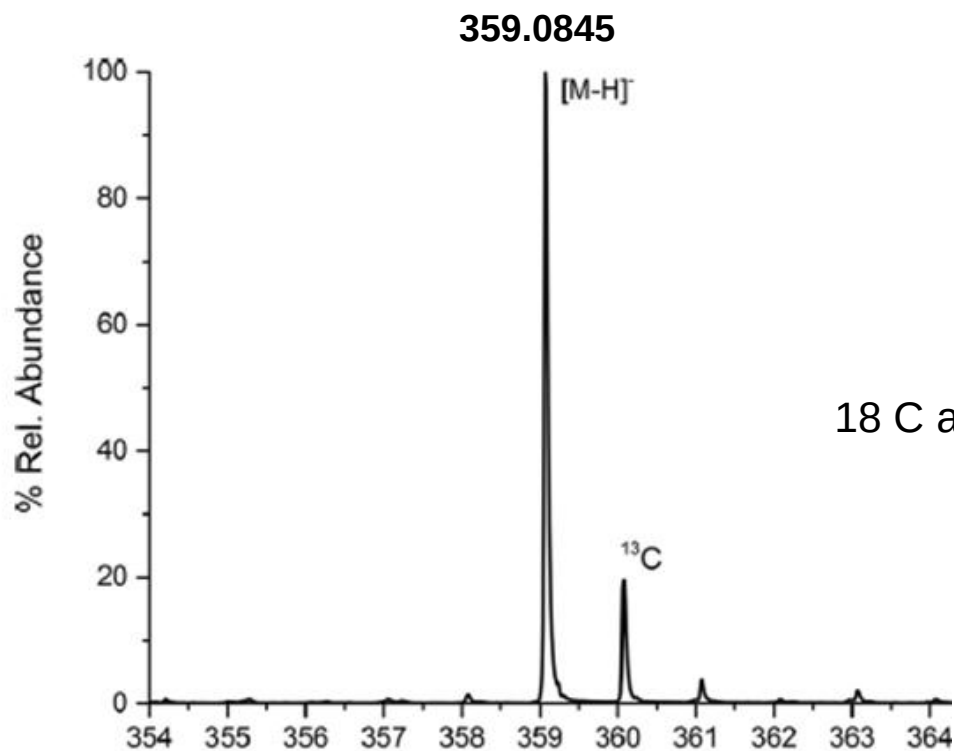
$$\left(0.9890 \times 12.000 + 0.011 \times 13.003355 \right)$$

7.5.2. High Resolution Mass Spectrometry

Element Atomic mass Nominal mass Isotopes Natural Abundance Isotopic mass

H	1,00794	1	^1H	99,985	1,0077825	X
			$^2\text{H} = \text{D}$	0,015	2,0141102	X + 1
Li	6,941	7	^6Li	7,5	6,015123	X – 1
			^7Li	92,5	7,016005	X
B	10,811	11	^{10}B	19,9	10,012938	X – 1
			^{11}B	80,1	11,009305	X
C	12,011	12	^{12}C	98,90	12,000000	X
			^{13}C	1,10	13,003355	X + 1
N	14,00674	14	^{14}N	99,634	14,003074	X
			^{15}N	0,366	15,000109	X + 1
O	15,9994	16	^{16}O	99,762	15,994915	X
			^{17}O	0,038	16,999131	X + 1
			^{18}O	0,200	17,999159	X + 2
F	18,998403	19	^{19}F	100	18,998403	X
Na	22,989768	23	^{23}Na	100	22,989770	X

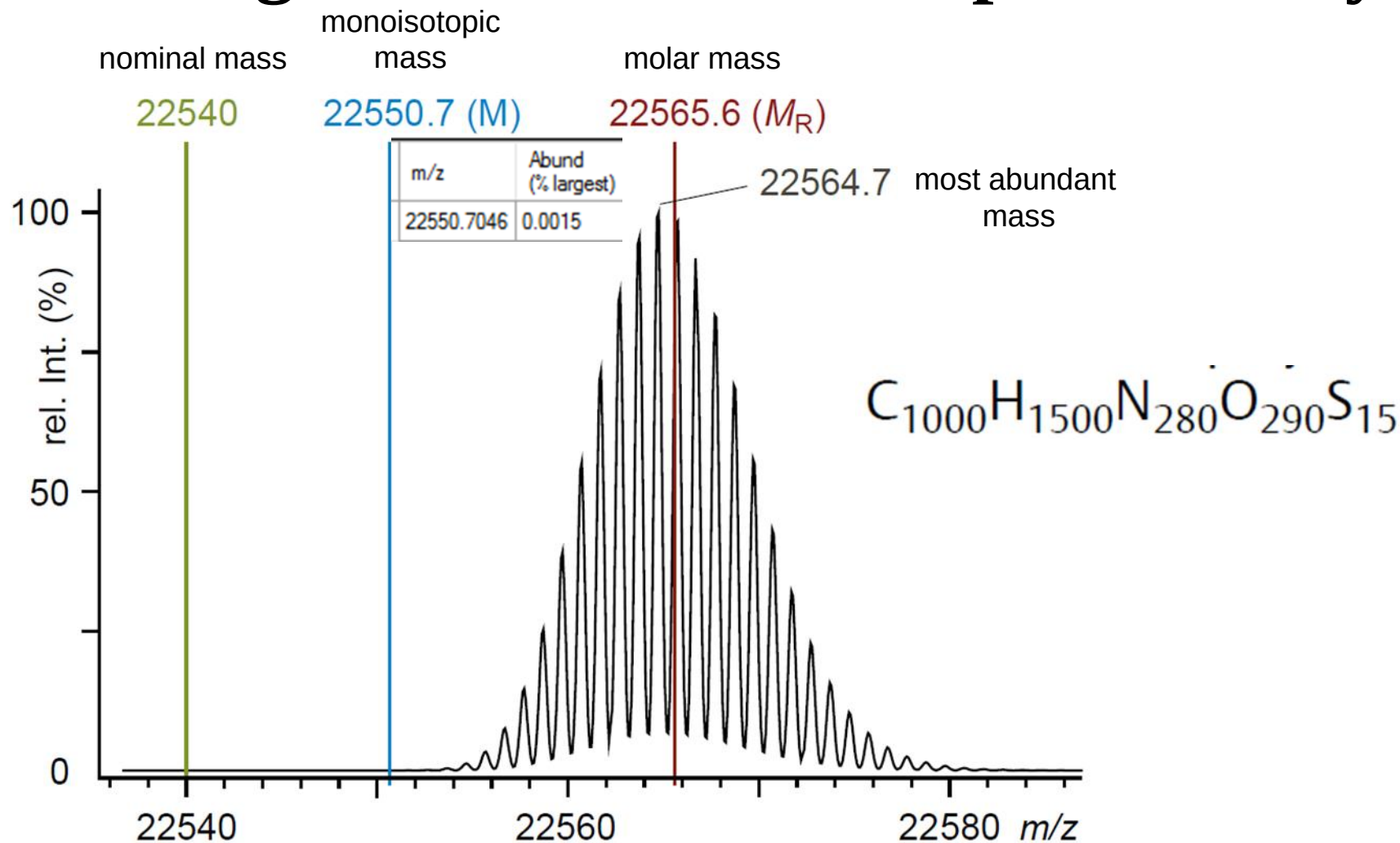
7.5.2. High Resolution Mass Spectrometry



18 C atoms ... ¹³C signal ~ 20% of ¹²C signal

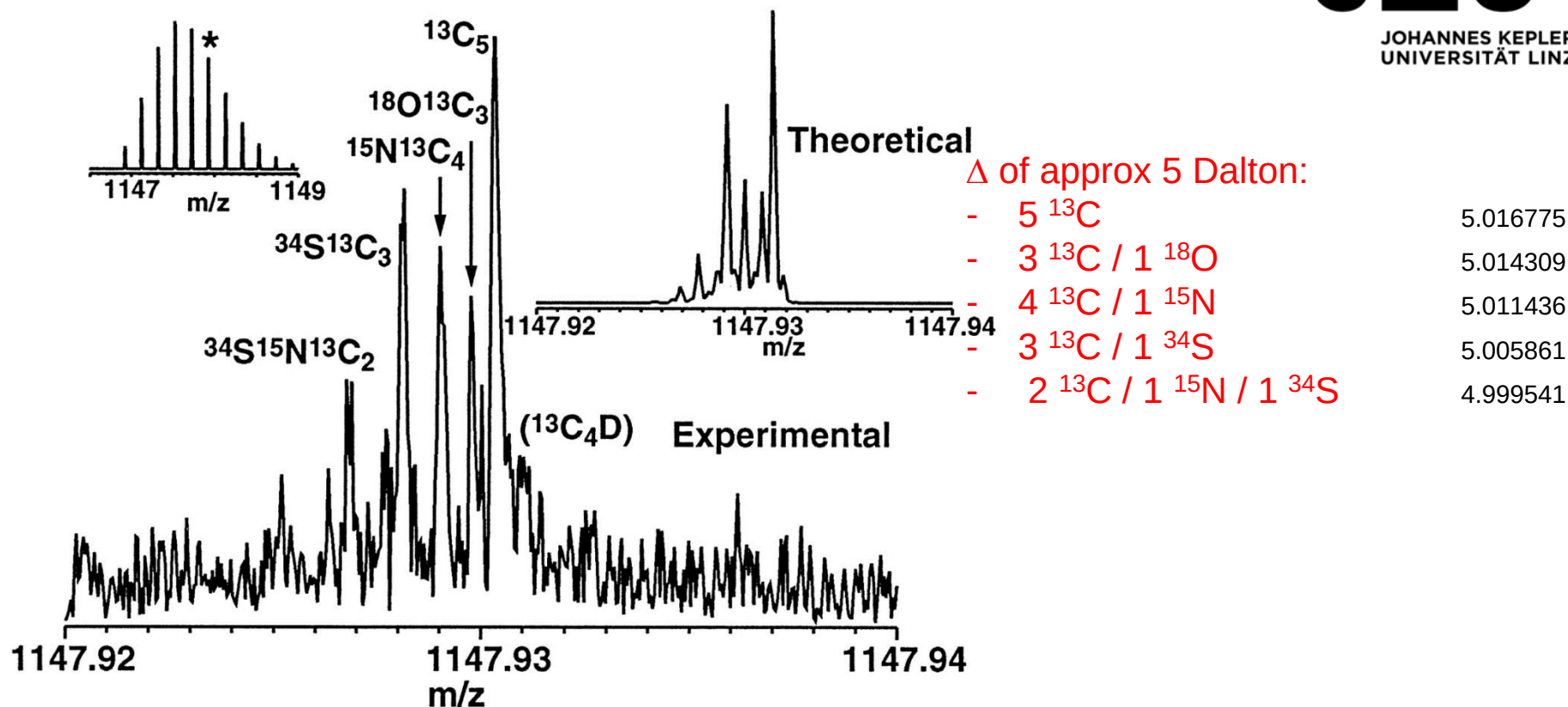
Determination of (monoisotopic) molecular mass – here: 359.0845 + 1.00778 = 360.0923

7.5.2. High Resolution Mass Spectrometry



7.5.2. (Ultra) High Resolution Mass Spectrometry

Insulin Isotopic Fine Structure



At ultrahigh resolution ($\sim 8 \times 10^6$) a „ ^{13}C isotopic peak“ is not always just a ^{13}C isotopic peak!

Counting individual sulfur atoms in a protein by ultrahigh-resolution Fourier transform ion cyclotron resonance mass spectrometry: Experimental resolution of isotopic fine structure in proteins

STONE D.-H. SHI*, CHRISTOPHER L. HENDRICKSON†, AND ALAN G. MARSHALL*†‡

7.5.2. High Resolution Mass Spectrometry

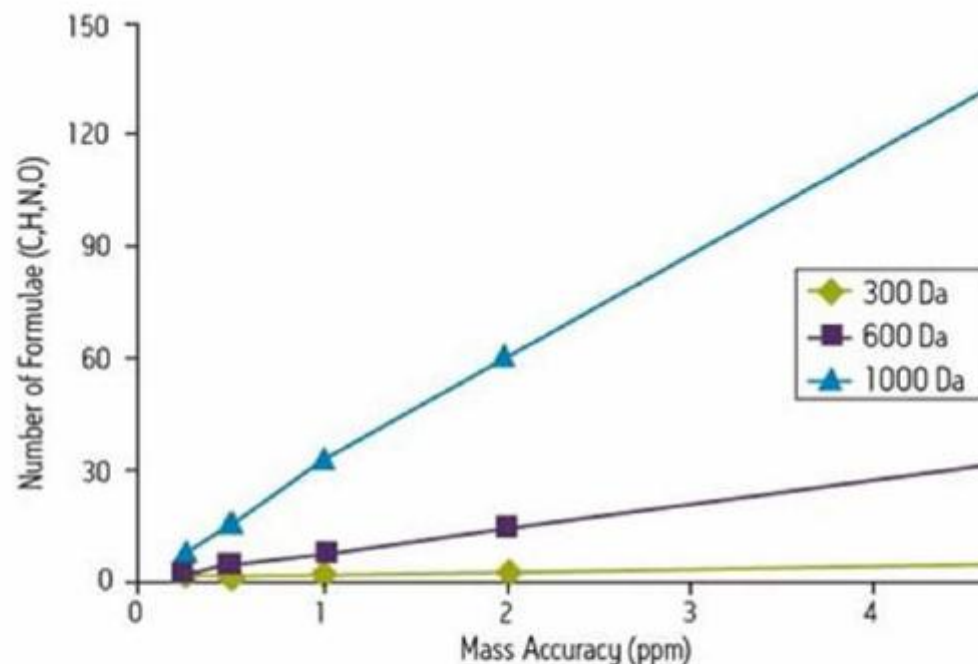
Mass Accuracy / Mass Error

Deviation from the theoretical monoisotopic mass in ppm

$$Error_{ppm} = \frac{m/z_{exp} - m/z_{theor}}{m/z_{theor}} * 10^6$$

Ion Trap, Quadrupol:	>100ppm (error of 0.1 @ m/z 1000)
TOF (with Reflectron):	1-5 ppm
Electrostatic Trap (Orbitrap):	~1 ppm (error of 0.001 @ m/z 1000)
FT-ICR:	< 1ppm

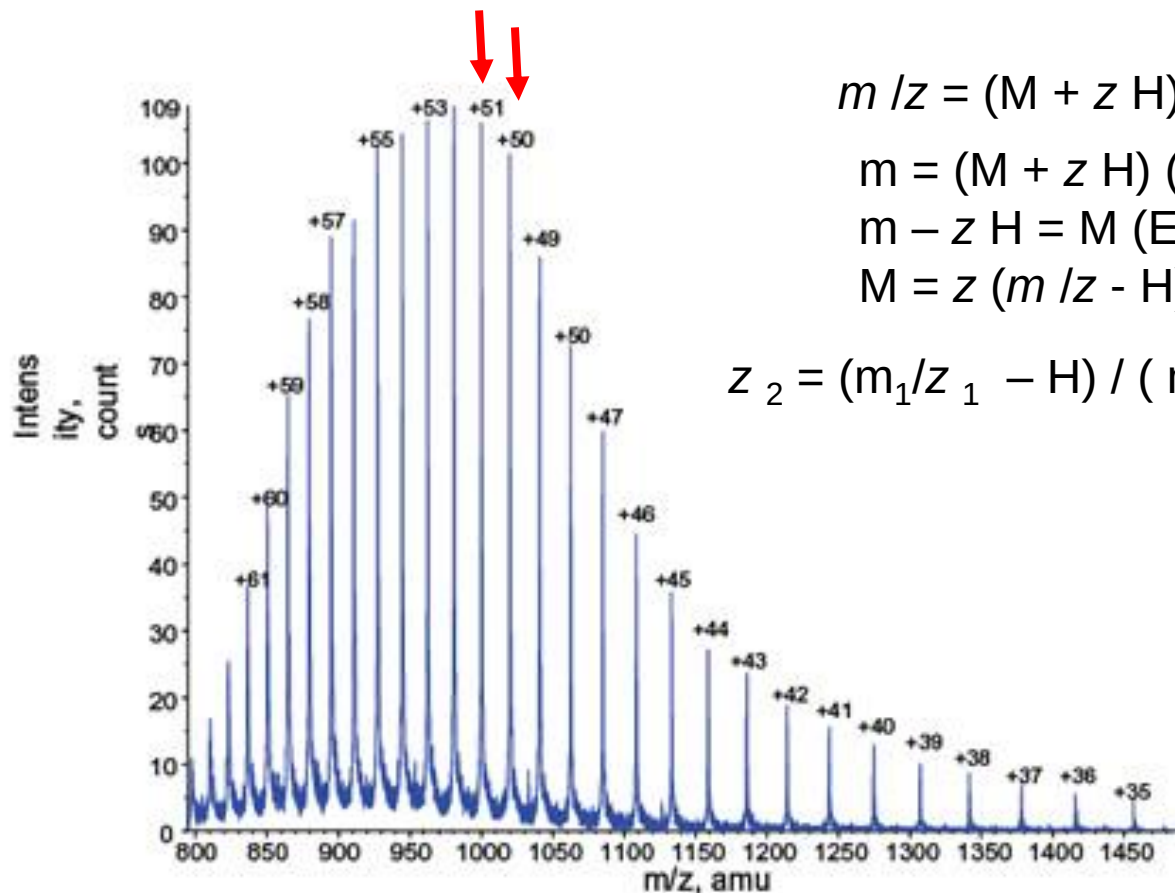
7.5.2. High Resolution Mass Spectrometry



The effect of increasing mass accuracy for unambiguous identification of compounds (Quenzer, T.L., Robinson, J.M., Bolanos, B., Milgram, E. and Greig, M.J., Automated accurate mass analysis using FTICR mass spectrometry, Proceedings of the 50th Annual Conference on Mass Spectrometry and Allied Topics, Orlando, FL, 2002).

Accurate mass determination with little error substantially enhances the chance of suggesting an unambiguous molecular formula solely based on mass spectrometric data

7.5.3. Multiple Charging-Deconvolution



$$m/z = (M + z H)/z \text{ (Eq. 1)}$$

$$m = (M + z H) \text{ (Eq. 2)}$$

$$m - z H = M \text{ (Eq. 3)}$$

$$M = z (m/z - H) \text{ (Eq. 4)}$$

$$z_2 = (m_1/z_1 - H) / (m_2/z_2 - m_1/z_1) \text{ (Eq. 5)}$$

Multiple Charging: ESI-Spectrum of a Protein

Signals refer to $[M + nH]^{n+}$

7.5.3. Multiple Charging-Deconvolution

Mathematical Charge Deconvolution (positive ion mode)

- For the charge states of neighboring peaks belonging to a pair at m/z_1 (higher value) and m/z_2 (lower value) we have $n_2 = n_1 + 1$
- Assuming all charges from protonation and using m_H for the mass of a proton, m/z_1 is determined by

$$m/z_1 = \frac{M_r + n_1 m_H}{n_1}$$

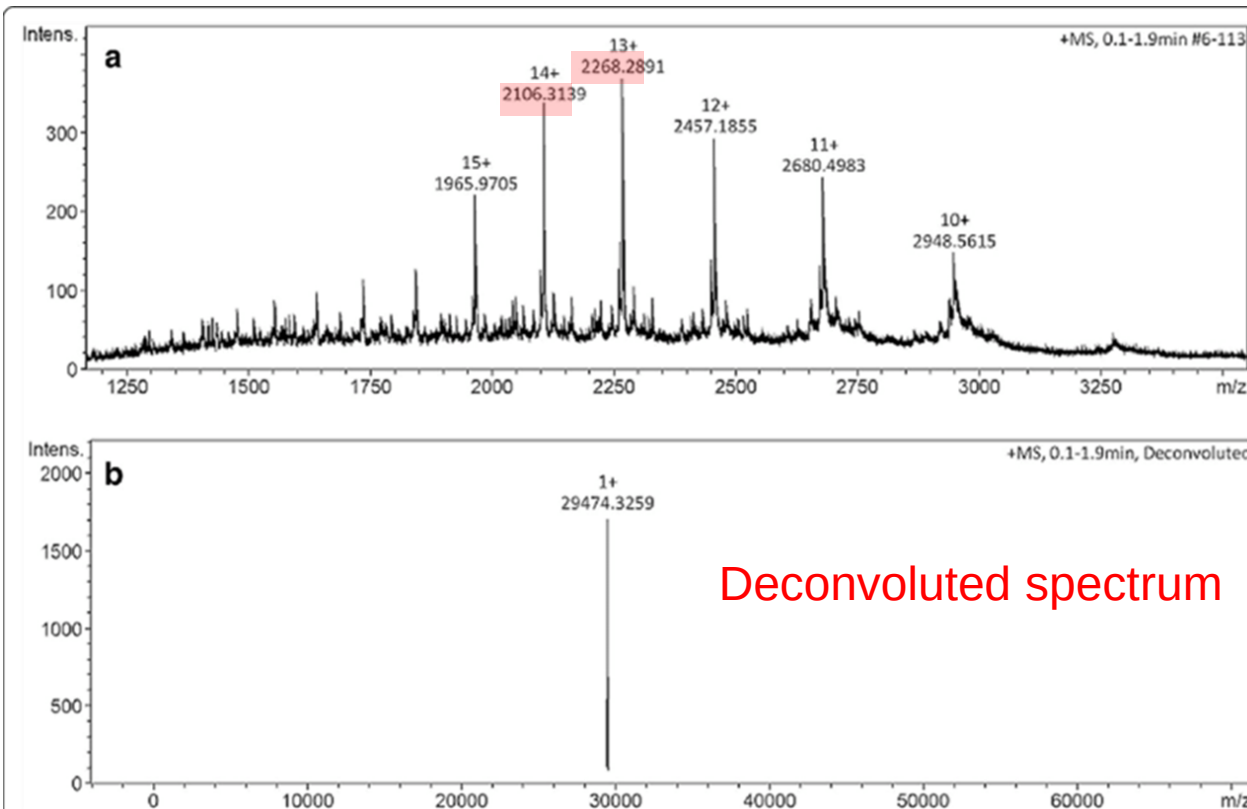
- and m/z_2 is given by $m/z_2 = \frac{M_r + n_2 m_H}{n_2} = \frac{M_r + (n_1 + 1) m_H}{n_1 + 1}$

- The charge state n_1 can then be obtained from $n_1 = \frac{m/z_2 - m_H}{m/z_1 - m/z_2}$

- Having calculated n_1 , M is given by $M_r = n_1 (m/z_1 - m_H)$

e.g. $m/z_1=1001$, $m/z_2=501$; $n_1=(501-1) / (1001-501) = 500/500 = 1$; $M = 1 \times 1001 - 1 = 1000$ The doubly protonated ion is detected at m/z 501 because $(1000 + 2) / 2 = 501$ – or-
 $m/z_1=667.67$, $m/z_2=626$; $n_1=(626-1) / (667.67-626) = 625/41.67 = 15$; $M = 15 \times 667.67 - 15 = 10.000$

7.5.3. Multiple Charging-Deconvolution



$$n_1 = \frac{m/z_2 - m_H}{m/z_1 - m/z_2}$$

$$n_1 = \frac{2106,3139 - 1,0079}{2268,2891 - 2106,3139} = 12.997$$

7.5.3. Multiple Charging-Deconvolution

Mass Spectrometry: Comparison ESI - MALDI

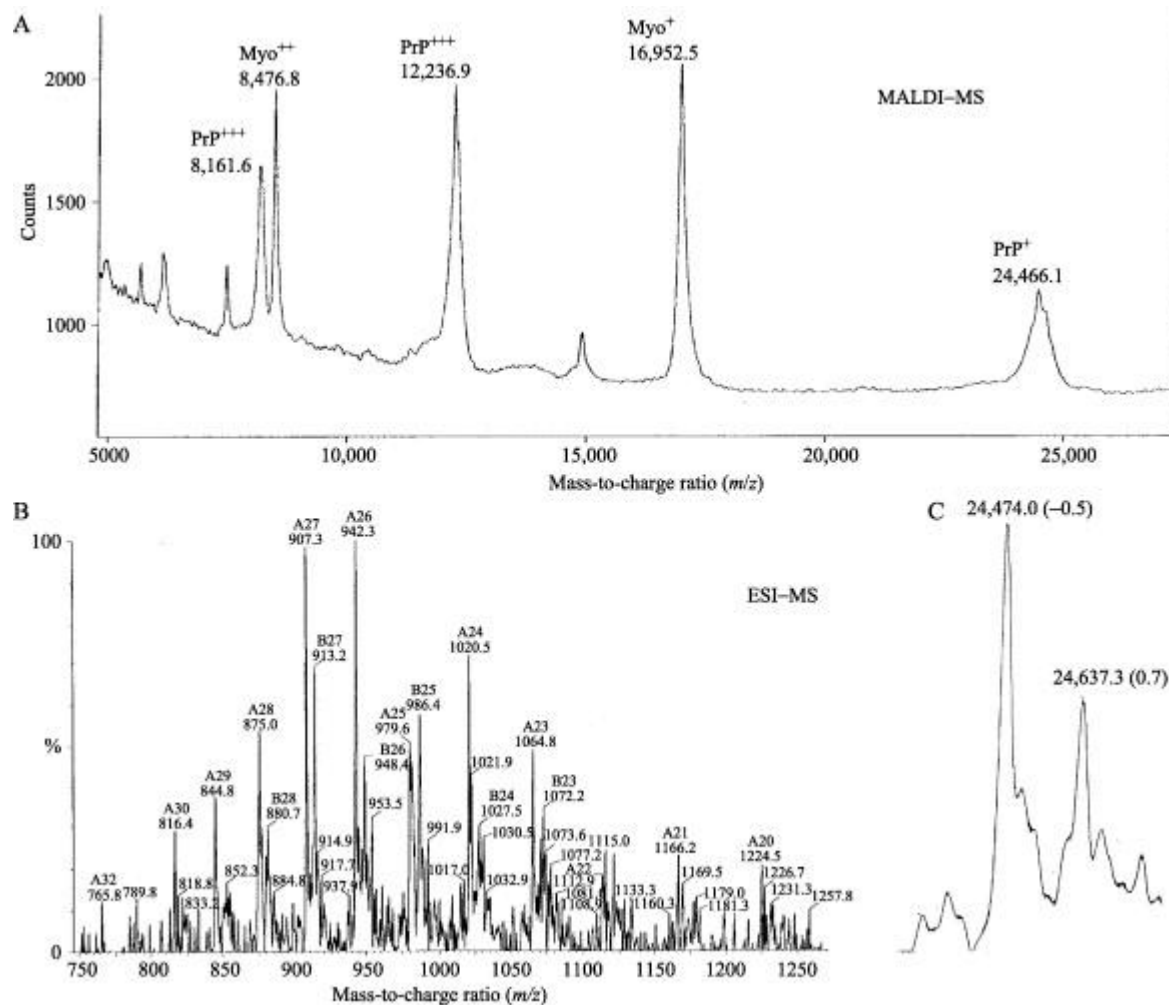
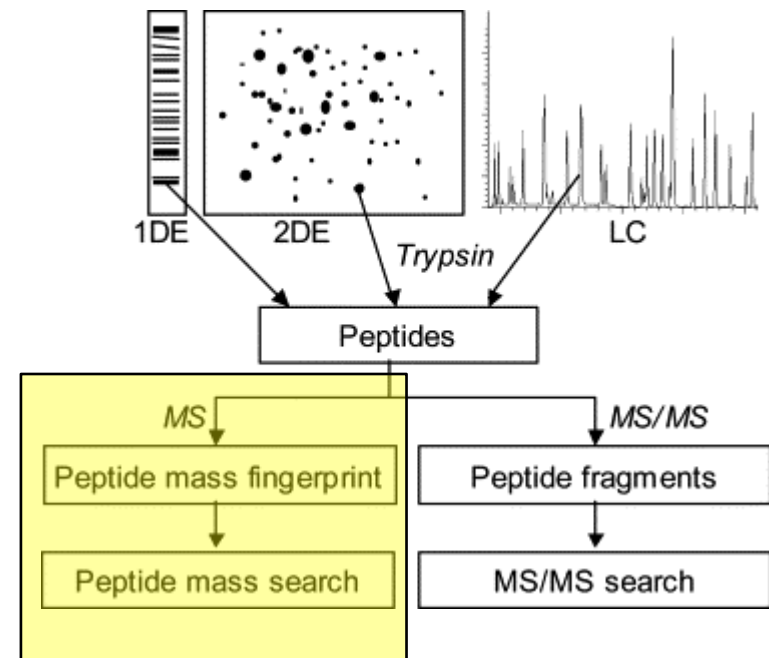
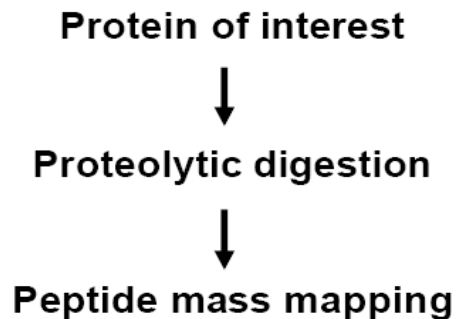


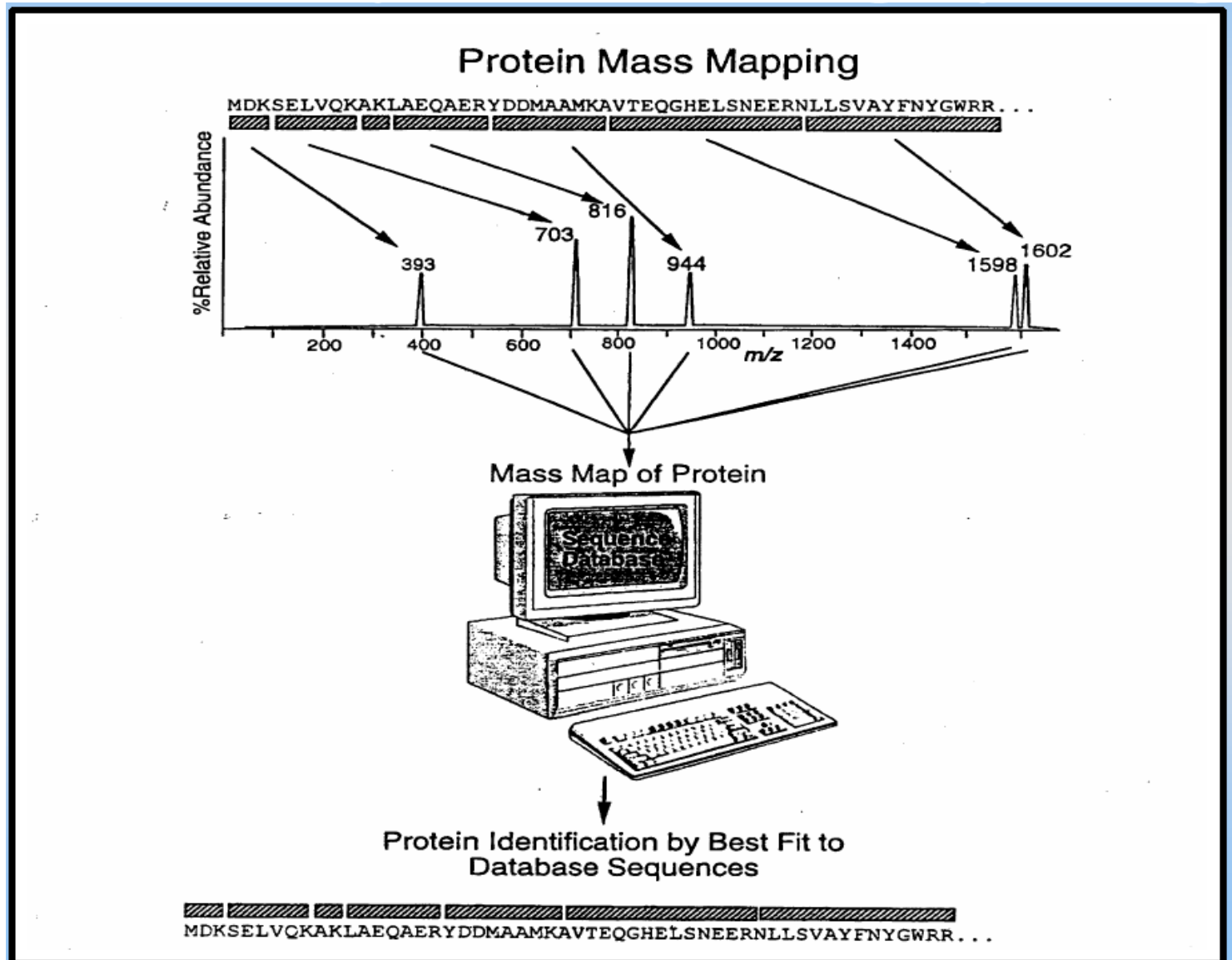
Fig. 7. Comparison of PrPC (PrionenProtein)spectra obtained by MALDI-MS and ESI-MS. (A) Mass spectra of PrPC from MALDI-MS. (B) Mass spectra of PrPC from ESI-MS. (C) The deconvoluted spectrum of raw data from [Fig. 7B](#).

7.5.4. Protein Identification

Protein Identification Protocol



7.5.4. Protein Identification



7.5.4. Protein Identification

ProFound - Peptide Mapping [Short Form] Version 4.10.5
The Rockefeller University Edition

General	Digestion
Sample ID Spot 189 Database NCBI nr (2003/09/01) Taxonomic Category . Bacteria Search for single protein only Protein Mass 0 - 100 kDa Protein pI 0 - 14 Report Top 10 Candidates Questions? Please write to ProFound What's new about ProFound?	Allow maximum 1 missed cleavages Enzyme Trypsin For user-defined cleavage, please click here . Modifications Complete Modification(s) 4-vinyl-pyridine (Cys) Acrylamide (Cys) Iodoacetamide (Cys) Iodoacetic acid (Cys) Performic acid (Cys+O3) Partial Modification ☒ Methionine oxidation For more partial modifications, please click here .
Masses	
Average Masses: Mass tolerance for average data: +/- 1 Tolerance unit: ☒ Da ☐ % ☐ ppm	Monoisotopic Masses: 2472.1 2484.2 2500.2 3031.3 Mass tolerance for monoisotopic data: +/- 0.4 Charge state: ☐ M ☒ MH+
Identify Protein Extra Settings Example Reset Form	

7.5.4. Protein Identification

ProFound - Search Result Summary

Version 4.10.5
The Rockefeller University Edition

Protein Candidates for search BBC64CAC-04D0-6748D1FE [554163 sequences searched]

Rank	Probability	Est'd Z	Protein Information and Sequence Analyse Tools (T)	%	pI	kDa	@
1	1.0e+000	2.28	T gi 6225128 sp P94798 CH60_FRATU 60 kDa chaperonin (Protein Cpn60) (groEL protein)	37	5.0	57.53	@
2	2.2e-009	-	T gi 10954815 ref NP_066750.1 probable virD3 gene [Rhizobium rhizogenes]	16	5.1	94.62	@
+3	6.3e-011	-	T gi 30261580 ref NP_843957.1 ATP-dependent DNA helicase RecQ [Bacillus anthracis str. Ames]	17	5.7	59.85	@
+4	4.5e-011	-	T gi 25527234 gb AA04557.1 FtsZ [Bacillus mycoides]	27	5.2	40.62	@
+5	3.9e-011	-	T gi 21231248 ref NP_637165.1 bifunctional aspartokinase/homoserine dehydrogenase I [Xanthomonas campestris pv. campestris str. ATCC 33913]	15	5.6	89.42	@
6	1.3e-011	-	T gi 29171494 ref NP_808678.1 type III effector HopPtoT1 [Pseudomonas syringae pv. tomato str. DC3000]	25	9.2	41.36	@
+7	4.3e-012	-	T gi 21684964 emb CAD22547.1 F1F0-ATPase subunit beta [Oenococcus oeni]	22	5.0	54.65	@
8	3.7e-012	-	T gi 23113742 ref ZP_00099092.1 hypothetical protein [Desulfitobacterium hafniense]	30	5.0	36.52	@
9	3.1e-012	-	T gi 17987207 ref NP_539841.1 XAA-PRO DIPEPTIDASE [Brucella melitensis]	21	6.0	36.75	@
10	3.1e-012	-	T gi 23120900 ref ZP_00103386.1 hypothetical protein [Desulfitobacterium hafniense]	30	5.8	39.31	@

7.5.4. Protein Identification

ProFound - Peptide Mapping [Short Form] Version 4.10.5
The Rockefeller University Edition

General	Digestion
Sample ID Spot 189 Database NCBI nr (2003/09/01) Taxonomic Category All taxa Search for possible two-protein mixture	Allow maximum 1 missed cleavages Enzyme Trypsin For user-defined cleavage, please click here .
Protein Mass 0 - 100 kDa Protein pI 0 - 14 Report Top 10 Candidates Questions? Please write to ProFound What's new about ProFound?	Modifications Complete Modification(s) 4-vinyl-pyridine (Cys) Acrylamide (Cys) Iodoacetamide (Cys) Iodoacetic acid (Cys) Performic acid (Cys+O3) Partial Modification ☒ Methionine oxidation For more partial modifications, please click here .
Masses	
Average Masses: 	Monoisotopic Masses: 2472.1 2484.2 2500.2 3031.3
Mass tolerance for average data: +/- 1 Tolerance unit: ☒ Da ☐ % ☐ ppm	Mass tolerance for monoisotopic data: +/- 0.4 Charge state: ☐ M ☒ MH+
Identify Protein Extra Settings Example Reset Form	

7.5.4. Protein Identification

ProFound - Search Result Summary							Version 4.10.5
Protein Candidates for search BAD9361D-058C-665CD112 [1473572 sequences searched]							The Rockefeller University Edition
Rank	Probability	Est'd Z	Protein Information and Sequence Analyse Tools (T)	%	pI	kDa	®
1	1.0e+000	1.55	T gi23272966 gb AAH37127.1 Atp5b protein [Mus musculus]	<u>41</u>	5.2	56.65	®
			T gi6225128 sp P94798 CH60_FRATU 60 kDa chaperonin (Protein Cpn60) (groEL protein)	<u>37</u>	5.0	57.53	
2	1.4e-004	0.29	T gi23272966 gb AAH37127.1 Atp5b protein [Mus musculus] T gi9229904 dbj BAB00625.1 cdc45 [Ciona intestinalis]	<u>41</u> <u>12</u>	5.2 5.5	56.65 66.29	®
3	4.2e-007	-	T gi23272966 gb AAH37127.1 Atp5b protein [Mus musculus] T gi27676754 ref XP_228318.1 similar to autophagy 5-like (S. cerevisiae); autophagy 5-like [Mus musculus] [Rattus norvegicus]	<u>41</u> <u>18</u>	5.2 9.9	56.65 55.54	®
4	3.6e-007	-	T gi23272966 gb AAH37127.1 Atp5b protein [Mus musculus] T gi15599716 ref NP_253210.1 probable chemotaxis transducer [Pseudomonas aeruginosa PA01]	<u>41</u> <u>11</u>	5.2 5.1	56.65 72.35	®
5	9.3e-008	-	T gi23272966 gb AAH37127.1 Atp5b protein [Mus musculus] T gi25989105 gb AAG01897.1 hyaluronan binding protein [Homo sapiens]	<u>41</u> <u>12</u>	5.2 4.4	56.65 56.32	®
6	3.2e-008	-	T gi23272966 gb AAH37127.1 Atp5b protein [Mus musculus] T gi15643365 ref NP_228409.1 hypothetical protein [Thermotoga maritima]	<u>41</u> <u>17</u>	5.2 5.2	56.65 62.09	®
7	1.9e-008	-	T gi23272966 gb AAH37127.1 Atp5b protein [Mus musculus] T gi15669827 ref NP_248641.1 glycogen phosphorylase (glgP) [Methanococcus jannaschii]	<u>41</u> <u>16</u>	5.2 7.3	56.65 61.02	®
8	7.0e-009	-	T gi23272966 gb AAH37127.1 Atp5b protein [Mus musculus] T gi29367646 gb AAO72667.1 centromere protein-like protein [Oryza sativa (japonica cultivar-group)]	<u>41</u> <u>21</u>	5.2 5.2	56.65 59.28	®

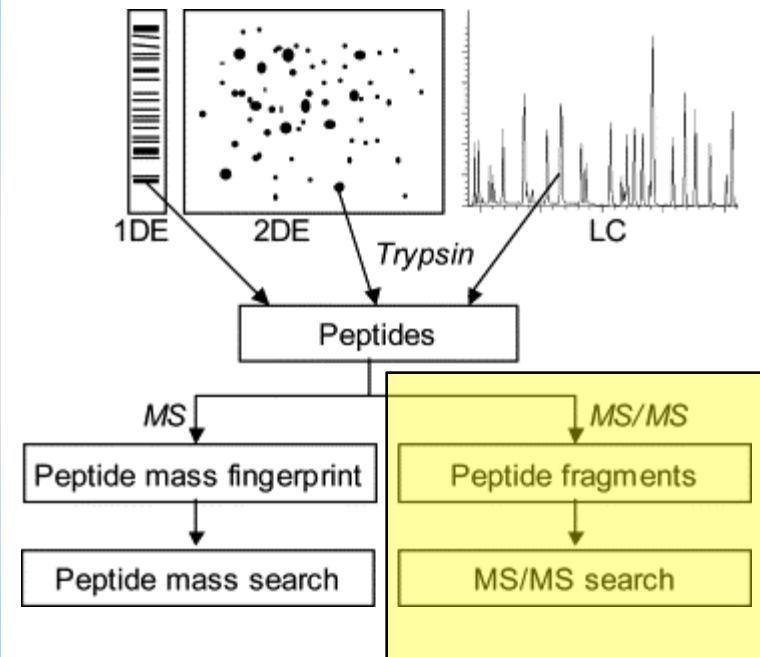
7.5.4. Protein Identification

Protein Identification Protocol

Protein of interest
↓
Proteolytic digestion
↓
Peptide mass mapping

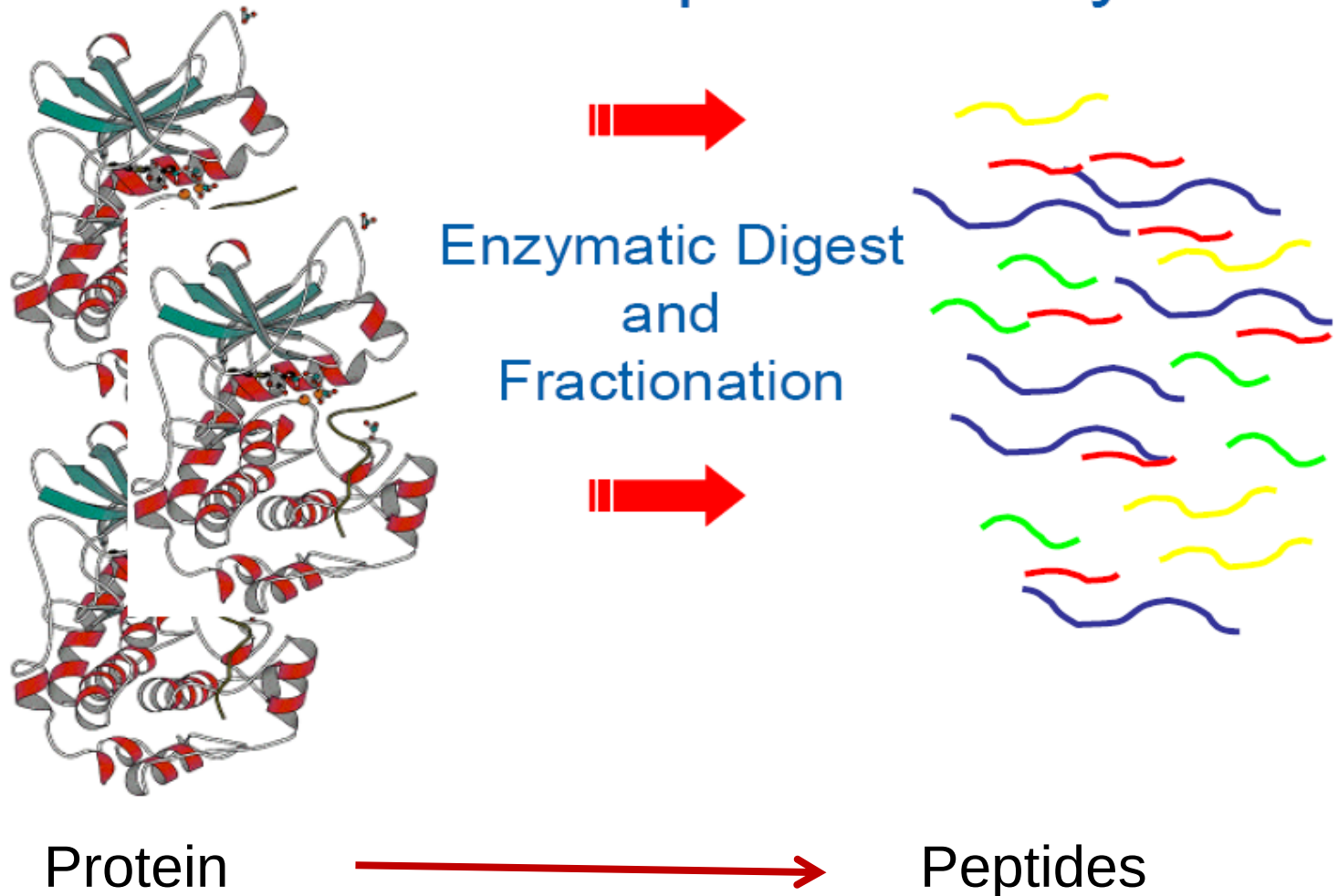
No Result

Peptide microsequencing (PSD, MS/MS)



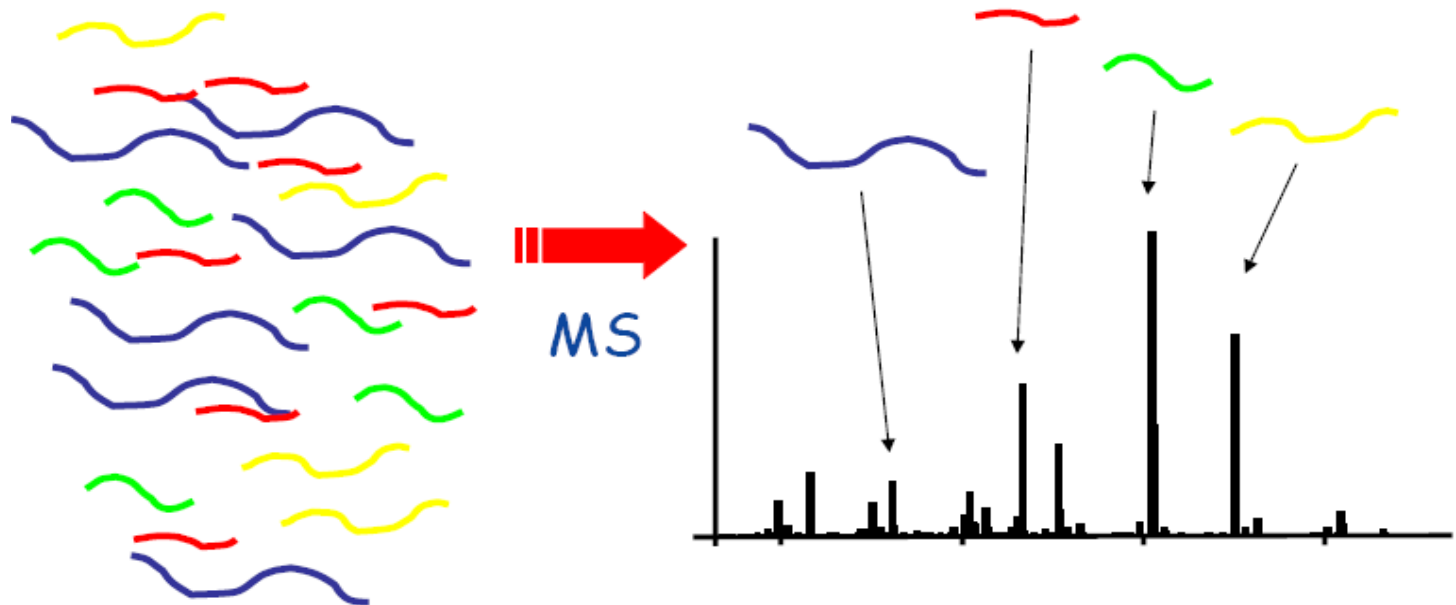
7.5.4. Protein Identification

Sample Preparation for Tandem Mass Spectrometry



7.5.4. Protein Identification

Single Stage MS



MS shows signals for peptides
formed during enzymatic digestion

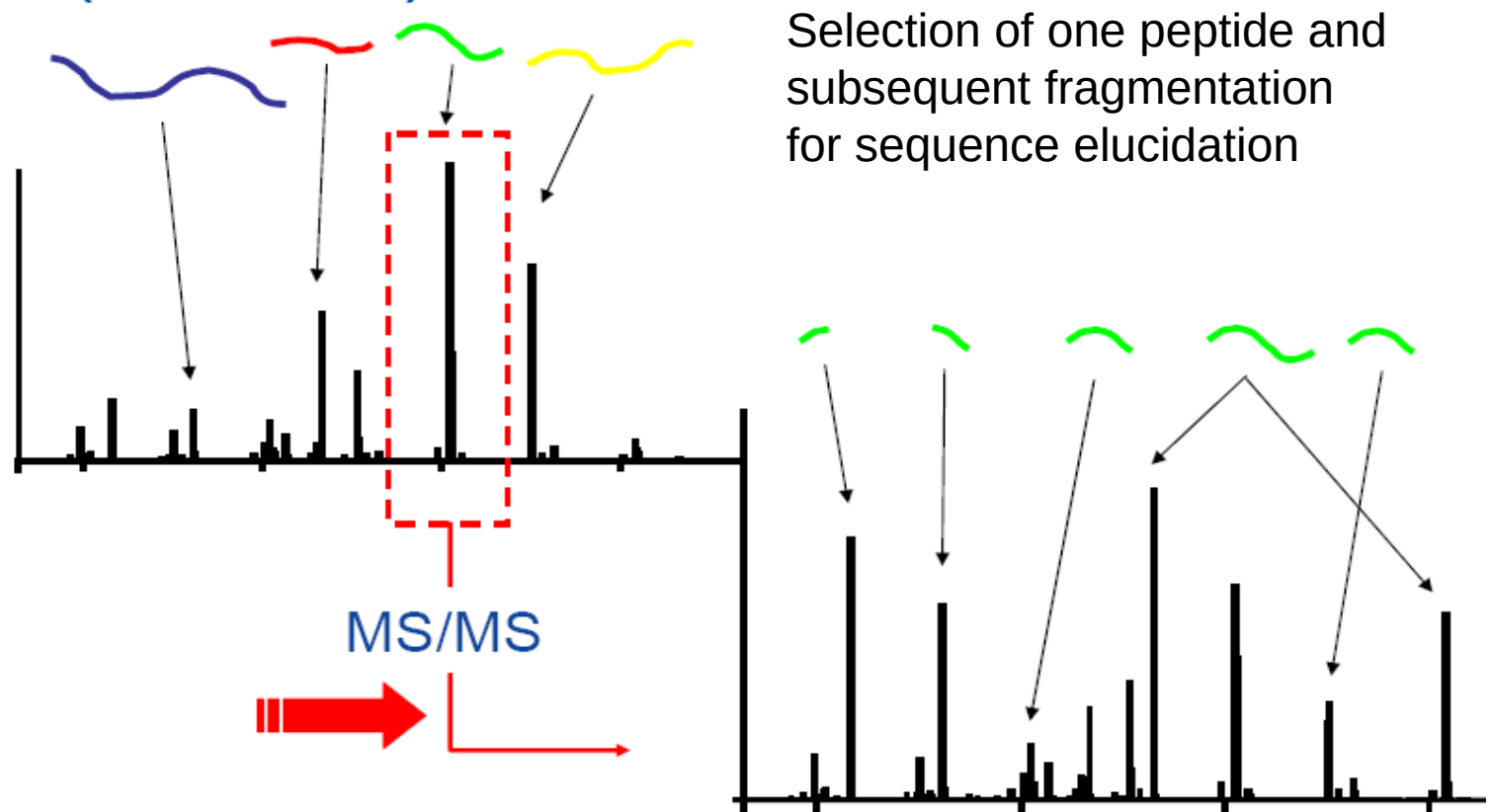
7.5.4. Protein Identification

Tandem Mass Spectrometry (MS/MS)

- Acquire mass spectrum of sample
- Select interesting ion by m/z value
- Fragment the selected “parent” ion
- Acquire mass spectrum of parent ion’s fragments

7.5.4. Protein Identification

Tandem Mass Spectrometry (MS/MS)

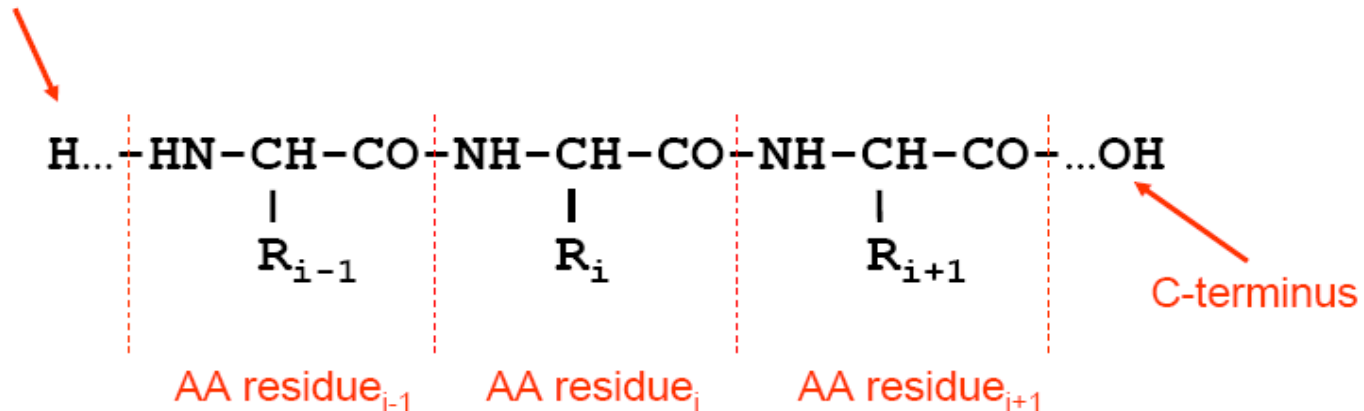


7.5.4. Protein Identification

Peptide Fragmentation

Peptides consist of amino-acids
arranged in a linear backbone.

N-terminus

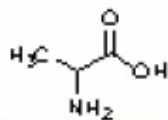


7.5.4. Protein Identification

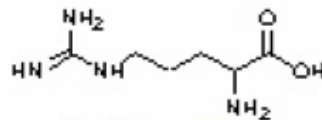
Peptide Fragmentation

Schematic diagrams of the 20 amino acids

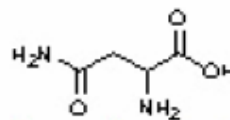
(picture taken from www.chemistry.pomona.edu)



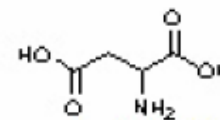
Alanine (Ala)



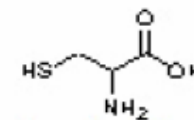
Arginine (Arg)



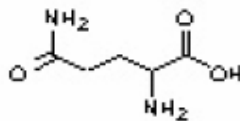
Asparagine (Asn)



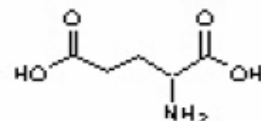
Aspartic Acid (Asp)



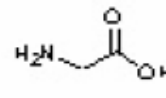
Cysteine (Cys)



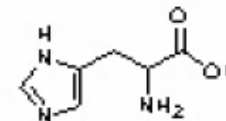
Glutamine (Gln)



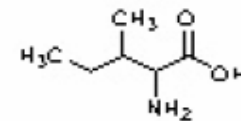
Glutamic Acid (Glu)



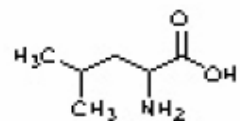
Glycine (Gly)



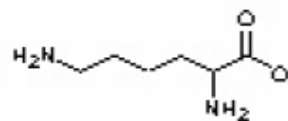
Histidine (His)



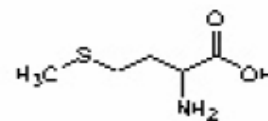
Isoleucine (Ile)



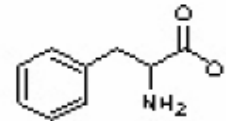
Leucine (Leu)



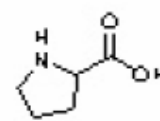
Lysine (Lys)



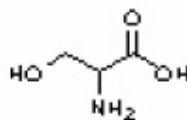
Methionine (Met)



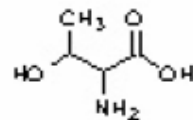
Phenylalanine (Phe)



Proline (Pro)



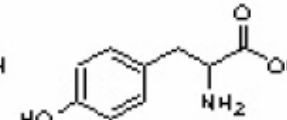
Serine (Ser)



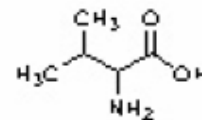
Threonine (Thr)



Tryptophan (Trp)



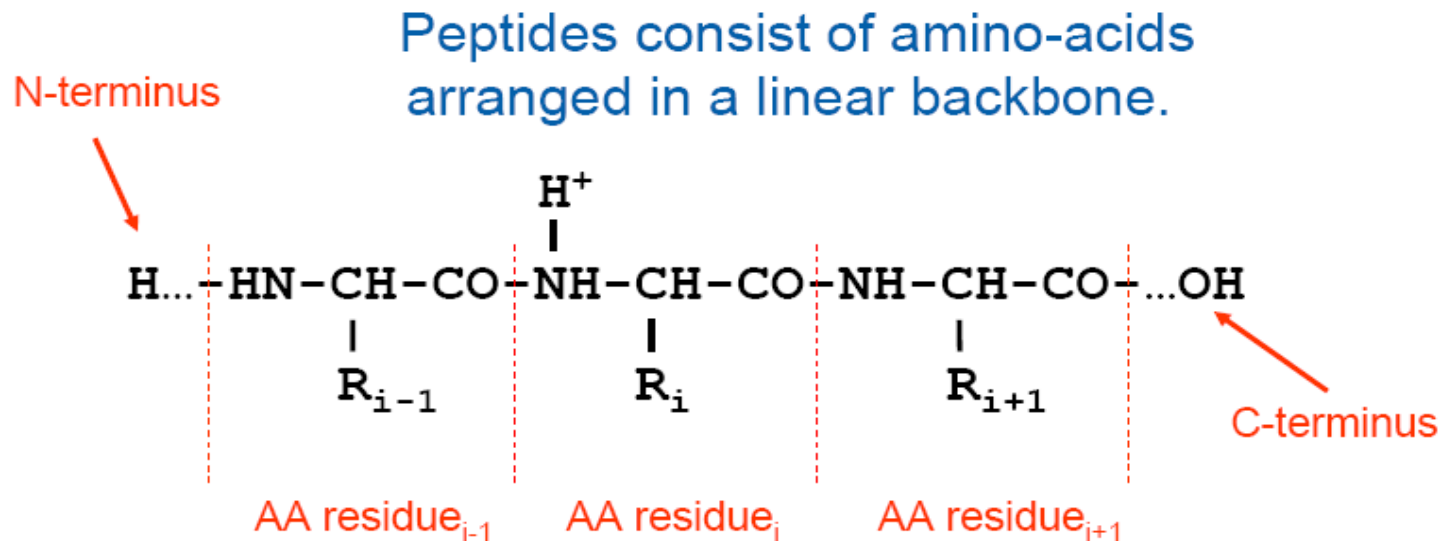
Tyrosine (Tyr)



Valine (Val)

7.5.4. Protein Identification

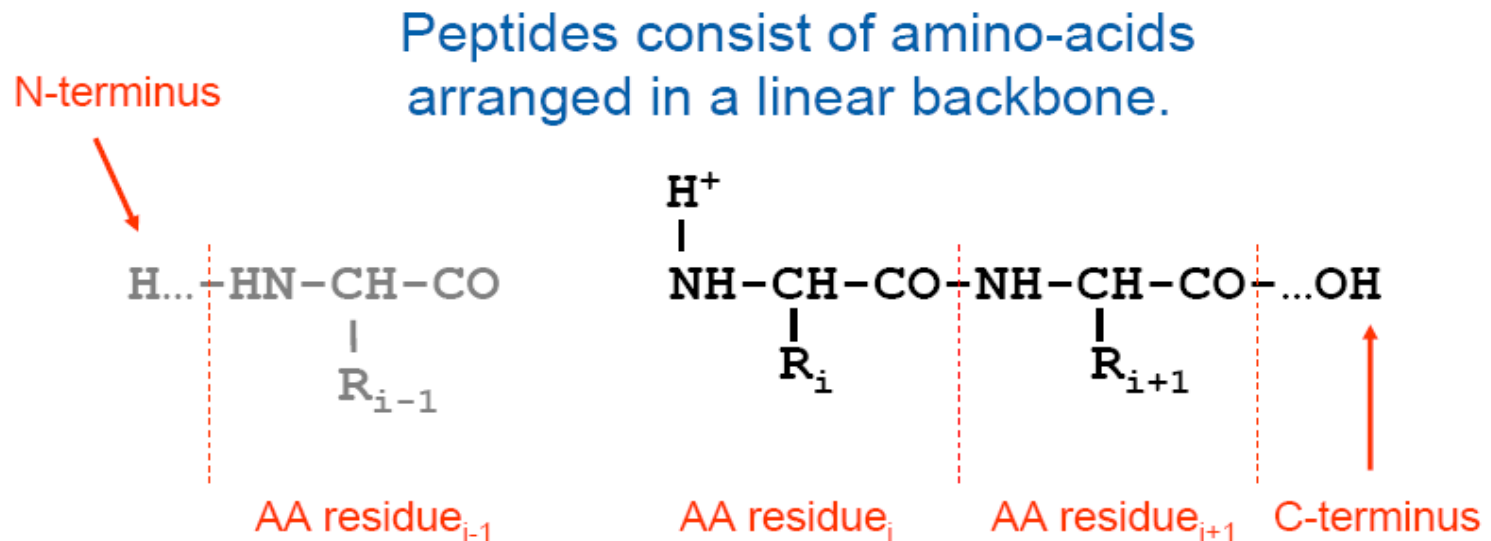
Peptide Fragmentation



Ionized peptide (addition of a proton)

7.5.4. Protein Identification

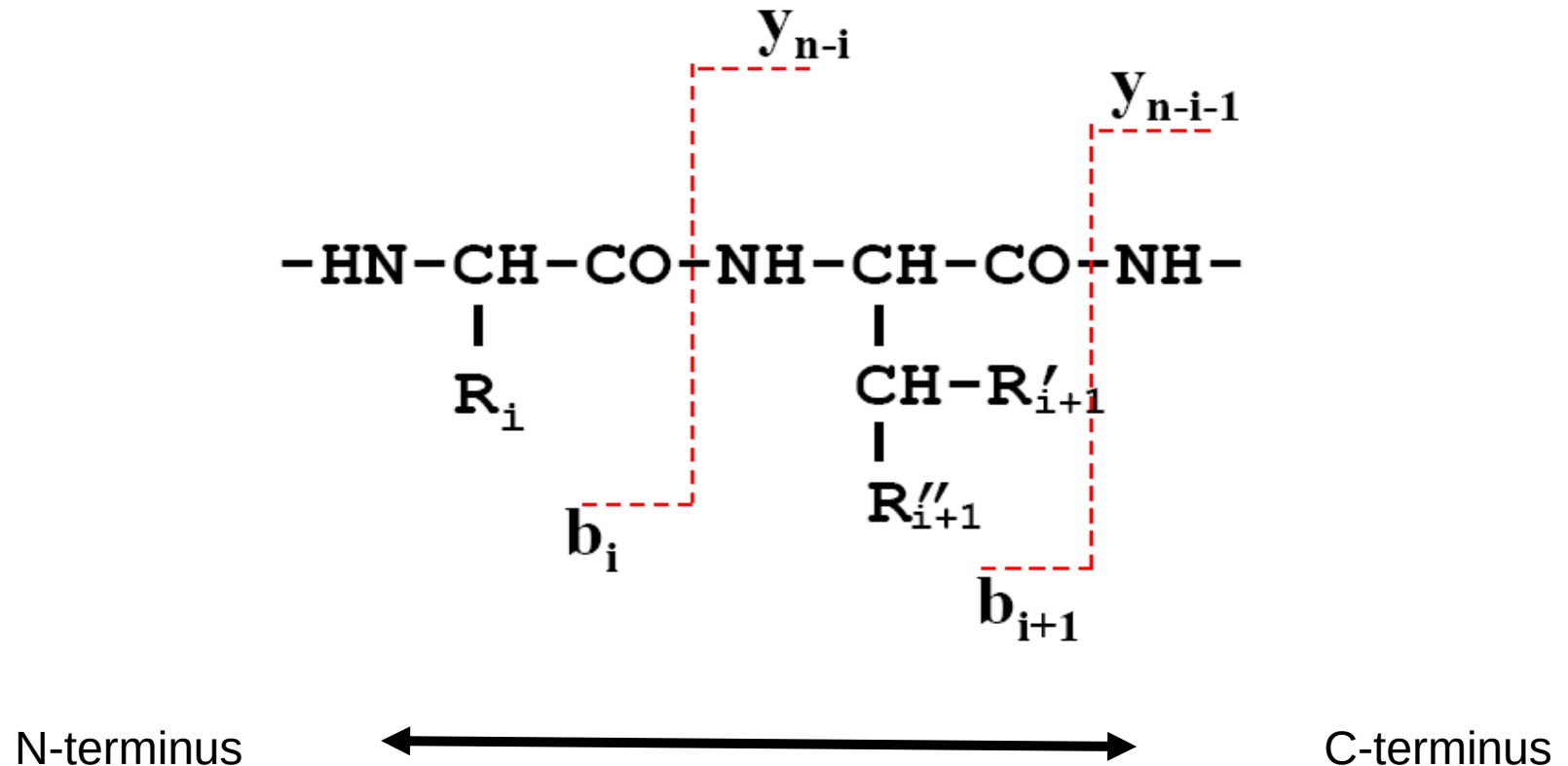
Peptide Fragmentation



Fragmented peptide
C-terminus fragment observed

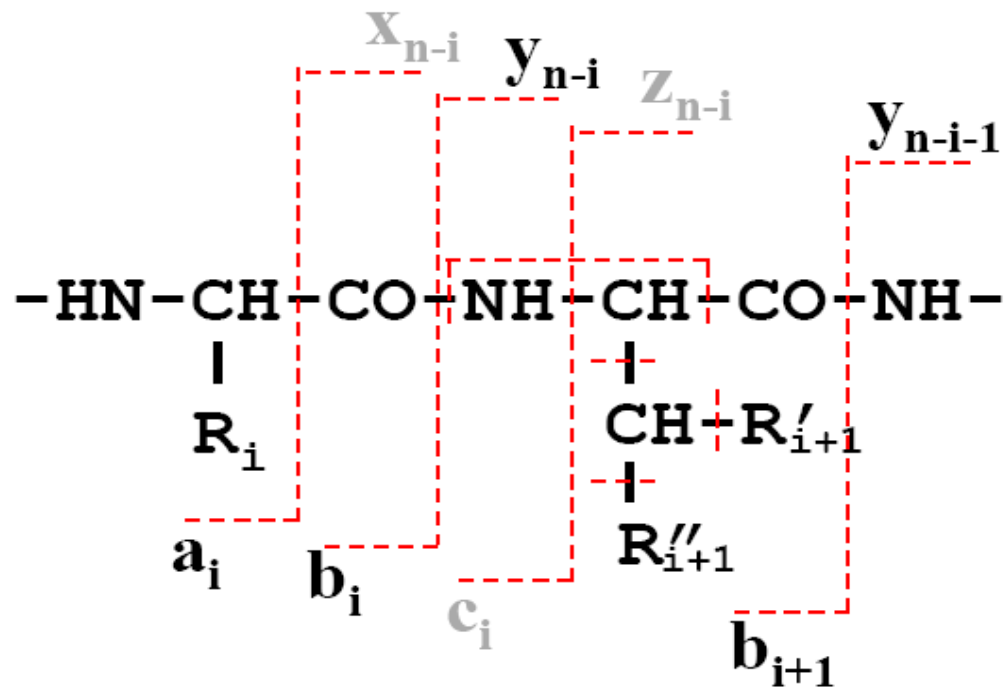
7.5.4. Protein Identification

Peptide Fragmentation

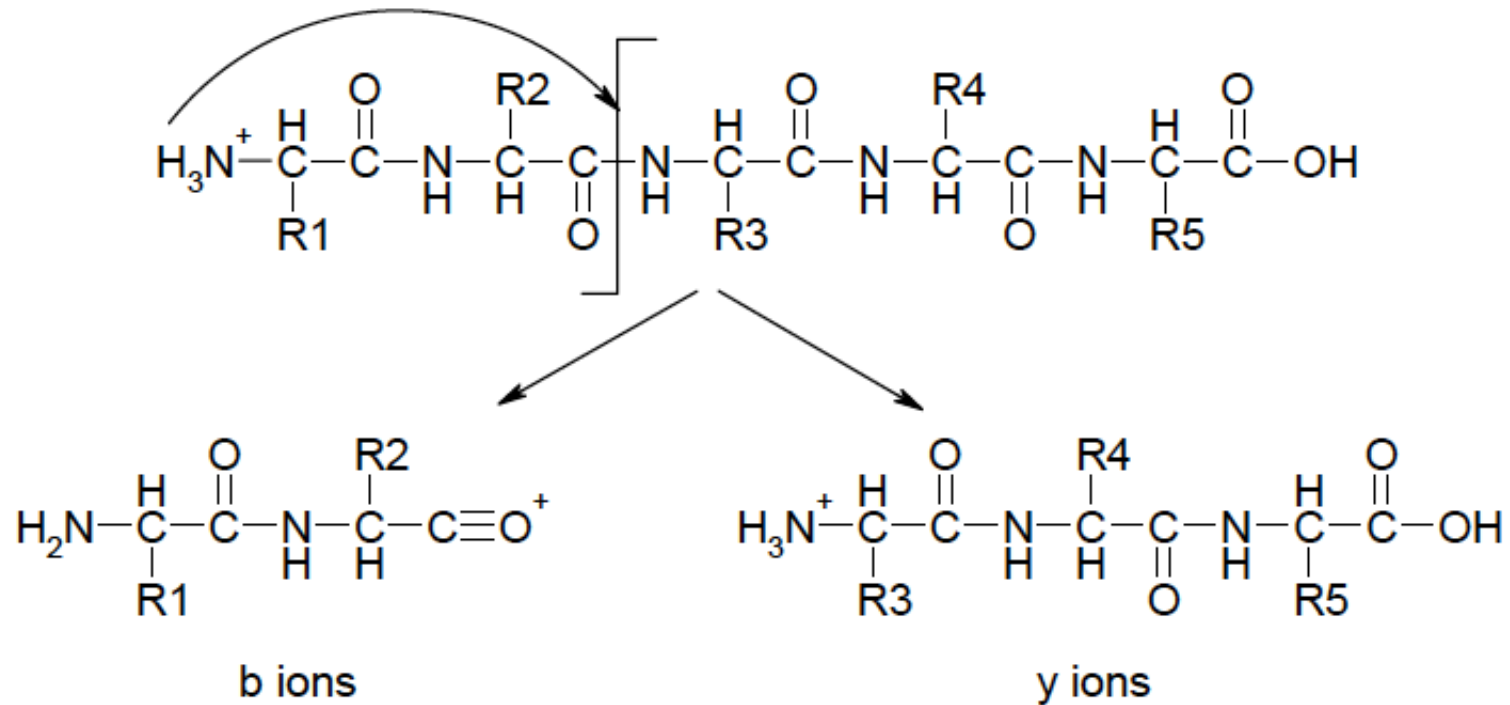


7.5.4. Protein Identification

Peptide Fragmentation



7.5.4. Protein Identification



7.5.4. Protein Identification

De Novo Interpretation

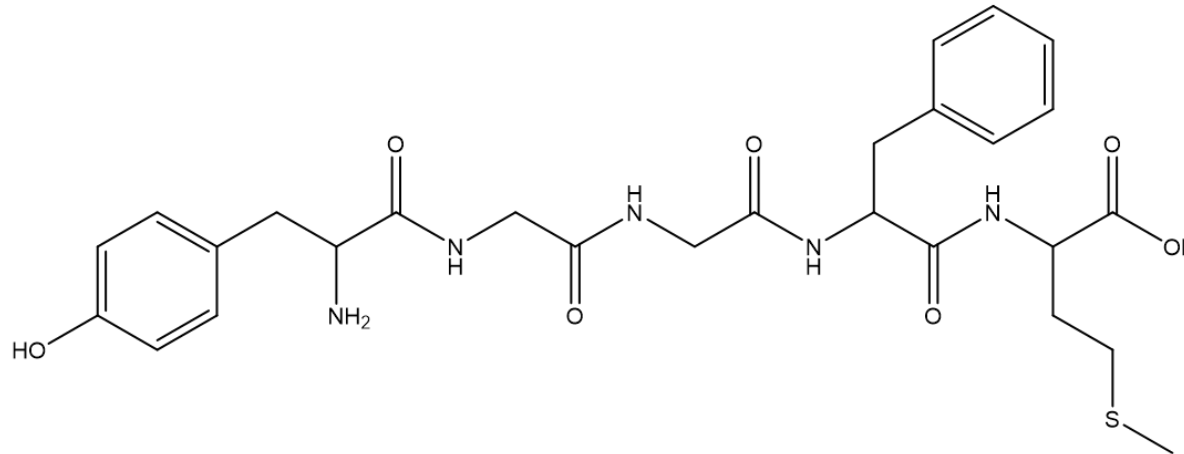
	Amino-Acid	Residual MW		Amino-Acid	Residual MW
A	Alanine	71.03712	M	Methionine	131.04049
C	Cysteine	103.00919	N	Asparagine	114.04293
D	Aspartic acid	115.02695	P	Proline	97.05277
E	Glutamic acid	129.04260	Q	Glutamine	128.05858
F	Phenylalanine	147.06842	R	Arginine	156.10112
G	Glycine	57.02147	S	Serine	87.03203
H	Histidine	137.05891	T	Threonine	101.04768
I	Isoleucine	113.08407	V	Valine	99.06842
K	Lysine	128.09497	W	Tryptophan	186.07932
L	Leucine	113.08407	Y	Tyrosine	163.06333

Residual MW = MW – H₂O (18)

7.5.4. Protein Identification

Met-enkephalin

Met-enkephalin is a pentapeptide with the amino acid sequence tyr-gly-gly-phe-met.



Fragment Ion Calculator Results

Sequence: YGGFM, pI: 5.52439

Fragment Ion Table, monoisotopic masses

Seq	#	B	Y	# (+1)
Y	1	164.07065	574.23304	5
G	2	221.09211	411.16971	4
G	3	278.11358	354.14825	3
F	4	425.18199	297.12679	2
M	5	556.22248	150.05837	1

7.5.4. Protein Identification

Sequence: YGGFM, pI: 5.52439

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De Novo Interpretation

	Amino-Acid	Residual MW		Amino-Acid	Residual MW
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K	Lysine	128.09497	W	Tryptophan	186.07932
L	Leucine	113.08407	Y	Tyrosine	163.06333

For y ions (e.g. y_3):

$$\begin{aligned}
 M &= 131^* \\
 F &= 147^* \\
 G &= 57^* \\
 H_2O &= 18^* \\
 \text{Proton} &= \underline{1^*} \\
 &= \mathbf{354^*}
 \end{aligned}$$

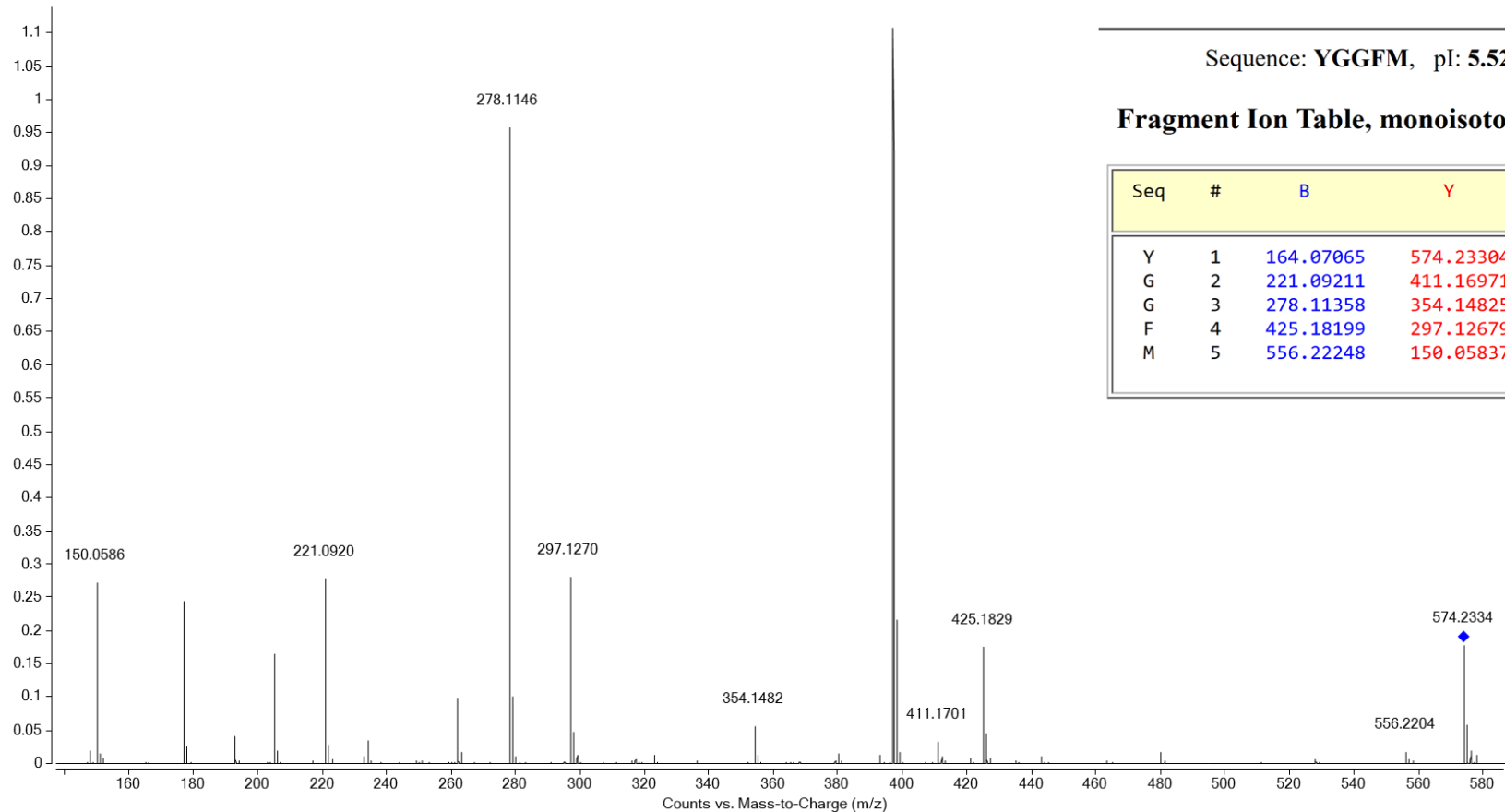
For b ions (e.g. b_3):

$$\begin{aligned}
 Y &= 163^* \\
 G &= 57^* \\
 G &= 57^* \\
 \text{Proton} &= \underline{1^*} \\
 &= \mathbf{278^*}
 \end{aligned}$$

* Calculated with unit mass only

7.5.4. Protein Identification

×10⁵ +ESI Product Ion CID@20.0 (574.2000[z=1] -> **



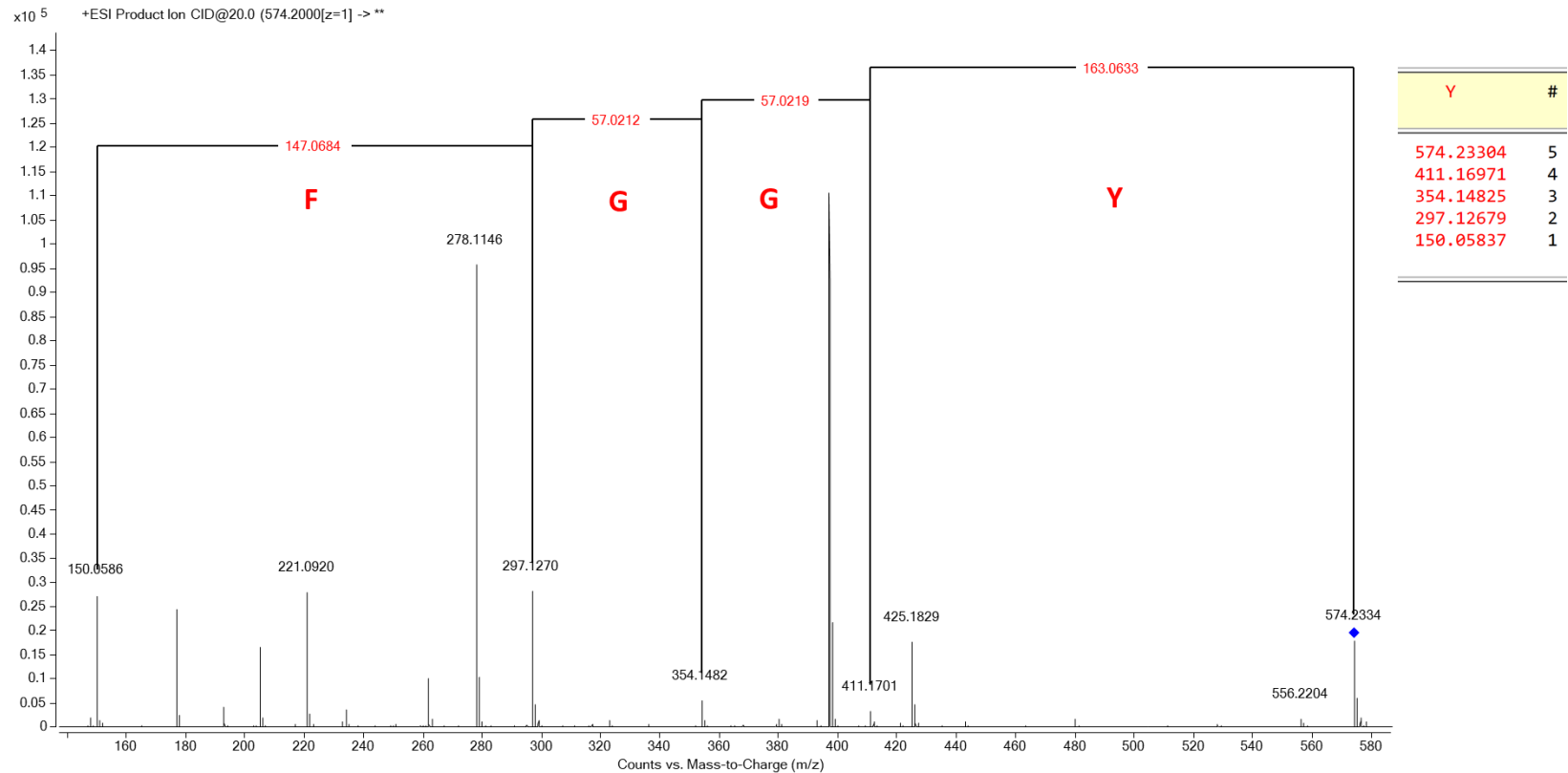
Fragment Ion Calculator Results

Sequence: YGGFM, pI: 5.52439

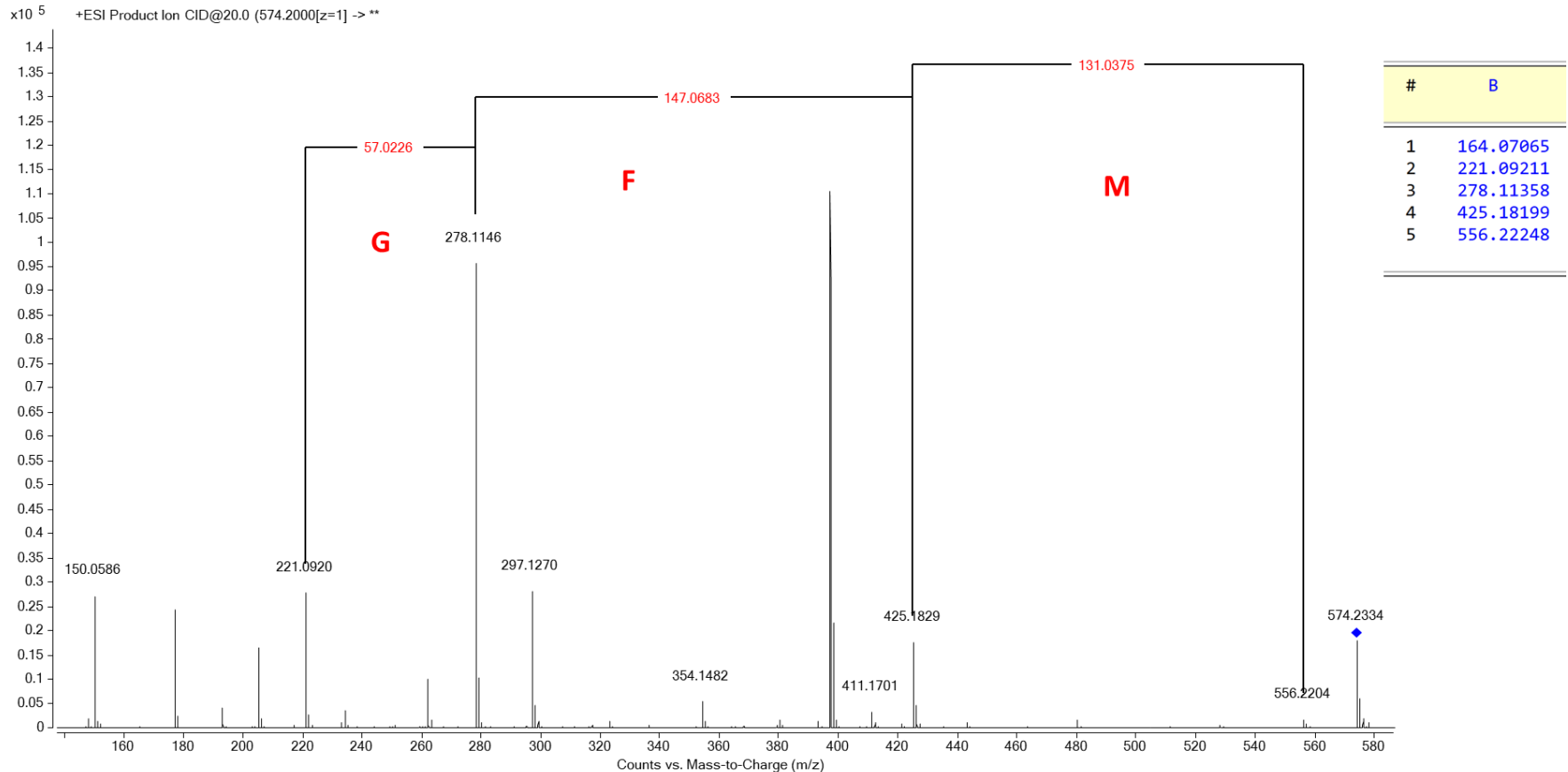
Fragment Ion Table, monoisotopic masses

Seq	#	B	Y	# (+1)
Y	1	164.07065	574.23304	5
G	2	221.09211	411.16971	4
G	3	278.11358	354.14825	3
F	4	425.18199	297.12679	2
M	5	556.22248	150.05837	1

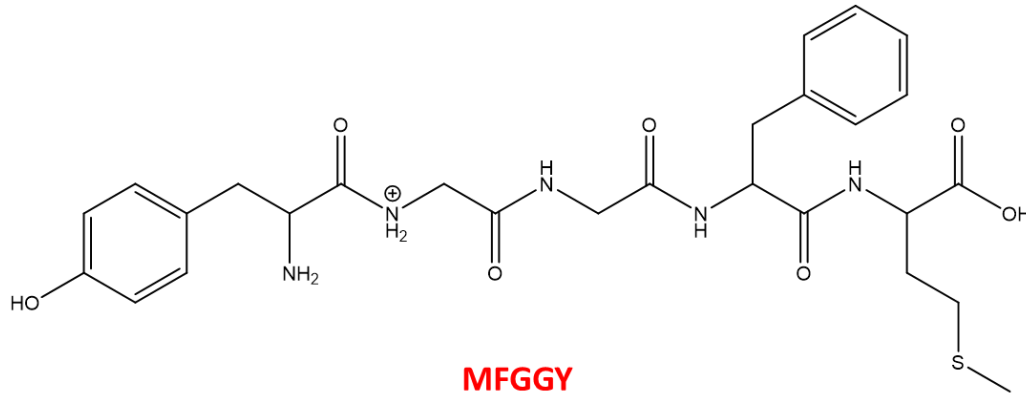
7.5.4. Protein Identification



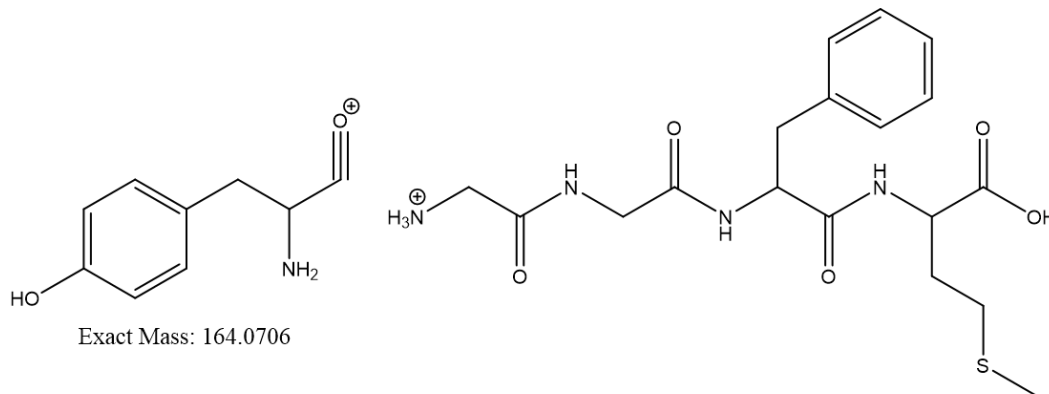
7.5.4. Protein Identification



7.5.4. Protein Identification



Exact Mass: 574.2330

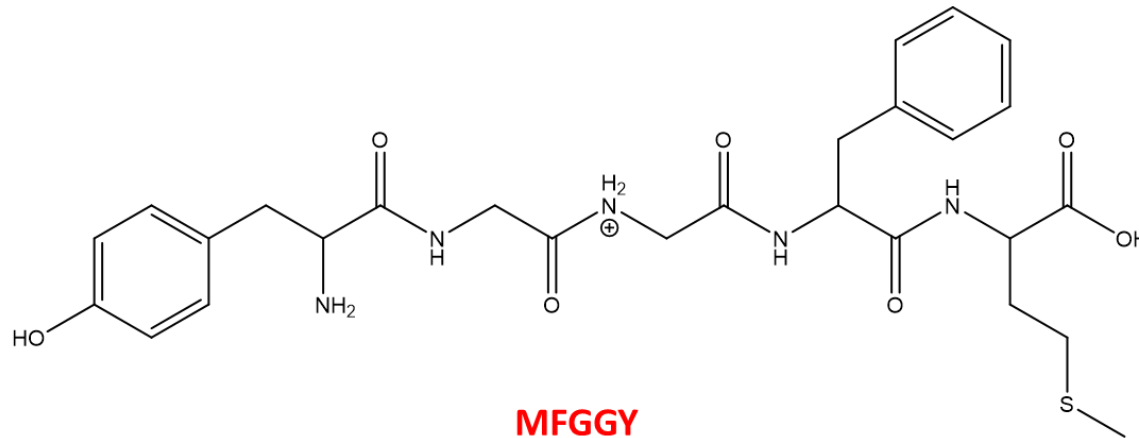


Y = b₁

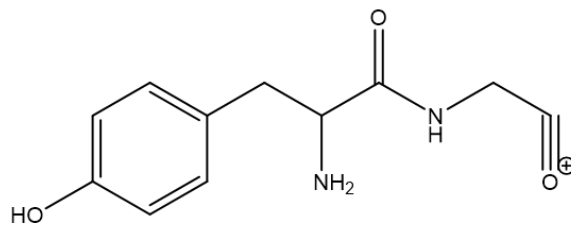
Exact Mass: 411.1697

MFGG = y₄

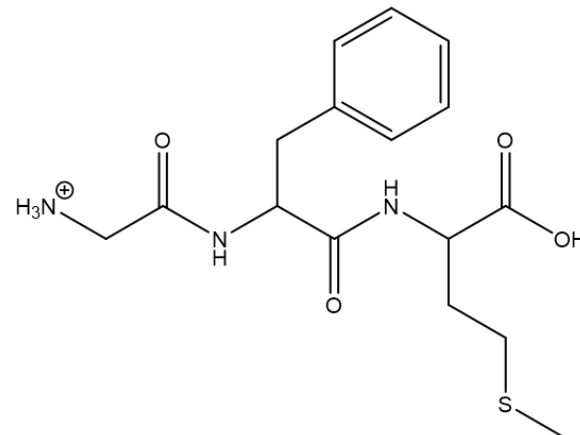
7.5.4. Protein Identification



Exact Mass: 574.2330

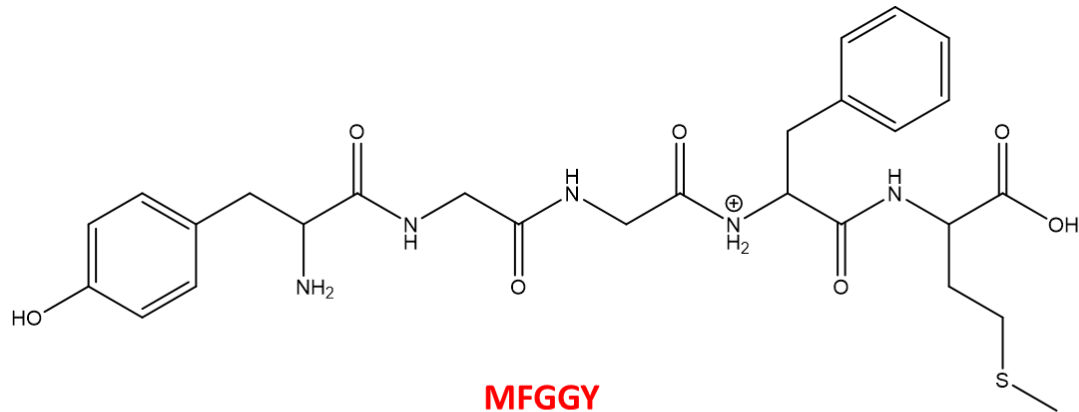


GY = b₂

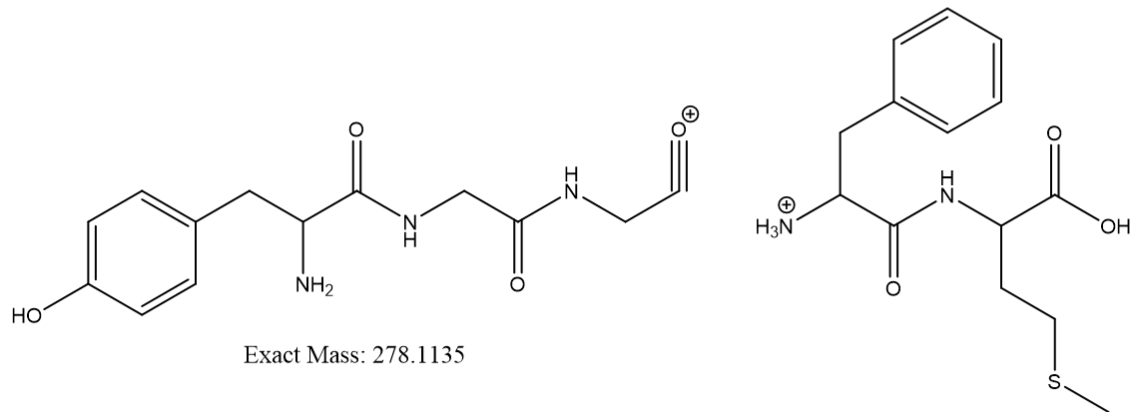


MFG = y₃

7.5.4. Protein Identification



Exact Mass: 574.2330



GGY = b_3

MF = y_2

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NC(=O)CCc1ccc(O)cc1NC(=O)CCNC(=O)CCNC(=O)CCc1ccccc1C(=O)[O+]

FGGY = b_4

$$M = y_1$$

Courtesy of Markus Himmelsbach

7.5.4. Protein Identification

Peptide Fragmentation

Ser – Gly – Phe – Leu – Glu – Glu – Asp – Glu - Lys

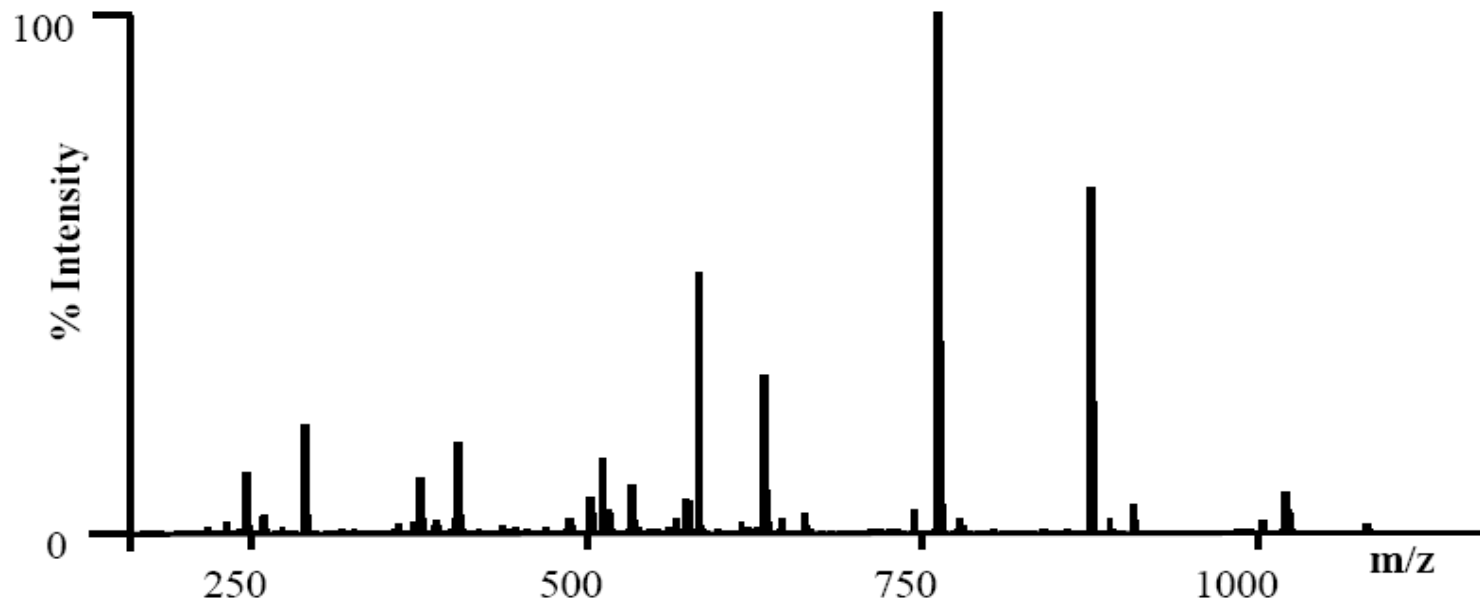
Peptide: S-G-F-L-E-E-D-E-L-K

MW	ion				ion	MW
88	b ₁	S	GFLEEDELK	y ₉	1080	
145	b ₂	SG	FLEEDELK	y ₈	1022	
292	b ₃	SGF	LEEDELK	y ₇	875	
405	b ₄	SGFL	EEDELK	y ₆	762	
534	b ₅	SGFLE	EDELK	y ₅	633	
663	b ₆	SGFLEE	DELK	y ₄	504	
778	b ₇	SGFLEED	ELK	y ₃	389	
907	b ₈	SGFLEEDE	LK	y ₂	260	
1020	b ₉	SGFLEEDEL	K	y ₁	147	

7.5.4. Protein Identification

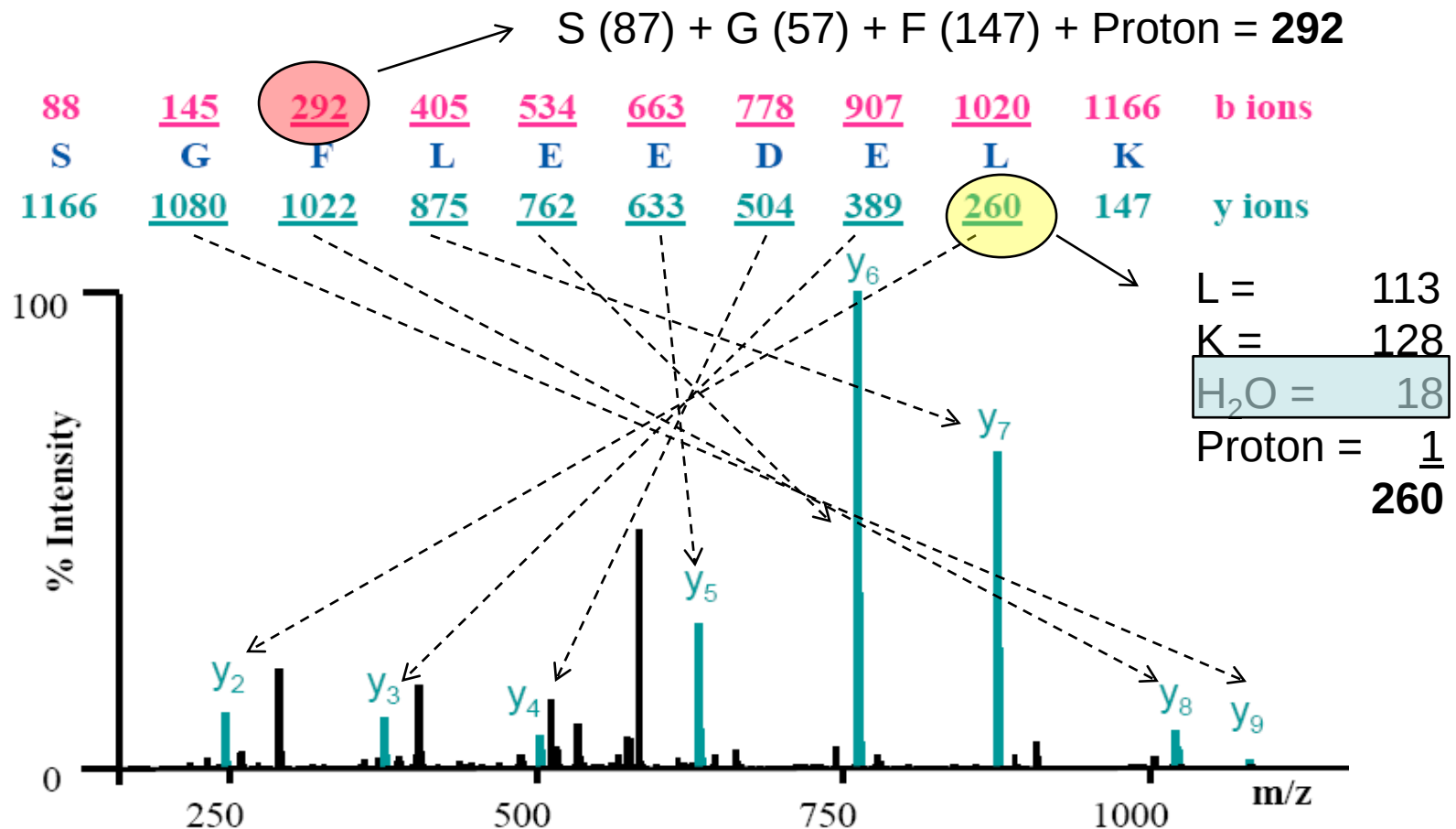
Peptide Fragmentation

<u>88</u>	<u>145</u>	<u>292</u>	<u>405</u>	<u>534</u>	<u>663</u>	<u>778</u>	<u>907</u>	<u>1020</u>	<u>1166</u>	b ions
S	G	F	L	E	E	D	E	L	K	
1166	1080	1022	875	762	633	504	389	260	147	y ions



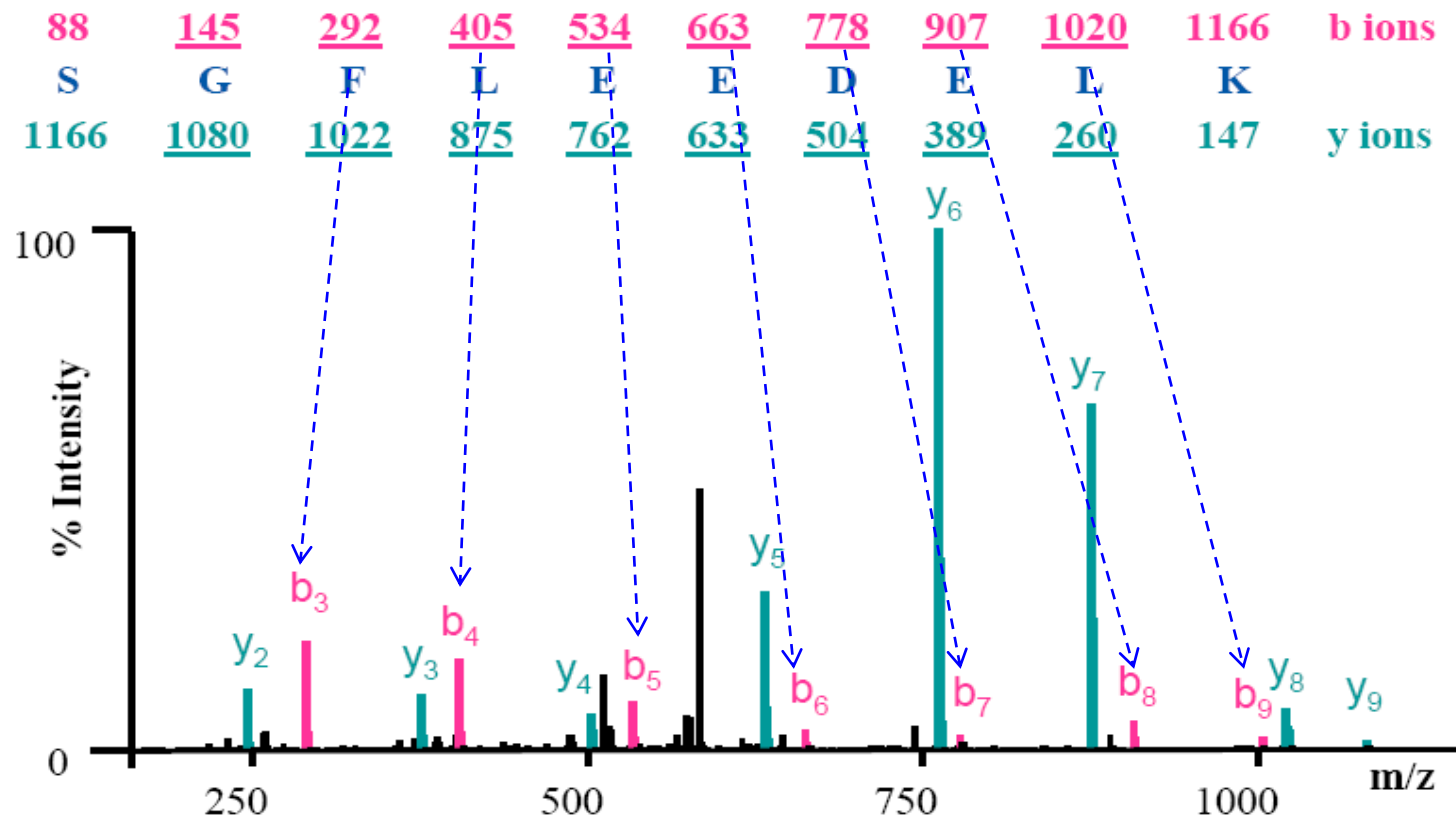
7.5.4. Protein Identification

Peptide Fragmentation



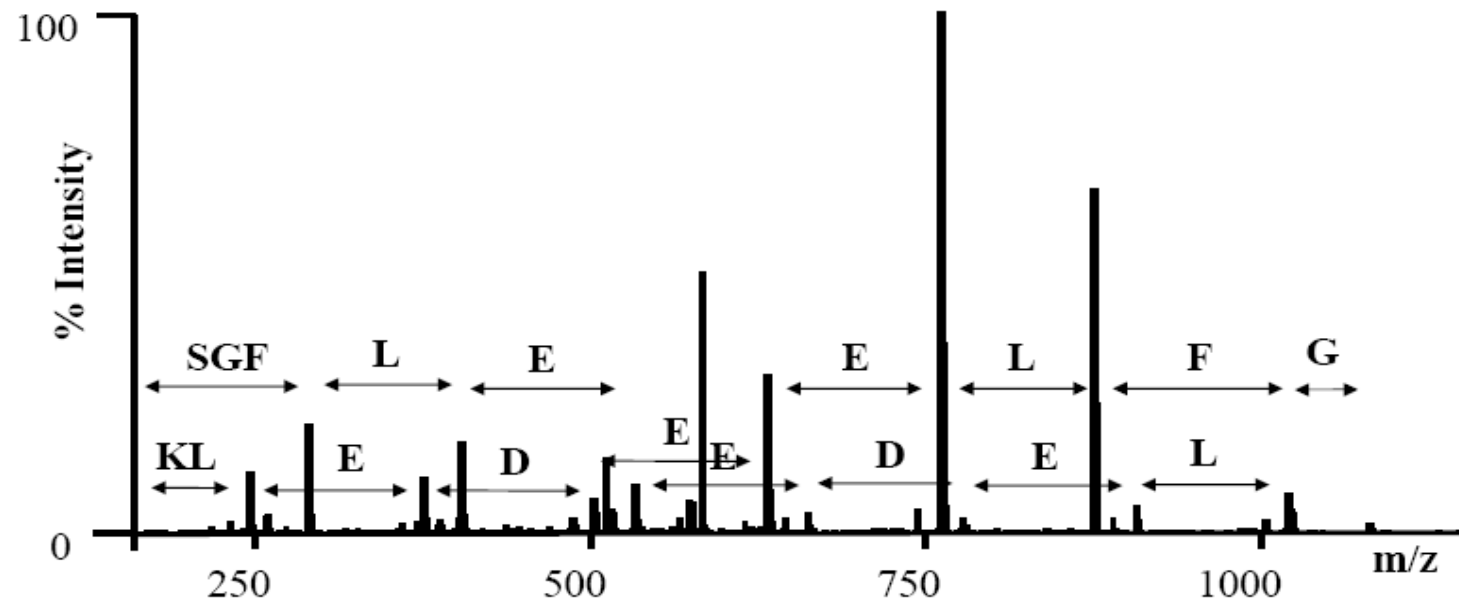
7.5.4. Protein Identification

Peptide Fragmentation



7.5.4. Protein Identification

De Novo Interpretation



8. Inductively Coupled Plasma Spectroscopy

8. Inductively Coupled Plasma Spectroscopy

Common techniques for the analysis of metals:

- Atomic absorption spectroscopy
 - Atomization by flame
 - Atomization by graphite furnace
- Atomic emission spectroscopy
 - Atomization and excitation by flame

These techniques have already been discussed in the „Analytical Chemistry“ lecture

- Atomic emission spectroscopy
 - Atomization and excitation (ionization) by an inductively coupled plasma

This technique will be discussed in this lecture

8. Inductively Coupled Plasma Spectroscopy

	Element	2000K	3000K	4000K
$\frac{N_J}{N_0}$	Cs	$4 \cdot 10^{-4}$	$7 \cdot 10^{-3}$	$3 \cdot 10^{-3}$
	Ca	$1 \cdot 10^{-7}$	$4 \cdot 10^{-5}$	$6 \cdot 10^{-4}$
	Zn	$7 \cdot 10^{-15}$	$6 \cdot 10^{-10}$	$1 \cdot 10^{-7}$

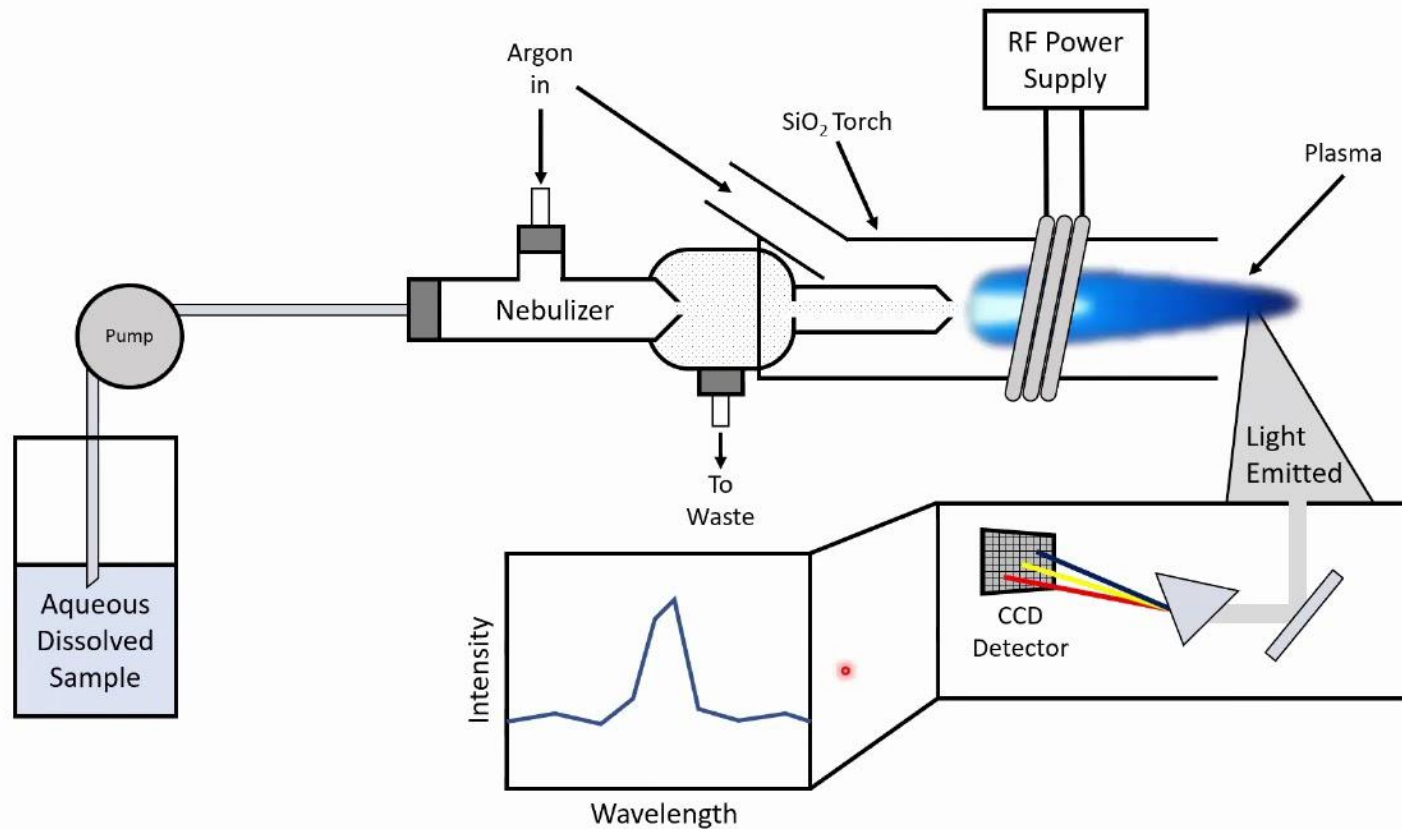


- Atomic Absorption Spectroscopy (AAS) is not well suited for multi-element analysis
- Atomic Emission Spectroscopy (AES) would be the better choice – but typical instrumentation using flame atomization only allows the excitation and measurement of a few elements, due to the low temperature (see table above)
- Instrumentation providing higher energies via higher temperatures would be the solution for this problem
- Inductively Coupled Plasma (ICP) Spectroscopy allows to excite almost all elements
- In addition elements can be ionized in ICP (something not desired in AAS)

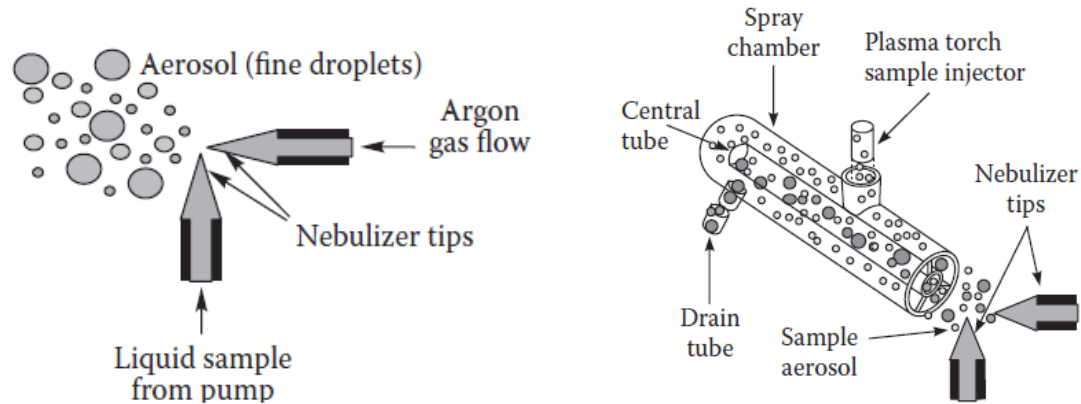
8. Inductively Coupled Plasma Spectroscopy

1	1																	18
1	H	2											13	14	15	16	17	He
2	3	4											5	6	7	8	9	10
	Li	Be											B	C	N	O	F	Ne
3	11	12											13	14	15	16	17	18
	Na	Mg											Al	Si	P	S	Cl	Ar
4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	55	56	57-71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
	Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	87	88	89-103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
	Fr	Ra	**	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Uut	Fl	Uup	Lv	Uus	Uuo
6	*	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71		
		La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu		
7	**	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103		
		Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr		
									</									

8.1. Inductively Coupled Plasma Spectroscopy

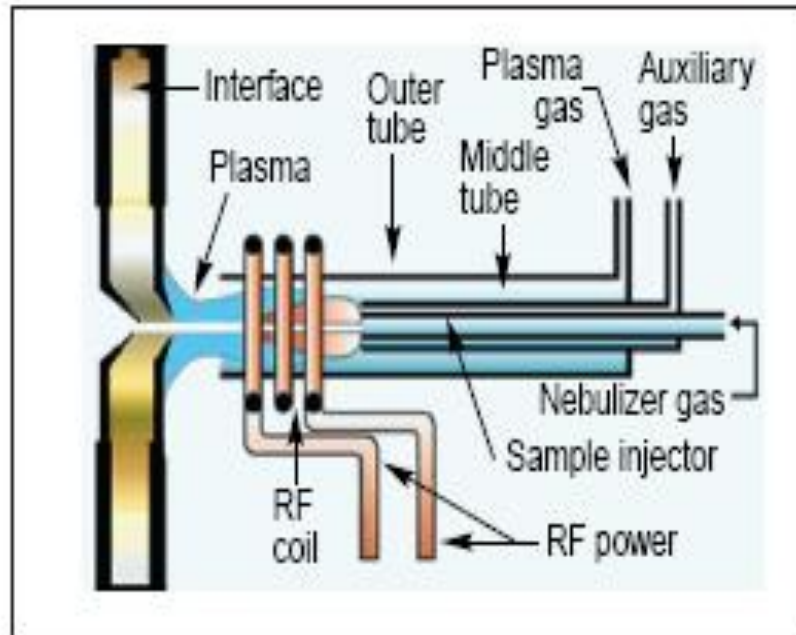


8.1. Inductively Coupled Plasma Spectroscopy



8.1. Inductively Coupled Plasma Spectroscopy **JKU**

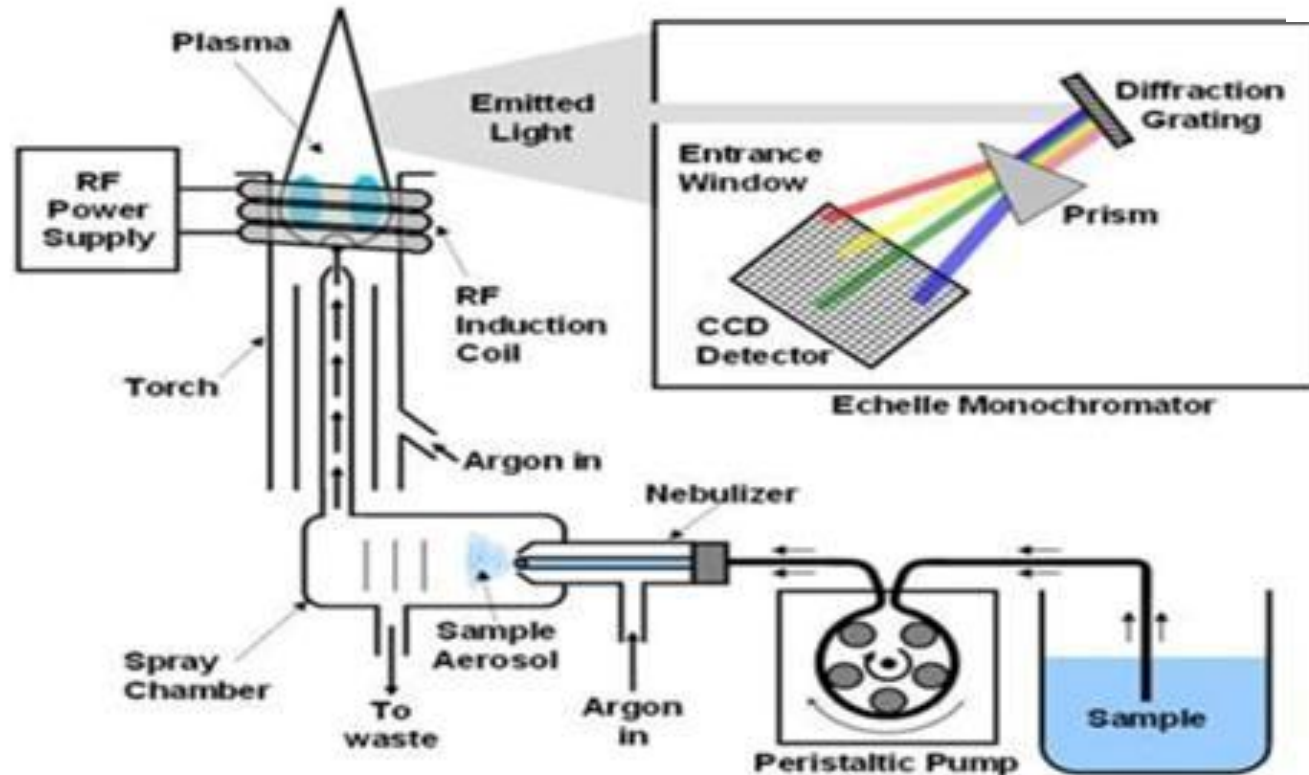
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- Ar gas moves between two quartz tubes
- via an RF-field a strong electromagnetic field is induced
- a HV spark generates free electrons
- free electrons are accelerated in the RF field and lead to ionized Ar gas via collisions
- at the end of the quartz tube an inductively coupled plasma is generated into which the sample is injected
- from the sample excited atoms and ions are formed – possibility to use ICP with an optical (ICP-AES) - or an MS detector

8.1. Inductively Coupled Plasma Spectroscopy **JKU**

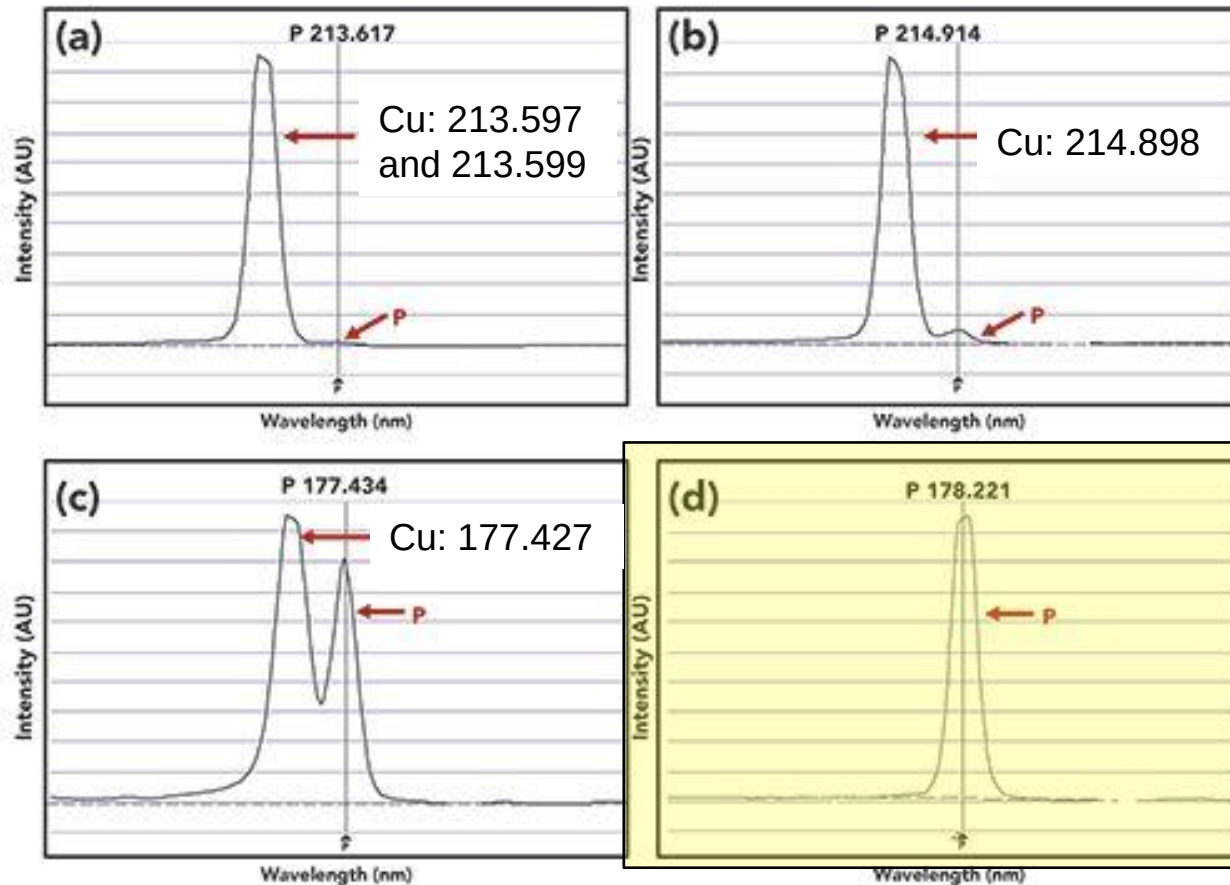
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ICP- with optical detection (ICP-AES):

- multi-element analysis
- excellent grating is crucial, as emission lines from different elements must be separated
- need for finding unambiguous line (without interference) for each element analyzed
- limited sensitivity (less than graphite furnace AAS)

8.1. Inductively Coupled Plasma Spectroscopy



ICP- AES:

- Analysis of phosphorous (10 ppm) in the presence of 200 ppm Cu
- Choice of most suitable wavelength to avoid spectral interference

8.2. Inductively Coupled Plasma Spectroscopy (with MS detection)

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ICP can also be seen as an ion-source for MS!

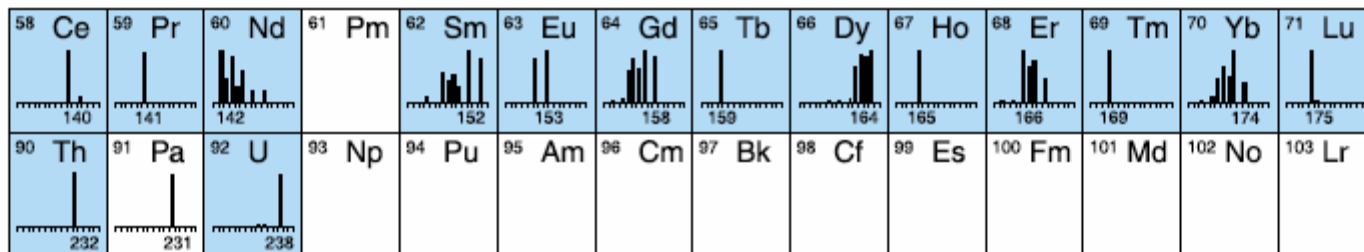
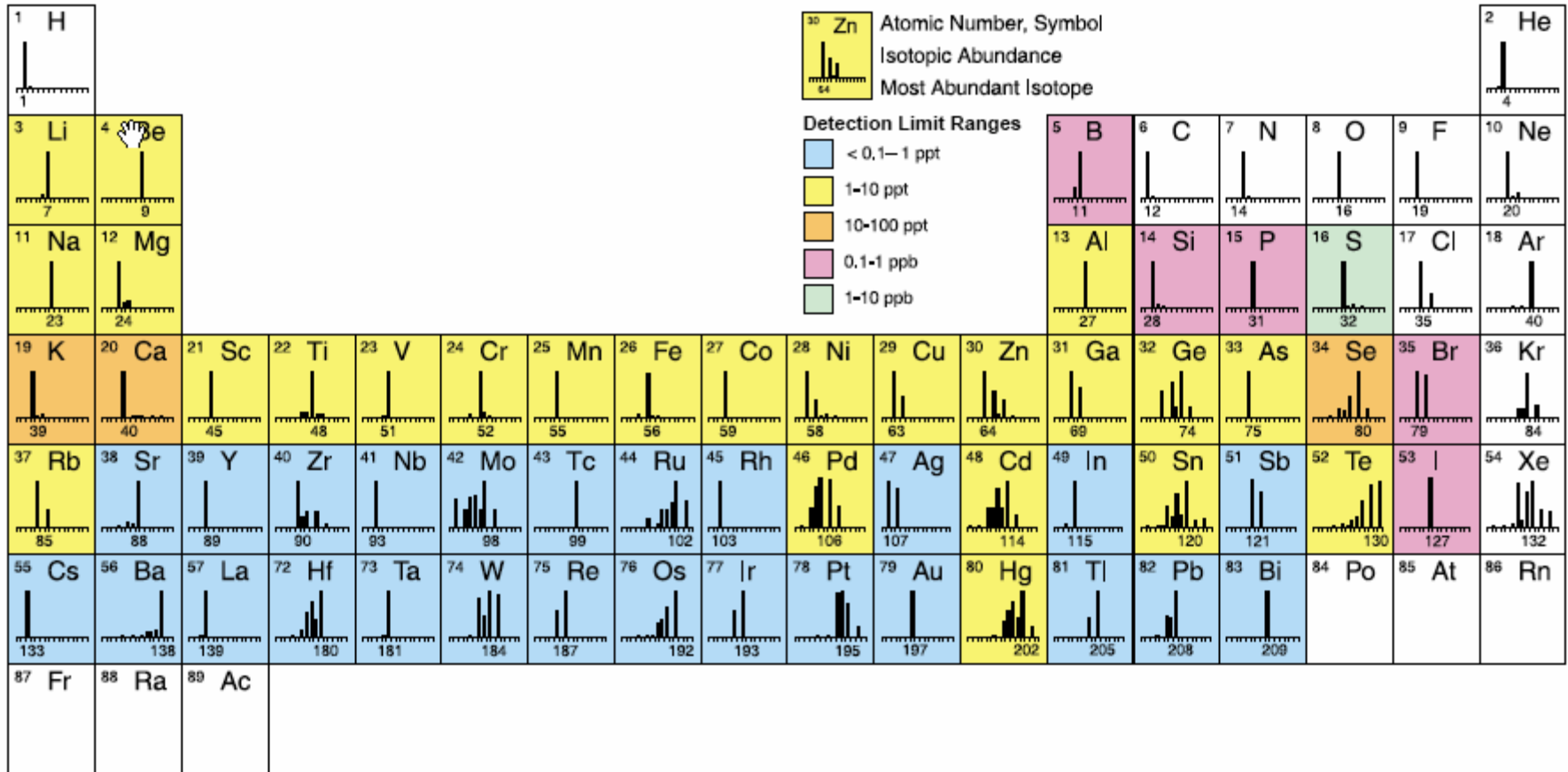
Ionization method	Typical Analytes	Sample Introduction	Mass Range	Method Highlights
Inductively Coupled Plasma (ICP)	Elements	Direct or liquid chromatography	Low (elements only)	Very high sensitivity (see next slide)

8.2. Inductively Coupled Plasma Spectroscopy

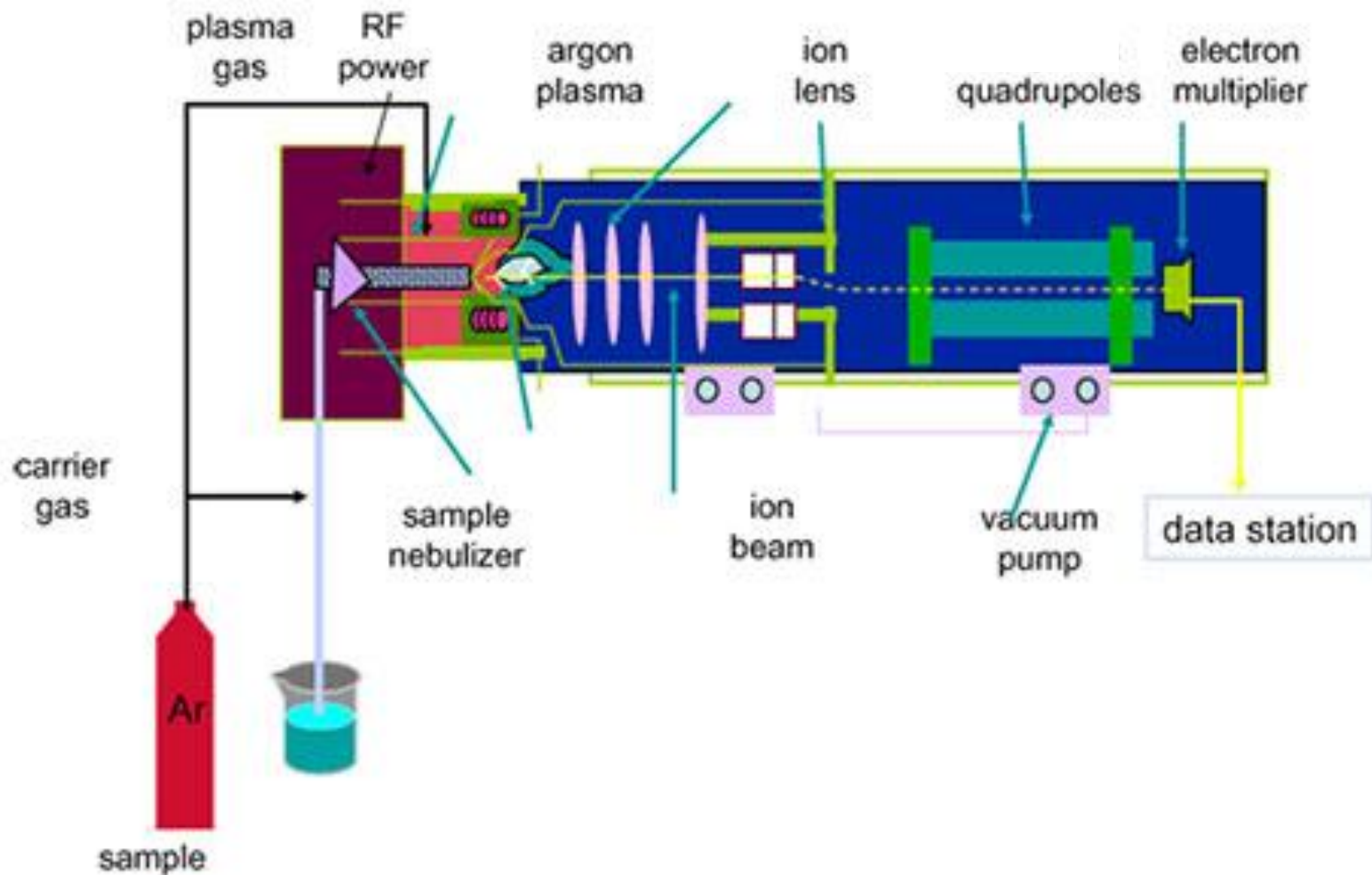
(with MS detection)



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8.2. Inductively Coupled Plasma Spectroscopy (with MS detection)



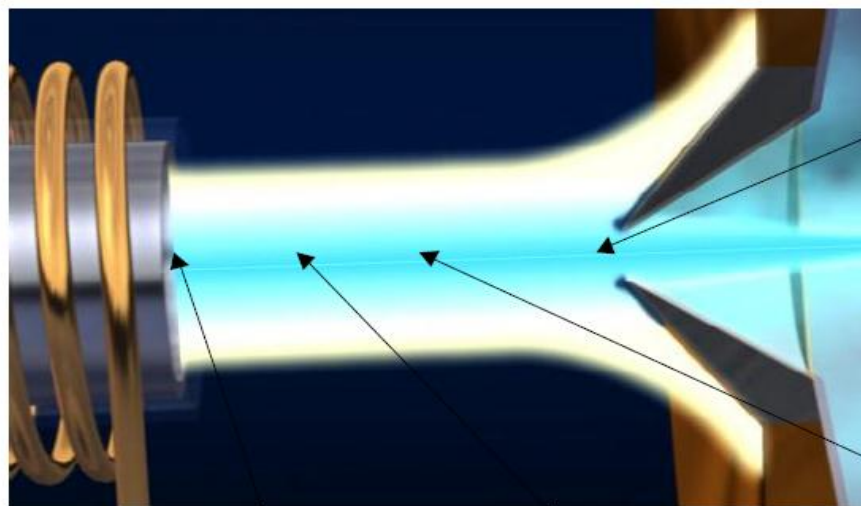
8.2. Inductively Coupled Plasma Spectroscopy (with MS detection)

Simplified Model of Plasma

Hottest part of
plasma ~ 8000K

Sample channel
is at ~6000K

Residence time is a
few milliseconds



By sample cone, analytes
present as M^+ ions

Highest M^+ population
should correspond to
lowest polyatomic
population

Aerosol is Dried

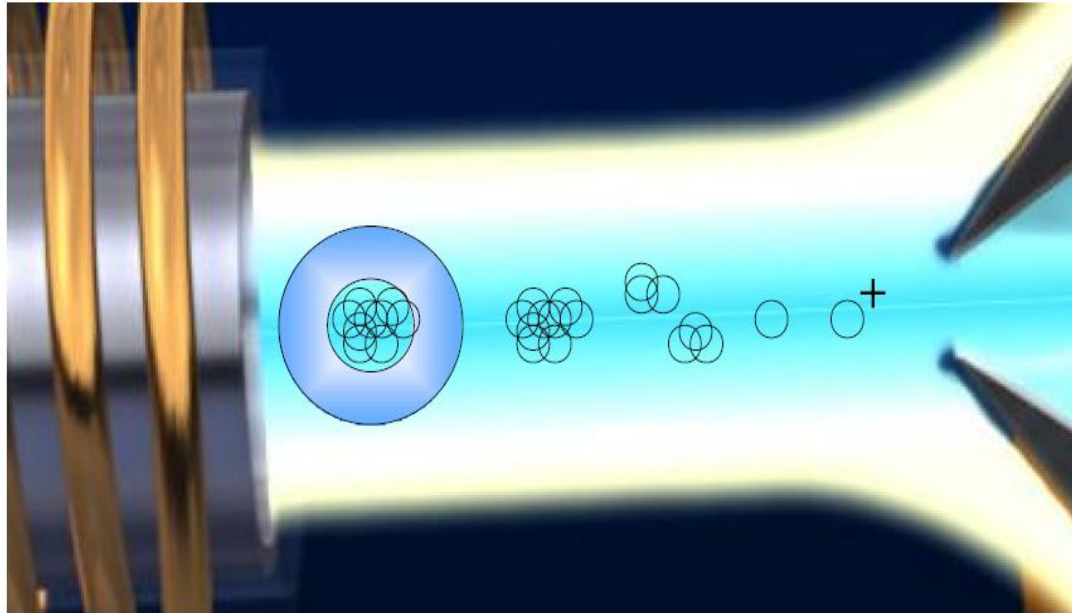
Particles are decomposed
and dissociated

Atoms are formed
and then ionized

8.2. Inductively Coupled Plasma Spectroscopy

(with MS detection)

Sample Decomposition

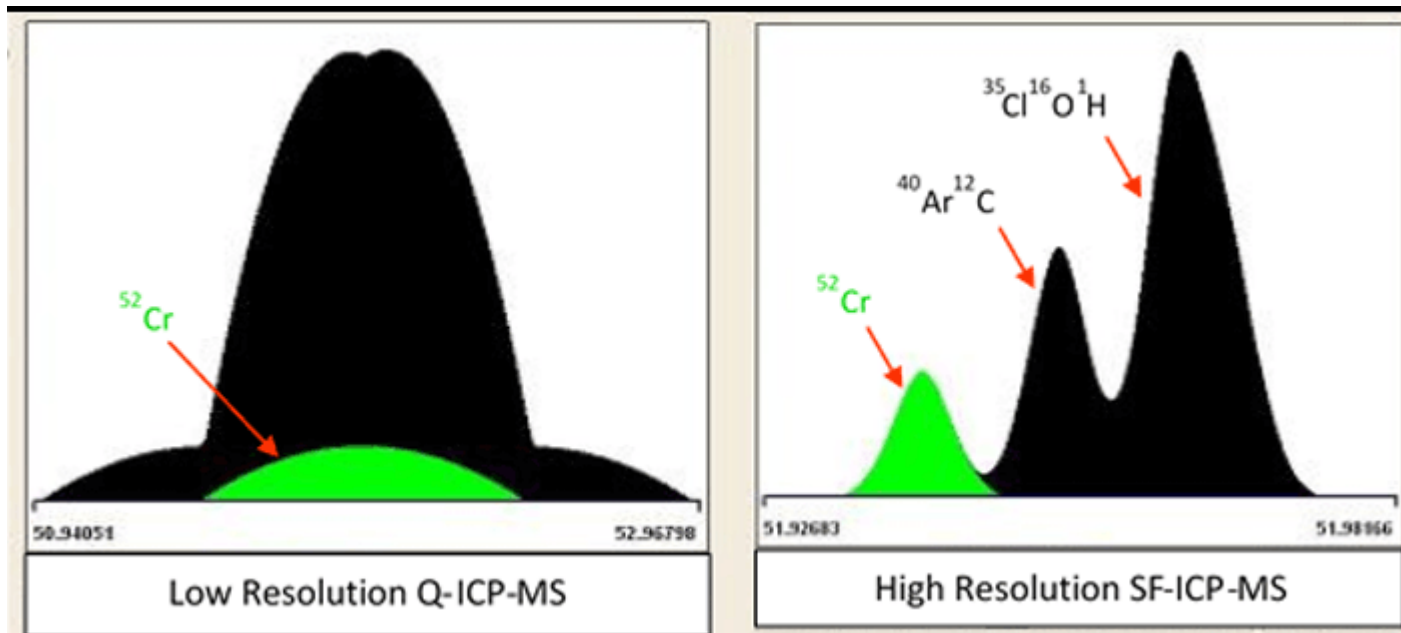


8.2. Interferences in ICP-MS

Typical Interferences:

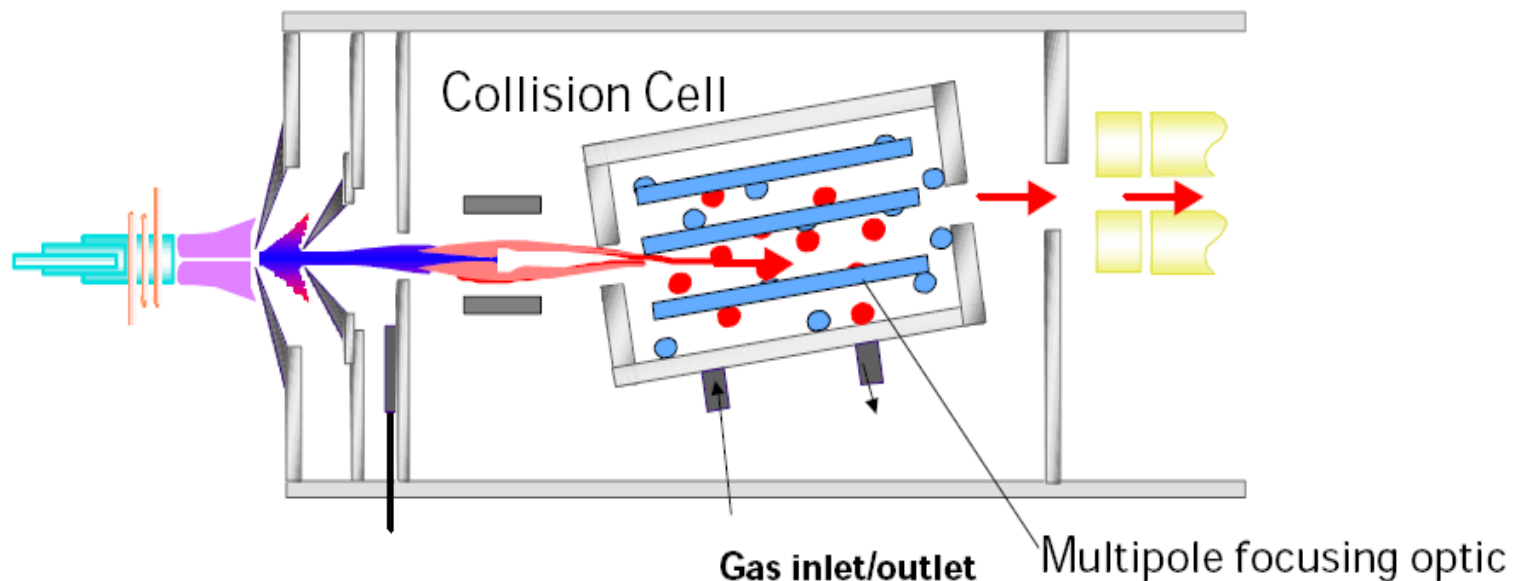
$^{40}\text{Ar}^{16}\text{O} - ^{56}\text{Fe}$
 $^{38}\text{Ar}^1\text{H} - ^{39}\text{K}$
 $^{40}\text{Ar} - ^{40}\text{Ca}$
 $^{40}\text{Ar}^{35}\text{Cl} - ^{75}\text{As}$
 $^{40}\text{Ar}^{12}\text{C} - ^{52}\text{Cr}$
 $^{35}\text{Cl}^{16}\text{O}^1\text{H} - ^{52}\text{Cr}$
 $^{35}\text{Cl}^{16}\text{O} - ^{51}\text{V}$

Resolved only in high-resolution ICP-MS



8.2. Collision Cell Technology in ICP-MS

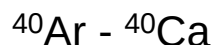
Principle of Collision Technology



Similar technology as used for fragmentation in QqQ-MS can also be employed for removing interferences in ICP-MS

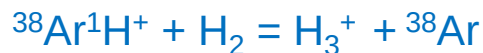
8.2. Collision Cell Technology in ICP-MS

Typical Interferences:



Solution:

eg.:



or.:

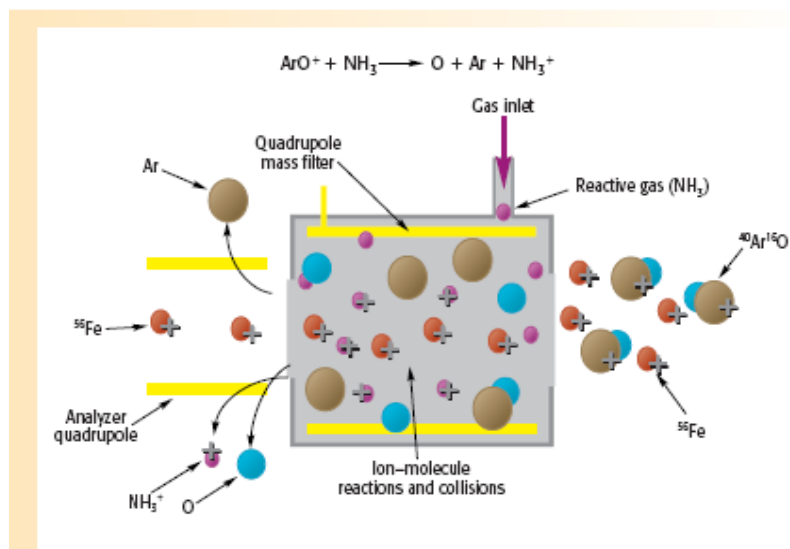


Figure 4. Elimination of the ArO interference with a dynamic reaction cell.

ICP-MS

8.2. Collision Cell Technology in ICP-MS

ICP-MS

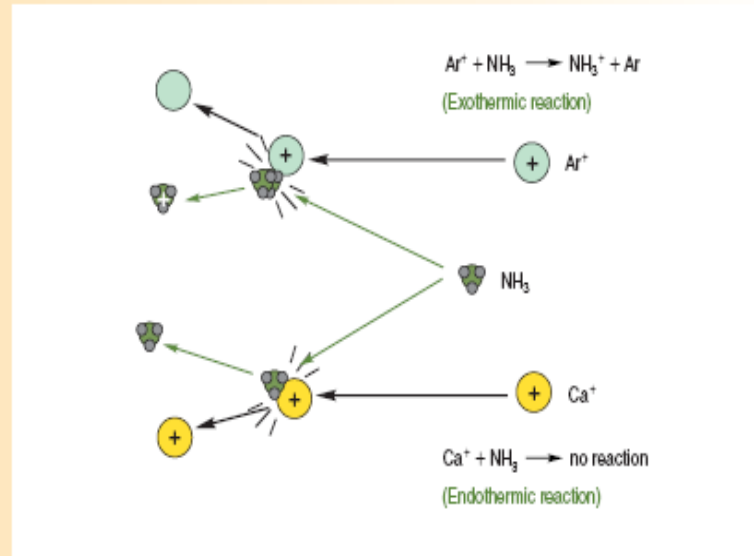


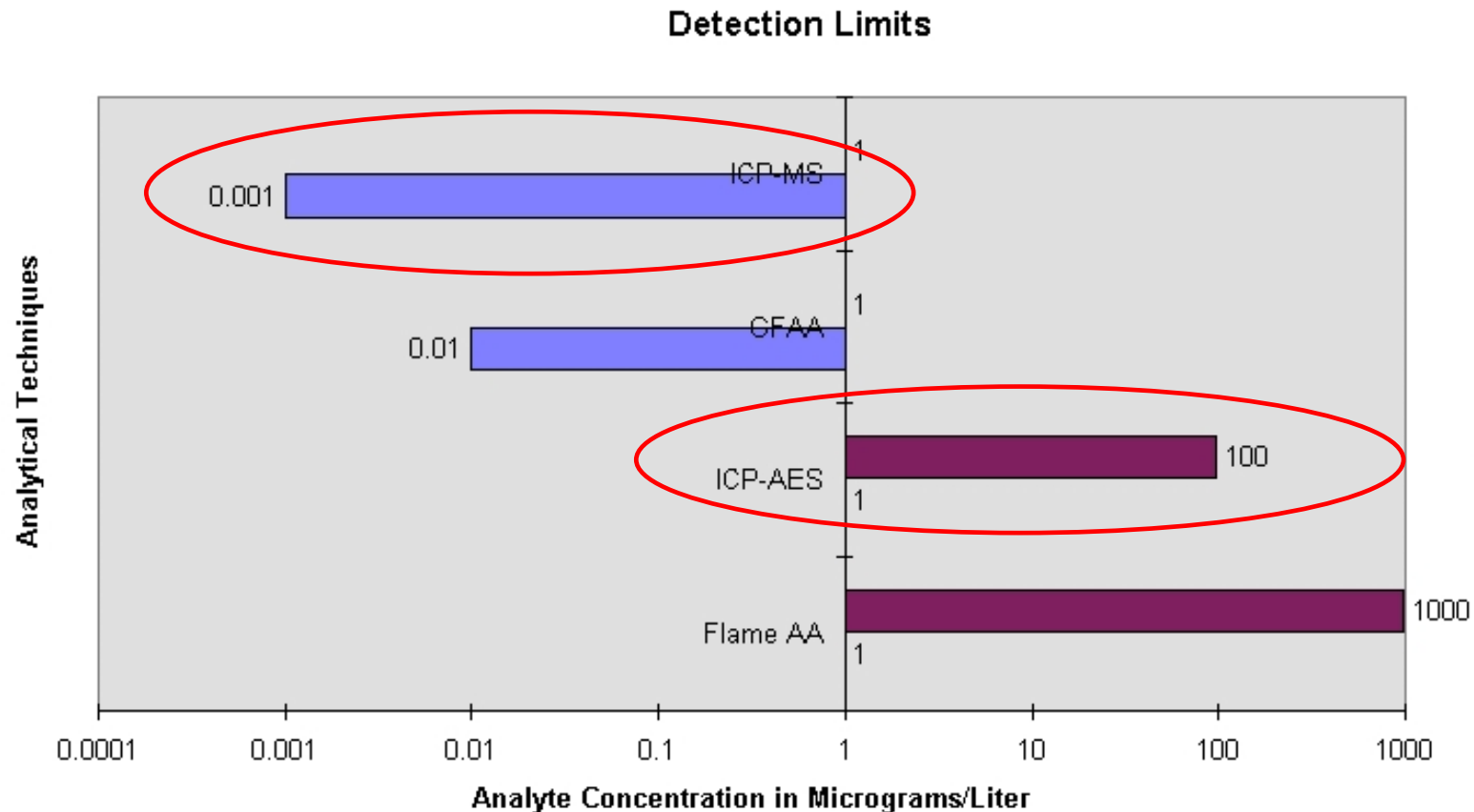
Figure 5. The reaction between NH_3 and Ar^+ is exothermic and fast, while there is no reaction between NH_3 and Ca^+ in the dynamic reaction cell.

Ionization Potentials:

Ca^+ :	6,1 eV
NH_3^+ :	10,2 eV
Ar^+ :	15,8 eV

8.3 Inductively Coupled Plasma Spectroscopy

Comparison of Methods



8.3 Inductively Coupled Plasma Spectroscopy

Comparison of techniques

AAS (flame):

- 😊 low cost
- 😞 low sensitivity
- 😞 large sample volume needed

AAS (graphite furnace):

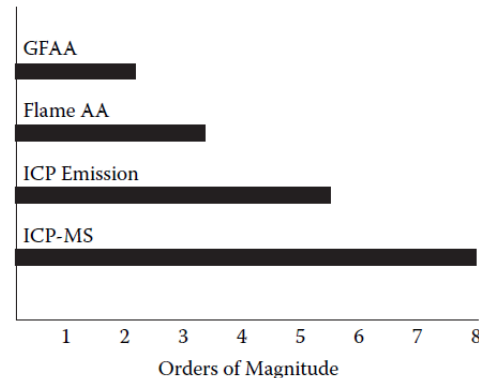
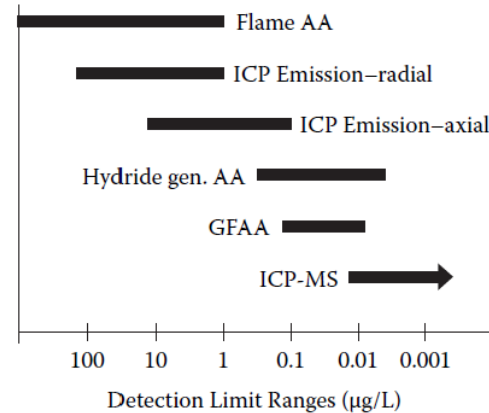
- 😊 low cost
- 😊 good sensitivity
- 😊 low sample volume needed
- 😞 only 1 element at a time

ICP (OES):

- 😊 low cost
- 😊 multi-element-analysis
- 😞 large sample volume needed

ICP (MS):

- 😊 multi-element-analysis
- 😊 few interferences
- 😊 (very) good sensitivity
- 😞 high cost



Comparison of Sample Throughput of the Four AS Techniques

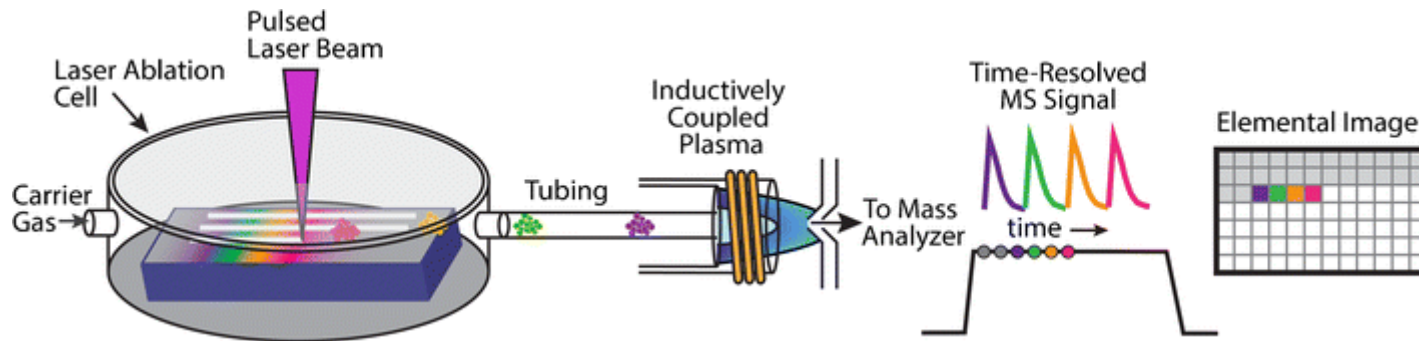
Technique	Elements at a Time	Duplicate Analysis (min)	Samples per Hour (1 element)	Samples per Hour (5 elements)	Samples per Hour (20 elements)
FAAS	1	0.3	180	36	9
GFAAS (single)	1	5	12	2-3	1
GFAAS (multi)	2-6	5	12	12	3
ICP-OES	Up to 70	3	20	20	20
ICP-MS	Up to 70	3	20	20	20

8.3 Inductively Coupled Plasma Spectroscopy

	ICP-MS	ICP-OES	Flame AAS	GF-AAS
Detection limits	Excellent for most elements	Very good for most elements	Very good for some elements	Excellent for some elements
Sample throughput	all elements 2-6 min	5-30 elements /min	15 seconds/ element	4 min/element
Linear dynamic range	10^5 (10^8 with range extension)	10^4 to 10^8	10^3	10^2
Precision:				
Short term (in-run)	1-3%	0.3-1%	0.1-1%	1-5%
Long term (4hrs)	< 5% *	< 3% *		
Isotopes	yes	no	no	no
Dissolved solids				
max. concentration	0.1-0.4	1-30	0.5-3	> 30
No. of elements	> 75	> 75	> 68	> 50
Sample usage	low	low	very high	very low
Semi-quant analysis	yes	yes	no	no
Isotope analysis	yes	no	no	no
Routine operation	easy	easy	easy	easy
Method development	skill required	skill required	easy	skill required
Unattended operation	yes	yes	no	yes
Combustible gases	no	no	yes	no
Operating cost	high	high	low	medium
Capital cost	very high	high	low	medium/high

8.4. Inductively Coupled Plasma Spectroscopy

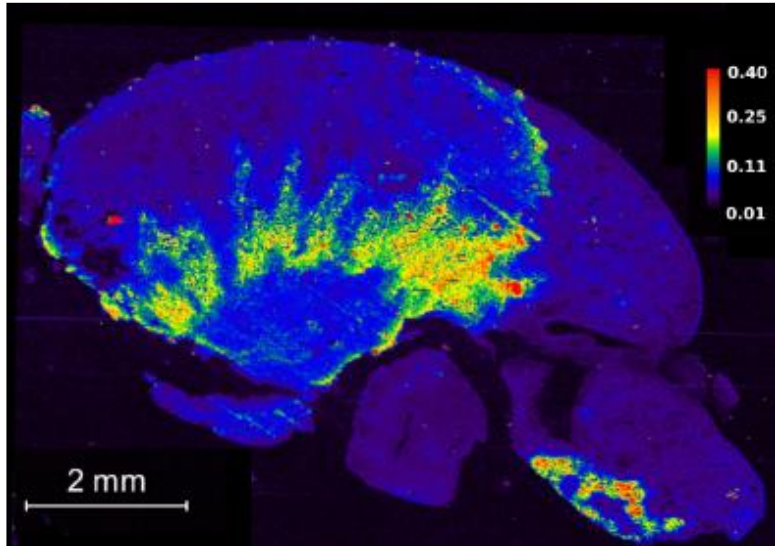
Laser Ablation (Imaging)



- A pulsed laser beam evaporates material from the surface
- This material is transported into the ICP-torch by a carrier gas
- This technique allows to scan (image) surfaces
- By firing at the same spot multiple times depth profiling is possible

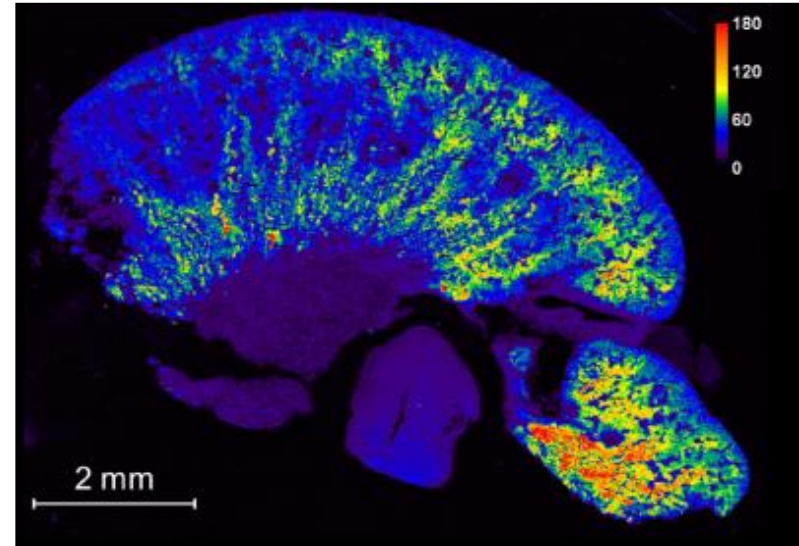
8.4. Inductively Coupled Plasma Spectroscopy

Laser Ablation Imaging



Cu intensity

and



Pt intensity

map of a rat kidney

8.4. Inductively Coupled Plasma Spectroscopy

Laser Ablation – Depth profiling

